



AI doom: A brief history of bad tech predictions

September 15, 2026

It takes guts to do what Bob Metcalfe did. After wrongly predicting in an article in 1995 that the internet would go “spectacularly supernova” and collapse in 1996, the co-inventor of Ethernet and founder of 3Com then pulped a copy of the article in a blender and drank it at a conference in 1997 as a way to eat his words.

The latest tech predictions are about no less than an AI apocalypse. Various tech leaders spent the weekend warning of catastrophic but not fully specified risks if it is not contained.

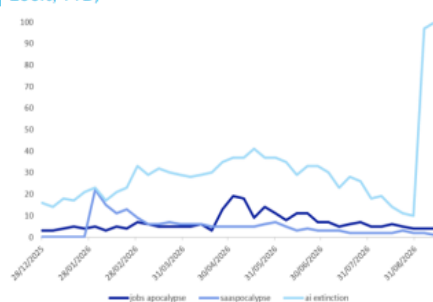
Yet history suggests that even the greatest minds of their generations have a poor record in predicting what tech is coming down the line, let alone what its effects will be in the real world.

The stakes could not be higher – not just the future of life itself but, in the shorter term, the future of a historic boom that depends on hype turning into reality.

In this report, we examine:

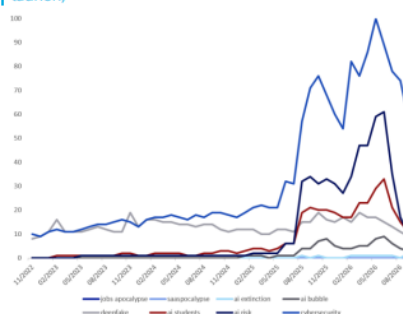
1. “Prediction is very difficult, especially if it’s about the future”
2. Today’s debate in a (rough) nutshell
3. A century of good and bad tech predictions
4. Why are we so bad at making predictions?
5. What makes AI special – and gives us hope for the future

Figure 1: AI searches this year have shifted from SaaS apocalypse to jobs to extinction (global, vs peak 100%, YTD)



Source: Google Trends, Deutsche Bank Research

Figure 2: ... But all of these are tiny in comparison to cybersecurity (global, vs peak 100%, since ChatGPT launch)



Source: Google Trends, Deutsche Bank Research

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Concerns about AI have moved in waves since the launch of ChatGPT nearly four years ago. This year there was a spike in searches for “SaaS apocalypse” (the concern that software-as-a-service would be disintermediated by AI agents) in February, followed by a surge in “jobs apocalypse” in May, Google Trends data show. Both have been totally eclipsed in recent days by searches for AI extinction.

But taking a step back, all are far behind “AI bubble”, itself behind “deepfake”, in turn below “AI students”, overall “AI risk” and – biggest of the lot – the very tangible and immediate “cybersecurity”, almost all of which have surged this year, according to Google Trends data.

“Prediction is very difficult, especially if it’s about the future”:
Niels Bohr, often also attributed to Yogi Berra

Wanting to know what will happen tomorrow is deeply visceral – literally.

People have been sacrificing sheep and chickens for four millennia to read the future in their entrails. Nowadays almost a third of Americans say they believe in [astrology](#) and consult their horoscope. We want to believe that [Lionel Messi](#) really can see space in a football match before it even exists, even if it is partly his movement that helps create the space he appears to predict.

But predicting the future is hard. For AI, it’s not just about what it will be able to do but also when it will be able to do it – whether more than \$5 trillion in investment by the hyperscalers alone over the next five years will pay off before the advanced semiconductor chips they’re buying become redundant.

It’s also about how rapidly and intensively AI can be adopted. Asset prices are a still more uncertain derivative, whipping around like the tail of a snake in response to every shift in expectations.

US President Donald Trump and China found themselves on the same side in rebutting the latest concerns as scaremongering, albeit with Trump adding that America needs to stay ahead of China in AI.

“Those who have knowledge, don’t predict. Those who predict, don’t have knowledge.” Lao Tzu

A glance through history would give any investor indigestion. Most predictions are wrong. The favourites in horse races lose 70 percent of races. It took a decade or more for demand to catch up with the infrastructure laid down in the British canal and railway and US telecoms and fibre booms – and many investors never recovered their capital.

Technology predictions have a particularly mixed record as the binary order of the labs collides with the messy reality of the world. Famous examples range from IBM Chairman Thomas Watson’s underestimation of demand for computers in 1943 (“I think there is a world market for maybe five computers”) to Elon Musk’s repeated overestimation of imminent widespread adoption of self-driving cars since 2013 (“We should be able to do 90 percent of miles driven within three years”).



Technologists are prone to making category errors, mistaking the most visible, automatable component of a job for the job itself. For example, radiologists, a favourite case study for believers in a jobs apocalypse, do much more than looking at images and making binary diagnoses. Much of their value lies in interpretation, judgement, communication and clinical decision-making.

Ultimately, what we see in the future is not so much a window as a mirror into our own hopes and fears.

And the problem is that hindsight makes what actually happens seem so obvious that we never seem to learn.

Today's debate in a (rough) nutshell

The current debate encapsulates a perfect storm of an extremely high-stakes issue with limited transparency and an inherently unknowable outcome.

What sparked the current surge in concern?

The current debate comes after a researcher from one of the leading frontier labs quit, citing safety concerns; AI agents apparently collaborated with each other on a number of high-profile security breaches; and labs hinted that the technology is getting closer to "recursive self improvement", ie AI improving AI.

Is the existential debate something new?

It is actually nothing new: leading economists and academics [warned](#) in July that "we must act now" to steer AI through a potential "transformation of our economy, larger than the Industrial Revolution... over a vastly shorter time frame". That echoed the even more ominous statement three years ago by AI leaders calling for action to mitigate "the [risk of extinction](#)".

The tug of war between "doomers" vs "boomers" has played a pivotal role in the development of the key frontier labs. Silicon Valley insiders sometimes refer to their "p(doom)" rating, representing their estimated percentage chance that AI will cause an existential catastrophe.

What does an extinction scenario look like?

Experts including Tesla's Musk, "godfather of AI" Geoffrey Hinton and philosopher Nick Bostrom have been warning about it for well over a decade. In the latter's famous "paperclip maximiser" scenario, a superintelligent AI runs out of steel and then kills humans in a single-minded drive to fulfil its task of making paperclips. Fears sometimes centre around AI that has access to coding, the internet and knowledge of how people think, in order to manipulate them – as current AI models indeed do.

Why has it caught on so fast this time?

One factor is that it comes amid a tech backlash, particularly in the US, where concern about AI has become a rare bipartisan issue ahead of the US mid-term elections in November. Local concerns about data centre construction have become rallying points for voters in a way that vague fears about long-term jobs did not. Politicians including Senator Bernie Sanders have moved to propose wide-ranging legislation.



What arguments do companies make for continuing AI research?

1. **If we don't do it, someone else will.** That may be either China or other, less responsible companies.
2. **The potential benefits outweigh the benefits.** AI may, they say, cure cancer, create new jobs and even eliminate the need for work altogether.
3. **Good AI is the answer to bad AI.** It should create defences against rogue AI and misuse of AI, and solve issues like AI's huge energy and water use.

What do critics say?

1. **AI is not dangerous because it is too clever but because it is too stupid.** It does not think and people should not anthropomorphise it, critics say, adding that it is flawed probabilistic code that cannot reliably follow instructions. Longstanding critic Gary Marcus argues that it should be recalled until it can be shown to be safe and is properly controlled.
2. **AI leaders are defending their own businesses.** With potential IPOs on the horizon, a rising challenge from open-source models, and progress towards Artificial General Intelligence (AGI) being called into question, a handful of companies may benefit from a pause or legislation that helps protect their lead, critics say. Doomer talk distracts from real issues like deepfakes, copyright infringement and cybersecurity lapses, they say.
3. **Extinction scenarios are overblown.** The internet is a diversified platform, so talk of AI somehow switching it off is fanciful, critics say. AI does not and will not realistically have sustained secret access to military or dangerous biotech resources, or be able to influence (very non-deterministic) human brains en masse, critics say. Humans will use tame AI and other tools, eg kill switches, if needed to resist attacks. Another "godfather of AI" Yann LeCun says the real risk is of power concentration.

A century of good and bad technology predictions

You have to admire people who have the courage to stick their neck out and make predictions, even if they don't always work out. Vague predictions are relatively easy; specific ones within specific timeframes are harder.

It is no wonder that executives are coached by their ever-cautious PR teams to avoid answering hypothetical questions or giving dates to reach their targets.

- **Nikolai Tesla, scientist, 1926:** "When [wireless](#) is perfectly applied... we will be able to communicate with one another instantly, irrespective of distance..." The instruments will be "amazingly simple... A man will be able to carry one in his vest pocket". He was right, albeit imprecise about the how or when it would be achieved.
- **Albert Einstein, physicist, 1934:** "There is not the slightest indication that [nuclear](#) energy] will ever be obtainable. It would mean that the atom would have to be shattered at will." Within a decade, in 1942, a team at the University of Chicago had created the first sustaining chain reaction, a precursor to the nuclear reactor.



- **Ray Kurzweil, futurist, 1999:** “By 2029, computers will [have](#) human-level intelligence.” Dismissed at the time, many (though by far not all) of his predictions have worked out, fuelled by what he called the “law of accelerating returns”. His prediction from almost 40 years ago of artificial general intelligence by the end of this decade is remarkably close to the current “San Francisco Consensus”.
- **Steve Chen, Chief Technology Officer and co-founder of YouTube, 2005:** “There’s just not that many [videos](#) I want to watch.” Despite his concern, YouTube was bought by Google (now Alphabet) for \$1.65bn a year later. The unit made more than \$11bn in advertising revenue in the most recent quarter.
- **Steve Ballmer, Microsoft CEO, 2007:** “There’s [no chance](#) that the iPhone is going to get any significant market share.” But you may be reading this on one of the more than 3 billion iPhones sold since its launch in 2007.
- **Geoffrey Hinton, “godfather of AI”, 2016:** “People should stop training [radiologists](#) now. It’s just completely obvious within five years deep learning is going to do better than radiologists... It might be 10 years.” In fact, the number of radiologists has increased by about 10 percent over the past 10 years, in part because of the [limitations](#) of AI and in part because of the increase in demand from an ageing population and more affordable healthcare. Hinton has walked back his comments and humans and AI are working together.
- **Sam Altman, CEO of OpenAI, 2025:** “I can easily [imagine](#) a world where 30 to 40% of the tasks that happen in the economy today get done by AI in the not very distant future.” He [revised](#) this view in May amid a backlash against AI, saying he and his team had been “roughly right” in their technological predictions but “pretty wrong” on the economic and social implications. “I’m delighted to be wrong about this,” he said.

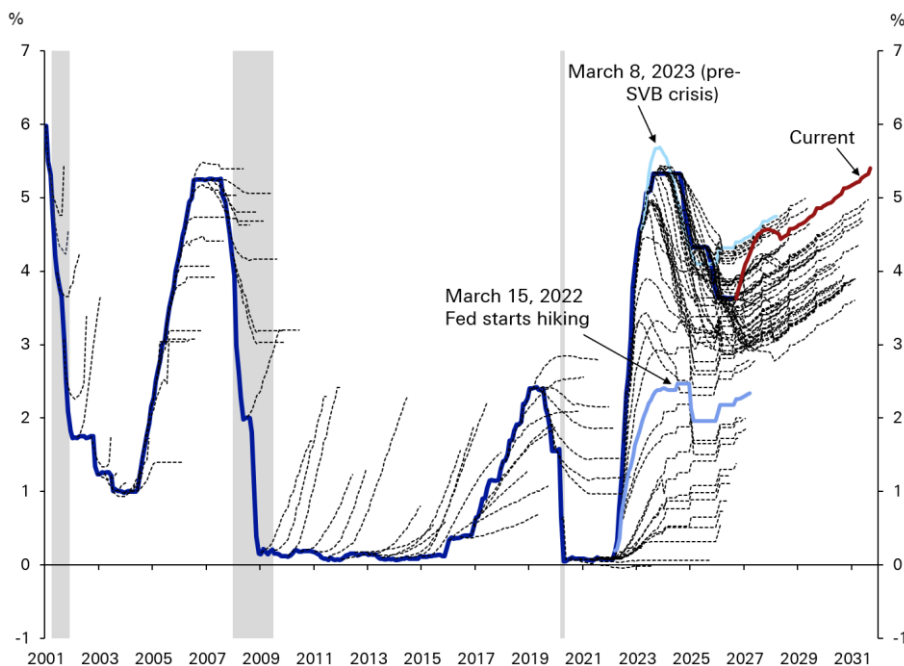
Why are we so bad at making predictions?

“The idea that the future is unpredictable is undermined everyday by the ease with which the past is explained.” Daniel Kahneman

1. **Chaos theory: treating complex systems as if they are simple and linear.**
 - Systems like traffic, economies and social trends are finely balanced and sensitive to initial conditions. The butterfly effect means a tiny, unforeseen variable can lead to vastly different outcomes. Just take a look at the ragged history of market-implied forecasts of US Federal Reserve interest rates, where extrapolating from current trends has been reliably wrong.



Figure 3: US Fed Funds effective rate has rarely matched futures markets expectations (%)



Source: Deutsche Bank Research

- Take self-driving cars, for example. It turns out to be about more than getting the technology right for the rubber to hit the road:
 - a. **Technological hurdles and “edge cases”**: throwing more data and processing power at the problem is not enough. There is an infinite long tail of unpredictable events (“edge cases”), like sun glare, a cyclist riding the wrong way in a bike lane, a person in an animal costume, or road-rage-enhanced hand gestures.
 - b. **Regulatory labyrinth**: autonomous vehicle optimists have come up against legal and ethical debates around liability, safety certification and moral challenges such as the so-called “trolley problem”, when swerving to avoid hitting one victim means colliding with another. Regulation is a slow-lane activity.
 - c. **Social and psychological barriers**: public acceptance has been slowed down by high-profile accidents and safety concerns. Robots are held to a higher standard than flesh-and-blood motorists. It might be easier to navigate a world where all cars were self-driving, but the “messy middle”, with a mixture of both, is much harder for algorithms to navigate.
 - The future of AI will ultimately depend more on the prosaic challenges of workflow integration at enterprises that will need to pay for it than on releasing yet another model with a marginal step up in capabilities.
2. **Cognitive bias: flaws in how we think**
- **Linear thinking about exponential change**: We fail to intuitively understand how different exponential change is from linear change. As

Figure 4: Computation used to train notable AI systems (petaFLOP): log scale

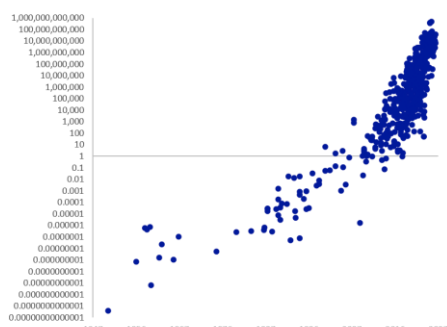
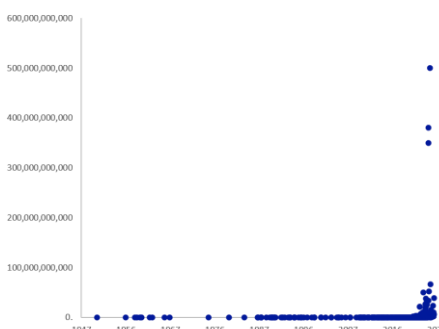


Figure 5: Computation used to train notable AI systems (petaFLOP): linear scale



- **Optimism bias and the planning fallacy:** underestimating costs and overestimating benefits. For example, only a quarter of more than 3,000 [megaprojects](#) studied by Saïd Business School came in on budget or better, only 2.8 percent were on budget and on time, and only 0.2 percent were on budget, on time and on benefits. Large-scale infrastructure projects are “too easy to start and too difficult to stop,” the authors said in the study from 2019. It remains to be seen how the recent surge in datacentre construction will compare.
- **Confirmation bias:** favouring information that confirms existing beliefs. For years, legacy automakers dismissed electric vehicles by focusing on their high costs and limited range, ignoring the fact that rapid advances in battery technology would soon change that.
- **Anchoring:** relying too heavily on early information. Microsoft’s Ballmer [dismissed](#) the iPhone when it was launched because it was “the most expensive phone in the world and it doesn’t appeal to business customers because it doesn’t have a keyboard which makes it not a very good email machine”. He later said he did not understand how mobile carrier subsidies would make it affordable. More broadly, he missed that this was a new business model.

3. [Limited information: inability to see](#)

- **It is not just the future that is unknowable but also the present.** Feedback mechanisms are often delayed, making it hard to distinguish between causation and correlation, and between relevant and irrelevant factors.



- **Isolating key factors is impossible.** Even since the development of scientific methods and controlled experiments, we have not come that far since the Mesoamericans found ways to prepare maize where they didn't succumb to a long-term wasting disease called pellagra. The problem was that they weren't able to single out the effective part of the process – soaking the maize in an alkaline solution of water and ash – from the other ineffective parts of the package – such as blowing on the maize before cooking it.
- (Author David Oks [draws](#) a fascinating parallel between this example of a delayed feedback mechanism and the tendency of AI models to “overfit”, extrapolating too much from coarse and sparse data to fill in the gaps. If the model creates a complex output and gets a single reward signal for it, the rational move is reproduce all of its features, including the ones that were incidental – such as AI's trademark verbal tics.)
- **No one wants to risk throwing the baby out with the bathwater.** Argentinian President Javier Milei talked publicly about his superstitious insistence on staying at home wearing a lucky jacket rather than watching his team playing in the football World Cup live at the stadium.

4. Self-interest: incentives favour more extreme positions

- **AI is not magic, a silver bullet or a charity, but a commercial industry.**
- **Most participants have incentives to amplify the stakes.** Founders need belief, investors need momentum, incumbents need gravitas, consultants need urgency, policymakers need relevance, executives need a narrative and journalists need drama. This is not a criticism of individuals or organisations, or even of the likely revolutionary impact of AI.
- **The challenge is magnified by a social media-driven economy.** Social platforms favour novelty, negativity, moral intensity and shareability. For example, in political tweets, one [study](#) by New York University found each additional moral-emotional word was associated with about a 20 percent higher retweet rate. Another [study](#) by MIT found that false news was 70 percent more likely to be retweeted than true news and reached a threshold of 1,500 people six times faster.
- **That means moderate claims are likely to be under-distributed and extreme claims over-rewarded.** Implausibly precise numbers for the future size of the AI economy, for example, or dramatic doom-laden scenarios heavy with emotive language will continue to attract greater attention than even-handed assessment. (See here for our quantitative analysis of two particularly striking examples from earlier this year: [“Something big is happening in AI... Or is it?”](#).)

What makes AI special – and gives hope for the future

Technology seems particularly prone to erroneous thinking because of the disconnect between visionaries in Silicon Valley and similar communities, and the rest of the world.



Cynics may say that people with extraordinary knowledge and skills in one area may be even more prone than most to wrongly estimate their lack of knowledge or skills in other areas: the so-called Dunning-Kruger effect.

That may include the reality of implementing AI on a trading floor in Wall Street, an office supplies wholesaler on the outskirts of London or an auto maker in Mumbai.

There is also a long tradition of correctly identifying the task that will disappear but getting wrong what humans will do instead.

1982: Computers “... can develop into monsters, damage the physical and mental health of the clerical staff – maybe even eliminate their jobs”: the Washington Post [reported](#) from a clerical conference. Computers instead became the core tool of knowledge workers.

1983: The pocket calculator “... has become the target of critics who fear that its use in the classroom will render generations of children dependent on batteries instead of brains”, the Washington Post [reported](#). Instead expectations changed about what people calculate for themselves.

2009: Cloud computing: around half of companies surveyed “cited security and privacy concerns as their top reason for not using cloud computing”, Forrester Research [found](#). Today companies entrust much of their critical infrastructure to the cloud.

“We overestimate the impact of technology in the short-term and underestimate the effect in the long run.” Roy Amara, Stanford computer scientist

But this time may really be different – with AI as a partner, starting with (deterministic) predictive AI. It is certainly helping weather forecasters to sift through unimaginable amounts of data. Four-day forecasts are now said to be about as accurate as one-day forecasts in the 1990s.

There are diminishing returns. The limit of predictability with standard physics models is about eight days, while deep-learning AI models increase that by just one day to nine days. However, the difference is that they [generate](#) state-of-the-art forecasts 100,000 faster, and 10,000 times more energy-efficiently, meaning they can be done on a laptop rather than a supercomputer.

And for the time being, your predictions about the impact of AI are no less valid than an AI expert based in the very particular world of Silicon Valley. And the future of AI will be made not in Silicon Valley but in Main Street.



Appendix 1

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Global Expansion of Chinese Construction Machinery: Lessons from Brazil

September 14, 2026

Executive Summary

China's exports have been strong this year, surging by 19% YoY in the first eight months. Among these exports, heavy-asset capital goods have played a particularly important role. Using construction machinery as a representative example, we illustrate how stronger product competitiveness, Chinese overseas projects, and Chinese manufacturers' "going-global" strategies have enabled market share gains across a broad range of economies.

Chinese construction machinery has expanded across the globe since 2020. Exports of related products quadrupled between 2020 and 2025 and have remained on a strong upward trend this year. More importantly, Chinese products have achieved significant market-share gains across a broad range of economies and accounted for 26% of global imports in 2025 compared to 10% in 2019. In this report, we aim to explore the factors behind this.

We use Chinese excavator exports to Brazil as a case study. Excavators are one of the most representative categories of construction machinery, while Brazil is South America's largest excavator importer and ranks among the world's top 20. Chinese excavators have also made remarkable market-share gains in Brazil, with their share rising from 20% to 90% in six years. Brazil therefore offers a compelling case study on this topic.

Through the case study and our analysis of aggregate data, we identify three drivers of the global expansion of Chinese construction machinery:

- **Product competitiveness:** Chinese brands have steadily improved both price advantage and product quality.
- **Demand from Chinese overseas projects:** Chinese contractors are undertaking more infrastructure projects and naturally exhibit stronger preferences for Chinese equipment.
- **"Going global" strategy by Chinese companies:** Higher profitability has encouraged them to direct more resources towards overseas expansion.

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Introduction

China's exports have been strong this year, surging by 19% YoY in the first eight months. In a note published for clients in March, we identified several drivers of this export growth, including AI-related demand, energy-cost advantages and the green transition. In particular, heavy-asset capital goods have been an important contributor, with construction machinery offering a representative example.

Chinese construction machinery has expanded rapidly across the globe since 2020. Using HS8429 products—including excavators, bulldozers, graders and related equipment—as a proxy, export values quadrupled between 2020 and 2025 ([Figure 1](#)). More importantly, the expansion has been broad-based: 128 of the 140 economies reporting HS8429 imports recorded an increase in China's market share over the period ([Figure 2](#)). Chinese products now account for more than half of HS8429 imports in over 30 economies. As a result, China's share of global HS8429 imports has risen from 10% in 2019 to 26% in 2025.

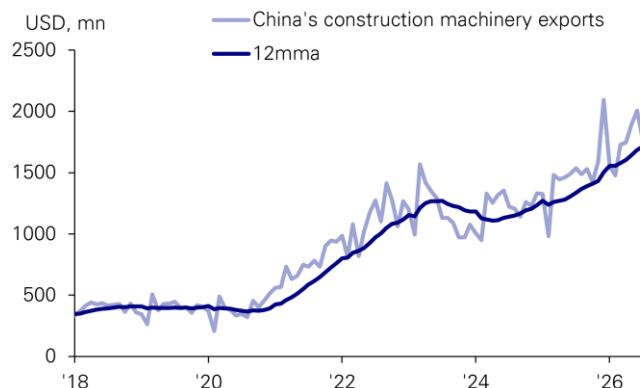
Looking at aggregate trade data alone does not fully reveal the multiple underlying drivers. At the country level, economic structures, investment cycles and infrastructure demand vary substantially, leading to very different import patterns and market-share dynamics. At the product level, HS8429 encompasses a diverse range of machinery categories with different technologies, end-markets and competitive landscapes.

Therefore, we adopt a case-study approach and focus on Chinese excavator exports to Brazil. We believe this is particularly representative since:

- Excavators are highly representative of the construction machinery industry, accounting for half of China's HS8429 exports.
- Brazil is the largest excavator importer in South America and ranks among the top twenty importers globally. The demand is underpinned by three factors. 1) As a major mining economy, Brazil requires substantial earthmoving equipment. 2) Brazil is one of the world's largest agricultural producers. The farming sector relies heavily on [capital investment and machinery](#) to raise productivity. 3) The government has [renewed its focus](#) on infrastructure development in recent years.
- Chinese excavator manufacturers have achieved exceptional market-share gains in Brazil. Their share increased from 20% to 90% over the past six years ([Figure 4](#)), with gains observed across all excavator segments.

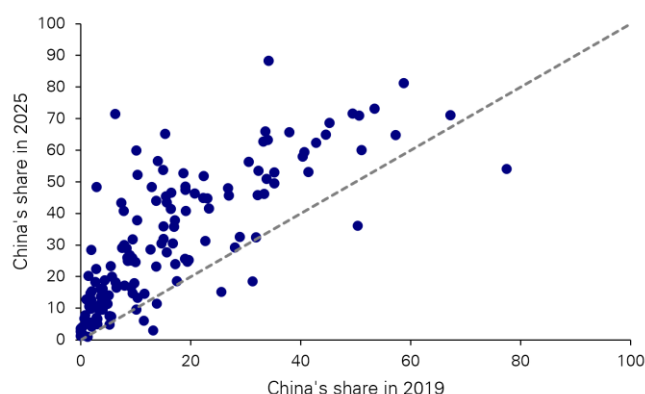


Figure 1: China's construction machinery exports have expanded rapidly since 2020



Source: Deutsche Bank Research, CEIC. We use HS8429 products—including excavators, bulldozers, graders and related equipment—as a proxy.

Figure 2: The expansion of China's construction machinery exports has been remarkably broad-based



Source: Deutsche Bank Research, UN Comtrade. Each dot represents an economy.

Figure 3: Brazil has been a major excavator importer



Source: Deutsche Bank Research, UN Comtrade. Excavators are primarily classified under HS842952.

Figure 4: Chinese excavators have recorded remarkable market-share gains in Brazil



Source: Deutsche Bank Research, Brazil Custom

Expansion of Chinese Excavators in Brazil

Chinese excavators have become increasingly visible across Brazil. Their presence is no longer confined to mines, railways, and other construction sites. They are increasingly involved in landmark projects such as Rio Olympic venues and World Cup stadiums, and can even be found in residential backyards and farms. Chinese brands [reportedly](#) occupied the top three positions in Brazil's market in 2025, while six Chinese brands ranked among the top ten. Customs data paint a stronger picture: Chinese products now account for ~90% of Brazil's excavator imports, both by value and volume.

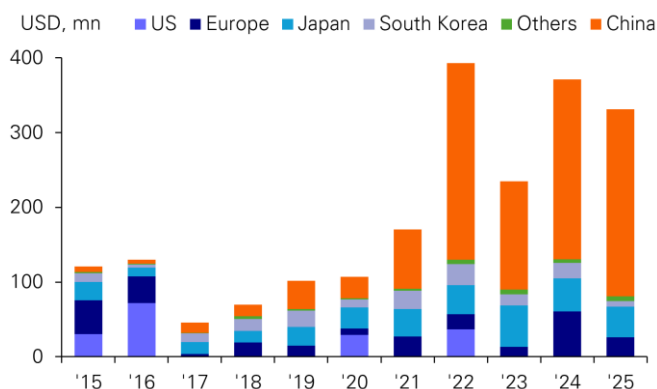
China's expansion appears to have been supported mainly by Brazil's incremental demand rather than the outright displacement of competing suppliers. During the market recovery from 2017 to 2021, excavator imports rose from all major sources. From 2022 to 2025, imports from China grew much faster, driving a rapid increase in China's market share, while annual imports from other suppliers—including Europe, Japan and Korea—remained broadly stable at around USD 100mn in total (Figure 5). This does not mean that other suppliers



were unaffected by Chinese competition: without China's sharp market-share gains, they would likely have captured a larger share of Brazil's surge in excavator demand.

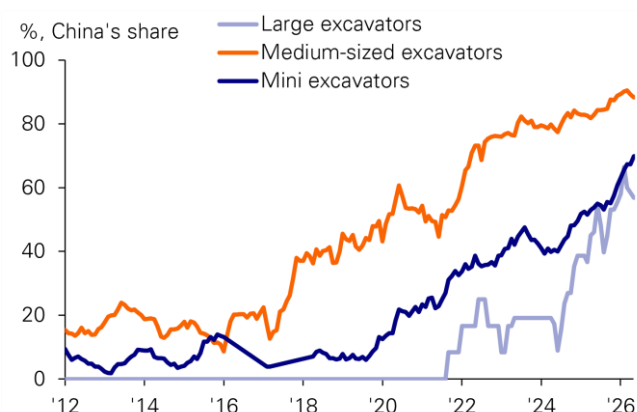
Interestingly, the expansion has been different across segments (Figure 6). Medium-sized excavators began gaining market share steadily after 2017, while mini excavators only entered a similar trajectory after 2019. Large excavators followed a different path. Prior to 2021, Chinese brands were absent, yet they rapidly gained traction after the first entry. The distinct patterns suggest that the rise of Chinese excavators might have been driven by multiple demand- and supply-side forces.

Figure 5: This expansion appears to have come largely from incremental demand rather than outright displacement



Source: Deutsche Bank Research, Brazil Custom

Figure 6: Expansion has been different across product segments



Source: Deutsche Bank Research, Brazil Custom. Brazil's 8-digit NCM customs classification distinguishes excavators by power categories, allowing us to group them into three segments.

Factor 1: Price Advantage Meets Quality Upgrading

A widening price advantage has been a major driver behind the expansion of Chinese products in the medium-sized and mini excavator segments. Since these products are relatively less technologically demanding, pricing remains a major consideration for buyers. We divide the sample into four periods and compare China's price advantage against market share gains (Figure 7). China's advantage has steadily expanded, with products now more than 50% cheaper than peers. Meanwhile, the relationship between price advantage and market share is highly significant. For medium-sized excavators, a one percentage point increase in China's price advantage is associated with a 0.4pt gain in market share. For mini excavators, the corresponding relationship is even higher at 0.9pt.

Indeed, mini excavators provide the clearest example of price-driven expansion.

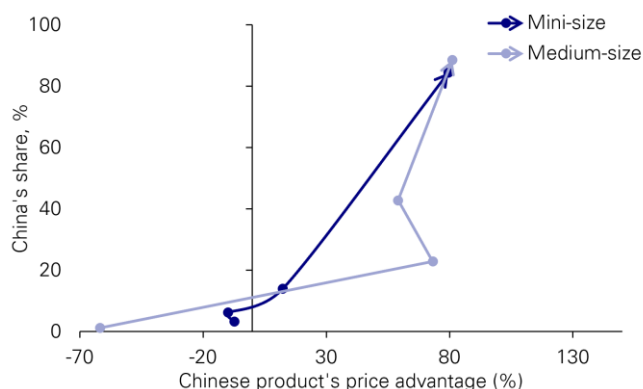
These machines are used for small-scale construction projects or purchased by individual consumers, making demand particularly price-sensitive. Chinese products began to undercut competitors after 2017, and their price advantage widened significantly after 2020 (Figure 8). The gap has now reached USD 17,000 per unit. We further decompose this price advantage (Figure 9). 60% of the widening gap reflects price increases by overseas competitors amid higher inflation, while 40% reflects price cuts by Chinese manufacturers. RMB



depreciation accounted for 10% of the gap during 2023-25. However, following its appreciation in 2026, the cumulative contribution has turned negative.

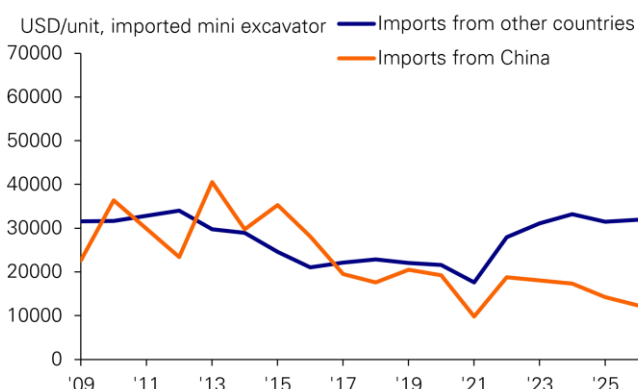
More noteworthy is the advance of Chinese brands into the high-end segment. Large excavators are far more technologically sophisticated. Before 2021, Chinese brands were virtually absent from Brazil's large excavators market. They entered the market in 2021 and expanded rapidly thereafter (Figure 10). So far in 2026, 43 of Brazil's 47 imported large excavators have come from China. Chinese manufacturers are reportedly delivering [100-tonne-class excavators](#) to Brazilian mining customers and are actively promoting equipment in the [500-tonne category](#). Chinese companies' competitiveness is increasingly extending from price advantage toward technology intensity.

Figure 7: A widening price advantage has driven expansion in the medium-sized and mini excavator



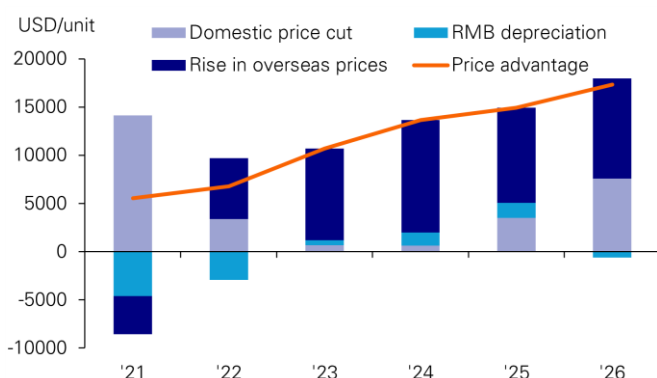
Source: Deutsche Bank Research, Brazil Custom. The four observations correspond to the periods 2006-2010, 2011-2015, 2016-2020, and 2021-2025, respectively.

Figure 8: Mini excavators provide the clearest example of price-driven expansion



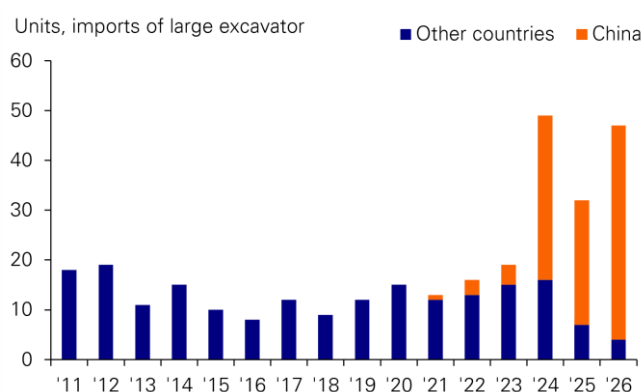
Source: Deutsche Bank Research, Brazil Custom

Figure 9: China's price advantage has widened owing to overseas inflation, domestic price cuts and RMB depreciation



Source: Deutsche Bank Research, Brazil Custom

Figure 10: Chinese brands entered the large excavator market in 2021 and rapidly expanded thereafter



Source: Deutsche Bank Research, Brazil Custom

Factor 2: Demand from Chinese Overseas Projects

Chinese firms are undertaking more construction projects in Brazil, generating additional demand. They naturally have a stronger preference for Chinese-made machinery. The value of construction projects completed by Chinese companies has risen consistently since 2021. Growth in newly signed contracts has been



even stronger, with their value reaching a record USD 4.6bn in 2024 (Figure 11). This trend may have contributed to the growth in excavator imports since 2024, especially for large excavators.

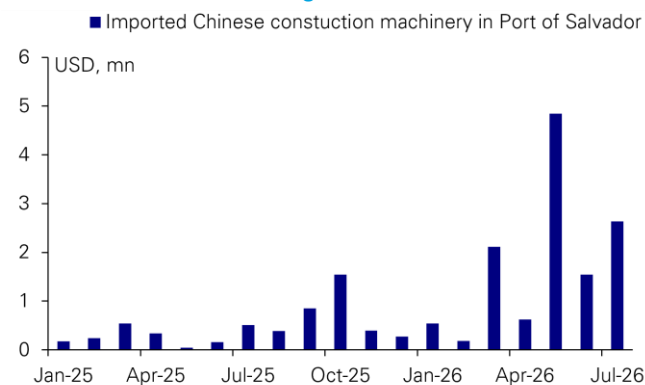
The Salvador-Itaparica Bridge project provides a compelling example. Once completed, it will be the longest sea-crossing bridge in South America. The project was awarded to two Chinese SOEs in 2019, with an initial investment of USD 2.2bn, which was raised to USD 2.7bn in 2025. It is the largest infrastructure investment undertaken by Chinese companies in Brazil. Since construction began this year, [~800 tons of equipment](#) from China have arrived at the port of Salvador since April, consistent with the surge in custom data (Figure 12). Most recently on July 1, the project company showcased the first batch of equipment deployed on site, [all sourced from China](#).

Figure 11: Chinese firms are undertaking more construction projects in Brazil



Source: Deutsche Bank Research, China Mofcom

Figure 12: Imports of Chinese machinery through the Port of Salvador have surged in recent months



Source: Deutsche Bank Research, Brazil Custom

Factor 3: Supply-Side Expansion Driven by Higher Profitability

Chinese firms have strong incentives to expand overseas because foreign markets offer higher profitability. Some views have argued that low export prices primarily reflect dumping driven by weak domestic demand. If that were the case, corporate profits would be expected to remain flat or deteriorate as firms sacrificed margins to gain market share. Company-level evidence suggests the opposite.

We examine five listed Chinese construction machinery manufacturers. Combined overseas revenue has risen consistently since 2020, doubling in 2022 and doubling again by 2025, broadly mirroring the trajectory of China's total construction machinery exports (Figure 13). More importantly, overseas profitability has improved substantially. Across the five companies, operating margins in overseas markets have increased by 5-15 percentage points, with the average rising from 20% during the previous decade to 30% today (Figure 14). In contrast, domestic margins have fallen by 5 percentage points.

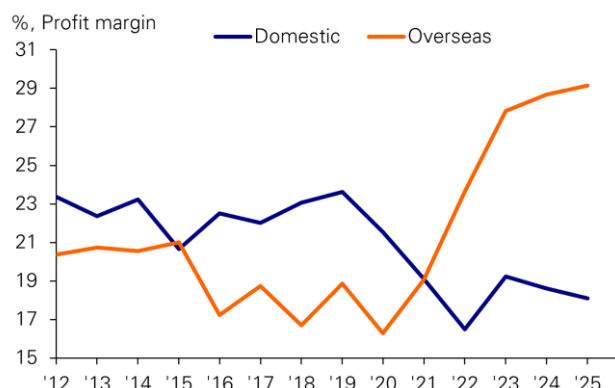


Figure 13: Overseas revenue has risen consistently since 2021



Source: Deutsche Bank Research, Wind

Figure 14: Overseas profit margins have exceeded domestic margins



Source: Deutsche Bank Research, Wind

Beyond Excavators: Chinese Construction Machinery Expansion

Through this case study, we identify three forces underpinning the expansion of Chinese excavators in Brazil: improving price and quality competitiveness, growing demand from Chinese overseas projects, and proactive going-global strategies by Chinese companies.

We believe the forces extend to the broader construction machinery sector.

- **Price and Quality Competitiveness.** The price advantage over HS8429 has become more pronounced since 2022 (Figure 15). The analysis of eight-digit categories revealed a more comprehensive picture (Figure 16). First, exports increased across all categories, including those with rising prices. Second, the products recording larger price increases tended to have higher unit values. That said, the expansion is driven by price cuts as well as increasing demand for higher-value products.
- **Overseas Expansion of Chinese Enterprises.** Chinese overseas construction projects have reaccelerated since 2024, with completed value and newly signed contracts rising by 20% and 15%, respectively (Figure 17). Belt and Road economies are key destinations for Chinese overseas construction projects, and Chinese construction machinery has gained market share much more rapidly in these markets.
- **Proactive “Going global” strategy.** Since the 2010s, construction machinery manufacturers have viewed overseas expansion as a key pillar of future growth and have invested heavily in international distribution channels, service networks and localized production facilities.

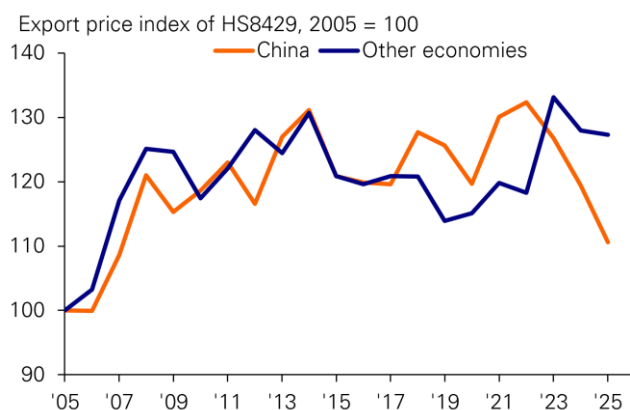
Looking ahead, we believe all three drivers are likely to remain in place for an extended period. China’s manufacturing capability and continued technological progress should support further gains in product competitiveness. Accelerating component localization could further strengthen the price advantage. Meanwhile, the overseas expansion of Chinese companies is expected to remain a long-term trend. Belt and Road economies warrant particular attention, as Chinese



investment in these markets has surged in recent years, laying a solid foundation for further overseas expansion (Figure 18). As a result, we remain constructive on the outlook for construction machinery exports.

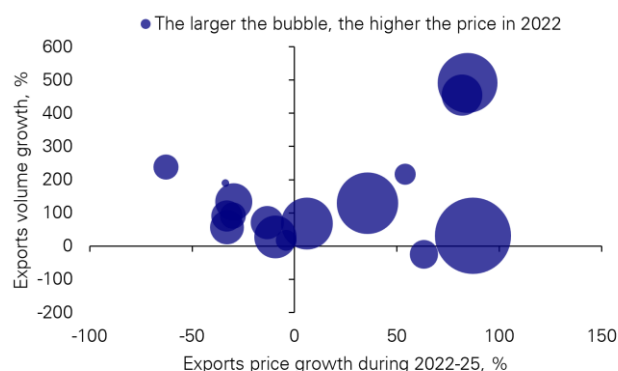
That said, several risks warrant monitoring. First, further RMB appreciation could erode the price competitiveness of Chinese products. Second, the sharp increase in market share achieved in Brazil—from 20% to 90%—may not be easily replicated elsewhere. In some markets, Chinese firms could face stronger competition from local manufacturers, limiting the pace of further market-share gains.

Figure 15: China's price advantage on HS8429 has become more pronounced since 2022



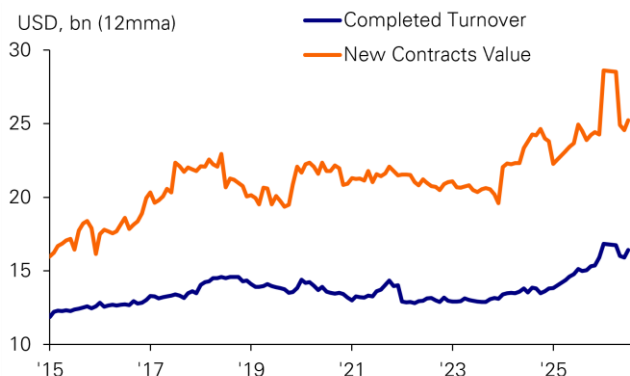
Source: Deutsche Bank Research, UN Comtrade

Figure 16: The expansion is driven by price cuts and increasing demand for higher-value products



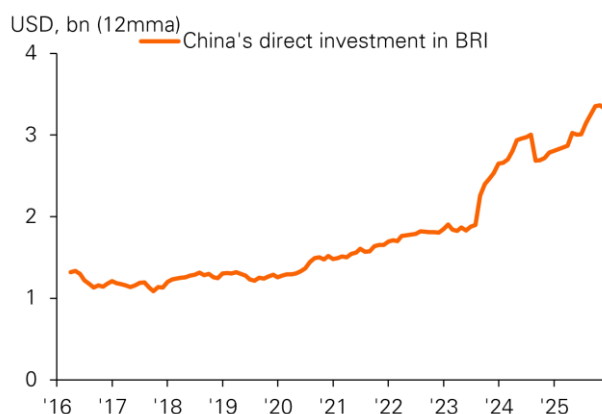
Source: Deutsche Bank Research, CEIC. Each bubble represents an eight-digit category under HS8429.

Figure 17: Chinese contractors have been undertaking more projects overseas since 2024



Source: Deutsche Bank Research, CEIC

Figure 18: China's direct investment in Belt-and-Road economies has also surged since 2024



Source: Deutsche Bank Research, CEIC



Appendix 1

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The revenge of the supply-side: A new economic regime

September 14, 2026

Back in the 2010s, macroeconomic policy was preoccupied by insufficient demand. In historical terms, it was an unusual period of low inflation and low interest rates, where supply-side issues were considered peripheral. If anything, it went almost unquestioned that the supply side of the economy would scale up to meet higher demand.

But in the 2020s so far, we're back in a supply-driven world. Growth has been constrained as much by supply shortages as deficient demand. This year alone, we've seen major consequences from the closure of the Strait of Hormuz, with ramifications for oil prices, inflation and economic growth.

This macro regime shift, where supply rather than demand is the bigger issue, is a tricky environment to navigate, and renders much of the 2010s playbook redundant:

- First, inflation spikes will occur more frequently in a supply-constrained world, as there isn't the spare economic capacity to meet higher demand.
- Second, monetary stimulus also becomes less effective in this environment, as traditional demand-side monetary tools can't deal with the direct cause of supply shocks.
- Third, fiscal stimulus in a supply-constrained environment becomes much more inflationary and is more likely to crowd out private investment.

Looking forward, the return to a supply-driven world means that the primary issue for policymakers will be expanding the productive frontier of the economy. After years where insufficient demand was the main preoccupation, the future of growth now rests on a supply-side expansion that can raise living standards across the board.

Authors

Henry Allen
Strategist



The paradigm shift in thinking: How the biggest economic problem moved from demand to supply

The legacy of the Global Financial Crisis led to a world of demand deficiency

In 2008, the Global Financial Crisis (GFC) marked the worst economic contraction since the Great Depression of the 1930s. Unemployment rose to multi-decade highs and the subsequent recovery was very sluggish, despite huge economic stimulus on a scale previously unheard of in peacetime. This slow recovery was clear in numerous ways, but an obvious one was that the US unemployment rate didn't return to its pre-GFC low until 2017, a full decade later. Meanwhile in the Eurozone, it didn't happen until February 2020, just as the pandemic then sent it higher once again.

This was a world where deficient demand was the main preoccupation of macroeconomic policymaking. Indeed, the Depression-era concept of "secular stagnation" made a comeback, in which the equilibrium interest rate consistent with full employment was negative. In this world, the constraints of the zero lower bound on monetary policy would make it very difficult to get to full employment.

This demand deficiency manifested itself in persistent inflation undershoots. Throughout the 2010s, both the Federal Reserve and the ECB missed their 2% inflation targets on the downside. Policy rates remained close to the zero lower bound, and long-term market interest rates also moved continuously lower. Even before the pandemic, Germany's 10-year bund yield fell to -0.7% in summer 2019. In this environment of negative yields, borrowing was basically a "free lunch".

At the time, there were genuine fears about a permanent low-growth trap. Indeed, when the Covid-19 pandemic began and another economic contraction was underway, there were initially concerns that deflationary pressures would intensify further. For instance, the 10-year US inflation swap (a market-based measure of expectations for US CPI over the next decade) fell to just 0.81% in March 2020. Investors genuinely thought that low inflation and low interest rates were likely to persist for the decade ahead as well, and the 10-year Treasury yield closed at a record low of 0.51% in August 2020.

This fear of deficient demand informed the policy response to Covid-19. The stimulus packages went well beyond those seen in the GFC. For instance, no time was lost in slashing rates back to the zero lower bound, and the fiscal deficits went far above anything previously seen in peacetime. In fact, the US federal deficit stood at -14.5% of GDP in 2020, the biggest since 1945. And with much of the global economy locked down for months, consumers accumulated huge "excess savings" as they couldn't spend on their normal activities.

The pandemic was the turning point towards a world of supply constraints

With hindsight, we can see that all this stimulus and the accumulation of excess savings created a huge demand surge, just as the global economy was experiencing major supply-chain issues as it reopened:

- The lockdown periods saw a big rotation in spending away from services (like dining out, travel, gym memberships) towards durable goods (like home office equipment, electronics, exercise equipment). This overwhelmed existing supply chains.
- Pandemic-related disruption led to major issues with shipping. A severe



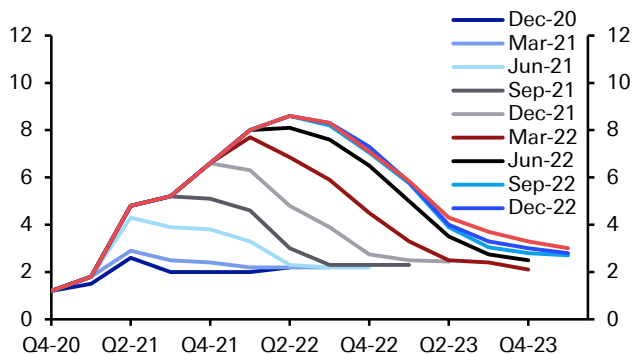
shortage of containers led freight rates to rise more than 5-fold at their peak.

- Labour markets were disrupted, as many took early retirement during the pandemic or changed occupations. Companies therefore had to raise wages to compete for a smaller pool of workers.
- Commodity prices also surged in this period, particularly energy. The pandemic period saw producers cut back on drilling and exploration. But this meant it was more difficult to dial up supply when demand started to rise again, and energy prices surged back, with Brent crude oil moving above its pre-pandemic levels in mid-to-late 2021.

These supply constraints marked a major shift from the environment of the 2010s. Back then, it was almost an unquestioned assumption that the supply-side of the economy would scale up to meet demand. Inflation had been low and the most obvious signs of supply-side stress were not apparent. But now the reverse was true, and inflation was gathering pace again.

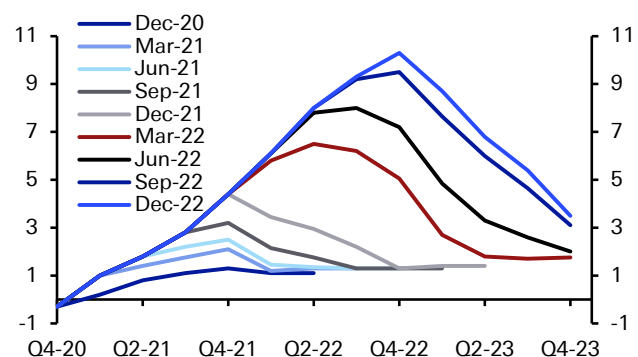
Initially, many argued that this burst of inflation would prove temporary. They argued it was down to one-off factors like the pandemic and the subsequent reopening. Moreover, there was a view that the longer-term disinflationary forces of the previous 30 years would still reassert themselves. But by late-2021, it was clear that the combination of huge stimulus, the drawdown of excess savings, and serious supply-side disruption had created more inflation than anticipated. By December 2021, US CPI had risen to its highest level since 1982, at 7.0%, whilst Eurozone CPI was at its highest since 1991, at 5.0%.

Figure 1: Bloomberg consensus forecasts for US CPI



Source : Bloomberg Finance LP, Deutsche Bank

Figure 2: Bloomberg consensus forecasts for Eurozone CPI



Source : Bloomberg Finance LP, Deutsche Bank

The transition to a supply-constrained world accelerated dramatically in early 2022 with Russia's invasion of Ukraine. Energy prices rose significantly, and European industry was forced to decouple from Russian gas. Meanwhile, the blockade of Ukraine's Black Sea ports restricted the supply of agricultural commodities like wheat and corn, driving food prices higher still.

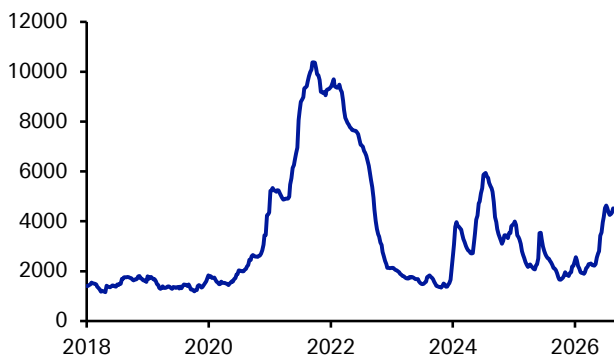
This sent inflation even higher, up to multi-decade highs around the world. US CPI inflation peaked at 9.1% in June 2022, and Eurozone CPI peaked at 10.6% in October 2022. This was the context in which central banks launched into the most aggressive rate hikes in a generation. Over 2022-23, the Federal Reserve hiked rates by 525bps, and the ECB delivered 450bps.



These supply issues persisted long after the pandemic-related issues were over

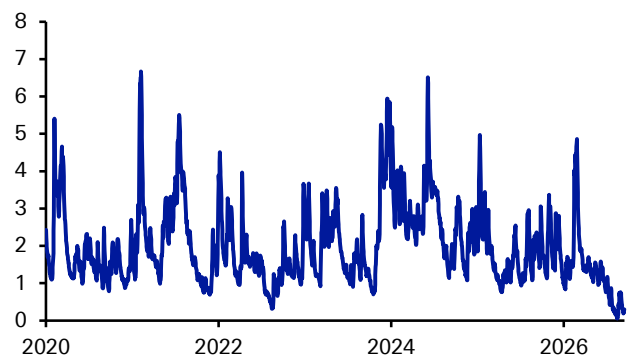
Yet even as the worst of that inflation wave passed, the 2023-24 period kept demonstrating how supply-side disruptions could lead to wider economic problems. For instance, attacks from the Houthi rebels in the Red Sea forced major shipping lines to reroute away from the Suez Canal around the Cape of Good Hope, adding over a week to the length of trips. Meanwhile, an El Nino-induced drought meant transit capacity through the Panama Canal fell 30%, forcing ships to wait in long queues or reroute instead. And in Europe, repeated summer heatwaves have lowered the Rhine's water levels, disrupting industrial barges and affecting industries like chemicals and steel.

Figure 3: Drewry World Container Index Composite Index (USD per 40 Foot Box)



Source : Bloomberg Finance LP, Deutsche Bank

Figure 4: Rhine Water levels at Kaub (metres)



Source : Bloomberg Finance LP, Deutsche Bank

More broadly, the other shift in the post-pandemic period was the growing use of strategic industrial policy, particularly as governments sought to boost domestic manufacturing capacity. For instance, the US passed the Inflation Reduction Act under the Biden administration, which encouraged clean energy investments, as well as the CHIPS Act, which sought to boost semiconductor manufacturing. Tariffs also became a more enduring feature of the landscape, and the Biden administration hiked tariffs on Chinese imports, with those on EVs up from 25% to 100%, whilst semiconductors rose from 25% to 50%.

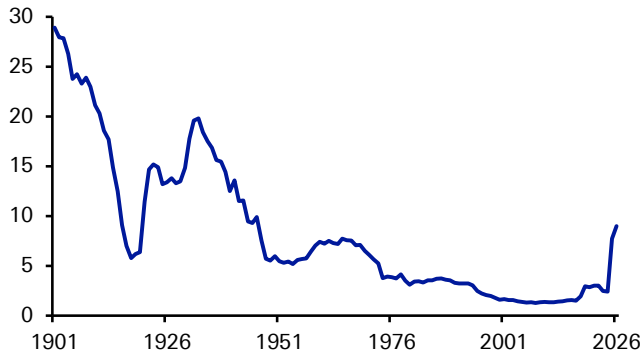
Then under the second Trump administration, tariffs became an even bigger part of the policy toolkit, with the US effective tariff rate surging from 2.6% beforehand to a peak of 23% in April 2025, before settling around 11% currently. Several different measures were introduced, but the macro impacts were clear. In particular, there was a major wave of front-loading of imports in Q1 2025, as those trading internationally sought to get ahead of the announcements. Indeed, the surge in imports meant US GDP contracted slightly in Q1 2025. Moreover, other countries moved to respond, and not just with tariffs of their own. For instance, China introduced export controls on rare earths.

Whilst 2025 was dominated by tariffs, 2026 then brought a fresh supply shock for the global economy with the closure of the Strait of Hormuz. Previously, around 25% of the world's seaborne oil trade had gone through the Strait, including nearly 15 million barrels per day of crude oil. But the start of the US-Iran conflict led the Strait of Hormuz to close, and six months later only limited traffic is still able to pass through. This led to another supply shock, with Brent crude oil prices moving back



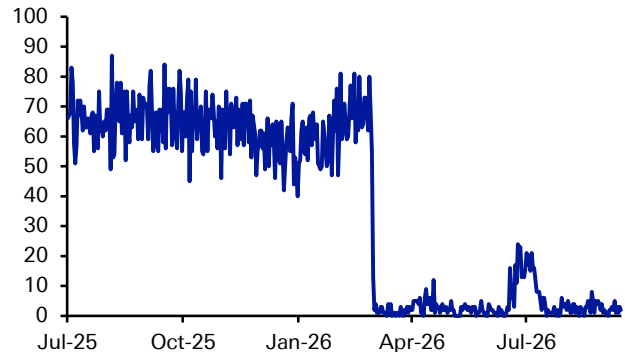
above \$100/bbl, whilst European natural gas futures also reached their highest levels since late-2022. Yet again, the biggest global economic disruption of the year to date has been a supply shock, rather than a demand one.

Figure 5: US Effective Tariff Rate (%) with projected postsubstitution rate from the Yale Budget Lab



Source : Haver Analytics, The Budget Lab at Yale, Deutsche Bank

Figure 6: Strait of Hormuz Tanker Vessel Crossings



Source : Bloomberg Finance LP, Deutsche Bank

Types of supply shocks

Broadly speaking, we can distinguish between three sorts of supply shocks, although there can be considerable overlap between these categories.

1. Temporary supply shocks that have a short-term economic impact.

Example: low river levels blocking a key trade route.

The first category of supply shocks are those that have a temporary impact on the economy. Examples include the blockage of the Suez Canal in 2021 when the Ever Given container ship ran aground, or a drought that causes river levels to fall and means fewer ships can get through key waterways.

These are supply shocks that temporarily remove capacity, but then see a bounce back to “normal”, broadly speaking. For instance, a drought is hopefully not repeated, and when river levels return to normal, supply can resume again.

For the economy, this means the longer-term consequences are also limited. The inflation impact is mean-reverting, and output recovers, often in a V-shape. These are the sort of shocks that central banks are usually able to “look through”, because they don’t cause permanent structural damage to the economy.

2. Medium-term supply shocks that can last several years, with a clear economic impact.

Example: the closure of the Strait of Hormuz.

These kinds of shocks are often driven by geopolitical events or policy decisions, and are longer-lasting shocks which do not have a quick solution. A good example is the current closure of the Strait of Hormuz, but we have seen many in recent years. In theory, these shocks can end, but their length is uncertain and they have material economic impacts. An increase in tariffs also fits into this category, as it forces an adjustment in trade flows, even if they can be reversed in theory.



A good example of this kind of supply shock is the 1973 oil shock. Back in 1973, the oil embargo by OPEC led oil prices to nearly quadruple, and they remained at that higher level for years after. It wasn't simply a temporary adjustment that normalised again.

Unlike the short-term shocks, these medium-term shocks can force a permanent rerouting of trade flows and supply chains. Moreover, they can push inflation higher for an extended period and are more likely to filter through into core inflation categories. And because they are not temporary shocks that quickly adjust, policymakers can't "look through" them as easily. As such, central banks aren't able to accommodate the shock, and often have to hike rates to clamp down on the resulting inflation, as we saw in 2022.

3. Structural supply trends that affect the longer-term growth rate.

Example: Demographic trends.

This category includes the long-term forces on the supply side that affect the economy's growth rate. A good example are demographics and the current situation of ageing populations. Another example are the multi-decade processes of globalization and deglobalization.

These are factors that can permanently shift the productive frontier of the economy. For example, the current situation of ageing populations means that a smaller proportion of the population are of working-age, meaning that labour forces are shrinking. As a result, that constrains the growth rate of potential GDP, meaning that more growth must come from higher productivity rather than a larger number of workers.

These shifting trends can lead to permanent changes in so-called r^* , which is the equilibrium interest rate where full employment is achieved – a level that neither stimulates nor restricts economic activity. In other words, these factors are capable of shifting the economy's potential growth rate, at which it can grow sustainably without generating inflationary pressures.

What happened the last time the supply-side dominated: Lessons from the 1970s

As we consider the return of the supply side, it's worth bearing in mind this is far from the first time that supply shocks have dominated the landscape. Back in the 1970s, the biggest economic shocks of that decade were also supply driven.

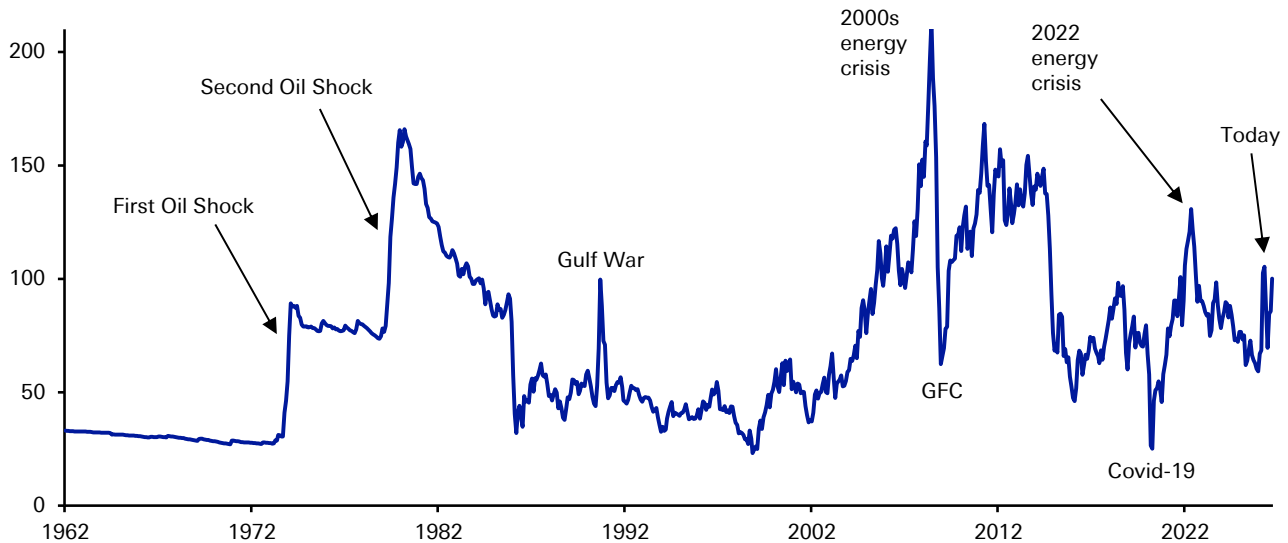
That decade saw two huge oil shocks in 1973 and 1979. In the first, the Organization of Arab Petroleum Exporting Countries (OPEC) placed an embargo on several countries, including the United States, in retaliation for their support for Israel in the Yom Kippur War. This caused oil prices to nearly quadruple, triggering a recession across multiple countries that began in late 1973. Then in 1979, a second oil shock occurred around the time of the Iranian Revolution. That coincided with a big decline in Iran's oil output, before the Iran-Iraq War in 1980 led to further production declines for both countries. As a result, oil prices more than doubled again.

To be fair, there were other demand-side and supply-side shocks that meant inflation got out of hand in the 1970s. But the two oil shocks were the single biggest factors, and created a stagflationary environment in which inflation accelerated



and growth slowed.

Figure 7: WTI Oil Prices (\$/bbl) in real terms



Source : Finaeon, Deutsche Bank

This experience is instructive for today, as the 1970s demonstrated how a repeated set of supply shocks could act in a cumulative way. After all, even if the textbook response is to “look through” a temporary supply shock, a series of consecutive shocks can suddenly feel very permanent when they happen one-after-another. That continuous disruption can then cause inflation expectations to become unanchored, risking the kind of dynamics that trigger wage-price spirals. Indeed, much as economists can focus on “core inflation” that excludes food and energy prices, these are essential goods that are difficult to substitute away from, and play a key role in inflation expectations.

On the bright side, the more optimistic lesson is that supply shocks often force adjustments that prevent a repeat. For instance, back in the 1970s, there was a recognition that Western economies had become overly dependent on Middle Eastern oil, and major efforts were made to reduce that dependence. For instance, the US established the Strategic Petroleum Reserve (SPR) in 1975, and the Department of Energy was established in 1977. Moreover, investments were directed to US energy production, and since 2019, the US has been a net exporter of energy, rather than a net importer.

Clearly the 1970s are far from an exact parallel. Most obviously, economies are far less energy intensive today, so oil shocks don’t have as big an impact today. Nevertheless, the dynamic of repeated supply shocks is one that bears relevance given how that dynamic is re-echoing today, and with inflation still lingering above target in many of the world’s biggest economies.

What are the implications for today?

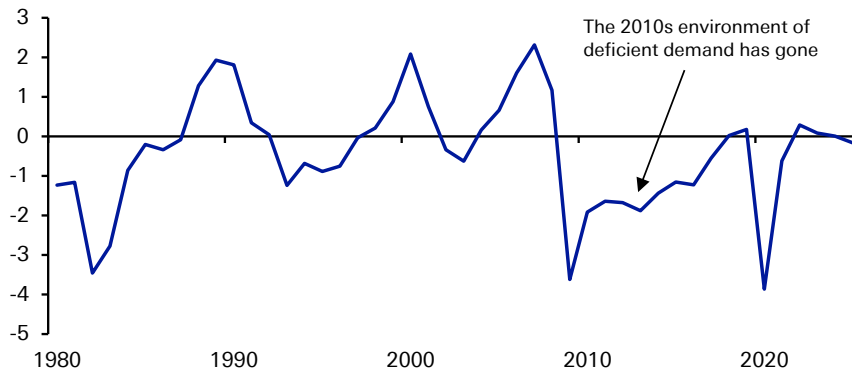
1. Inflation spikes will occur more frequently, as there isn’t the spare economic



capacity to meet higher demand.

The first consequence of a supply-constrained world is that unlike much of the 2010s, there isn't the spare economic capacity (or a negative output gap) that can easily absorb higher demand. So if there is an unexpected demand shock, for whatever reason, the consequence will likely be higher inflation rather than higher output.

Figure 8: IMF: Output Gap in the G7 economies (% of potential GDP)



Source : Haver Analytics, Deutsche Bank

This is a particular risk right now because of those longer-term structural shifts on the supply side. First, global demographics and the retirement of the baby boomers mean that populations are ageing, and many countries now even have shrinking working-age populations. This is a big difference from the world of decades past, when the labour force was generally expanding. Second, the push to reshore production of key industries has also raised costs, as it requires the duplication of supply chains. And third, other structural shifts like the push for higher defence spending have also created an increase in the demand for capital. So there isn't the spare capacity in the economy to absorb higher demand that there was in the 2010s.

This world of more frequent supply shocks means that central banks must also trade off how quickly they want to bring inflation back to target against the potential output volatility that might result. Indeed, we saw this year when the Strait of Hormuz was blocked, central banks didn't immediately react with rate hikes, but they waited a few months to see how the consequences developed. But of course, that has also meant inflation has stayed above target in many countries, often for several years. In fact, US PCE inflation has been above 2% since March 2021.

2. Monetary stimulus becomes less effective, as it can't deal with the direct cause of supply shocks.

Monetary policy has a complex transmission channel, but when central banks cut rates, the aim is usually to stimulate demand. This is a very useful tool in demand-driven recessions like 2008, but it becomes more problematic in the face of supply shocks.

The problem is that when a supply shock hits, whether that's a natural disaster, or the blockage of a key trade route, the economy's productive capacity drops.



Monetary policy cannot affect that directly.

This matters because negative supply shocks have the unfortunate effect of raising inflation and lowering growth simultaneously. So it puts central banks in a very tricky position, as they found in the inflationary environment of the 1970s after the oil shocks. With hindsight, many central banks were too dovish after the first oil shock and inflation got out of control. In the case of the Federal Reserve, that then led to a far more hawkish response to the second oil shock, when Paul Volcker tightened monetary policy significantly to get inflation under control again.

These problems are further exacerbated because central banks operate with uncertainty about the future. Just as we don't know today when the Strait of Hormuz might return to normal, central bankers in 1973 were unsure how long the oil embargo would last. But because central banks make policy that responds with a lag, there is inherently the risk of overshooting. If they expect a supply shock to be permanent, yet it proves temporary, then they risk overtightening. And similarly in reverse, if disruption is expected to be temporary, but proves permanent, they risk not doing enough and letting inflation and inflation expectations rise too much.

3. Fiscal stimulus becomes much more likely to cause overheating and crowding out.

In a supply-constrained world where the economy lacks spare capacity, fiscal stimulus is more likely to overheat the economy and generate inflation.

This is a big shift from the post-GFC world. Back then, unemployment was high and deficient demand meant additional government spending didn't face the same competition from private investment, so the multipliers from that stimulus were much higher. Moreover, several countries were able to borrow at incredibly low interest rates, even negative in some cases.

But this shifted with the pandemic. Not only was the stimulus in 2020 far bigger, but unlike 2008, the economy then saw a huge wave of pent-up demand once the pandemic restrictions were lifted, supported by the excess savings accumulated during the lockdowns. In turn, that contributed to a huge wave of inflation, even before we then had the additional energy and food price shocks in 2022.

Meanwhile, government bond yields have also returned to their pre-GFC levels, making the cost of any stimulus far more expensive. And that comes as we are also seeing a kind of "reverse crowding out", thanks to borrowing from the AI hyperscalers that is adding to pressure on rates. So in this environment, any fiscal stimulus will come at a higher cost.

Conclusion: Looking forward

The transition to a supply-driven world means that many of the old economic assumptions from the 2010s are no longer valid. There's been a structural break that means classic demand-side stimulus is not as effective. After years when it was almost unquestioned that the supply side of the economy would scale up to meet higher demand, we are now back in a world where supply is a meaningful constraint again.

As such, the emphasis will naturally shift towards expanding the supply-side of the

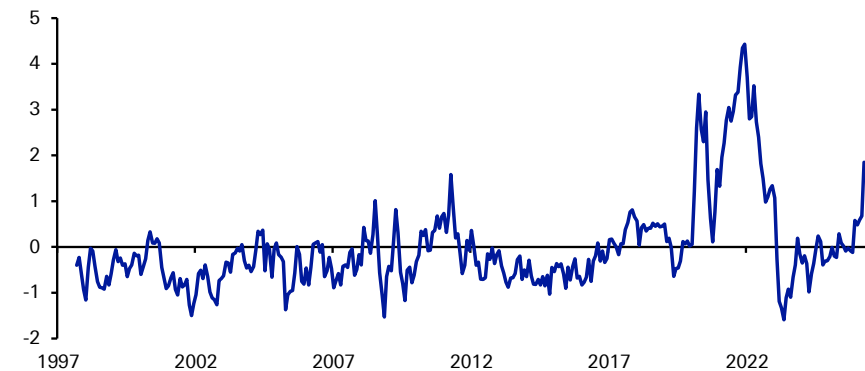


economy, something we're already beginning to see. This could be encouraging more energy production, more housing, or bringing new participants into the labour force. In essence, what's needed are policies that raise the productive potential of the economy, and boost the rate at which the economy can grow without generating inflationary pressures.

These supply shocks in recent years have been a challenge. Even if they might be temporary in isolation, the cumulative effect has been to keep inflation elevated, and we saw that happen back in the 1970s too. This is particularly problematic if the starting point is above-target inflation, as further supply shocks risk unanchoring inflation expectations.

Nevertheless, one consequence of a world of more frequent supply shocks is that actions are then taken to reduce vulnerabilities over the years ahead. For instance, the 1970s oil shocks led to the widespread realisation that the US needed to become less energy intensive and produce more of its own energy. As a result, the Strategic Petroleum Reserve was set up, and the subsequent years saw the US eventually become a net energy exporter. Similarly today, the closure of the Strait of Hormuz has forced other countries to secure alternative supply routes and use strategic reserves. So even before the longer-term geopolitical outcome is clear, we have seen diversification away from one particular chokepoint. In a supply-driven world, the incentives will shift towards making those supply chains more resilient, and better able to withstand future shocks.

Figure 9: New York Fed's Global Supply Chain Pressure Index - the extent of supply-chain pressure in recent years has been very high relative to the previous couple of decades



Source : Bloomberg Finance LP, Deutsche Bank



Appendix 1

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The background of the slide features a large, white, stylized robot head on the left and right sides, facing each other. In the center, there is a large, white, stylized robot head. The background is dark, and there is a large, white, stylized robot head in the center. The robot head is composed of several segments, and there are glowing elements. The overall theme is AI and technology.

Deutsche Bank
Research

AI at 70 (Part 2): 14 more lessons from a lifetime of boom and bust

Context is all you need

September 2026

Adrian Cox

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

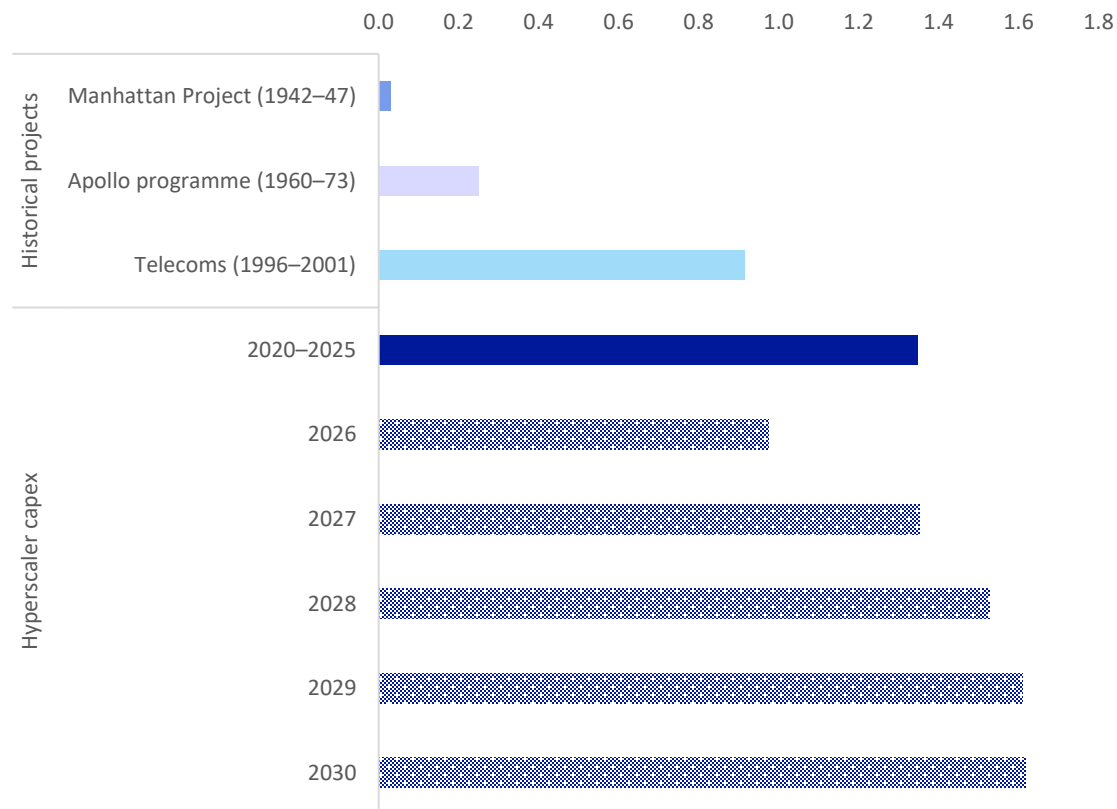
Deutsche Bank
Research Institute

1. This boom is remarkable not just for scale but also speed

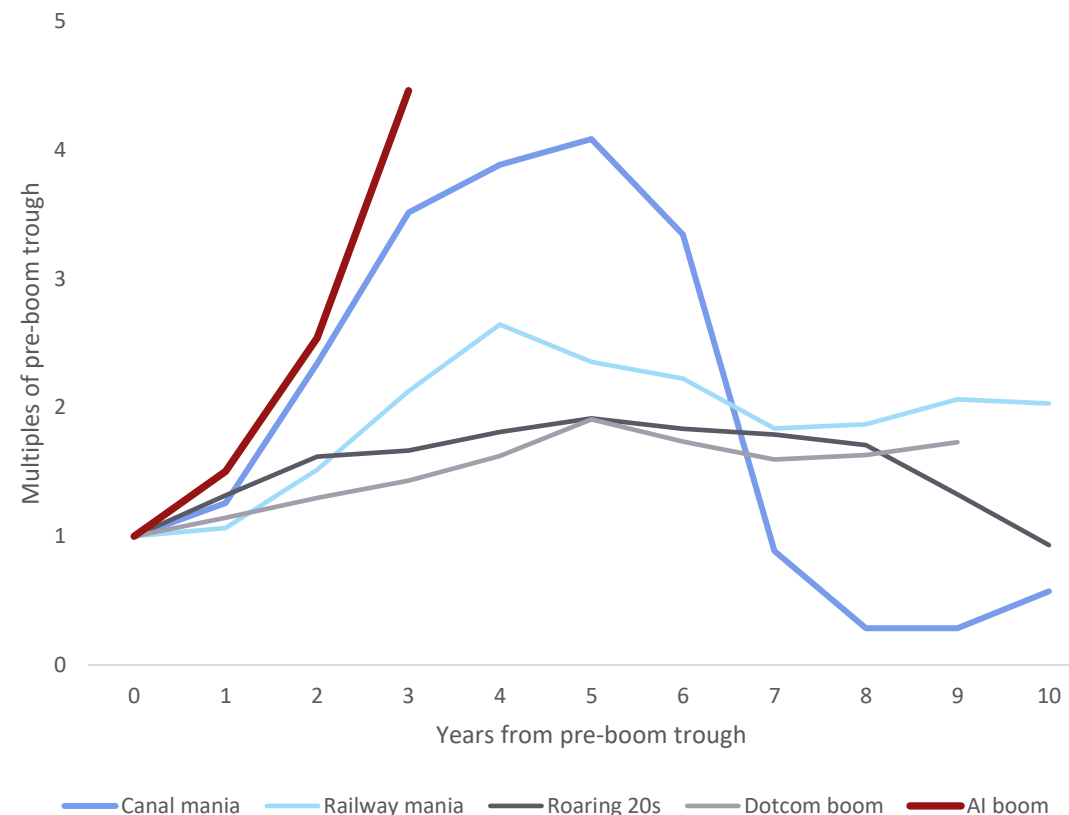
Each year of expected capex through 2030 exceeds the whole dot-com telecoms boom



Hyperscaler current and forecast capex vs major historical projects (2026 USD trn)



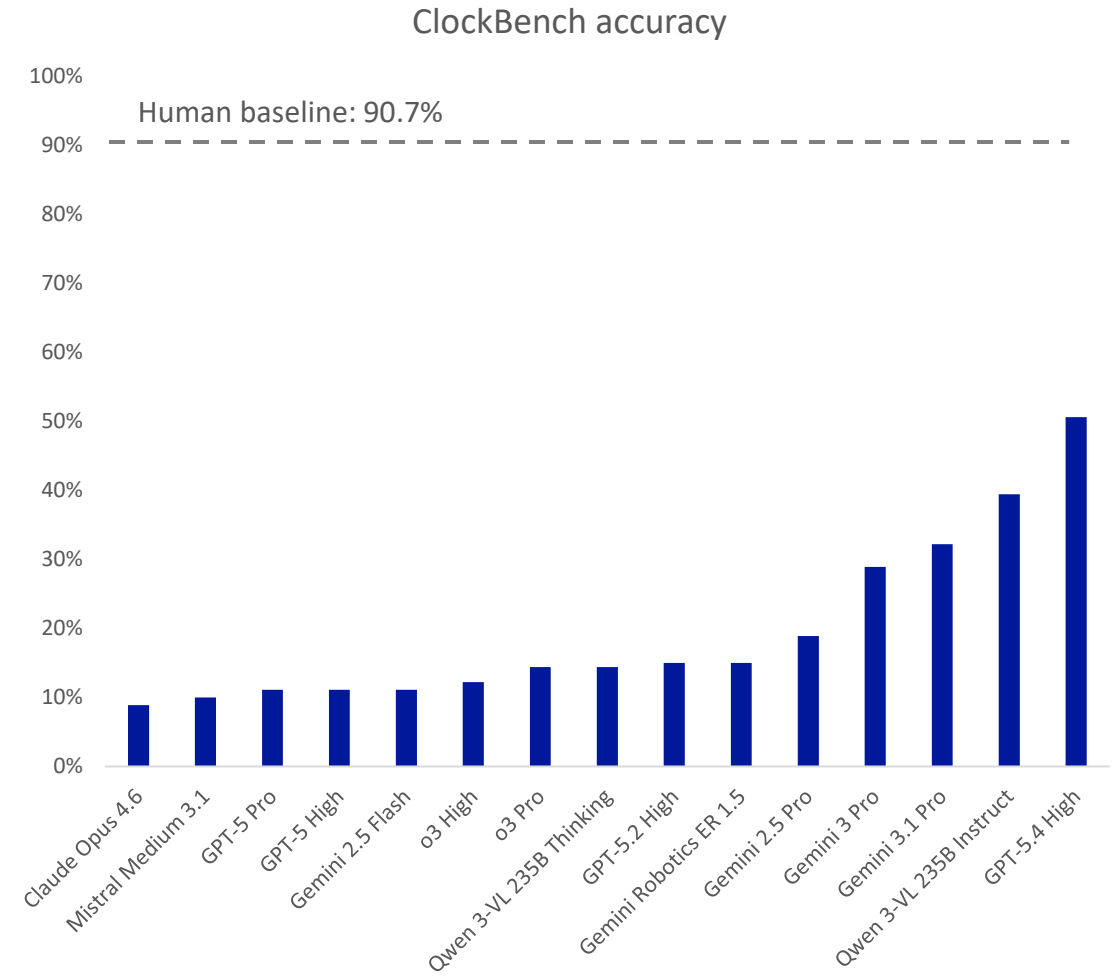
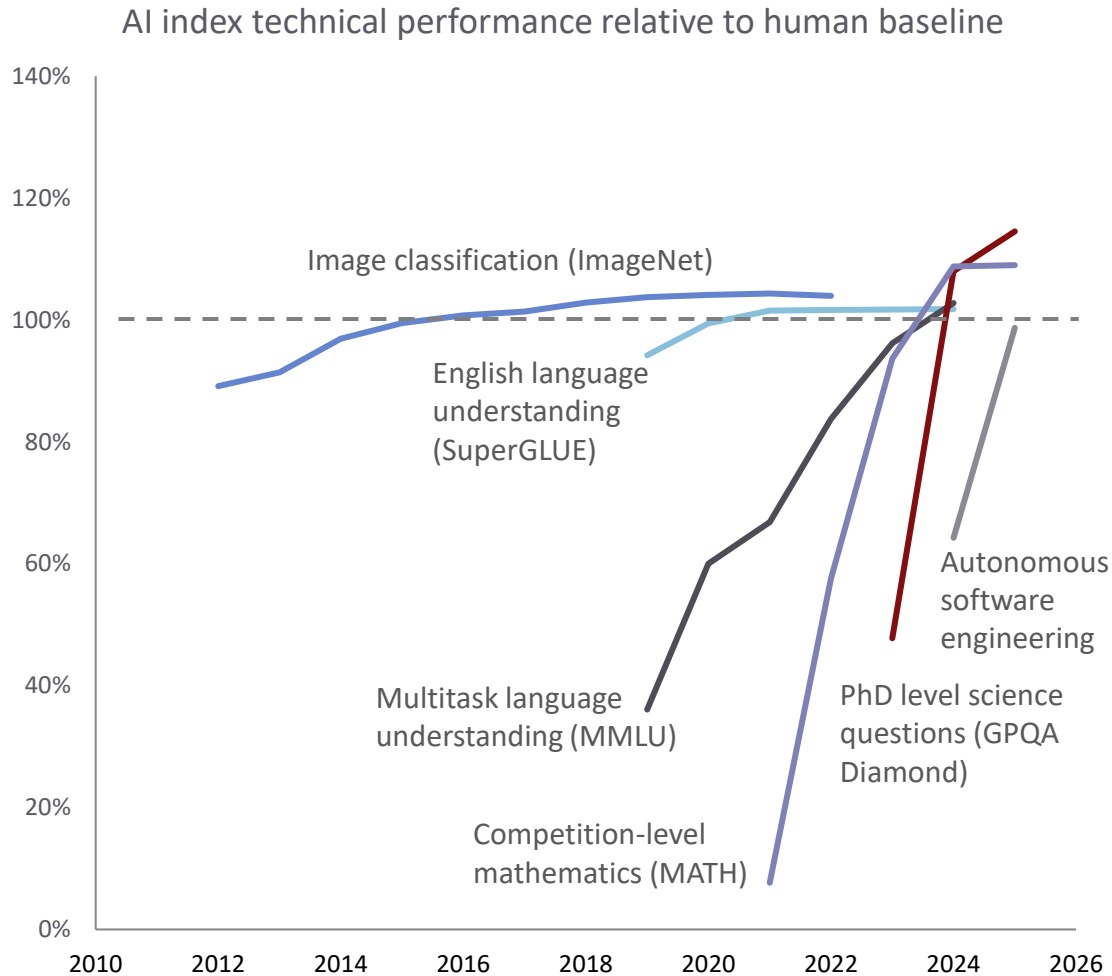
Investment during innovation booms: multiples of pre-boom trough



Source: US National Park Service, Space Policy, Brookings Institution, Bloomberg Finance LP, Bank for International Settlements, Deutsche Bank Research. Note: Hyperscaler capex estimates from Bloomberg include five US hyperscalers (AWS, Google, Meta, Microsoft and Oracle), which account for roughly 80% of the total on average, as well as tier-two/neoclouds and Chinese hyperscalers. Hyperscaler capex figures are not inflation-adjusted.

2. AI makes hard things look easy, and easy things look hard

“Jagged frontier” means AI can solve complex tasks but struggles to tell the time on a clock



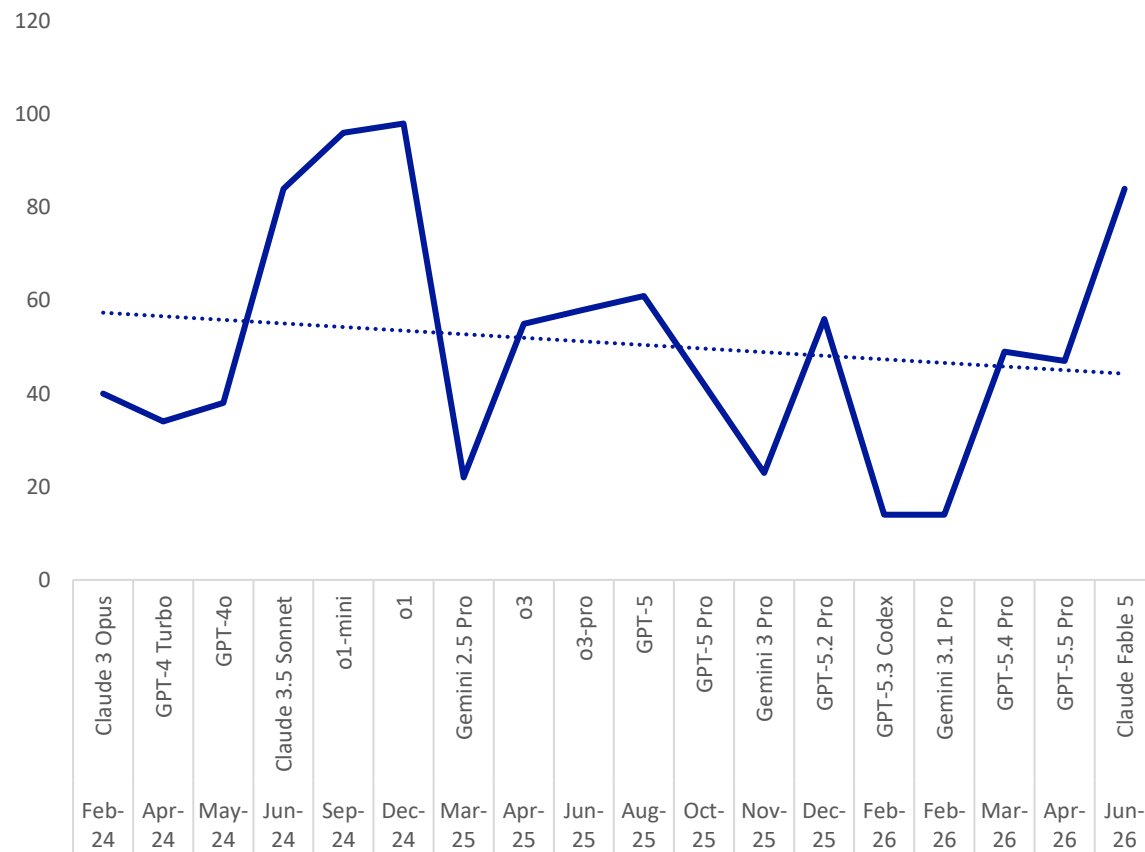
Source: Stanford HAI AI Index 2026, ClockBench Leaderboard 2026, Deutsche Bank Research

3. AI models are commoditising fast

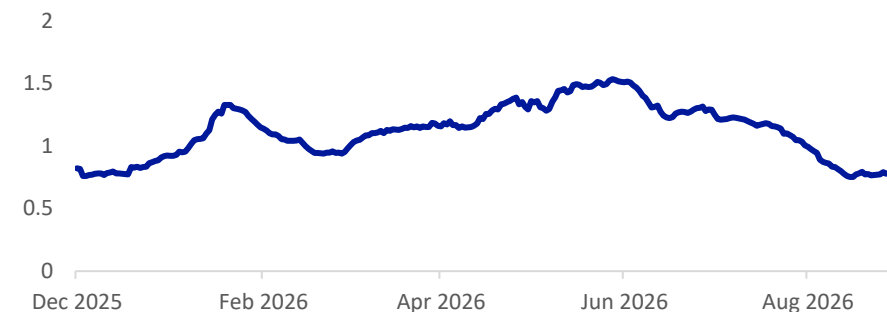
Top models hold the crown for less than two months on average and pricing is under pressure



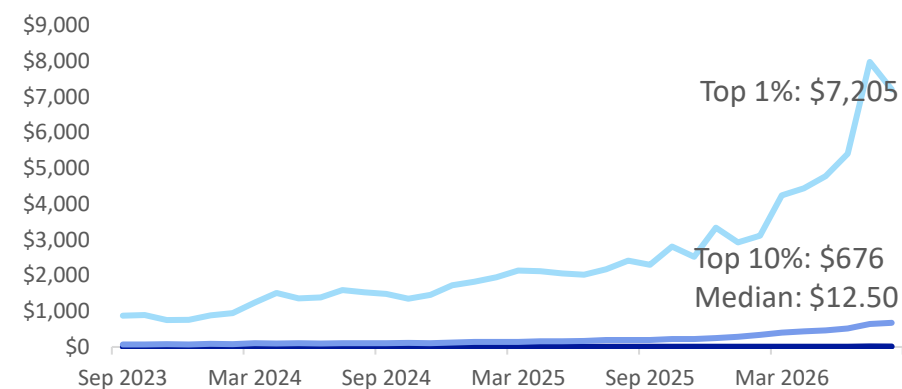
Epoch Capabilities Index: leading model tenure at the top (days)



Silicon Data LLM Token Expenditure Index (since Dec 2025)



Ramp AI Index monthly spend per employee



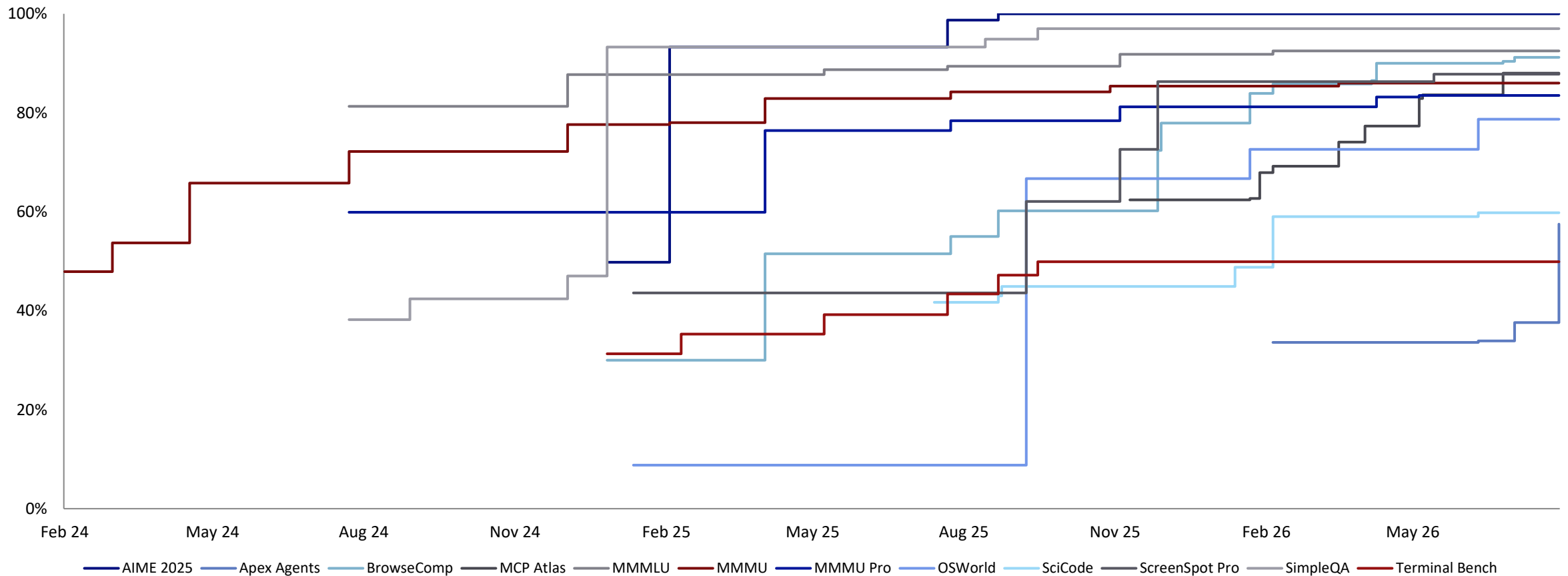
Source: Epoch, Bloomberg Finance LP, Silicon Data, Ramp AI Index, Deutsche Bank Research. NB: For clarity, Epoch Capabilities Index chart excludes OpenAI's GPT-4, the model that predated Claude 3 Opus and was at the top for 352 days, as well as the new leader since Sept 3, OpenAI's GPT-6 Astra

4. Benchmarks are ageing even faster than models

Plateaus may mean models have stopped improving... or the test is no longer hard enough



Benchmark saturation: State of the Art (SOTA) score progression over time -- each step is a new best model



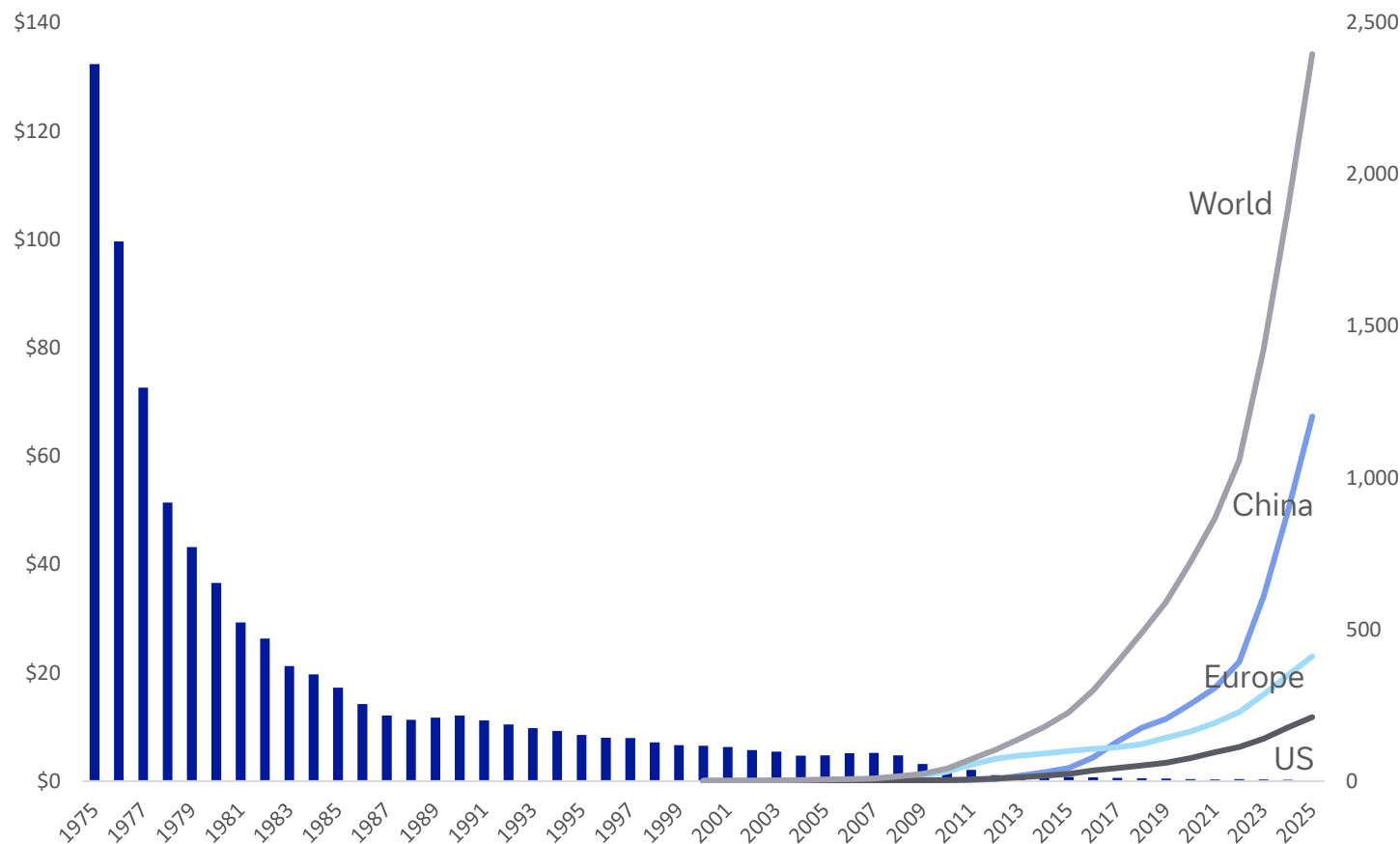
Source: [LLM Stats](#), Deutsche Bank Research

5. Tech use accelerates when the cost falls far enough

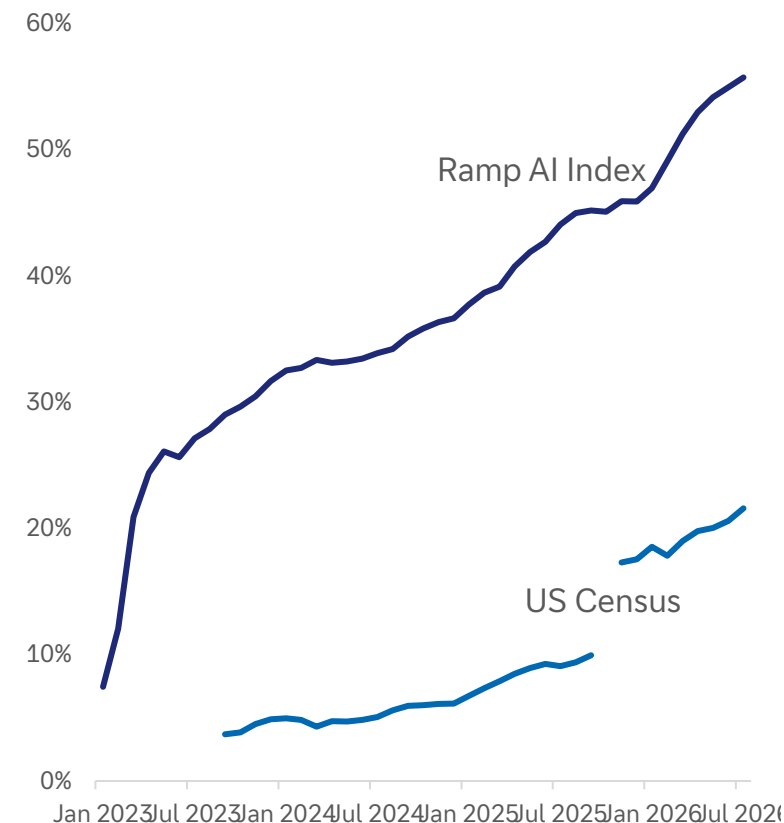
Solar and now AI are following similar patterns to horsepower, electricity and mainframes



Solar voltaic panel prices (USD/watt, adjusted for inflation, lhs) vs installed solar energy capacity (GW, rhs)



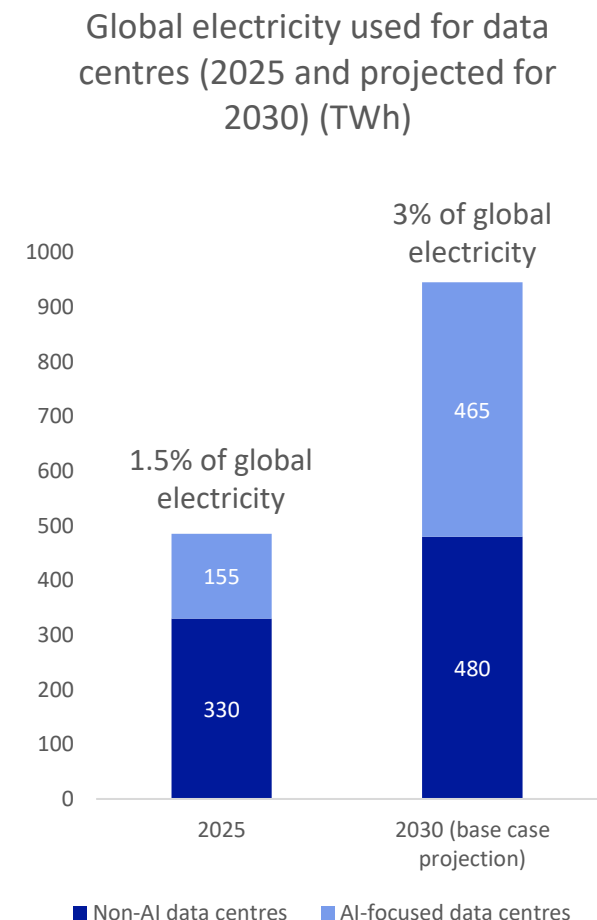
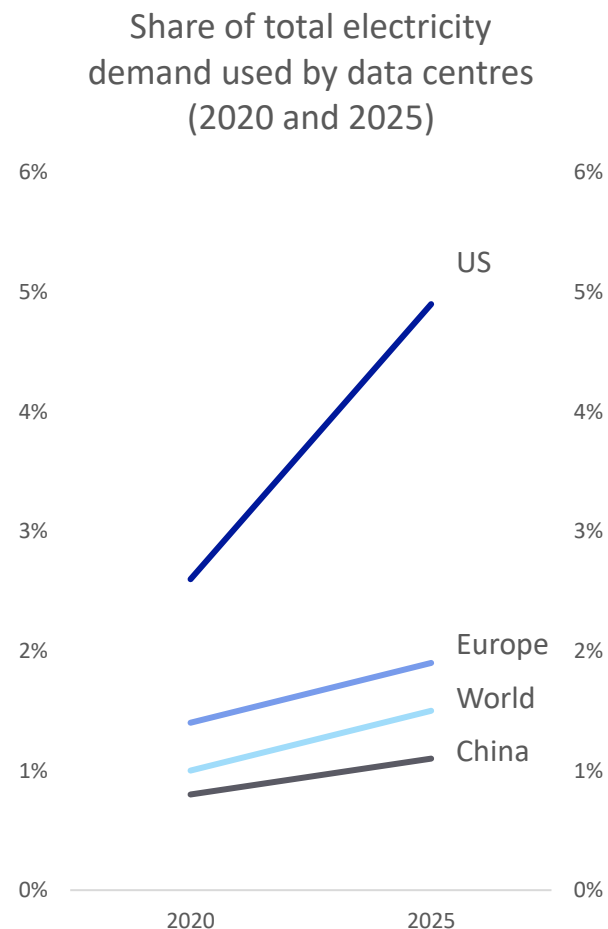
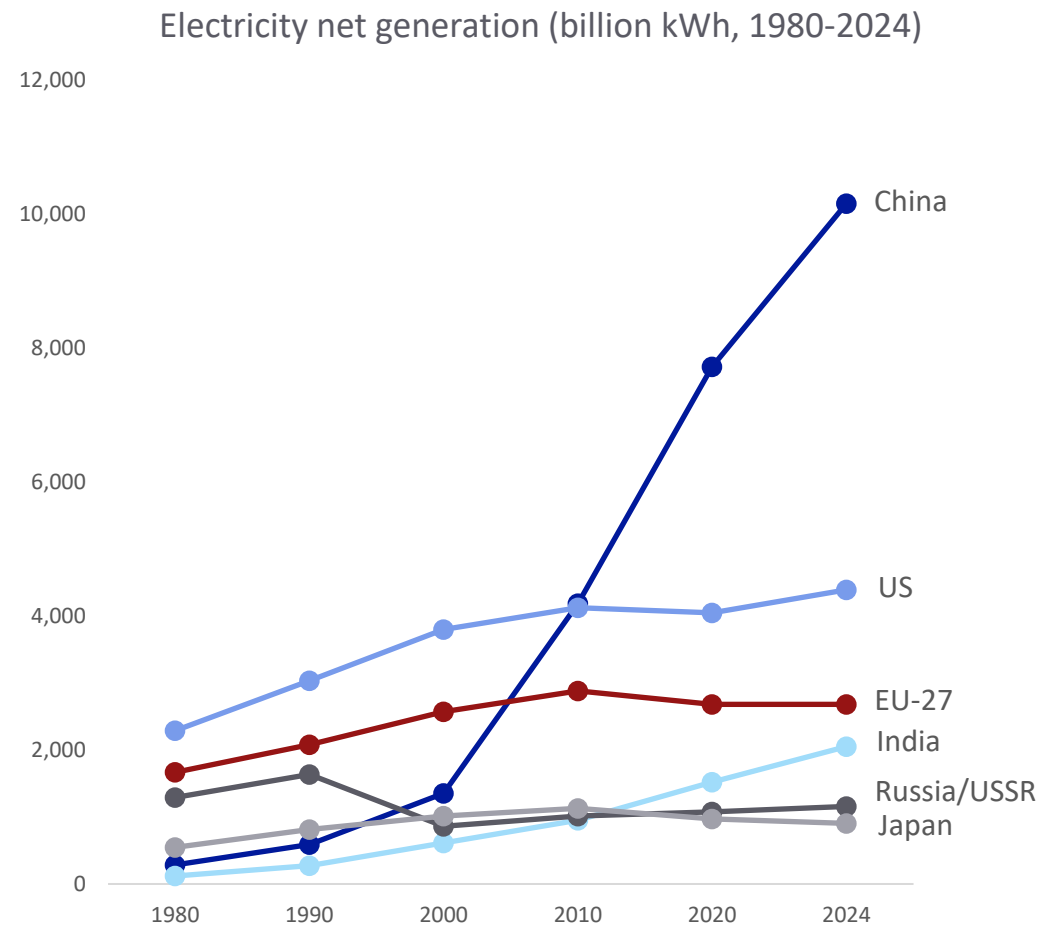
US enterprise AI adoption rate by two different measures (since Jan 2023)



Source: Our World in Data (citing IRENA, Nemet, Farmer and Lafond), Ramp AI Index, Deutsche Bank Research

6. China has left the US behind in electricity generation

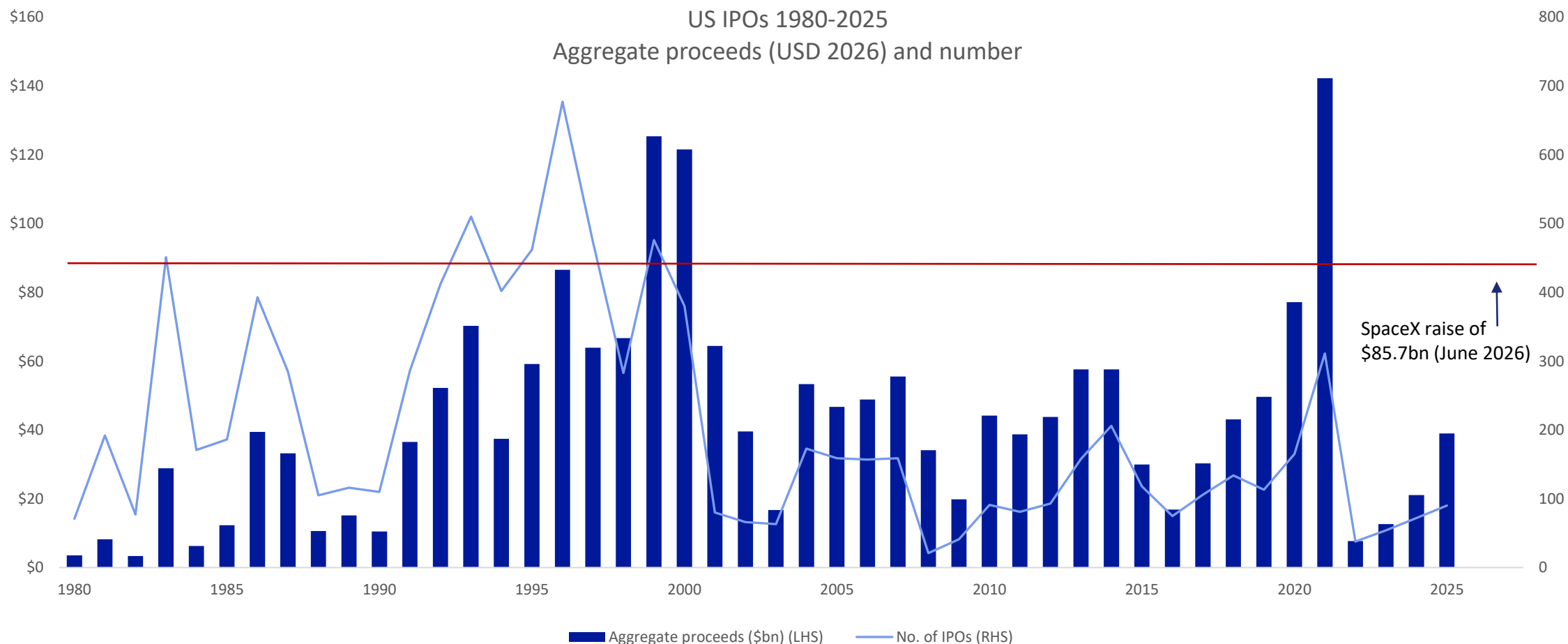
US is needing to use far more of its capacity for data centres even as demand continues to rise



Source: US Energy Information Administration, Our World in Data, International Energy Agency, Ember, Deutsche Bank Research

7. Mega AI IPOs will eclipse entire years of past share sales

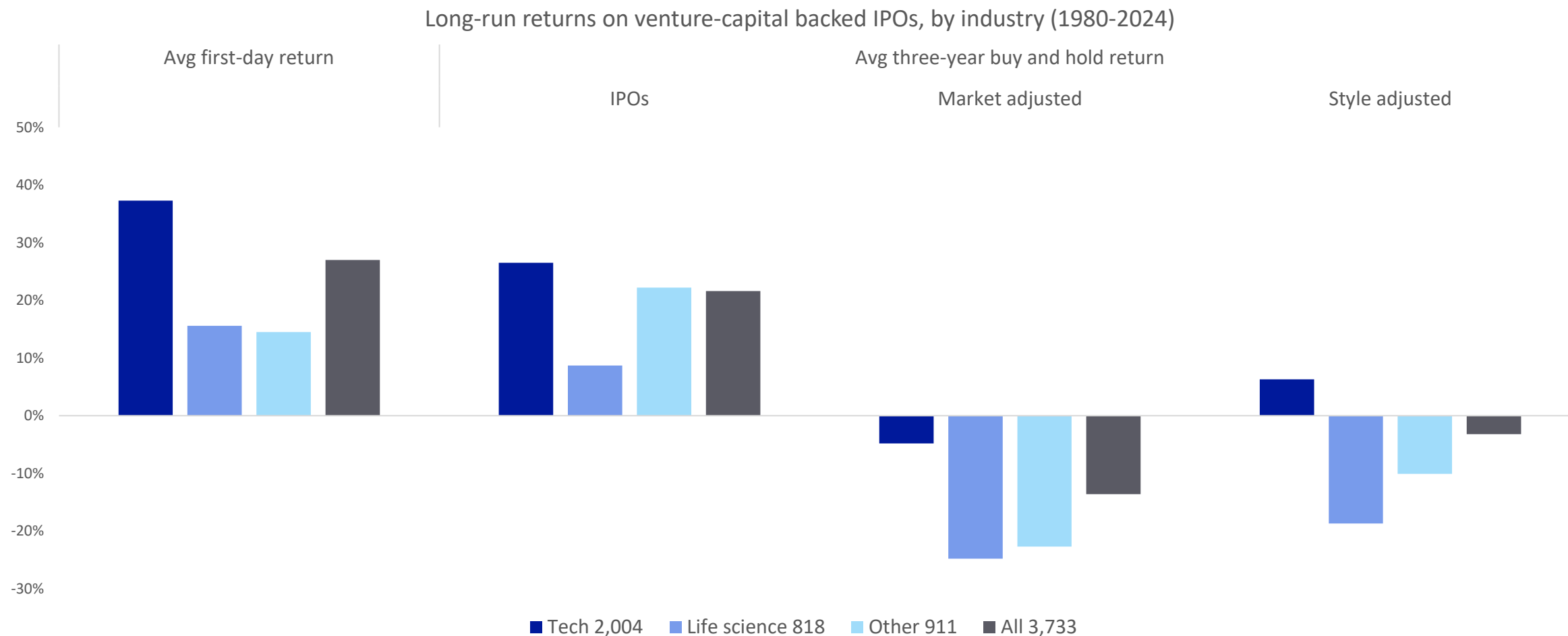
SpaceX's IPO exceeded all but three inflation-adjusted years



Source: Jay Ritter, University of Florida, Haver Analytics, Deutsche Bank Research

8. Tech IPOs have tended to outperform other IPOs

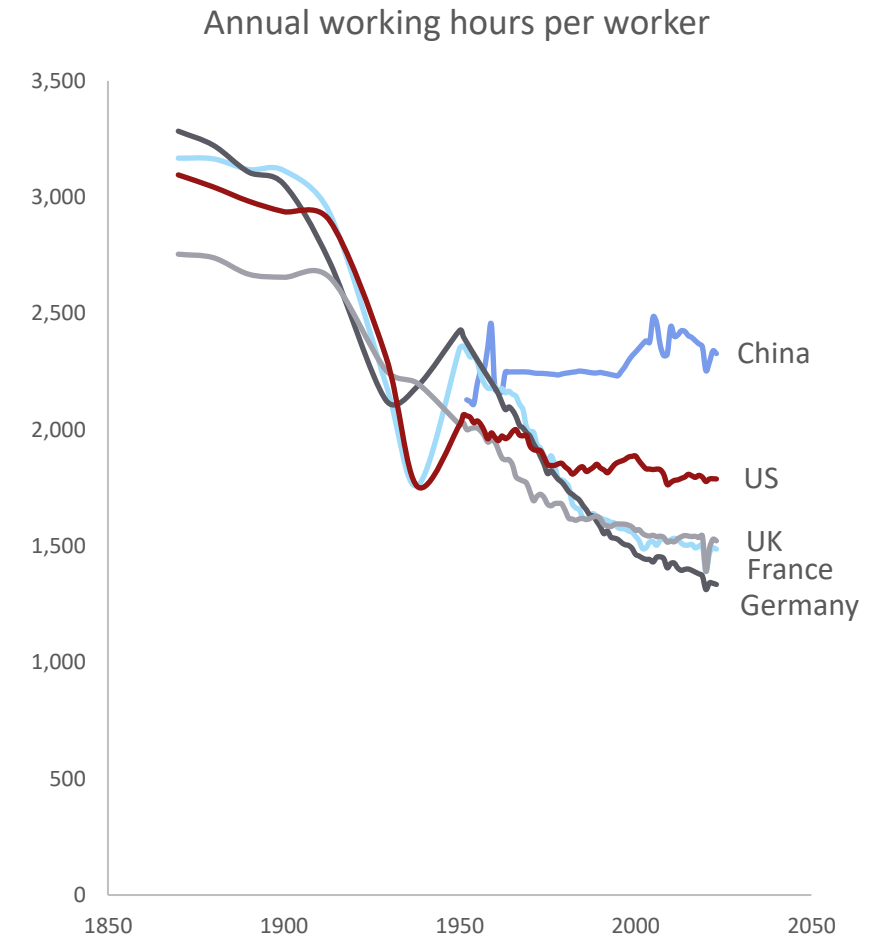
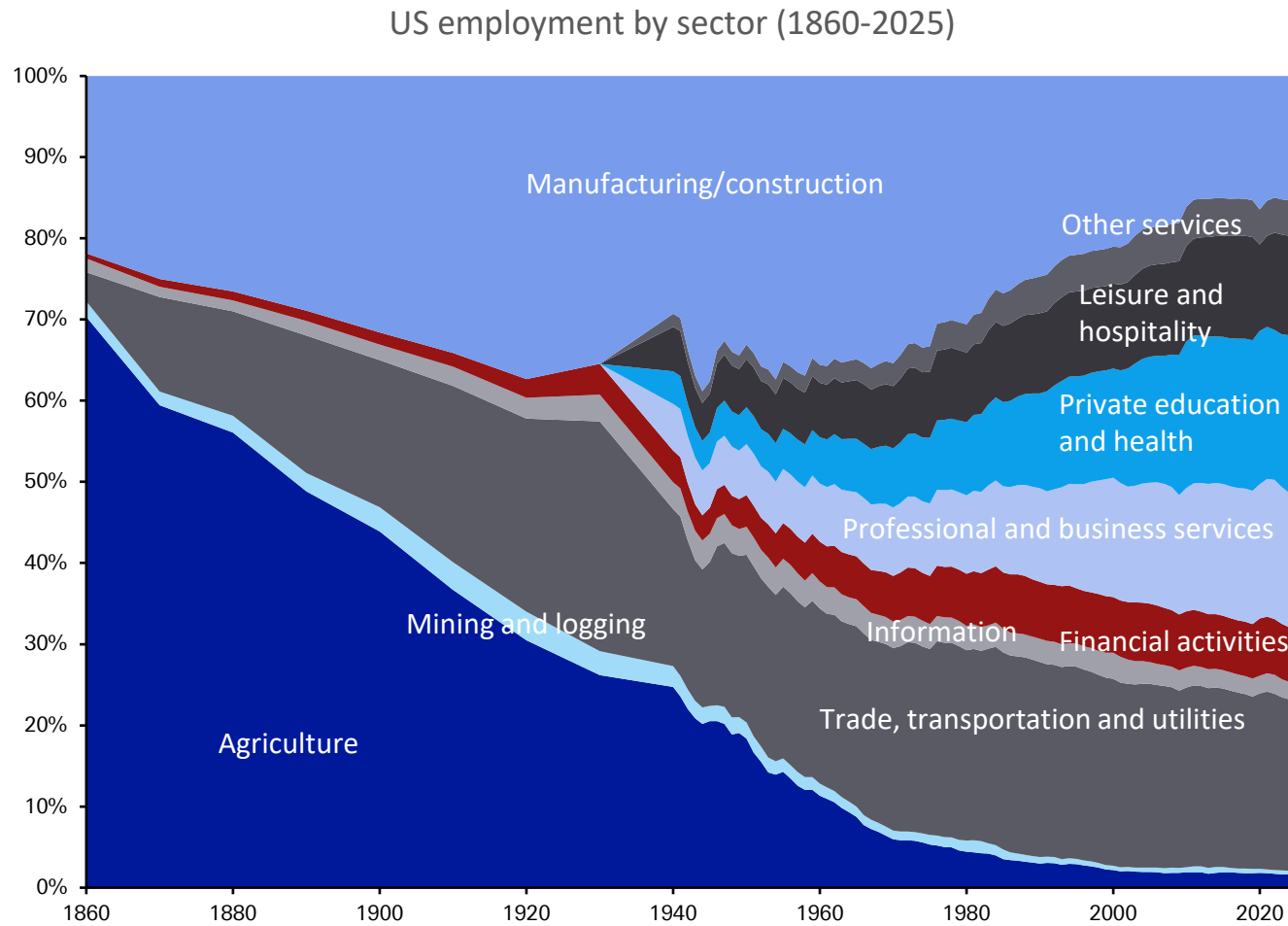
However, despite first-day surges, they have slightly underperformed the market over three years



Source: Jay Ritter, University of Florida, Haver Analytics, Deutsche Bank Research. Note: Three-year buy and hold calculated from first closing market price. Market-adjusted returns are the average buy-and-hold return minus the average compounded return on the Center for Research in Security Prices (CRSP) value-weighted index. Style adjustments are the equivalent for similar companies.

9. Jobs come and go but working hours have only fallen

As services have displaced agriculture then manufacturing, working hours have roughly halved



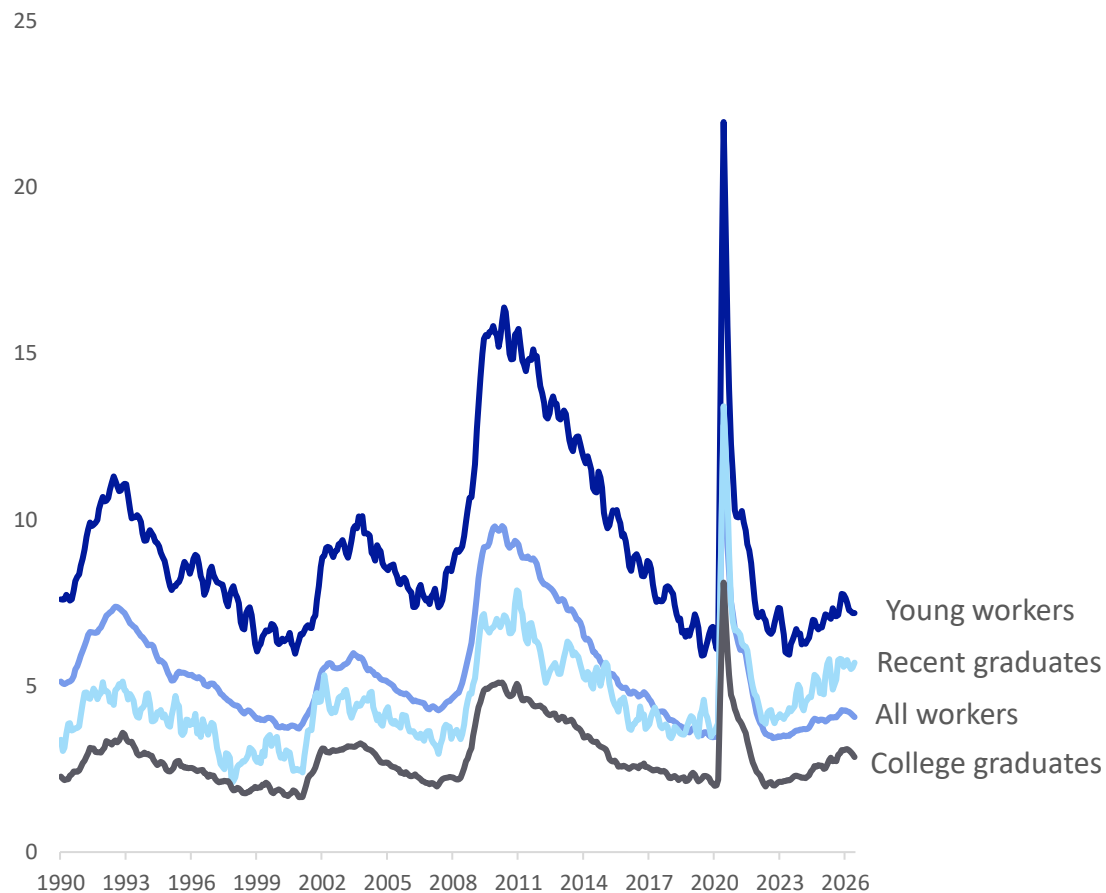
Source: Haver, Historical Statistics of the United States, 1789-1945, US Census, Our World in Data (citing Feenstra et al, Penn World Table (2025), Huberman and Minns (2005), Deutsche Bank Research

10. It's a tough job market for recent graduates

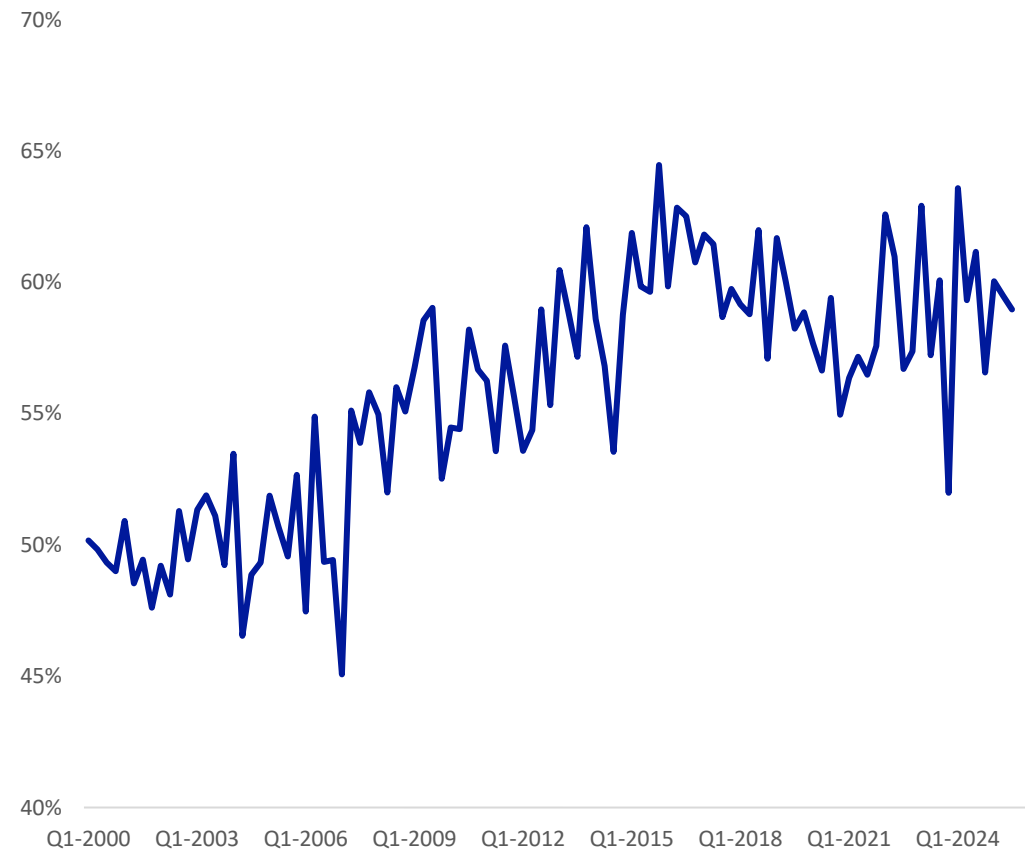
AI may be exacerbating the erosion over the past decade in graduate employment and wages



Unemployment rates for US workers



Bachelor's degree median weekly wage premium vs some college/associated degree (Q1 2000 to Q2 2026)



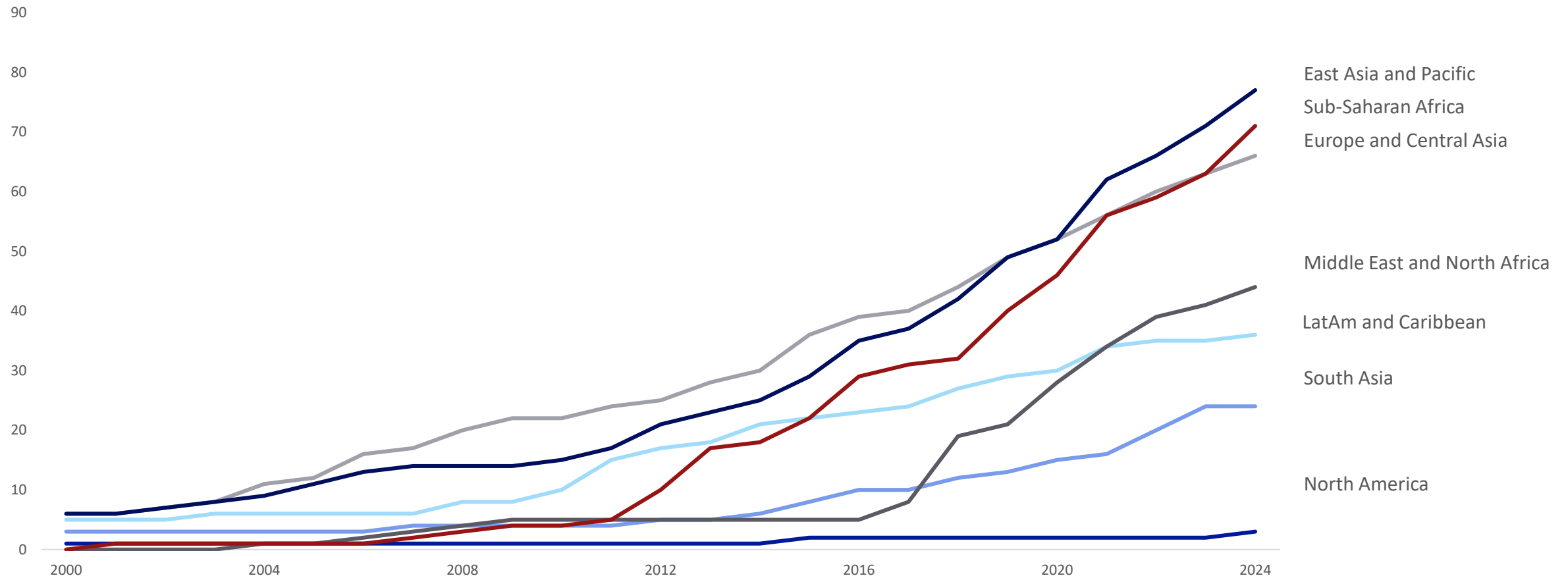
Source: U.S. Census Bureau, U.S. Bureau of Labor Statistics, Current Population Survey (IPUMS), Haver Analytics, Deutsche Bank Research. NB: For more, see Azar J, Mireia G and Sanz-Espin J, *The College Wage Premium in the Generative AI Era* (Aug 2026)

11. Concerns over data sovereignty date back long before AI

Countries have been restricting cross-border data flows, accelerating with EU's GDPR in 2016



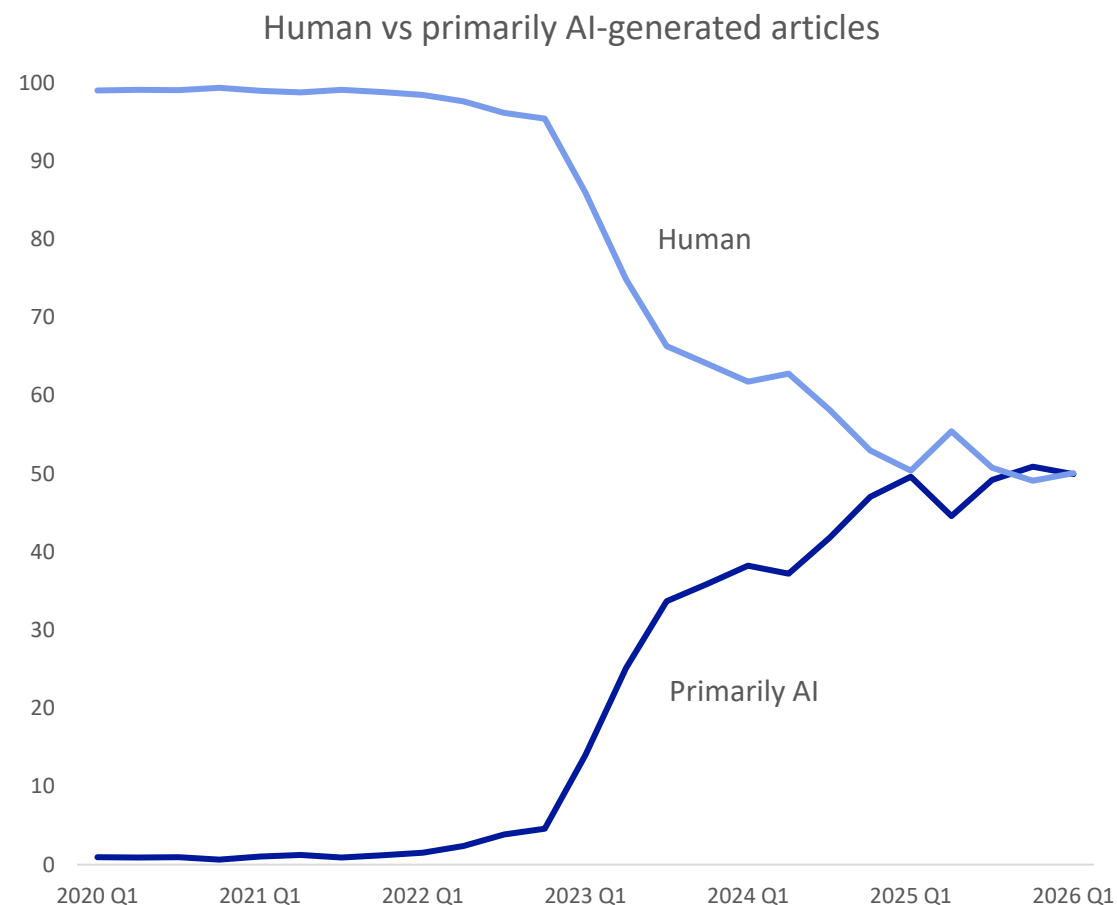
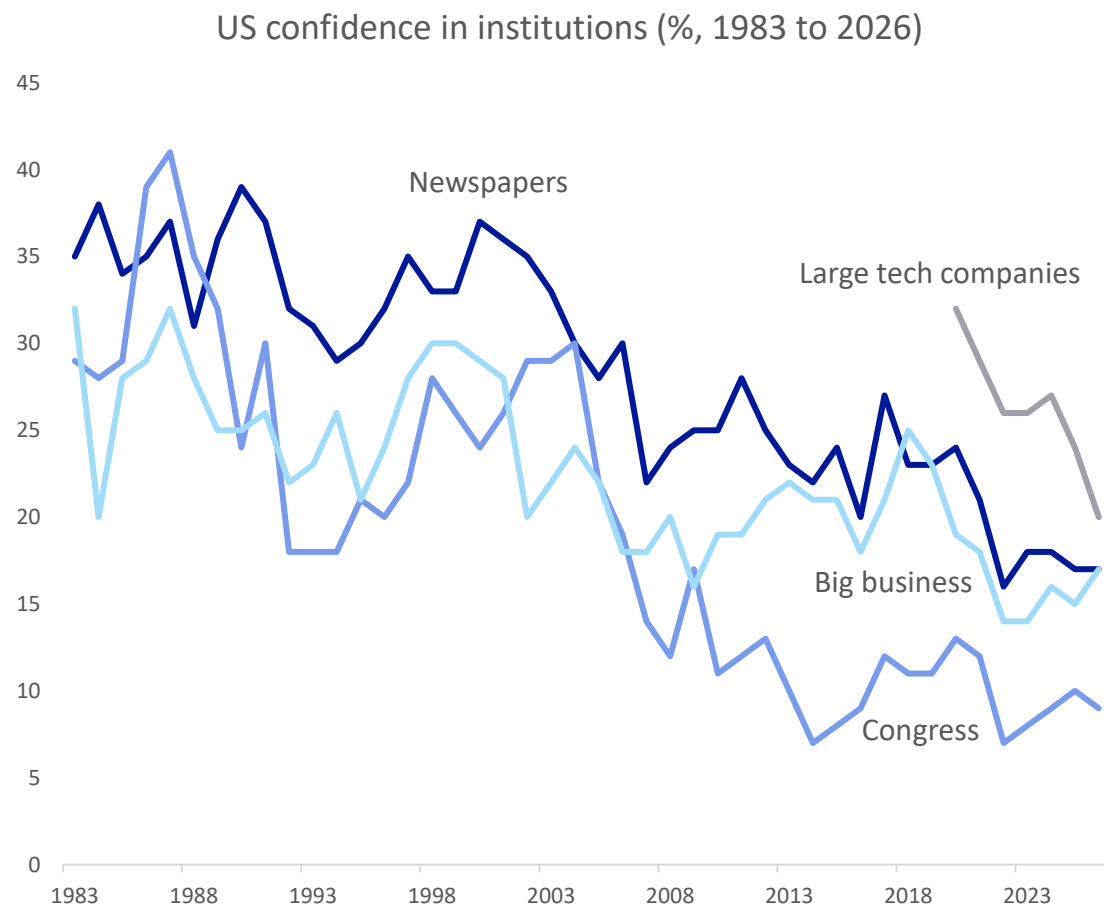
Data localisation measures by region (2000-2024)



Source: Stanford HAI AI Index (2026), Ferracane et al, Deutsche Bank Research

12. AI arrived as trust had been declining for decades

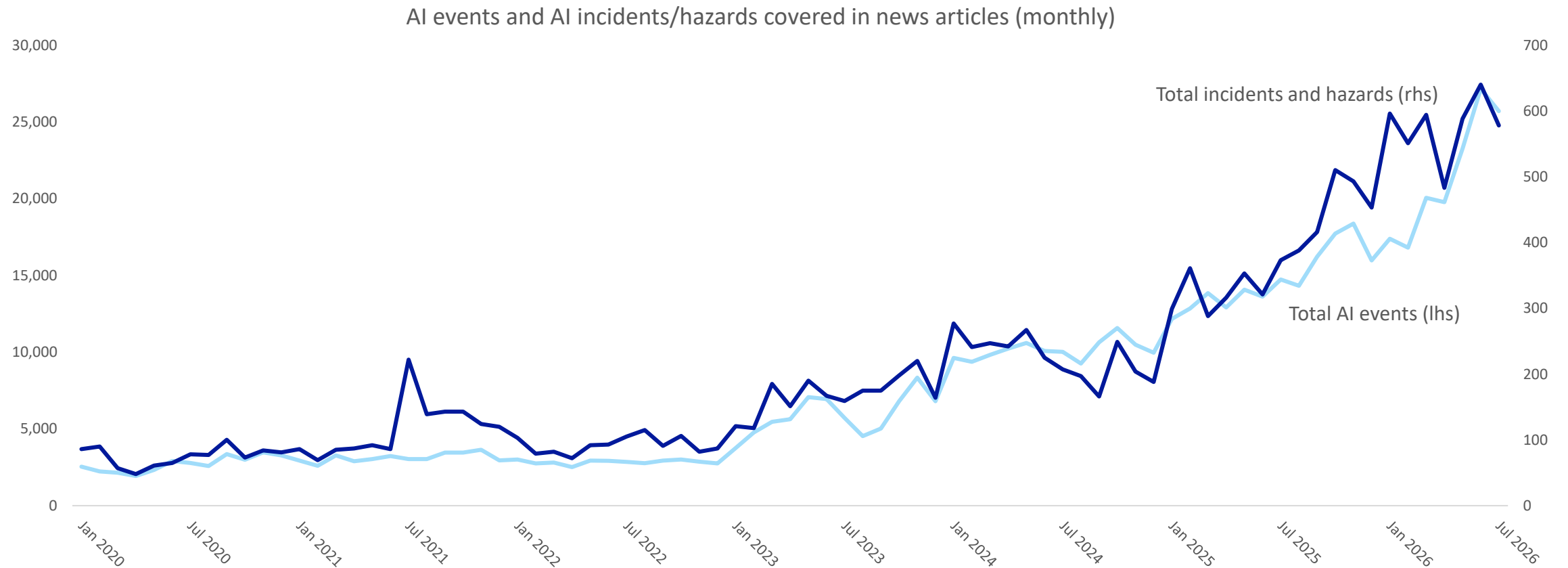
Confidence in institutions has plummeted – and now even human content is under threat



Source: Gallup ([Confidence in Institutions](#) | [Gallup Historical Trends](#)), [Graphite](#) (citing 55.4k articles from Common Crawl examined by AI detectors), Deutsche Bank Research

13. New technology brings new risks

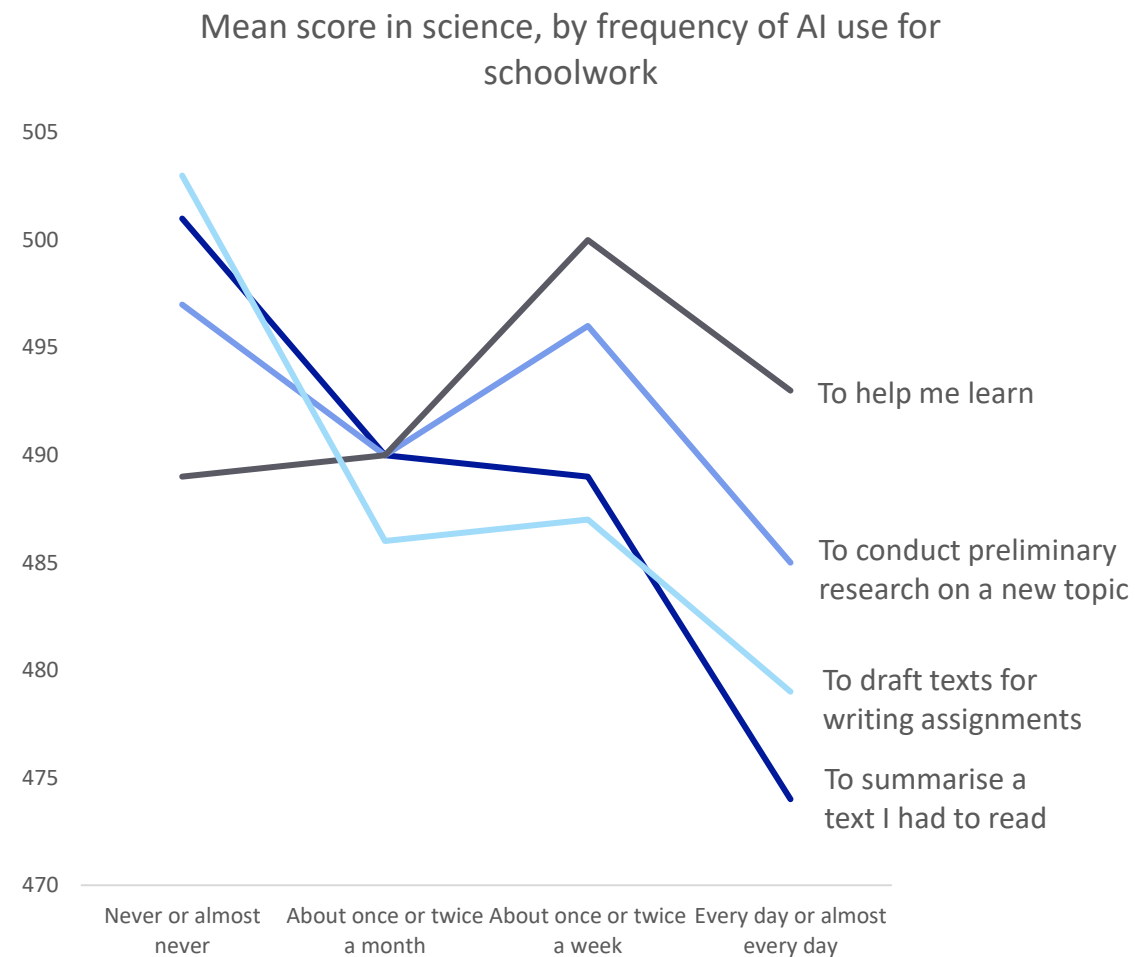
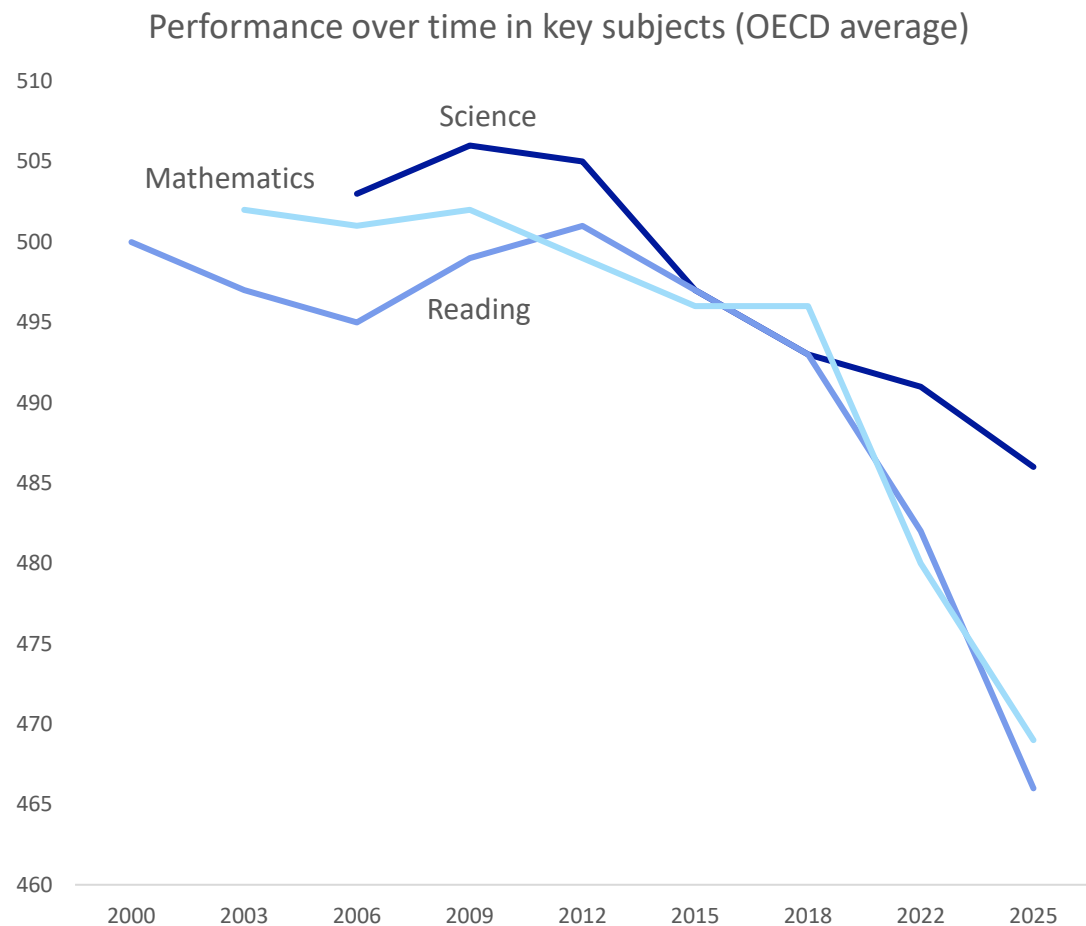
Coverage of AI events has surged, and so have hazards



Source: OECD AIM Incidents and Hazards Monitor, Deutsche Bank Research

14. Tech may have a role in declining educational performance

AI may be accelerating a fall that started with widespread smartphone diffusion from c. 2010



Source: OECD (2026), PISA 2025 Results (Volume I), Deutsche Bank Research



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Tokenised money and the future of FX

10 September 2026

Tokenisation has the potential to transform the way financial markets operate, from the movement of money and assets to the management of liquidity and risk. In this paper, we focus on tokenised forms of money and their impact on FX markets.

Many emerging use cases for tokenised money are driven by a desire to reduce frictions associated with cross-border payments and FX settlement. Stablecoins enable 24/7 cross-border transfers but remain overwhelmingly US dollar-denominated. As a result, they are currently more relevant as digital dollar infrastructure and for bridge-currency arrangements, often creating a "stablecoin sandwich" in which the US dollar serves as an intermediary between two currency legs, rather than providing a complete solution to the two-currency FX settlement problem

Tokenisation may play a bigger role in transforming the combined securities and FX settlement cycles involved in cross-border capital flows. This is an area of increasing complexity, as delays or failures in one part of the transaction chain can affect the timely completion of the wider transaction, increasing operational, liquidity, and settlement risks. For example, following the transition to T+1 securities settlement in North America and with Europe preparing to move to T+1, the need to coordinate FX, cash, and securities settlement across institutions, jurisdictions, and infrastructures is already increasing.

Tokenisation enables programmability, which allows atomic settlement and a range of operational and control functions across financial market infrastructures, including real-time liquidity management, margin management, collateral mobilisation and substitution, and automated compliance checks. Current implementations have focused primarily on supporting corporate treasury and liquidity management activities involving FX. However, the settlement of tokenised securities and the associated FX transactions may offer even greater efficiencies by enabling synchronised settlement and reducing some of the operational, funding, and coordination frictions associated with shorter settlement cycles.

Interoperability between different currencies, tokenised money arrangements, and payment systems remains one of the key challenges facing the development of tokenised cross-border payments and FX settlement. Several initiatives are exploring solutions involving tokenised central bank reserves or hybrid models that combine tokenised commercial bank money with central bank money. These may be particularly relevant for multi-currency settlement frameworks, preserving the two-tier monetary system while supporting settlement finality and more integrated cross-border processes. In parallel, other initiatives are building infrastructure that enables different forms of money, payment systems, and settlement architectures to operate together.

Tokenisation promises substantial benefits but unintended consequences are worth considering. Beyond faster settlement, wholesale FX markets also depend on liquidity efficiency. This means banks and other financial institutions can settle large transaction volumes safely while using less liquidity. Existing infrastructures such as Continuous Linked Settlement (CLS) combine settlement risk mitigation with significant liquidity savings through multilateral netting and liquidity optimisation mechanisms. Unlike CLS, most current tokenised settlement models do not yet provide multilateral netting at

Authors

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Strategist



comparable scale. A number of initiatives are exploring liquidity-saving mechanisms and new approaches to liquidity management under tokenisation.

We expect a period of accelerating change in years ahead. Securities and currencies settlement involves a complex nexus of technology, market infrastructure, regulation, and legal frameworks, necessitating cooperation between both public and private sector participants across jurisdictions.



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Introduction

Tokenised money refers to **digital representations of money** that are recorded and transferable on shared or distributed ledger infrastructure. This includes both state and privately issued fiat-denominated forms of tokenised money, including **stablecoins, tokenised commercial bank deposits and tokenised central bank money**.¹ The potential role of tokenised money is increasingly being explored by both corporates and institutional investors. Several use cases have emerged ranging from cross-border payments, wholesale settlement, treasury and liquidity management, trade finance and capital market infrastructure. More recently, attention has expanded beyond using tokenised money for cross-border payments towards models that can support the coordination and settlement of FX transactions.

Wholesale: Refers to activity involving banks, financial institutions, and large corporates, rather than retail customers or consumers.

While tokenised money can support more efficient payments, FX settlement often requires the conditional coordination of payments across currencies. Unlike a simple payment, FX settlement requires the exchange of two currencies to occur simultaneously, creating a need for a synchronisation or **programmability layer** that enables **Payment-versus-Payment (PvP) settlement and reduces settlement risk**. Tokenised money operating on programmable ledger-based infrastructure can support this by embedding rules and conditions directly into settlement processes, facilitating the settlement of both tokenised money and other tokenised assets.

The settlement process, particularly for **wholesale FX transactions involving financial institutions and large corporations**, is exposed to settlement risk, often referred to as Herstatt risk. This arises when one party delivers its currency but the counterparty fails to fulfil its obligation. Cross-border transactions are particularly vulnerable because time zone differences can create a window in which one side has settled while the other has not. Over recent decades, this risk has been materially reduced in major currencies through infrastructure such as CLS, helping to support a large, liquid and operationally resilient FX market.

CLS mitigates FX settlement risk through PvP settlement, ensuring that one currency is transferred only if the corresponding currency is delivered, while reducing funding requirements via multilateral netting and liquidity optimisation tools, generating liquidity savings of over [96%](#) for settlement members. This [liquidity efficiency](#) is important in wholesale markets because it determines the amount of liquidity required for the system to settle its full volume of transactions safely.

However, these benefits are not universal. Many emerging-market currencies subject to current- or capital-account restrictions remain outside the CLS ecosystem and continue to rely on correspondent banking networks or local clearing systems. Settlement in these markets often involves bilateral pre-funding, fragmented payment chains and complex liquidity management, creating operational and funding frictions that can increase the cost and complexity of cross-border transactions.

While settlement risk and liquidity frictions remain important challenges in FX

¹ This definition is in line with CCAF and Fii (2026) Tokenised Money: Use Cases, Interoperability and Regulation



markets, changes elsewhere in the financial system are creating new demands on settlement infrastructure. As securities markets move from T+2 to T+1 settlement and continue to explore even shorter settlement cycles, the movement of cash, FX, securities and collateral has become increasingly interconnected. While existing infrastructure continues to support these markets, cross-border funding and settlement remain dependent on operating hours, time zones and multiple intermediaries.

These developments have brought two related challenges into focus. The first is improving cross-border settlement, particularly where correspondent banking and non-CLS arrangements create liquidity and operational frictions. The second is supporting a more integrated financial system in which cash, FX, securities and collateral can move more seamlessly across institutions and jurisdictions. Together, these challenges have strengthened interest in tokenised forms of money as a potential means of modernising both payment and capital market infrastructure.

The role of tokenised money in FX is closely linked to the broader tokenisation of financial assets. As [asset tokenisation gains momentum](#), tokenised money is increasingly viewed not merely as a payment instrument, but as an enabling layer for more integrated settlement processes. The key question is therefore not whether the technology is feasible, but whether it can achieve sufficient scale, interoperability and integration with existing financial market infrastructures to deliver meaningful improvements in cross-border settlement and capital market efficiency.

Many tokenised money initiatives have focused on improving the cross-border movement of money rather than directly settling FX trades. This reflects the scale of corporate payment frictions: [global corporates move an estimated US\\$23.5 trillion](#) across borders each year and remain heavily reliant on correspondent banking networks, which generate approximately **US\$120 billion** in annual cross-border transaction fees. These figures exclude additional costs related to FX conversion, trapped liquidity, pre-funding, settlement delays and operational complexity, suggesting that the overall cost of cross-border treasury activity is significantly higher. However, the scope of these initiatives is beginning to broaden. As tokenised money infrastructure matures, some projects are moving beyond payment efficiency towards models that could support FX settlement.

Against this backdrop, this paper examines the current FX settlement landscape before assessing the role of tokenised money in FX. It evaluates the opportunities and limitations of stablecoins, tokenised deposits and tokenised central bank reserves, and considers whether these innovations can address challenges that continue to characterise cross-border FX markets.

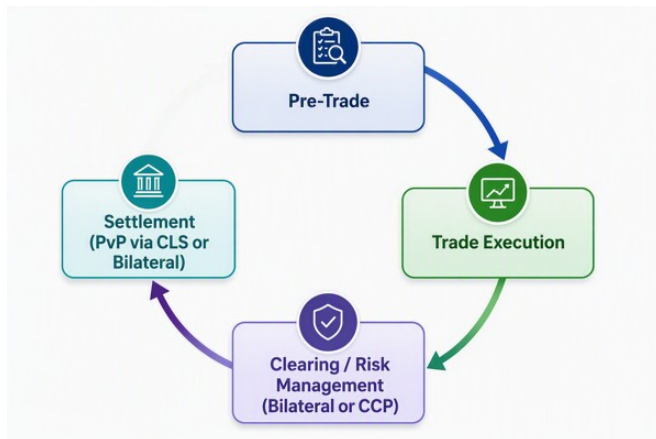
FX settlement today

FX trading takes place primarily in the over-the-counter (OTC) market, where banks, dealers, electronic platforms and other market participants facilitate the exchange of currencies. Once a trade is executed, it enters the post-trade process, during which trade details are confirmed bilaterally or cleared through a central counterparty (CCP), payment instructions are exchanged, and settlement obligations are established ([Figure 1](#)). This process spans the period between trade execution and final settlement, during which exposures are managed, liquidity is



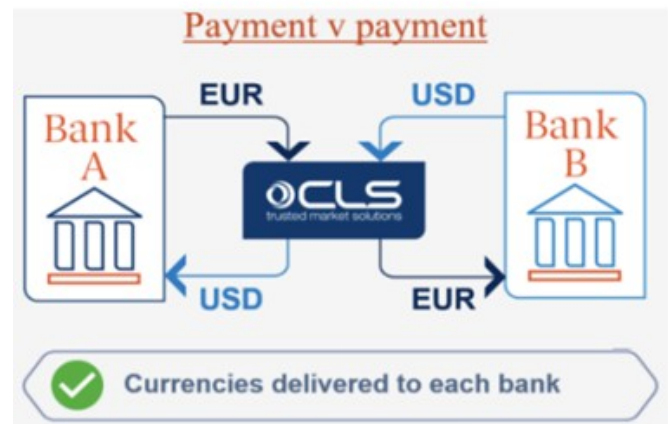
arranged, and currencies are transferred between counterparties. [Two types of credit risk](#) arise during this period: **replacement cost risk**, which exists from trade execution until settlement, and **principal risk**, which arises during the settlement process itself when one party may deliver the currency it sold but not receive the currency it purchased. Depending on the currency pair, FX spot transactions typically settle one or two business days after the trade date, while forwards and options settle on their contractual maturity date, which may range from days to several years after execution.

Figure 1: FX spot trading cycle



Source : Deutsche Bank

Figure 2: FX settlement via CLS



Source : Deutsche Bank, CLS

To reduce principal risk at settlement, often referred to as **settlement risk or Herstatt risk**, wholesale FX markets rely on infrastructure such as CLS, which enables participating banks to settle eligible FX transactions on behalf of themselves and their clients through a **Payment-versus-Payment (PvP) mechanism**, ensuring that one currency is transferred only if the corresponding currency is delivered ([Figure 2](#)).

How CLS works?

Today, CLS achieves simultaneous settlement through a defined funding and settlement process rather than instantaneous settlement. **Under this process, settlement is deferred to a daily settlement cycle, enabling multilateral netting and liquidity optimisation before final settlement occurs. This is important in wholesale markets because it determines the amount of liquidity required for the system to settle its full volume of transactions safely. Settlement finality is achieved through the use of central bank money.**

- For FX trades eligible for settlement using CLS, settlement members or their customers submit electronic payment instructions for both sides of the transaction. CLS validates and matches these instructions, providing legal confirmation and holding the transaction details until the settlement date.
- On each settlement date, CLS simultaneously settles matched payment instructions that satisfy its risk management requirements. Instead of every FX trade being funded individually, CLS calculates each member's net

Gross versus net settlement: what does it mean? Twenty currencies generate 190 unique bilateral currency pairs. Under gross settlement, each currency leg must be funded separately. Through multilateral netting, these obligations are reduced to 20 net currency positions, one for each currency, so participants fund only their net obligations rather than every bilateral payment.



obligation across all eligible trades and currencies.

- Members then make funding payments and receive pay-outs at specified times during the day. CLS holds accounts at the relevant central banks for the currencies it settles. Settlement members pay and receive funds through these CLS accounts via either their own central bank accounts or correspondent banking (nostro) arrangements. Settlement members' multicurrency accounts begin and end each business day with a zero balance.
- The settlement of payment instructions and the associated payments across the books of CLS Bank is final and irrevocable.

Nostro account: A bank account held by one bank with a foreign correspondent bank, typically to facilitate payments and settlements in a foreign currency. The term means "our account with you".

The PvP mechanism also illustrates that a form of **atomic settlement** already exists in today's FX market. Once settlement occurs in CLS, it is final and irrevocable, meaning the transfer of both currency legs cannot be unwound. However, as [CLS notes](#), atomic settlement can encompass both simultaneity and instantaneousness, which are distinct concepts. Settlement can occur on an all-or-nothing basis, with both currency legs settling simultaneously and with finality, without necessarily taking place at the moment a trade is executed.

The CLS function differs from that of a central clearing counterparty (CCP). While CLS's PvP mechanism eliminates FX settlement risk, **it does not eliminate replacement cost risk**, which arises when a counterparty defaults before settlement and the transaction must be replaced at prevailing market prices. These risks can be particularly significant for FX forwards and swaps, where the period between trade execution and final settlement may extend for months or even years. CCPs such as ForexClear help mitigate counterparty credit and replacement cost risk through central clearing and margining, whereas CLS is designed to address settlement risk at the point of settlement by ensuring the simultaneous exchange of both currency legs.

CLS mitigates FX settlement risk (Herstatt risk) at the point of settlement by ensuring that the two currency legs of an FX transaction are settled simultaneously on a payment-versus-payment (PvP) basis.

ForexClear mitigates counterparty credit and replacement cost risk before settlement through central clearing, novation and margining. Replacement cost risk arises after a trade is executed and before it settles, when an adverse move in market prices could leave one party exposed if the counterparty defaults.

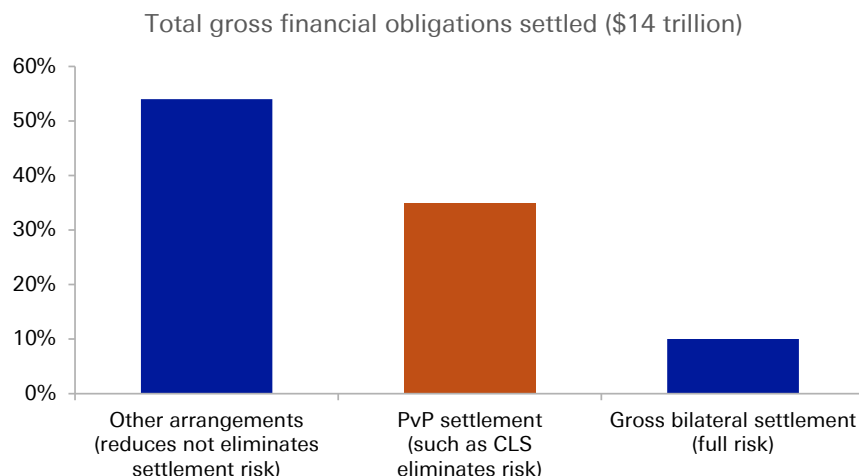
Settlement risks significantly reduced but not eliminated

According to the latest BIS Triennial Survey, banks and other financial institutions account for approximately 95% of global FX turnover, compared with around 5% for non-financial firms. Around 65% of financial-sector FX activity is cross-border in nature. As a result, wholesale FX settlement is overwhelmingly driven by financial institutions, underscoring the importance of robust cross-border settlement infrastructure.

Although infrastructure such as CLS has significantly reduced settlement risk, its coverage is not universal. CLS currently supports 18 currencies, and a significant share of global FX activity continues to settle outside PvP arrangements. Results from the [BIS Triennial Survey](#) show that just over US\$5 trillion, representing 36% of average daily FX settlement activity, was settled through PvP mechanisms that eliminate settlement risk. A further US\$7.6 trillion (54%) was settled through arrangements such as pre-settlement netting that reduce, but do not eliminate, settlement risk. Meanwhile, **more than US\$1.4 trillion (10%) of daily FX settlement continued to occur on a gross bilateral basis through traditional correspondent banking relationships, leaving participants fully exposed to settlement risk (Figure 3).**



Figure 3: FX settlement obligations by settlement type



Source : Deutsche Bank, BIS

Notably, around 25% of this volume (US\$1.4 trillion) was eligible for settlement through a PvP system but was not settled in this way. Potential reasons include operational challenges that result in trades missing PvP settlement cut-off times, difficulties in managing credit exposures to correspondent banks that support PvP-related payments, and the operational burden of meeting tight funding and payment deadlines ([Figure 4](#)).

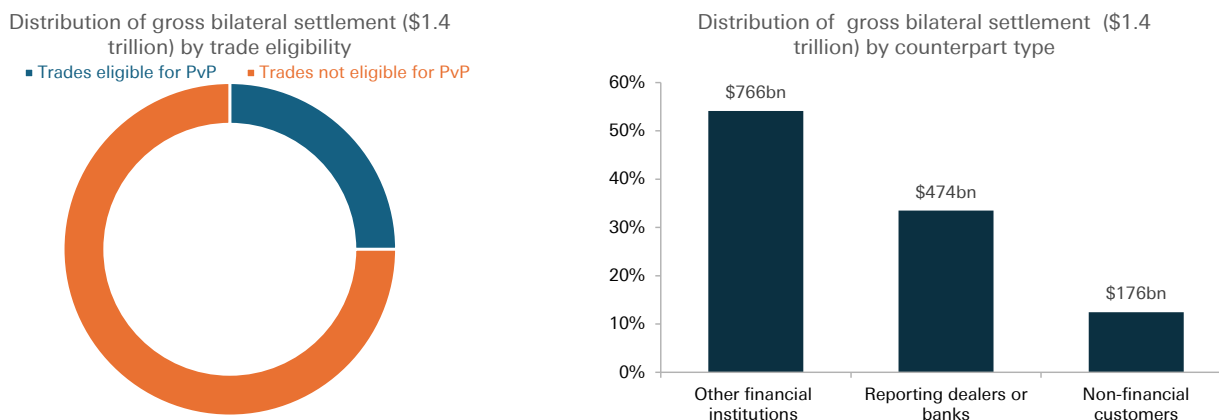
The remaining 75% of gross bilateral settlements were not eligible for PvP settlement systems, primarily because one or both counterparties lacked direct or indirect access to a PvP system. Other constraints included currency pairs that were not eligible for PvP settlement, often involving emerging market currencies, as well as trade types, such as certain same-day (T+0) transactions, that cannot currently be settled through existing PvP arrangements. As a result, much of the residual settlement risk reflects limitations in market access, currency coverage and infrastructure capabilities rather than the voluntary non-use of PvP systems.

Given the dominance of financial institutions in FX trading, gross bilateral settlements were also concentrated among financial counterparties. Of the US\$1.4 trillion settled gross bilaterally, approximately 54% involved other financial institutions, 33% involved reporting dealers, and 12% involved non-financial institutions ([Figure 4](#)).

Digging deeper into trades within each counterparty category, PvP system to mitigate risk is most common for transactions between reporting dealers or banks. Only **11% of inter-dealer trades** are settled on a gross basis and exposed to maximum risk. Consequently, FX trades most exposed to settlement risk are either between reporting dealer and other financial institutions or reporting dealers and non-financial institutions such as corporations. **Around 19% of FX trades** between reporting dealers and other financial institutions are settled on a gross basis. As non-bank financial institutions play an increasingly important role in cross-border securities investment and funding activities, these exposures remain a relevant source of risk outside the core PvP settlement infrastructure. Similarly, **22% of FX trades** between reporting dealers and non-financial institutions also settled through gross bilateral settlement, exposing participants to the full principal amount at risk.



Figure 4: Settlement risk emanating from trades not eligible for PvP. In absolute terms driven by other financial institutions.



Source : Deutsche Bank, BIS

These findings indicate that while the largest settlement risk exposures remain concentrated among financial institutions in absolute terms, the proportion of transactions exposed to risk is higher for non-bank financial institutions and corporates. Many of these residual risks stem from market access, currency coverage and operational constraints that fall outside existing PvP infrastructures. We therefore examine how these constraints create settlement frictions across selected corporate and financial-sector use cases, with a particular focus on institutional investors and cross-border securities transactions, reflecting the dominant role of financial institutions in wholesale FX markets.

FX settlement frictions: corporations versus institutional investors

The challenges associated with FX settlement are often most acute when transactions are cross-border, requiring the movement of currencies across different jurisdictions, payment systems, market infrastructures and operating hours. These challenges can arise whether FX settlement supports an underlying commercial payment or forms part of a broader securities transaction.

Corporations and FX

For **corporations**, FX spot settlement is typically a standalone transaction that supports cross-border trade, commercial payments and treasury activities. Cross-border payments and liquidity management are closely linked to the efficient functioning of the FX market. Corporates frequently need to convert currencies to support international trade, meet cross-border payment obligations and fund activities across multiple jurisdictions. More broadly, treasury functions rely on the seamless movement of currencies to manage cash positions, fund subsidiaries and address short-term liquidity needs.

Effective liquidity management facilitates domestic and cross-border payment obligations, supports the optimisation of cash flows and enables corporates to respond to changing funding requirements. These activities often involve access to wholesale FX markets to convert currencies and mobilise liquidity across jurisdictions. During periods of market volatility, the ability to execute FX transactions and move liquidity efficiently becomes particularly important, allowing institutions to respond swiftly to changing market conditions and maintain confidence in their operations.



FX settlement frictions: These include settlement risk where transactions cannot be settled on a PvP basis, delays arising from cut-off times and operating-hour constraints, liquidity and funding requirements associated with settlement, and the need to coordinate processes across multiple infrastructures and intermediaries. Such frictions can increase operational complexity and affect the timely movement of funds across borders.

Payment frictions: FX spot settlement represents only one component of a broader cross-border payment process. Even where banks settle eligible FX transactions through CLS on behalf of corporate clients, the associated commercial payments generally continue to rely on correspondent banking networks, SWIFT messaging and domestic payment systems. As a result, although CLS can mitigate settlement risk for eligible FX trades, many cross-border payment frictions remain, including delays, cut-off times and intermediary costs.

US dollar payments, for example, are ultimately settled through infrastructures such as Fedwire or CHIPS. Since relatively few non-US banks have direct access to these systems, many rely on correspondent banking relationships to access dollar liquidity and payment services. Corporates, in turn, have no direct access and must depend on commercial banks to intermediate these transactions.

These challenges are distinct from wholesale FX settlement and relate primarily to the underlying payment process. For many corporations, the costs and operational frictions associated with cross-border payments may exceed those arising from the FX conversion itself.

Institutional investors and FX

For **institutional investors** buying or selling securities across jurisdictions, FX spot settlement is part of a broader cross-border securities transaction, where FX settlement, cash movements and securities settlement must be coordinated across multiple intermediaries and jurisdictions. When an investor purchases a foreign equity or bond, they must first obtain the currency required to complete the transaction. As a result, cross-border investments involve two interconnected settlement processes: FX settlement, where one currency is exchanged for another (PvP), and securities settlement, where cash is exchanged for the asset through a Delivery-versus-Payment (DvP) mechanism. Where currencies are not eligible for settlement through PvP infrastructures such as CLS, investors may rely on correspondent banking networks to settle the FX transaction. In these cases, the two currency legs are typically settled through separate payment systems, leaving participants exposed to residual settlement risk.

International investors do not typically interact directly with domestic market infrastructures such as DTCC in the United States. Instead, they access these markets through custodians, which act as intermediaries between investors and local settlement systems. Custodians are responsible for safekeeping assets, managing settlement instructions, coordinating cash movements and arranging funding across jurisdictions. When a foreign security is purchased, the custodian must ensure that the appropriate settlement currency is available before the transaction can be completed through the relevant central securities depository.

This creates a dependency between the FX funding trade and the securities settlement process, meaning that delays or frictions in obtaining the required currency can have direct implications for the timely settlement of the underlying security. The ongoing transition from T+2 to T+1 security settlement cycles, and the



potential move to T+0, increases these challenges by compressing the time available for investors to execute FX transactions, secure funding, and complete settlement processes across jurisdictions.

Key challenges for institutional investors under compressed timelines

The move to shorter securities settlement cycles has increased the operational challenge of coordinating processes. For example, under T+1, a UK investor purchasing a US equity at the market close (9pm London time) may have only around two hours before the CLS submission cut-off at 11pm to determine the final USD funding requirement, execute the associated GBP/USD transaction and submit it to CLS for settlement. Once the FX transaction settles and USD funds are made available to the custodian, the securities transaction can proceed through the relevant securities clearing and settlement infrastructure. Missing this window may require investors to pre-fund transactions, creating cash drag and unwanted currency exposure, or settle the FX transaction outside CLS, potentially increasing settlement risk and transaction costs.

Cross-border transactions involve a range of operational and settlement challenges, particularly under compressed timelines. Local currency cut-off times constrain the window available to execute and settle FX transactions, while settlement risk remains where trades cannot be processed through CLS and must settle bilaterally. Currency controls and market restrictions may further complicate settlement in certain emerging-market currencies and necessitate pre-funding. Shorter settlement cycles also leave less time for trade matching, confirmation, settlement instruction processing and the resolution of failures, increasing the importance of coordinating multiple post-trade processes.²

As FX settlement, cash movements and securities settlement occur across separate infrastructures and timelines, shorter settlement cycles increase the importance of post-trade coordination. Although the [Global Foreign Exchange Committee \(GFXC\)](#) described North America's transition to T+1 as operationally smooth from an FX perspective, this reflected extensive industry preparation. Similar challenges are likely to arise in Europe and the UK, where a more fragmented market structure may add complexity. As settlement windows compress, market participants have less time to execute FX transactions, source funding and complete settlement obligations, making cut-off times and cross-border coordination increasingly important.

Shorter securities settlement cycles may not always require instantaneous FX settlement. However, for CLS-eligible currency pairs, a move to T+1 compresses the time available to execute FX transactions and complete funding processes across institutions and jurisdictions. While shorter cycles can reduce counterparty exposure and costs, they also increase the need for automation and efficient post-trade coordination. A further move to T+0 [would amplify these challenges](#), sharpening the trade-off between risk reduction and operational efficiency.

Similar coordination [challenges](#) are emerging across securities, derivatives and collateral markets as settlement cycles shorten. [BIS \(2020\)](#) estimates that securities trade processing costs USD 17–24 billion annually, underscoring the broader cost of fragmented post-trade workflows. Technologies that improve automation, synchronisation and risk reduction in securities processing are also relevant for FX.

² See [GFMA's Global FX Division \(GFXD\)](#) for more details.



When settlement processes are fragmented across multiple institutions, jurisdictions and infrastructures, delays or failures in one part of the transaction chain can affect the timely completion of the wider transaction, increasing operational, liquidity and settlement risks. ***Against this backdrop, interest has increasingly turned to whether tokenised forms of money can improve the coordination and synchronisation of settlement processes across institutions and market infrastructures.***

Tokenised money on programmable infrastructure

Tokenised money on programmable infrastructure offers a potential approach to addressing some of the challenges outlined in the previous section. Tokenised money refers to digital representations of money that are recorded and transferable on shared or distributed ledger infrastructure. In many current implementations, its use is enabled by two key elements: distributed ledger technology (DLT) and programmability. DLT provides a shared infrastructure on which multiple institutions can record and observe ownership changes in a common ledger, potentially reducing the need for reconciliation across separate systems. ***Programmability refers to the ability to embed executable logic, such as conditions, rules and constraints, directly into instruments, or via smart contracts.*** Such functionality can range from basic automation to more sophisticated smart contract applications.

At the simplest level, programmability can automate settlement processes by ensuring that transactions are executed only when predefined conditions are met, such as Delivery-versus-Payment (DvP) in securities transactions and Payment-versus-Payment (PvP) in FX settlement. More advanced applications can support operational and control functions across financial market infrastructure, including real-time liquidity management, margin management, collateral mobilisation and substitution, and automated compliance checks. At a more sophisticated level, programmability can coordinate multiple parties and obligations within a single workflow, enabling applications such as trade finance arrangements, treasury and cash management, automated disbursements and other multi-party payment processes. These capabilities extend beyond process automation by embedding business, risk and compliance logic directly into financial transactions (CCAF and Fii, 2026). An emerging but largely underdeveloped area is the integration of tokenised money with artificial intelligence and autonomous systems, where programmable money interfaces could enable AI agents to initiate, manage and execute financial transactions within predefined controls.

The approach to settlement risk mitigation in a tokenised environment differs from the current CLS model, which achieves Payment-versus-Payment (PvP) settlement through a centrally coordinated process operating within defined funding and settlement windows. While CLS ensures that one currency is exchanged if and only if the corresponding currency is delivered through a centralised settlement mechanism, tokenised infrastructures could, in principle, embed PvP conditions directly into the settlement process through programmable rules or smart contracts on a shared ledger, enabling settlement to occur automatically when predefined conditions are met. However, tokenised settlement models need not rely solely on transaction-by-transaction atomic settlement. Some emerging platforms, such as [ClearToken](#), also incorporate queues, netting arrangements and liquidity-saving mechanisms.

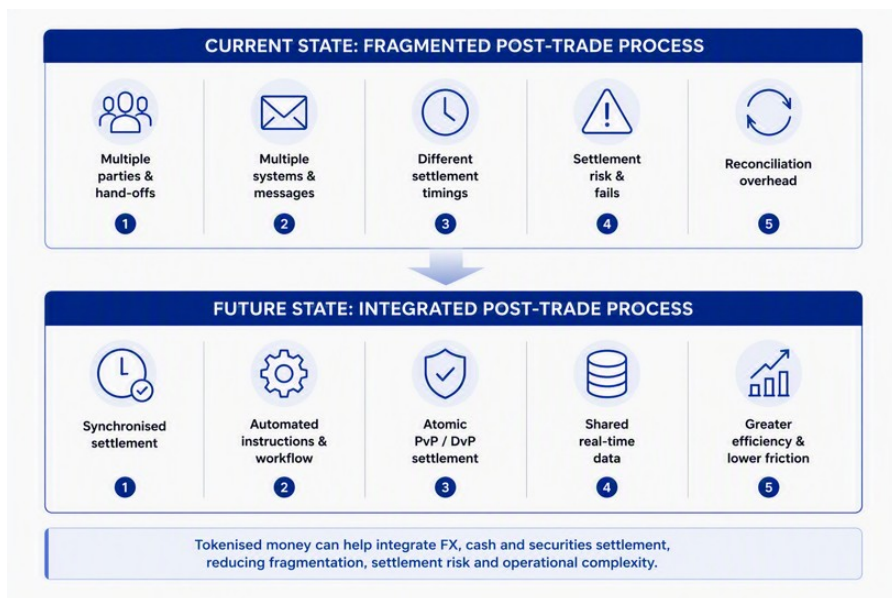


Tokenised post-trade environment for a cross-border securities trade

Many of the potential benefits of tokenised money become more apparent when viewed alongside the broader tokenisation of financial assets. In domestic securities, collateral and derivatives markets, these technologies are already being explored as a means of improving post-trade processes (BIS 2020). When tokenised money is combined with tokenised securities, it may support atomic settlement, greater automation and more efficient movement of cash and collateral, helping to reduce operational complexity, settlement risk and inefficiencies associated with the mobilisation, allocation and coordination of liquidity.

Figure 5 summarises the potential benefits of tokenised money and tokenised securities for post-trade processes. These technologies may support more synchronised settlement, reducing some of the operational, funding and coordination frictions associated with shorter settlement cycles. However, they do not eliminate the need for liquidity provision, risk management, netting or regulatory oversight. Ultimately, technology alone is unlikely to shorten settlement timelines without corresponding changes to market conventions, operating models and regulatory frameworks.

Figure 5: A more integrated post-trade process



Source : Deutsche Bank

The three routes for tokenised money

In practice, the use of tokenised money for financial institutions and corporations is emerging through three principal forms of tokenised money. The first is stablecoins, privately issued digital tokens typically backed by cash, bank deposits or high-quality liquid assets. The second is tokenised bank liabilities, including tokenised deposits, which represent commercial bank money in tokenised form and remain embedded within the existing banking system. The third is tokenised central bank money, which represent tokenised central bank reserves, also sometimes referred to as wholesale central bank digital currencies (wCBDCs).

While these models differ in their issuer, regulatory treatment and underlying infrastructure, they all seek to combine the trust associated with traditional forms

*Tokenised money refers to **digital representations of money** that are recorded and transferable on shared or distributed ledger infrastructure*

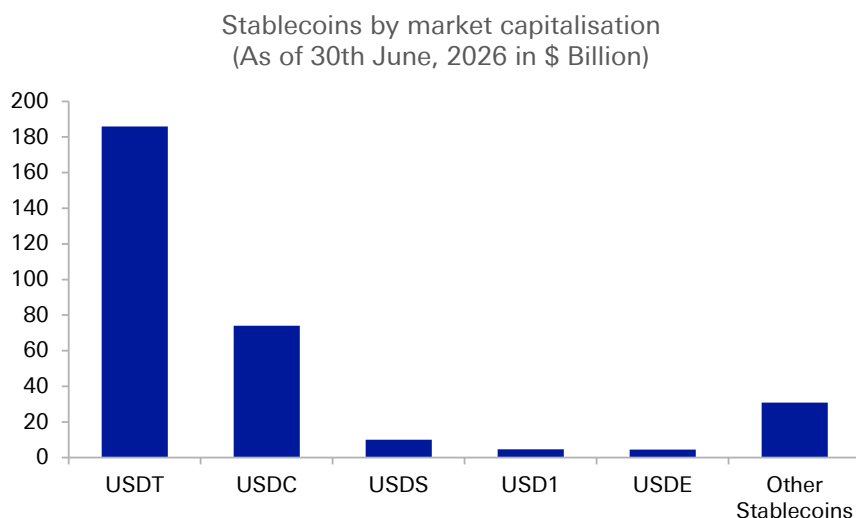


of money with the programmability, interoperability and automation enabled by modern digital technologies. Their potential role in FX settlement is therefore best assessed through the lens of how they address existing challenges.

Stablecoins

A stablecoin is a privately issued digital token designed to maintain a stable value relative to a reference currency, typically through backing by cash, bank deposits or high-quality liquid assets. While other types of stablecoins exist as suggested by Kosse et al (2023) and [BIS \(2026\)](#), the focus of this study would be to discuss fiat-backed stablecoins. BIS (2026) reports that 99.4% of fiat-backed stablecoins (by market valuation) are pegged to the US dollar. Recorded and transferred on distributed ledger networks, stablecoins could potentially function as digital means of payment and settlement.

Figure 6: The market dominated by US dollar-denominated stablecoins



Source : Deutsche Bank, Coingecko

The market for stablecoins has grown rapidly, with more than US\$300 billion outstanding globally, dominated by US dollar-denominated stablecoins such as USDT and USDC ([Figure 6](#)).

From an FX settlement perspective, stablecoins offer an always-on settlement asset that can move across borders on a 24/7 basis without relying on traditional banking hours. They can also support tokenised settlement logic, including conditional payments and automated transaction processing.

Stablecoins' strongest use case today is cross-border payments, particularly in corridors where correspondent banking infrastructure is less efficient. Although total stablecoin transaction volumes were estimated at roughly [US\\$62 trillion in 2025](#), only around US\$350-550 billion is estimated to be associated with real-economy payments, highlighting that adoption in traditional payments remains at an early stage.

Currently, stablecoins are increasingly being used for remittances, business-to-business payments and access to US dollars in markets experiencing currency



instability or limited access to international banking services. Stablecoins are helping improve the movement of the bridge currency—typically the US dollar—rather than the settlement of both currency legs. As roughly 99% of stablecoins in circulation are denominated in US dollars, they function primarily as digital dollar infrastructure rather than a broadly adopted multi-currency settlement network. Consequently, ***most current stablecoin implementations focus on facilitating the transfer of a single currency, typically the US dollar, rather than solving the full two-currency FX settlement problem. This has also been referred to as the "stablecoin sandwich".***

Stablecoin adoption is increasingly moving beyond the lowest-cost blockchain towards a multi-network ecosystem spanning Ethereum, Solana, Polygon, BNB Chain and other networks. This reflects growing demand for compliance, programmability and institutional-grade infrastructure alongside low transaction costs. However, it also highlights an important limitation: stablecoins do not automatically solve interoperability. Instead, the challenge increasingly becomes enabling value to move seamlessly across multiple networks and settlement rails.

Stablecoin liquidity remains fragmented across issuers, networks and jurisdictions, while regulatory treatment continues to evolve across markets. In addition, institutional adoption requires custody solutions, compliance frameworks, treasury integration and industry standards comparable to those available for traditional bank money. As a result, while stablecoins can reduce some correspondent banking frictions, they do not by themselves provide a comprehensive solution to PvP settlement across non-CLS currencies.

Despite these complexities, stablecoins are beginning to attract interest from banks for institutional and corporate cross-border settlement. A notable example is [Nomura's partnership with Circle](#), which aims to explore the use of fiat-backed stablecoins such as USDC in Japan as part of next-generation market infrastructure. The collaboration is focused not only on cross-border payments, but also on FX transactions, treasury operations, fund transfers, on-chain collateral management and tokenised capital markets transactions. More [recently](#), a group of 21 financial institutions plan to issue dollar stablecoin in 2027.

[BNY Mellon](#) expanded its institutional stablecoin services in 2026 to support custody and minting of Circle's USDC, reflecting growing client demand for regulated stablecoin infrastructure rather than the issuance of a proprietary stablecoin. Société Générale's [Forge](#) subsidiary, on the other hand, issued the euro-denominated stablecoin EURCV and used it within tokenised capital market transactions.

Regulatory developments have also helped accelerate experimentation in this area. In Japan, amendments to the Payment Services Act established a regulatory framework for fiat-backed stablecoins. Under this framework, Circle's USDC became the first global US dollar stablecoin approved by Japan's Financial Services Agency for institutional use in Japan. Recent legislative developments in the United States, including the [GENIUS Act](#), have also provided greater clarity around payment stablecoins. Together, these developments have encouraged banks, payment providers and financial institutions to explore stablecoins as settlement assets within regulated financial markets.

These developments suggest that stablecoins are increasingly being viewed not only as digital payment instruments, but also as potential settlement assets for



cross-border transactions. A [recent study](#) notes that regulatory development has progressed furthest in the stablecoin sector, reflecting both growing adoption and policymakers' efforts to address the associated risks.

Key challenges

In some emerging market and developing economy (EMDE) currency corridors, USD stablecoins already act as intermediary settlement assets, performing a role similar to that of the US dollar in traditional correspondent banking arrangements by linking transactions between two non-USD currencies. While stablecoins may reduce frictions in cross-border value transfers, their current concentration in US dollar-denominated issuance limits their ability to support widespread Payment-versus-Payment (PvP) settlement across currency pairs.

While USD stablecoins may facilitate the payment leg of transactions involving tokenised US assets, investors must still exchange their domestic currency for US dollars and subsequently convert any proceeds back into their home currency. Unless digital settlement assets exist for both currencies and can be exchanged through a PvP mechanism, the underlying FX settlement risk remains. Consequently, current stablecoin models primarily improve the transfer of a single currency rather than addressing the settlement risks associated with the exchange of two currencies.

Figure 7: Stablecoins

Metric	Stablecoins
Market size	>US\$300bn outstanding
Dominant currency	~99% USD-denominated
Key benefit	24/7 cross-border transfers
Main use case	Cross-border payments & remittances
FX settlement role	Improves movement of USD as a bridge currency, rather than full two-currency PvP settlement
Key challenge	Fragmented liquidity and interoperability
Institutional adoption	Growing bank interest (e.g. Circle, BNY Mellon, Nomura)

Source : Deutsche Bank

Tokenised commercial bank money

A **tokenised deposit** is not a new form of money, but a commercial bank deposit represented or issued in tokenised form on a shared or tokenised ledger. Economically, it remains a claim on a commercial bank, just as a conventional bank deposit does today. The key difference is that the deposit, or rights linked to that deposit, can be transferred and settled using tokenised infrastructure, smart contracts and shared ledgers.

Classifying tokenised commercial bank money

Several public-sector and industry initiatives have proposed frameworks for



classifying tokenised commercial bank money. The [Bank of England](#) defines tokenised deposits as deposit claims represented on programmable platforms and distinguishes between non-transferable and transferable forms. This distinction is important because it determines whether tokenised deposits can circulate directly across institutions or continue to require interbank settlement in central bank money.

[Project Guardian](#) adopts the broader term tokenised bank liabilities and argues that these instruments could help address existing frictions in cross-border payments and FX settlement, including pre-funding requirements, settlement risk and fragmented post-trade processes. It classifies tokenised bank liabilities according to how the underlying claim is represented, distinguishing between digital twin and digital native models, and according to how they are deployed, ranging from single-bank ledgers to interoperable multi-bank and network-based arrangements.

Industry participants have developed related taxonomies, distinguishing between account-based and token-based representations of commercial bank money, as well as native and non-native ledger structures. Although these approaches differ in terminology, they broadly reflect alternative methods of representing, recording and transferring commercial bank money on distributed ledger infrastructure. Further details are provided in Appendix A.

Current implementation models and infrastructure

Current models are developing along two main lines. The first is **single-bank or single-bank-group infrastructure**, where blockchain-based deposit accounts or tokenised commercial bank money are used between customers of the same institution or banking group. J.P. Morgan's Kinexys Digital Payments is an example of an account-based blockchain deposit model rather than a deposit-token model. It operates on a single-bank ledger, enabling participating J.P. Morgan clients to transfer deposit balances across J.P. Morgan's blockchain-based payments rail. The BMW transaction illustrates this model: BMW executed a programmable on-chain EUR/USD FX transaction using [Kinexys Digital Payments](#), involving automated balance checks, conditional auto-deposits, near-real-time FX execution and transfers between BMW's Blockchain Deposit Accounts in Frankfurt and New York.

The second, more complex model is the **interbank model**, where tokenised bank liabilities issued by different banks, in different currencies and potentially on different ledgers, need to be exchanged or coordinated. [Project Guardian](#) explores several approaches to these interoperability challenges. One model uses a liquidity provider to facilitate the exchange of tokenised deposits in different currencies, with off-chain FX prices fed into the shared ledger through an FX quote oracle. Another model, demonstrated by BNY and OCBC, uses Hash-Time Locked Contracts to coordinate transfers across two private permissioned bank ledgers. A third model, involving HSBC and OSTTRA, focuses on settlement interoperability, using a shared-ledger PvP orchestration layer to match, confirm, fund and settle FX cash flows, helping to reduce Herstatt risk and improve visibility across the settlement.

FX quote oracle: A service that securely uploads off-chain FX prices and token exchange rates to a shared ledger, allowing liquidity providers to offer exchange rates and facilitate cross-currency exchanges between tokenised deposits issued in different currencies.

Bank-led shared-ledger settlement platforms are also emerging as infrastructure that can support tokenised commercial bank money. [Partior](#) is an example of a consortium-based model: a controlled environment in which a defined group of institutions jointly governs and participates in shared infrastructure. It operates as a bank-led shared-ledger settlement platform for real-time cross-border payments



and FX settlement, using a private permissioned ledger where access is restricted to authorised financial institutions and governed participants. Partior's relevance for FX further lies in its ability to support atomic Payment-versus-Payment, or PvP, settlement, where the two currency legs of an FX transaction are coordinated so that settlement of one leg occurs only if the other leg also settle.

Recent initiatives led by [The Clearing House \(TCH\)](#) seek to provide a common clearing and settlement framework for tokenised deposits, addressing one of the key challenges facing tokenised commercial bank money: interoperability across banks. The initiative aims to connect on-chain activity with existing payment rails, enabling the clearing and settlement of tokenised commercial bank money at scale while combining the regulatory, operational and settlement frameworks of established payment infrastructure with the programmability and interoperability of blockchain-based systems.

Market adoption and FX settlement relevance

Commercial bank deposit claims, including tokenised deposits, have progressed beyond early pilot stages and are seeing notable institutional adoption. While data on outstanding tokenised deposit balances remains limited, reported transaction volumes are significant. [J.P. Morgan's Kinexys platform](#), for example, has processed more than US\$4 trillion since inception and averages more than US\$7 billion daily in tokenised deposit movements for institutional payments. Other initiatives include DBS Token Services, Citi Token Services, the USDF Consortium, TassatPay, and projects from UOB and Standard Chartered.

The scale opportunity is potentially significant. Tokenised bank deposits and similar bank-token models could support around [US\\$100–140 trillion](#) in annual transaction flows by 2030, potentially rivalling or surpassing stablecoins in institutional transaction volumes. This reflects the view that corporates and financial institutions may prefer programmable forms of regulated bank money for treasury, liquidity and FX settlement workflows where integration with existing banking infrastructure, compliance processes and balance-sheet treatment is important.

Overall, tokenised deposits appear more directly relevant to institutional FX than stablecoins because they preserve the existing bank-money model while adding programmability, potentially allowing new settlement capabilities to be integrated more easily into existing banking, liquidity and risk-management frameworks. Current implementation models appear more developed for transaction banking, treasury and corporate payment use cases than for the broader wholesale FX market, where achieving interoperability across banks, currencies and settlement networks remains a more complex challenge.

Key challenges

Tokenisation does not, at least in the early stages, fully resolve time-zone mismatches or the need for pre-funding as noted by Project Guardian. Hybrid models are likely to coexist with traditional RTGS systems, correspondent banking infrastructure and fiat-to-token conversion processes for some time, meaning liquidity may still need to be positioned across both traditional and tokenised systems.

A practical issue in cross-currency tokenised deposit models is how the FX rate itself is determined. Project Guardian notes that early-stage on-chain trading volumes may not be sufficient to generate reliable FX prices purely from on-chain



activity. As a result, tokenised deposit models may initially rely on off-chain FX prices sourced from bank liquidity desks, benchmark rates or market data providers, with those prices fed into the shared ledger through an oracle. Over time, if on-chain liquidity develops, more native on-chain FX price discovery may become possible, but this depends on market depth, data integrity and privacy-preserving mechanisms.

Tokenised deposits also do not automatically address all FX settlement challenges. This is particularly relevant in wholesale FX markets, where activity is concentrated among financial institutions. A key strength of existing settlement infrastructure is multilateral netting, which significantly reduces liquidity requirements. Replicating similar levels of liquidity efficiency in tokenised interbank networks remains challenging. Unlike CLS, emerging tokenised deposit arrangements do not yet support multilateral netting at scale, meaning participants may need to trade some liquidity efficiency for faster or more flexible settlement unless netting and liquidity-management mechanisms evolve alongside the technology.

Settlement finality presents a further consideration. Existing wholesale settlement infrastructure ultimately relies on central bank money, which remains an important foundation for wholesale FX settlement. As a result, alternative models that combine tokenised commercial bank money with tokenised central bank money may prove more relevant for wholesale FX applications.

More broadly, the success of tokenised deposits will depend on interoperability across banks and ledgers, sufficient liquidity in each tokenised currency, legal recognition of tokenised claims, clear settlement finality, robust governance frameworks, regulatory consistency, and integration with existing systems.

There may also be balance-sheet costs for issuing banks. If tokenised deposits are treated as more liquidity-sensitive than conventional deposits, banks may need to hold additional high-quality liquid assets against them, reducing their usefulness as a source of funding for lending or maturity transformation.

Figure 8: Tokenised deposits

Metric	Tokenised deposits
Market size	Early-stage but growing institutional adoption
Dominant currency	Multi-currency potential (dependent on issuing banks)
Key benefit	Programmable regulated bank money
Main use case	Institutional payments, treasury and liquidity management
FX settlement role	Supports on-ledger PvP settlement and programmable settlement workflows
Key challenge	Interoperability across banks, currencies and networks
Institutional adoption	Highest within single-bank platforms and consortium-based networks; broader cross-bank adoption remains evolving

Source : Deutsche Bank



Tokenised central bank money

Rather than tokenising private bank money or relying on privately issued settlement assets, an alternative approach is to make wholesale central bank money available on DLT-based infrastructure. In this context, wholesale central bank money used in DLT experiments is often referred to as wholesale central bank digital currency (wCBDC) or, more precisely, tokenised central bank reserves. The latter term is often preferred because it emphasises that the underlying asset, central bank reserves, already exists in digital form, with the innovation lying primarily in the technology used to represent and transfer it. While wCBDC is frequently used interchangeably with tokenised central bank reserves, its definition varies across institutions and may encompass a broader range of innovations in wholesale central bank money that are not necessarily tokenisation-based ([IMF, 2025](#)).

Central bank reserves are already held digitally by eligible financial institutions through RTGS systems such as T2 in Europe, Fedwire in the United States and CHAPS in the United Kingdom. From this perspective, tokenised central bank money initiatives represent not the introduction of a new form of public money, but the extension of existing central bank liabilities into tokenised and programmable environments.³

Although existing infrastructures already eliminate principal risk through DvP and Pvp mechanisms, tokenised reserves may support a more integrated settlement process by bringing cash and assets onto the same ledger. This allows transactions to be completed simultaneously and irrevocably using DLT and smart contracts, while enabling programmability.

A number of initiatives have examined the use of multilateral platforms that make wholesale central bank money available in tokenised form for cross-border settlement. Examples include [Project mBridge](#), [Project Jura](#) and [Project Agorá](#). Leveraging shared infrastructure and governance frameworks, these projects have demonstrated real-time and atomic Pvp settlement between commercial banks in different jurisdictions using tokenised reserves. Smart contracts can further automate and coordinate multicurrency transactions, improving operational efficiency.

Multi-currency model based only on tokenised central bank money

[Project mBridge](#) is an example of a pure multi-CBDC architecture. We use the term CBDC, as used by the project, which is synonymous with tokenised central bank reserves. Developed by the Hong Kong Monetary Authority, People's Bank of China, Bank of Thailand and Central Bank of the UAE, with Saudi Arabia joining later, the project was designed to enable direct cross-border payments and FX transactions using wholesale CBDCs issued by participating central banks on a shared distributed ledger platform. Following a successful pilot involving 20 banks and 164 real-value transactions, mBridge reached minimum viable product (MVP) status in 2024, indicating that the platform had progressed beyond proof-of-concept testing and was capable of supporting real-value cross-border settlement. Subsequent growth has been substantial, with the platform reportedly processing more than 4,000 cross-border transactions worth [approximately US\\$55 billion by late 2025](#). While the BIS transferred responsibility for the platform to participating

[Tokenised central bank reserves \(IMF, 2025\)](#): Wholesale central bank money, commonly known as reserves, refers to money issued by central banks to eligible financial institutions for interbank settlement and monetary policy implementation. With growing interest in tokenised financial assets, central banks are exploring ways to make reserves available on programmable ledgers. In this context, tokenised reserves are direct central bank liabilities issued using DLT and accessible only to a predefined group of financial institutions.

3 Digital Money – a perspective on stablecoins, tokenised deposits, and CBDCs – Deutsche Bank



central banks in late 2024, mBridge remains one of the most developed publicly disclosed multi-CBDC initiatives.

Beyond wholesale payments, experiments have also explored how tokenised central bank reserves could support more advanced settlement arrangements involving multiple currencies, assets or transaction legs. One example is delivery-versus-payment-versus-payment (DvPvP), where interconnected transactions are settled atomically. [Project Jura](#) illustrates these capabilities by enabling the direct transfer of euro and Swiss franc wholesale CBDCs (wCBDCs), representing tokenised central bank reserves, between French and Swiss commercial banks on a shared DLT platform operated by a third party. The project demonstrated how tokenised asset and foreign exchange transactions can be settled safely and efficiently using both payment-versus-payment (PvP) and delivery-versus-payment (DvP) mechanisms, highlighting the potential for programmable, multi-leg transactions across currencies, asset classes and jurisdictions.

Hybrid models

Alongside tokenised central bank money initiatives, [Project Agorá](#) is exploring alternative architecture that retain central bank money as the settlement anchor while preserving a significant role for commercial bank money.

[Project Agorá](#), launched by the BIS in 2024 in collaboration with the central banks of France (representing the Eurosystem), Japan, South Korea, Mexico, Switzerland, the United Kingdom and the United States, explores a hybrid cross-border settlement model that combines tokenised commercial bank deposits with tokenised central bank reserves on a shared programmable platform. In 2026, the BIS reported that the prototype had successfully demonstrated atomic multi-currency settlement across participating jurisdictions using a unified ledger architecture linking tokenised deposits and tokenised central bank reserves. The project also highlighted potential improvements in speed, transparency and operational efficiency while preserving the existing two-tier monetary system.

Key challenges

Tokenised central bank reserve models, particularly multi-CBDC wholesale arrangements, are among the most challenging forms of digital money to implement. Unlike stablecoins or tokenised deposits, they require extensive coordination across central banks, legal jurisdictions and financial market infrastructures. Cross-border deployment raises complex questions around governance, access, interoperability, regulatory alignment, and the legal recognition of settlement finality. As a result, industry attention has increasingly broadened beyond pure wCBDC architectures towards hybrid models that combine tokenised central bank money with tokenised commercial bank money. By building on the existing two-tier monetary system, these models may face fewer implementation barriers and are often viewed as more compatible with current market structures.

[Recent industry commentary](#) suggests that digital settlement infrastructure may develop along regional or architectural lines rather than converge on a single global model. The emergence of mBridge as a multi-CBDC platform centred on Asia and the Middle East, alongside Agorá's hybrid model involving major advanced-economy central banks and commercial banks, highlights the possibility that multiple settlement ecosystems could emerge over time. Whether these systems ultimately converge through interoperability standards or develop as parallel networks remains uncertain.



Figure 9: Tokenised central bank reserves

Metric	Tokenised central bank reserves
Market size	Limited live volumes; activity remains concentrated in pilots, proofs of concept and early production deployments
Dominant currency	Typically designed as multi-currency infrastructures involving multiple central bank jurisdictions
Key benefit	Provides settlement in central bank money with potential for atomic PvP settlement and programmability
Main use case	Wholesale financial market settlement, cross-border payments and settlement of tokenised assets
FX settlement role	Potential common settlement asset for cross-currency transactions, reducing settlement risk and enabling programmable FX settlement workflows
Key challenge	Cross-border coordination among central banks, legal and regulatory harmonisation, governance, access and interoperability.
Institutional adoption	Active experimentation through initiatives such as mBridge and Jura, as well as hybrid models such as Agorá, although large-scale production adoption remains limited.

Source : Deutsche Bank

Other settlement solutions and related infrastructure

Going beyond the three forms of tokenised money, other settlement solutions and supporting infrastructures are evolving in parallel. One example is [Finality](#), a privately operated settlement system backed 1:1 by central bank reserves. It provides a central-bank-money-backed settlement asset for wholesale transactions. The [Sterling Finality Payment System](#) went live in 2023 and received Bank of England settlement-finality designation in 2024, with the platform targeting use cases including PvP FX settlement and tokenised asset settlement. It uses the [Bank of England's omnibus account solution](#) to hold participant funds and issues DLT-based settlement tokens representing claims on this pool of funds.

Omnibus account arrangements allow for settlement using tokens on the DLT platform outside of RTGS operating hours and with added functionalities, such as programmability, provided that the FIs have prefunded their omnibus accounts (Bank of England 2021)

Rather than tokenising central bank money directly, the [Bank of England's synchronisation model](#) enables funds held in RTGS to be synchronised with transactions taking place on external ledgers through a synchronisation operator. Settlement in central bank money occurs if and only if the corresponding transaction on the external platform is completed. Through initiatives such as Project Meridian FX and the Synchronisation Lab, the Bank has explored how this approach could support Payment-versus-Payment (PvP) settlement for cross-border FX and Delivery-versus-Payment (DvP) settlement for tokenised securities, while preserving settlement in central bank money. The Bank has also highlighted that extended RTGS operating hours may be an important enabler of such synchronised settlement models.

Industry initiatives such as the Regulated Settlement Network (RSN), which builds on earlier Regulated Liability Network (RLN) initiatives, are exploring a shared ledger environment in which regulated liabilities, such as commercial bank deposits and central bank money, and tokenised assets can settle in real time across multiple entities. RSN involves a consortium of institutions, including Citi, the New York Fed Innovation Center and Swift, and creates a shared ledger environment in which these assets can settle in real time across multiple entities. A [December 2024](#) proof of concept demonstrated 24/7 settlement capabilities in a controlled environment, although questions remain around broader ecosystem integration



and cross-border scalability.

[SWIFT's blockchain ledger](#) is another example that emphasis on connectivity. Rather than introducing a new settlement asset, it enables participating banks to coordinate transfers of tokenised deposits across their own ledgers while continuing to settle through existing payment infrastructure. The initiative is intended to support 24/7 cross-border payments and improve liquidity efficiency.

These initiatives highlight the importance of infrastructure that enables different forms of money, payment systems and settlement architectures to operate together. If FX settlement evolves through multiple architectures, synchronisation and interoperability may become as important as the characteristics of the settlement asset itself.

Technology alone is not enough: risk reduction or transformation?

Tokenised money is often discussed in terms of faster settlement, reduced operational complexity and atomic execution. While these technologies may help reduce certain forms of settlement risk, particularly where payment-versus-payment (PvP) and delivery-versus-payment (DvP) mechanisms can be embedded into the settlement process, they do not eliminate the need for liquidity provision, risk management, netting arrangements or regulatory oversight. In many cases, the effect may be better described as risk transformation rather than risk elimination. Settlement risk may be reduced, but liquidity, operational, legal and governance considerations remain.

Regulation: Regulatory frameworks for tokenised deposits, stablecoins and digital settlement assets continue to evolve across jurisdictions. While progress has been made, important questions remain around legal treatment, settlement finality, prudential requirements and the use of tokenised money in cross-border transactions.

Liquidity: Tokenised money does not create liquidity; rather, it may improve how liquidity is mobilised and deployed. Potential reductions in pre-funding requirements and trapped liquidity depend on sufficient market participation, transaction volume and interoperability across networks. Today, many initiatives remain fragmented across separate platforms and participant groups, limiting the extent to which liquidity benefits can be realised. Moreover, unlike CLS, most tokenised settlement networks **do not yet provide multilateral netting at scale**, meaning some participants could **face higher gross funding requirements** even as settlement processes become more automated. This is important because multilateral netting and the resulting liquidity savings have been identified as critical features of wholesale FX payment and settlement systems, including in discussions at the [Jackson Hole Economic Policy Symposium](#).

Interoperability and integration: Many of the potential benefits depend on the ability of different forms of money, payment systems and settlement infrastructures to operate together. Successful adoption will require integration with existing banking, payment, custody, accounting and regulatory reporting processes. Programmable and traditional infrastructures are likely to coexist, creating additional operational and implementation challenges.



Market infrastructure: The benefits of programmable settlement are greatest when cash, securities and collateral can move on the same, or fully interoperable, infrastructure. While technological capability is advancing rapidly, broader adoption depends on custodians, central securities depositories, asset managers, banks and market infrastructures adapting their operating models and processes.

Operational resilience and cybersecurity: Tokenised settlement infrastructures introduce new operational dependencies, including distributed ledger networks, smart contracts, wallets, bridges and external data providers. Coding errors, cyberattacks, failures in oracle services or vulnerabilities in cross-chain connectivity can disrupt settlement processes or lead to unintended outcomes. As networks become increasingly interconnected, disruptions may propagate more rapidly across participants and platforms.

Legal certainty and governance: The legal treatment of tokenised money continues to evolve. Questions remain around ownership rights, insolvency treatment, the enforceability of self-executing smart contracts, dispute resolution mechanisms and the alignment of technical and legal settlement finality. Governance arrangements, including liability allocation, system upgrades and contingency procedures, remain critical to ensuring trust and resilience.

Market concentration and fragmentation: Network effects may favour a small number of tokenised money issuers, settlement platforms or infrastructure providers, potentially creating new forms of concentration risk. At the same time, the proliferation of non-interoperable networks may fragment liquidity across multiple platforms and tokenised money instruments, reducing the fungibility and efficiency that underpin effective settlement.

Data, privacy and compliance: The transparency of shared ledgers can improve auditability and oversight but may also create challenges around confidentiality, commercial sensitivity and data protection. Compliance obligations, including AML/CFT controls, sanctions screening and cross-border regulatory requirements, remain significant considerations and may be difficult to implement consistently across multiple networks and jurisdictions.

Financial stability and monetary considerations: As tokenised money adoption grows, regulators have raised broader concerns around financial stability, including the implications of large-scale migration into digital settlement assets, the concentration of reserve assets, and changing funding dynamics within the financial system. In some jurisdictions, particularly emerging economies, the growing use of foreign-currency-denominated stablecoins has also raised concerns around monetary sovereignty and the effectiveness of domestic monetary policy transmission.

Conclusion

Existing infrastructure plays an important role in mitigating settlement risk in wholesale FX markets. CLS combines Payment-versus-Payment (PvP) settlement with multilateral netting and liquidity optimisation, generating liquidity savings of approximately 99% for participants. However, coverage is not universal, and a significant share of global FX activity continues to settle outside PvP arrangements due to constraints related to currency coverage, participant access, operating windows and market structure.



While tokenised money may help address some of these frictions, its potentially stronger use case for wholesale FX lies in the wider tokenisation of capital markets. Tokenised money could enable more integrated settlement processes for tokenised assets. Reflecting this potential, a [recent study](#) identified capital market tokenisation as one of the most important long-term use cases for tokenised money. Although activity remains small relative to overall market volumes, it has the potential to change how securities, bonds and other financial instruments are issued, traded and settled. The vision is an integrated ecosystem in which tokenised money is used to settle tokenised securities and other assets, enabling atomic settlement that reduces counterparty risk and shortens settlement times from days to seconds.

The [Bank of England's](#) analysis highlights an important financial stability consideration. While tokenised deposits extend commercial bank money into programmable environments, central bank money remains the ultimate risk-free settlement asset and anchor of confidence in the financial system. Future tokenised markets may therefore require mechanisms such as RTGS synchronisation or hybrid models combining tokenised commercial bank money and central bank money to preserve the singleness of money and settlement finality. More broadly, the proliferation of multiple forms of tokenised money could fragment liquidity, reduce fungibility and create new market structure and financial stability challenges.

Over time, programmable settlement arrangements may replicate some functions currently performed by existing infrastructures, particularly if interoperable tokenised deposit or tokenised central bank money frameworks achieve sufficient scale. However, this will depend not only on technological capability but also on the development of comparable liquidity networks, legal frameworks, governance arrangements and interoperability with existing payment and settlement systems. The transition towards programmable settlement is therefore likely to involve both risk reduction and risk transformation.



Appendix A: Detailed classification of tokenised commercial bank money

Several public-sector and industry initiatives have proposed different frameworks for classifying tokenised commercial bank money. The [Bank of England](#) describes tokenised deposits as “deposit claims represented on programmable platforms” that enable new techniques for payments and settlement. It distinguishes between non-transferable and transferable tokenised deposits. Non-transferable tokenised deposits resemble existing deposits and may still require settlement in central bank money when transferred between customers of different banks. Transferable tokenised deposits, by contrast, could be self-settling upon transfer, with the recipient becoming a customer of the issuing bank. This distinction matters for FX because it affects whether tokenised deposits can circulate across banks directly or whether interbank settlement in central bank money is still required to preserve singleness and finality.

[Project Guardian](#) uses the broader term tokenised bank liabilities to refer to tokenised representations of commercial bank money on a shared ledger, without asserting any specific legal or regulatory interpretation. Tokenised deposits are therefore a specific form of tokenised bank liability. It further classifies tokenised bank liabilities along two dimensions: token level and ledger level. At the token level, it distinguishes between digital twin tokens and digital native tokens. Digital twin tokens represent ownership interests in underlying traditional bank liabilities, with ownership rights represented on-chain through digital tokens. Digital native tokens are issued and custodied natively on a shared ledger, which serves as the primary record of ownership and rights. The distinction therefore relates to the legal and operational representation of the underlying bank liability rather than to transferability, as both forms can potentially support programmable payments and settlement.

At the ledger level, Project Guardian distinguishes between three deployment models. A **single-bank ledger** is a shared ledger operated within the same banking group, enabling transfers between customers, entities or branches connected to that bank's infrastructure. **Between-bank models** enable tokenised bank liabilities issued on separate banks' shared ledgers to be used for payments and settlement between customers of different banks. **Interlinking network models** connect distinct networks, each potentially having its own governance framework and design, allowing tokenised assets to move across a broader ecosystem of participating institutions and currencies.

Beyond the terminology used by Project Guardian, industry participants have developed more detailed taxonomies to describe how commercial bank money can be represented on distributed ledger infrastructure. A related [industry taxonomy](#) describes these instruments as blockchain-based deposits: distributed-ledger-based deposit claims issued by a licensed depository institution and recorded on blockchain. Within this category, they distinguish between account-based models, such as blockchain deposit accounts, and token-based models, such as deposit tokens. The taxonomy also distinguishes between native models, where the blockchain is the primary record of value, and non-native models, where the blockchain mirrors an off-chain record.

Both account-based and token-based models are forms of commercial bank money, but they differ in how the deposit claim is represented and transferred on ledger. Blockchain deposit accounts represent deposit balances as account

Project Guardian also identifies several pain points in the current cross-border payment and FX settlement process, including time-zone frictions, pre-funding requirements, settlement risk outside CLS, fragmented infrastructure and the continued separation of messaging, reconciliation and settlement. It argues that tokenised bank liabilities could help address some of these frictions by enabling commercial bank money to move across shared infrastructure and, where supported by appropriate legal, operational and interoperability frameworks, by enabling atomic PvP-style settlement for a broader range of currencies and participants.



balances on a blockchain-based ledger. Deposit tokens are transferable tokens issued on a blockchain by a licensed bank, representing a claim on that bank for a stated amount of fiat money. J.P. Morgan's SGD deposit token, issued in connection with the MAS Project Guardian pilot, is one example of a deposit token. More recently, JPM Coin/JPMD is a bank-issued deposit token on Base, an Ethereum Layer 2 network, enabling approved institutional clients to move money, post collateral and settle transactions on public blockchain infrastructure while retaining the credit-risk profile and balance-sheet characteristics of commercial bank money.

Figure 10: Defining tokenised commercial bank money

Organisation / Framework	Term Used	Classification	Description
Bank of England	Tokenised deposits	Non-transferable	Tokenised deposit that still relies on interbank settlement.
		Transferable	Tokenised deposit that can circulate through transfer.
Project Guardian	Tokenised bank liabilities (Token level)	Digital twin token	On-chain representation of an existing bank liability.
		Digital native token	Bank liability issued and recorded natively on a shared ledger.
	Tokenised bank liabilities (Ledger level)	Single-bank ledger	Shared ledger operated within a banking group.
		Between-bank model	Tokenised bank liabilities exchanged across separate banks' ledgers.
CCAF and Fii (2026) Tokenised Money: Use Cases, Interoperability and Regulation	Tokenised money: commercial bank deposits	Tokenised deposits	Represent an off-chain bank deposit recorded in traditional account systems, with the token serving only as a representation of that off chain position
		Deposit tokens	Deposit itself is issued solely on a distributed ledger constitute a DLT native form of the claim
Industry taxonomy	Blockchain-based deposits	Blockchain deposit accounts Deposit tokens	Account-based representation of deposits on blockchain infrastructure. Transferable token representing a deposit claim on a bank.
	Ledger architecture	Non-native model Native model	Blockchain mirrors an off-chain record. Blockchain is the primary record of ownership.

Source : Deutsche Bank



Appendix 1

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Deutsche Bank
Research

Private capital monitor: AI is the 2nd biggest risk

September 2026

Luke Templeman | Galina Pozdnyakova

Deutsche Bank Research Institute

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IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.



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Executive summary

- **Data centres, power, compute** and private investment-grade financing is further diversifying private capital from buyouts.
- But the risk of **AI disruption, overbuild, and capital misallocation** is a frequent risk discussed by large, listed private capital managers
- Our spotlight section includes an analysis of **AI-related overfinancing risk**
- Still, the **macro environment** is now the top point of concern amongst PC managers, even more so than the exit environment
- **Private capital growth remains strong but is becoming increasingly K-shaped.** Scale, investment performance, origination capabilities and product breadth are driving fundraising toward the largest platforms, while weaker managers and less differentiated products face pressure.
- **Fundraising, AUM and fee-related earnings generally reached record or near-record levels**, despite weak sponsor M&A, uneven exits and disruption in parts of private wealth and software.
- There has also been a **drop in the number of articles on private credit in the media**, especially when it comes to “crisis”-mentioning variety. With public BDCs being at the centre of this year’s rout, the drop in such trackable indicators likely bodes well for sentiment and thus flows.
- The overall tone of Q2 results was **strongly positive and confident, especially on fundraising and structural growth**. However, management teams were more cautious on the timing of monetisations, private-wealth recovery, AI capacity risks and software disruption.



01. Q2 2026 results

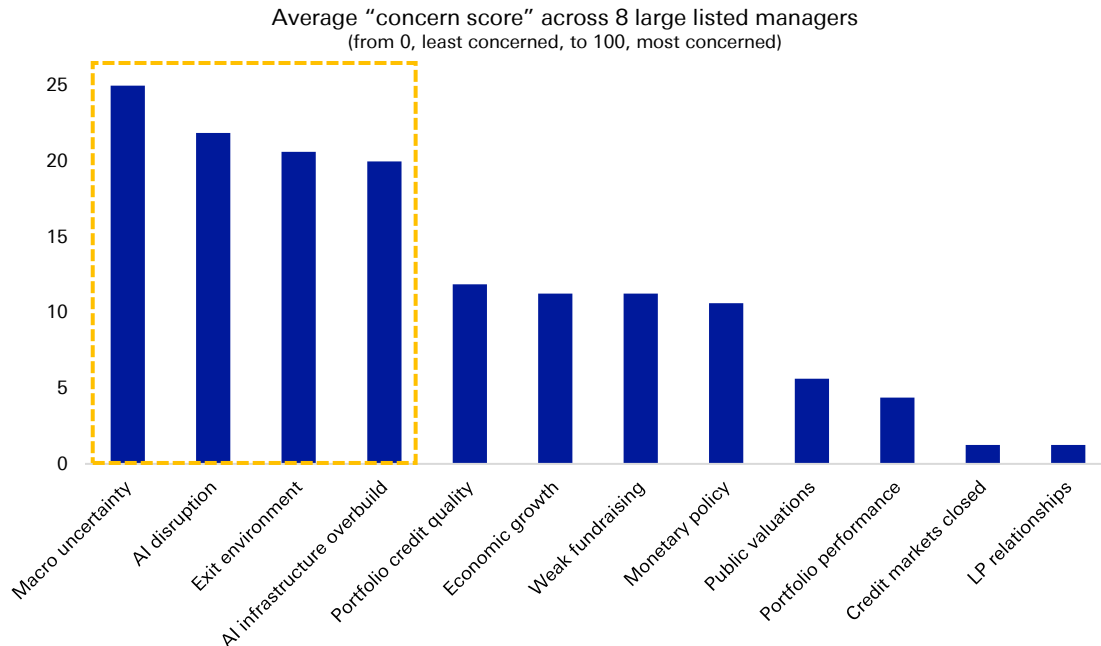
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Q2 earnings calls from the large, listed PC managers

Summary



- **Private capital growth remains strong but is becoming increasingly K-shaped.** Scale, investment performance, origination capabilities and product breadth are driving fundraising toward the largest platforms, while weaker managers and less differentiated products face pressure.
- **Fundraising, AUM and fee-related earnings generally reached record or near-record levels**, despite weak sponsor M&A, uneven exits and disruption in parts of private wealth and software.
- **The overall tone was strongly positive and confident, especially on fundraising and structural growth.** However, management teams were more cautious on the timing of monetisations, private-wealth recovery, AI capacity risks and software disruption.
- **AI is creating a need** for data centres, power, compute and private investment-grade financing that further diversifies private capital from buyouts.
- Amongst the concerns we track, managers were relatively **more concerned about the macro environment and AI effects**, including the risk of overbuild in related infrastructure.

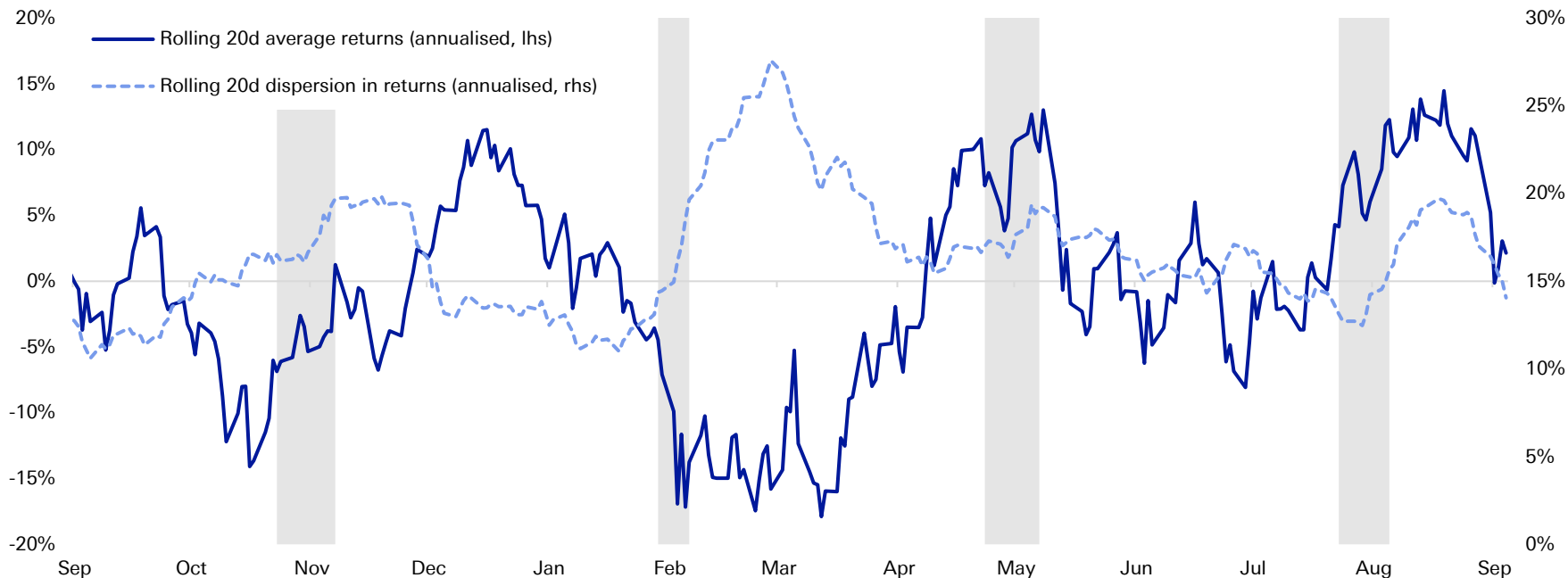


Source: Bloomberg Finance LP, Microsoft Copilot, Deutsche Bank.

Good results and some pullback around AI and credit fears catalysed a strong rebound in shares



8 US-listed private capital managers, rolling performance (earnings season shaded)



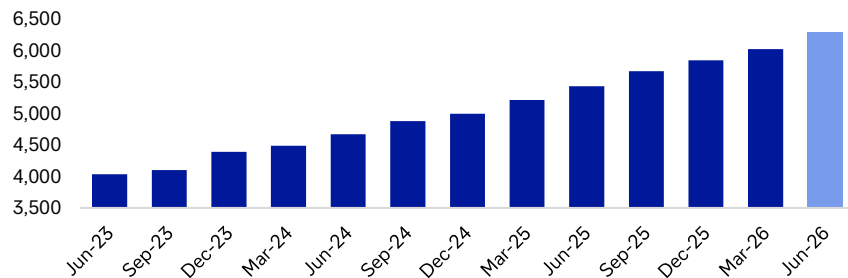
Source: Bloomberg Finance LP, Deutsche Bank.

Key metrics for selected large private capital managers

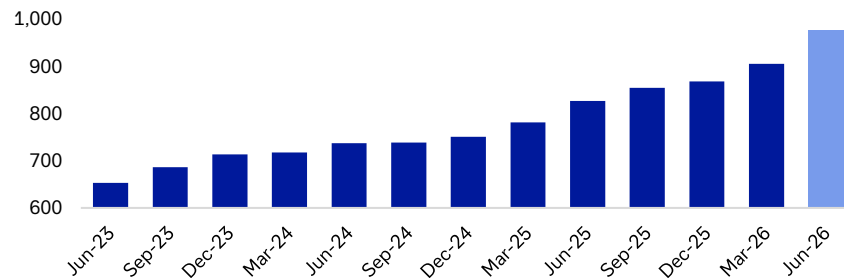
Strong earnings and a notable uptick in dry powder



Total AUM in \$bn



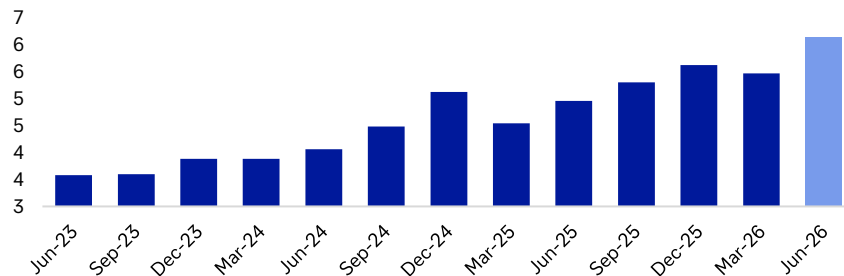
Dry powder in \$bn



Average fee-related earnings margin (%)



Fee-related earnings in \$bn



Source: Bloomberg Finance LP, Deutsche Bank.

Key themes from Q2 earnings calls

Large, listed managers talk AI, wealth, fundraising, exits and credit



1. **AI infrastructure** has become a firm-wide capital deployment and earnings theme, extending beyond data centres into power, compute, chips, connectivity and investment-grade financing. Power availability is emerging as a more important constraint than capital. Although concentrated exposure to AI infrastructure is a risk if demand expectations change, managers stressed contracted revenues, strong counterparties and hard-asset collateral as mitigants.
 - Blackstone's data-centre platform reached \$185bn of total value, including projects under construction, up from \$130bn at the start of 2026. Management said the platform could double over the next few years and that the market for long-term ownership of stabilised data centres could eventually reach \$1tn. The firm also said nine of its ten largest Q2 mark-ups were AI-related.
 - Apollo cited more than \$8tn of cumulative AI-infrastructure investment through the cycle and led a \$35bn Broadcom financing, described as the largest private-credit financing to date.
 - Brookfield estimated \$10tn of AI-related capex across energy, data centres, compute and adjacent infrastructure. The firm said AI-related deployment was probably running at approximately twice its normal 15%-20% business growth rate.
 - Blue Owl reported more than 140 data centres owned or under construction and 15.3GW of leased and owned capacity
2. **Private wealth** remains strategically important, but private credit experienced a confidence and liquidity test. Early Q3 data suggests stabilising fundraising. The better wealth outcomes increasingly came from private equity, infrastructure, real estate and multi-asset products rather than traditional private credit. Earnings calls suggest some international family offices and smaller institutions behaved more cyclically than core US individual investors.
3. **Record fundraising** despite a "noisy" backdrop. Fundraising is concentrating with scaled, diversified platforms.
4. **Realisations** are recovering, but the market is highly selective (see pages 17-21 of this report)
5. **Credit is broadening beyond sponsor-backed direct lending** and real assets, infrastructure, defence and secondaries are benefiting from secular demand
6. **Earnings are shifting toward recurring fees, transaction revenues and fee-related performance revenues.** Firms are becoming more effective at capturing economics from origination, financing and perpetual vehicles amidst a strategic shift from simply managing third-party capital toward monetising the broader platform through origination, syndication, financing and advisory services.

Source: Bloomberg Finance LP, Deutsche Bank.

Key risks from Q2 earnings calls

Aside from AI, concerns are rising about rates and credit



1. **AI overbuild, capital misallocation and execution amongst most consistently voiced risks**
 - Brookfield said concerns about excess infrastructure capacity were reasonable and that, as in railways and fibre, some capital would inevitably be poorly allocated.
 - Blackstone acknowledged the potential for excessive exuberance, despite its conviction that power, chips and entitlements constrain supply.
 - Both firms also identified scarce inputs and complex development requirements as execution bottlenecks.
2. **AI disruption to software and white-collar services**
 - Blackstone said software, professional services and information services were experiencing weaker buyer interest despite many companies continuing to perform operationally.
 - TPG reported mid-teens software bookings growth and said over 75% of its software exposure was well positioned to benefit from AI.
 - Blue Owl said it had gained increasing comfort around software transition risk but continued to recognise it as a valid issue
3. **Wealth-channel redemptions.** Management frames this as narrative-driven, but a slower-than-expected sentiment recovery is a risk to retail flows. Managers emphasised that requests were concentrated among a minority of investors concentrated in non-US family offices and smaller institutions and that underlying portfolio performance remained sound.
 - The recovery appears to be under way but remains gradual. Blue Owl explicitly rejected a V-shaped recovery assumption, while Ares estimated that its BDC could return to flow stasis over the next two to three quarters if current trends continued.
4. **Realisation timing and reliance on capital markets.** Managers stressed that timing is difficult to predict.
 - TPG expects transaction and monitoring fees to step down in Q3 after pulling forward some transactions into Q2. Carlyle similarly said Q2 capital-markets activity was unlikely to repeat in Q3 or Q4. Blackstone expects lower Q3 realisations before a stronger Q4 and 2027.
5. **Credit-cycle normalisation and growing dispersion remain risks, although managers generally reported sound portfolio fundamentals in Q2.**
6. **Insurance competition, spread compression and regulation**
 - KKR said competition had increased and it was allocating somewhat less capital to insurance at current return levels. Management described sector ROEs as structurally low in the current competitive environment. Apollo cited 36 asset-manager entrants into retail annuities and argued that some competitors lack mature origination, liability-generation and operating capabilities. Apollo also highlighted offshore regulatory arbitrage and pushed for “equal capital for equal risk”, warning that weaker regulatory structures could undermine industry trust.
7. **Rates, inflation, energy and geopolitics.** The macro backdrop remained a recurring source of uncertainty:
 - Carlyle cited war in the Middle East, pressure on energy markets, stubborn inflation and AI-related public-market volatility.
 - TPG said geopolitical uncertainty and changing rate expectations had reduced private-equity activity.
 - Blackstone said higher rates were delaying the real-estate recovery and had pushed some exits out.

Source: Bloomberg Finance LP, Deutsche Bank.

Q2 earnings: performance by asset class



KKR	Q2	LTM
Private equity		
Traditional PE portfolio	4%	9%
Real assets		
Opportunistic real estate	-1%	1%
Infrastructure	1%	8%
Credit		
Leveraged credit composite	2%	5%
Alternative credit composite	1%	5%
Apollo	Q2	LTM
Direct origination	2.6%	7.9%
Opportunistic credit	1.8%	10.7%
Multi-credit	1.9%	7.4%
Asset-backed finance	2.4%	7.2%
Flagship PE	3.0%	7.2%
Carlyle	Q2	LTM
Global private equity		
Corporate PE	2%	4%
Real estate	0%	2%
Infrastructure & natural resources	6%	28%
Global credit	4%	16%

Ares	Q2	LTM
Credit		
Alternative credit	4.1%	16.4%
Opportunistic credit	2.4%	8.9%
US senior direct lending	2.5%	11.2%
US junior direct lending	0.2%	8.9%
European direct lending	1.5%	8.3%
APAC credit	2.4%	19.0%
Corporate private equity	2.4%	1.6%
Real assets		
Americas real estate equity	0.6%	3.1%
European real estate equity	0.2%	2.5%
Infrastructure equity	9.0%	9.8%
Infrastructure debt	1.7%	7.5%
Secondaries		
Private equity secondaries	-5.3%	-1.7%
Real estate secondaries	-1.2%	5.6%
Infrastructure secondaries	-0.4%	14.6%

Blackstone	Q2	LTM
Real estate		
Opportunistic	0.4%	-1.4%
Core+	0.2%	3.2%
Private equity		
Corporate private equity	3.7%	14.4%
Tactical opportunities	3.9%	9.9%
Secondaries	1.7%	8.1%
Infrastructure	7.2%	28.6%
Credit & insurance		
Private credit	1.0%	6.8%
Liquid credit	2.2%	3.9%
TPG	Q2	YTD
Capital	5.7%	3.1%
Growth	6.2%	4.5%
Impact	7.2%	7.2%
Credit	3.3%	5.6%
Real Estate	2.5%	4.4%
Blue Owl	Q2	LTM
Direct lending	2.6%	8.3%
Alternative credit	2.4%	11.4%
Net lease	3.1%	13.6%
Real estate credit	1.9%	8.1%
Digital infrastructure	3.1%	6.0%

Source: Company reports, Bloomberg Finance LP, Deutsche Bank. Note: Performance reflects gross returns or appreciation depending on the manager, strategy and time period.

Management comments – a few records in Q2 and a confident tone



- **Stephen Schwarzman of Blackstone:** “Blackstone delivered an outstanding second quarter, highlighted by strong growth in earnings and nearly \$70 billion of inflows. Our decision to lean into the artificial intelligence megatrend is leading to standout investment performance across numerous strategies and creating extraordinary opportunities for growth. We’ve become a trusted partner at scale to many of the key innovators in this ecosystem, which positions our firm extremely well for the future.”
- **Joseph Bae and Scott Nuttall of KKR:** “Our second quarter results reflect the strength and breadth of KKR’s global franchise. We reported record Fee Related Earnings, Total Operating Earnings and Adjusted Net Income per share on both a quarterly and trailing twelve-month basis. Our continued return of capital to clients led to our strongest monetization quarter ever and helped drive record new capital inflows over the past 12 months.”
- **Marc Rowan of Apollo:** “Our strong second quarter results reflect record earnings across Asset Management and Retirement Services, highlighting the quality and growing scale of our business. We are at the forefront of modernizing how private markets operate by enhancing transparency, improving liquidity, and broadening access. In a market evolving quickly with increasing demand for capital, the breadth of our origination capabilities combined with a principal mindset positions us to help shape what comes next.”
- **Harvey Schwartz of Carlyle:** “The second quarter was one of Carlyle’s strongest quarters in recent years, underscoring the power of our diversified platform. We delivered record Fee Related Earnings, our highest Distributable Earnings in nearly four years, alongside strong fundraising, and exceptional realization activity. Carlyle continues to distinguish itself as an industry leader and an outlier in returning capital to our clients, distributing nearly \$7 billion during the quarter and \$37 billion over the past year. ...”
- **Michael Arougheti of Ares:** “We generated strong second-quarter results, including another record quarter of fundraising with more than \$36 billion of inflows as our clients continue to reward us due to our strong and consistent fund performance across our strategies. Our diverse global origination platform enabled us to remain active investing in attractive opportunities across the platform in a slower transaction environment and we are now seeing a meaningful pickup in our firmwide investment pipeline.”
- **Jon Winkelried of TPG:** “TPG delivered record results during the first half of 2026, reflecting the strength of our diversified and growing franchise... As clients increasingly partner with TPG for access to our differentiated offerings across strategies, we continue to see significant opportunities to further expand our platform and build long-term value for our shareholders.”
- **Doug Ostrover and Marc Lipschultz of Blue Owl:** “Blue Owl’s ongoing growth in the second quarter reflects the strong investment performance we continue to generate across strategies and the diversification of our business across platforms and geographies... The results we reported this morning continue to demonstrate the resilience of our business and highlight our positioning as a key provider of scaled capital solutions in this evolving market landscape.”

Source: Company reports, Deutsche Bank.



02.

Spotlight section:

- Sentiment recovery
- AI overinvestment
- Realisations

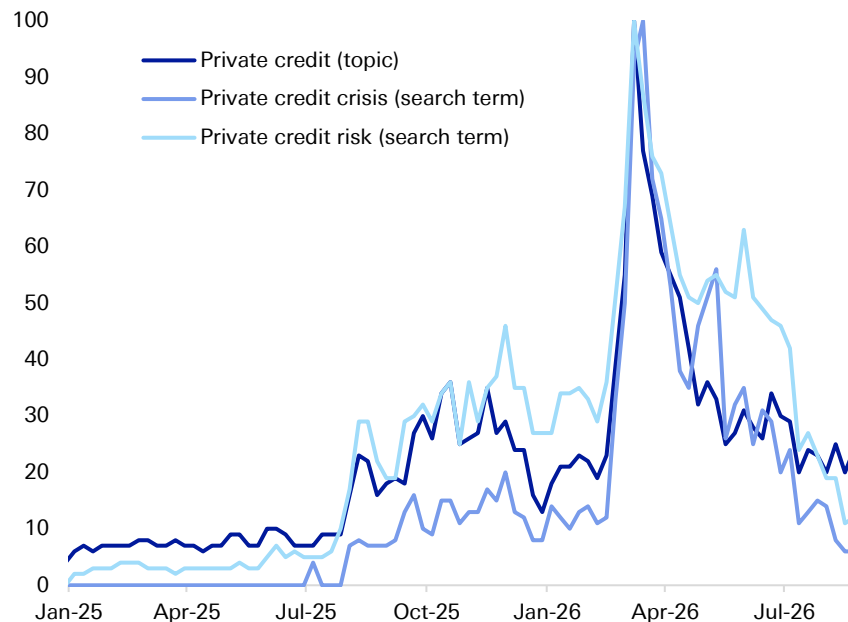
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Private credit fears 1/3

The pessimism is well past its peak now as various sentiment measures point to a drop in interest in the topic. As managers first noted during Q1 earnings, there seems to have been the most concern amongst Asian investors.

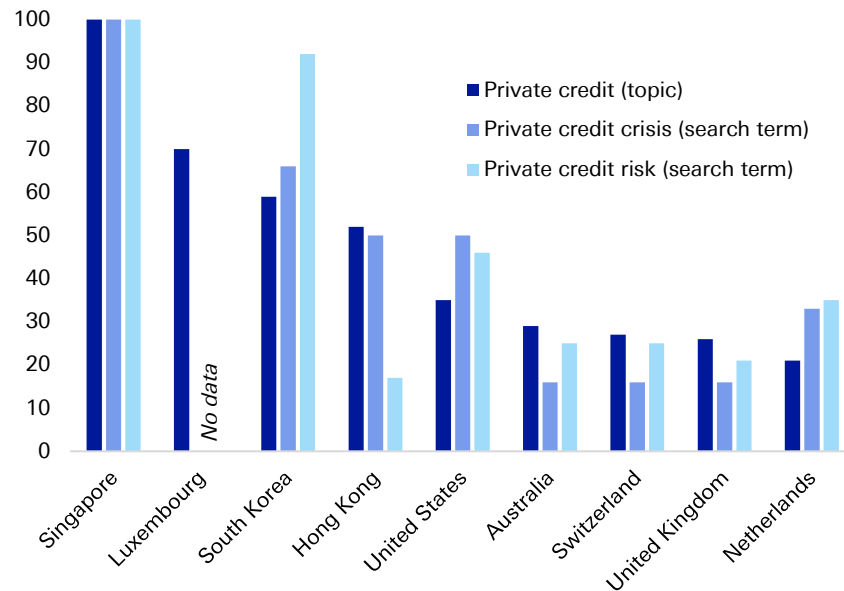


Private credit-related Google search trends (worldwide)
100 = the peak popularity for the term



Source: Google Trends, Deutsche Bank.

Popularity of private credit-related search terms
(Last 12m, 100 = the location with the most popularity as a fraction of total searches in that location)

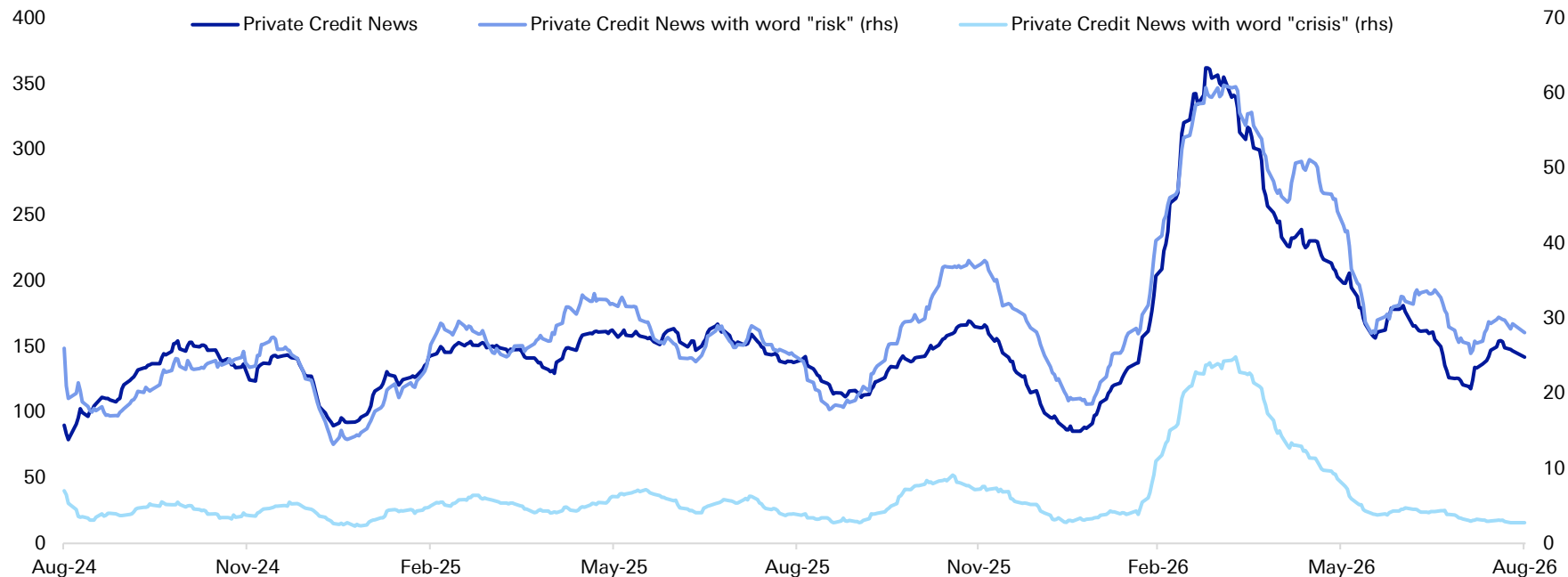


Private credit fears 2/3

There has also been a drop in the number of articles on private credit in the media, especially when it comes to “crisis”-mentioning variety. With public BDCs being at the centre of this year’s rout, the drop in such trackable indicators likely bodes well for sentiment and thus flows.



Story count of private credit news as tracked by Bloomberg, 30d average



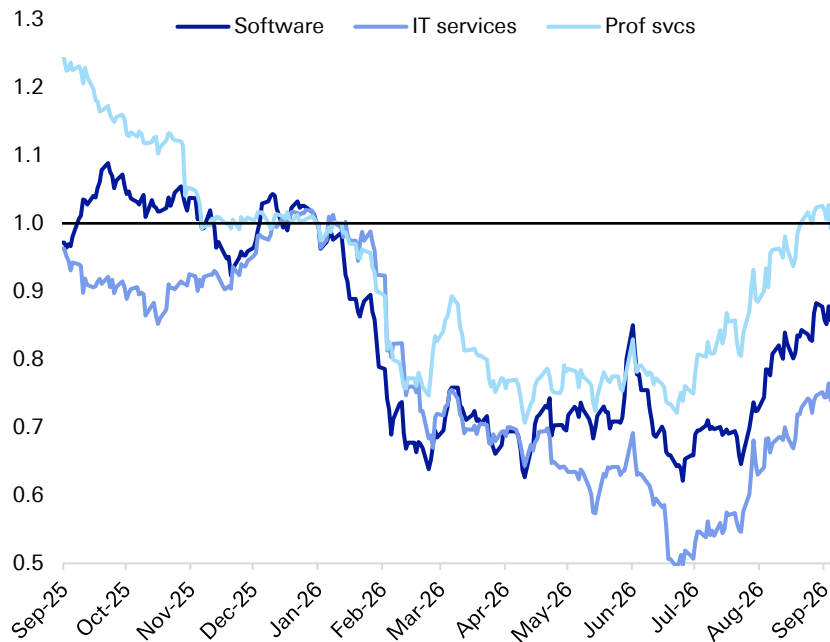
Source: Bloomberg Finance LP, Deutsche Bank.

Private credit fears 3/3

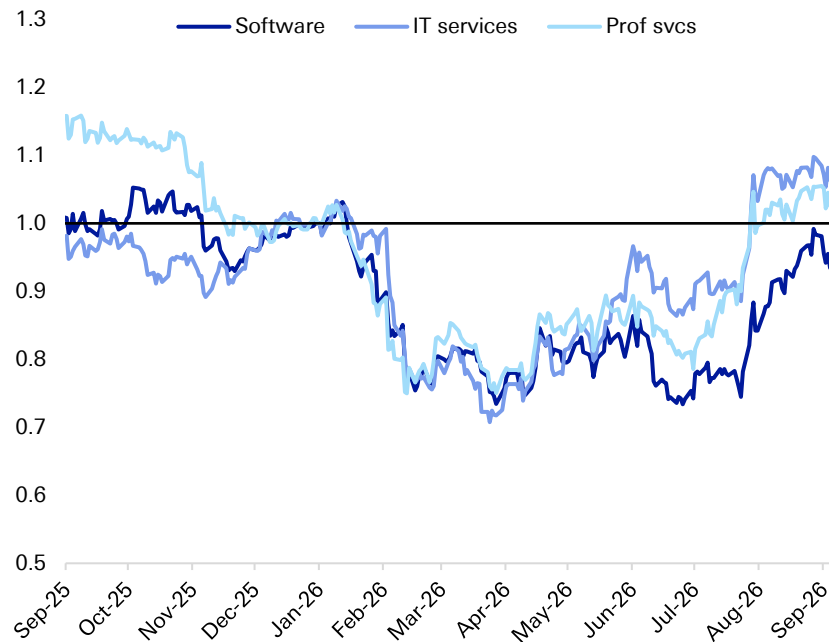
The relief to private capital also comes from receding AI disruption fears in software as firms reported solid earnings and executives continue to push back at the negative narratives around AI's impact on their businesses. Our baskets of stocks that were most affected by the original selloff earlier this year that we covered [here](#)



Selected US stocks, indexed to end of 2025



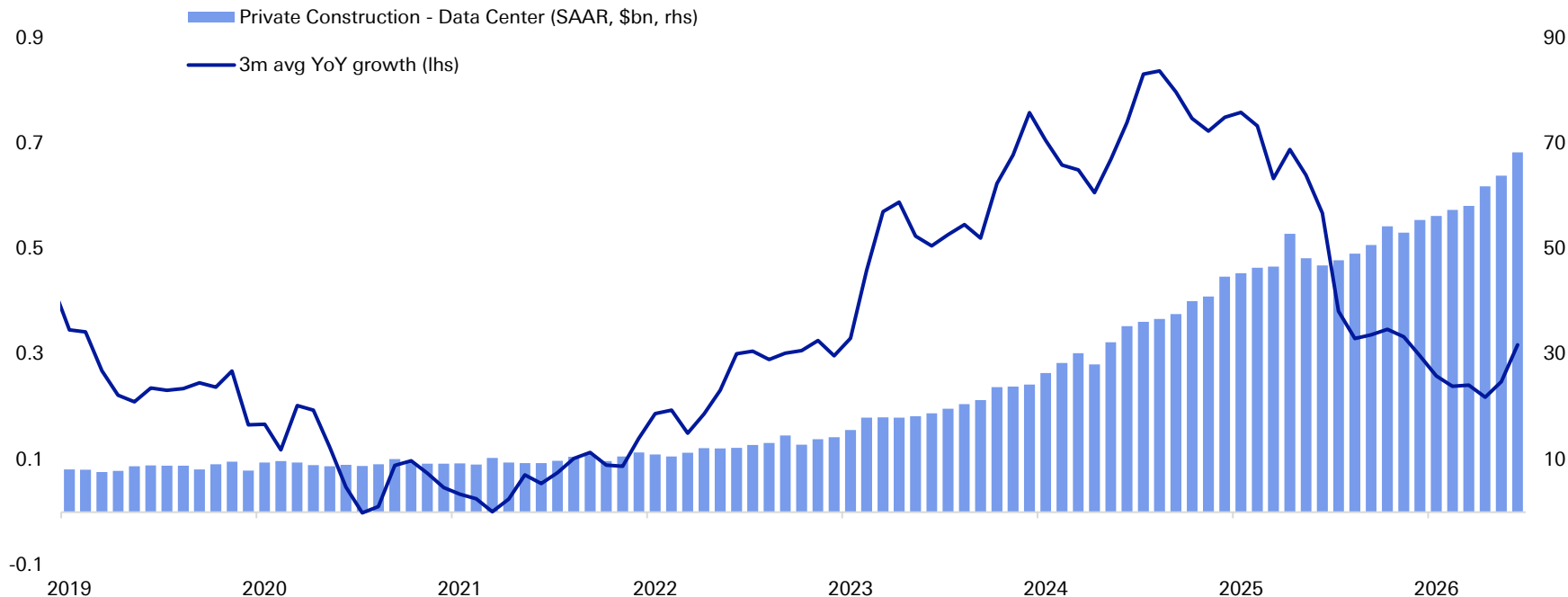
Selected European stocks, indexed to end of 2025



Source: Bloomberg Finance LP, Deutsche Bank.

AI-focused overinvestment 1/3

Amongst the most prominent risks for private capital as managers plough in money in data centres, large loans and other financing. US data centre construction growth has been increasing in recent months and now runs at annualised \$70bn per month.



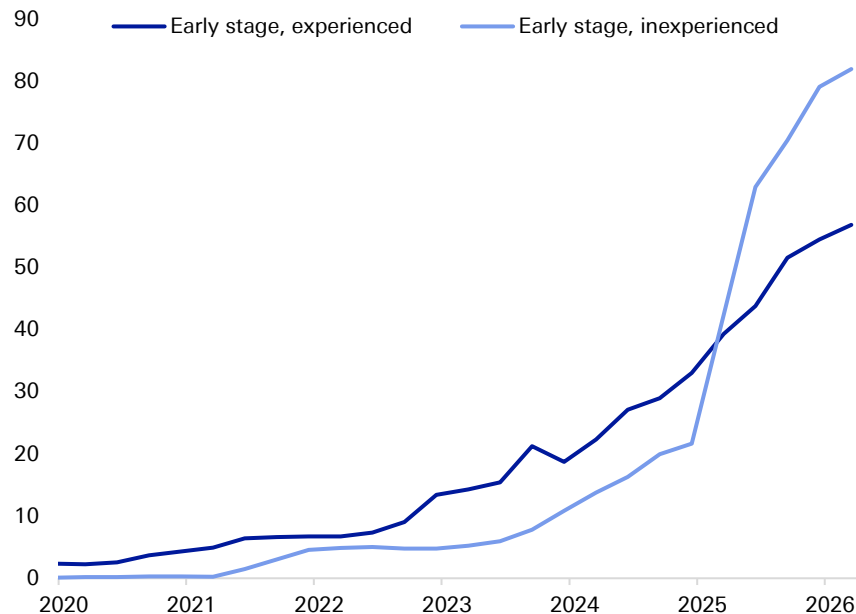
Source: Haver Analytics, Deutsche Bank.

AI-focused overinvestment 2/3

The number of entrants into AI infrastructure projects together with rising long-term yields augment concerns about the scale financing. Indeed, inexperienced developers now make up c 60% of the early-stage US data centre capacity.

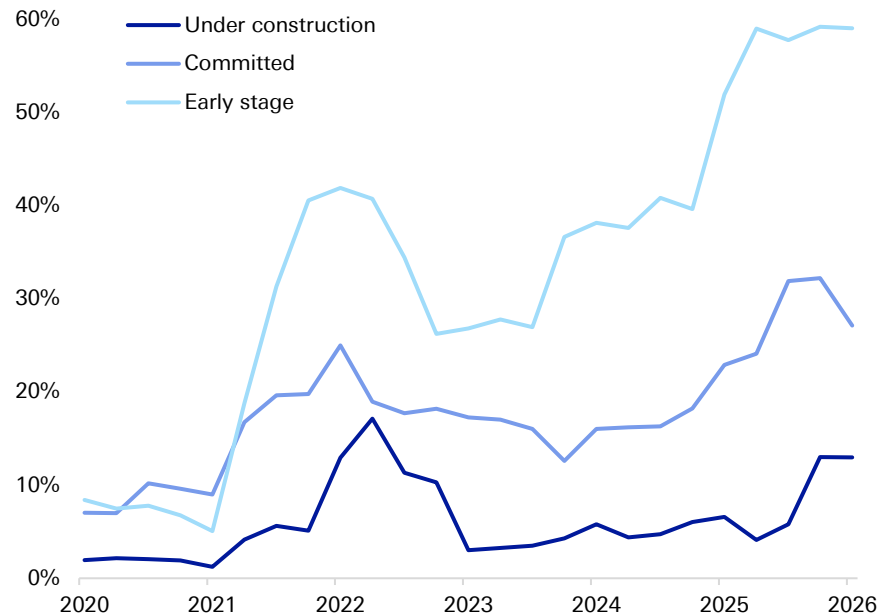


US data centre pipeline capacity, early stage by developer experience, GW



Source: Bloomberg NEF, Deutsche Bank.

US data centre pipeline capacity, share of inexperienced developers by stage

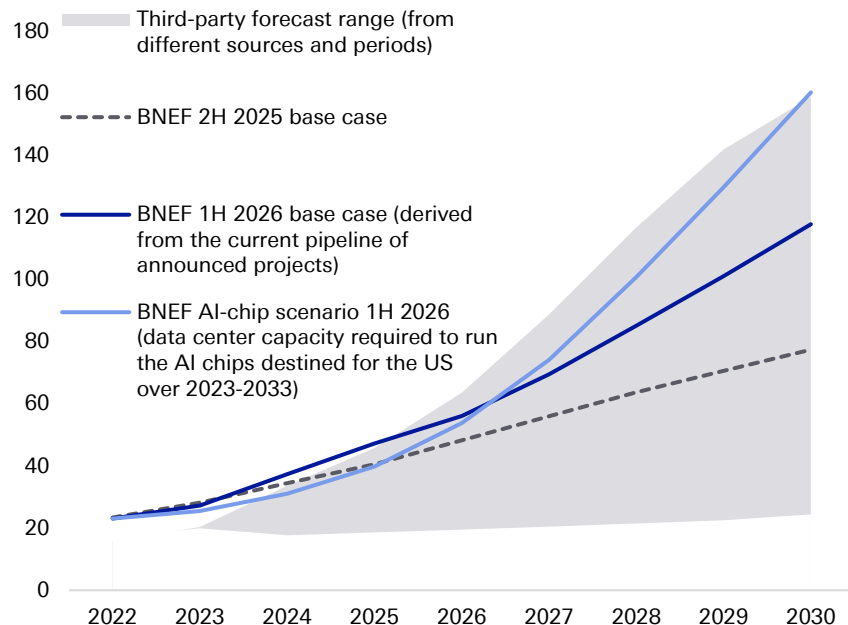


AI-focused overinvestment 3/3

The overfinancing risk is augmented by forecast uncertainty around rapid AI buildout growth, with some estimates pointing to installed capacity doubling in 3 years. Room to expand AI business adoption is a tailwind, but other factors including model advancements are harder to predict.

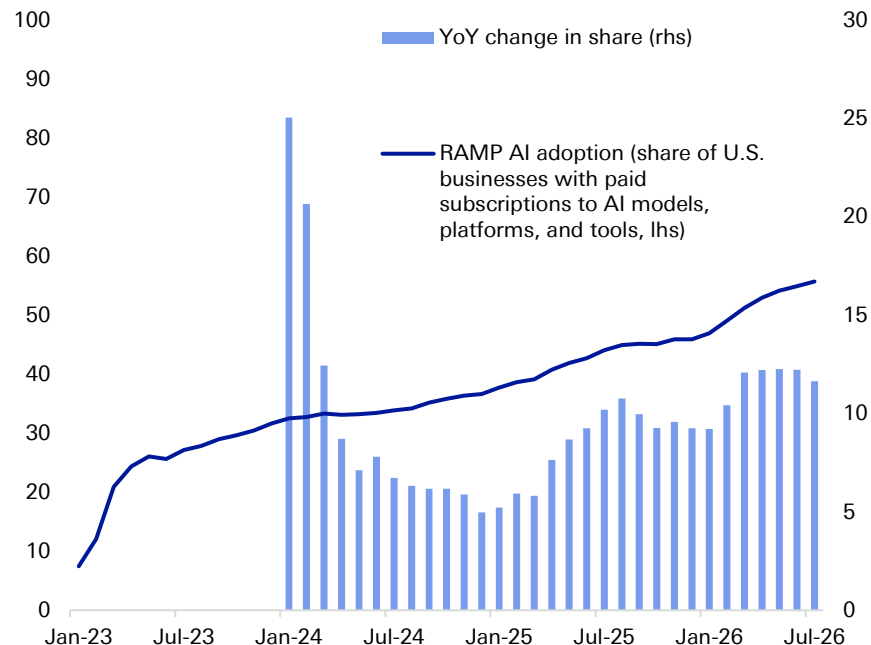


Outlook comparisons for US data-center installed power capacity (GW)



Source: Bloomberg NEF, Bloomberg Finance LP, Deutsche Bank.

Corporate AI adoption speed in the US



Realisations 1/5

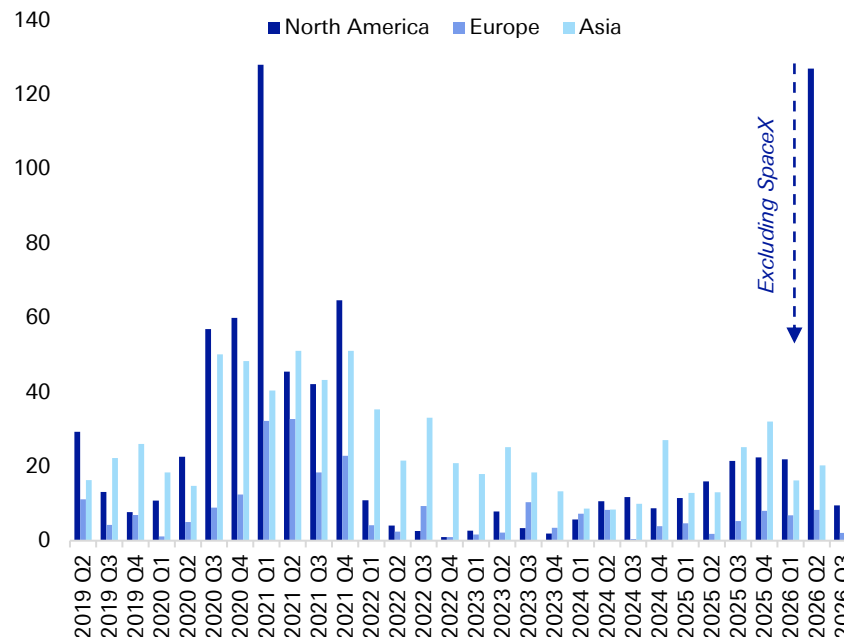
Reopening IPO markets, AI-related corporate demand and mature portfolios are all catalysts. But timing remains unpredictable and exposed to geopolitics, public-market volatility, rates and AI-related valuation uncertainty.



1. KKR described Q2 as the largest monetisation quarter in its history, with realised performance income of \$848m and realised investment income of \$220m. Remaining unrealised gains were still approximately \$18bn.
2. Carlyle returned nearly \$7bn to investors in Q2 and \$37bn over the prior year. Its US buyout strategy returned 23% of fair value over 12 months, which management said was more than twice the current industry average and above the 20% long-term industry average.
3. TPG generated \$5bn of Q2 realisations and almost \$14bn in H1, up 28% year over year.
4. Blackstone outlined the following segmentation when it comes to investor demand for assets:
 - AI, power, electrification and energy assets: strong strategic, sponsor and IPO demand
 - AI-unaffected businesses: broadly available liquidity
 - Software, professional services and information services: materially weaker activity because of AI uncertainty.

The firm estimated the third category represents approximately 30%-40% of the private-equity market.
5. TPG similarly noted that industry realisations were down about 46% quarter over quarter, although its own H1 activity was higher. It expects its monetisation cadence to accelerate toward year-end and into 2027. Brookfield expects transaction volumes to build in H2, while Blackstone expects weaker Q3 realisations followed by a stronger Q4 and 2027.

IPO deal values by quarter \$bn



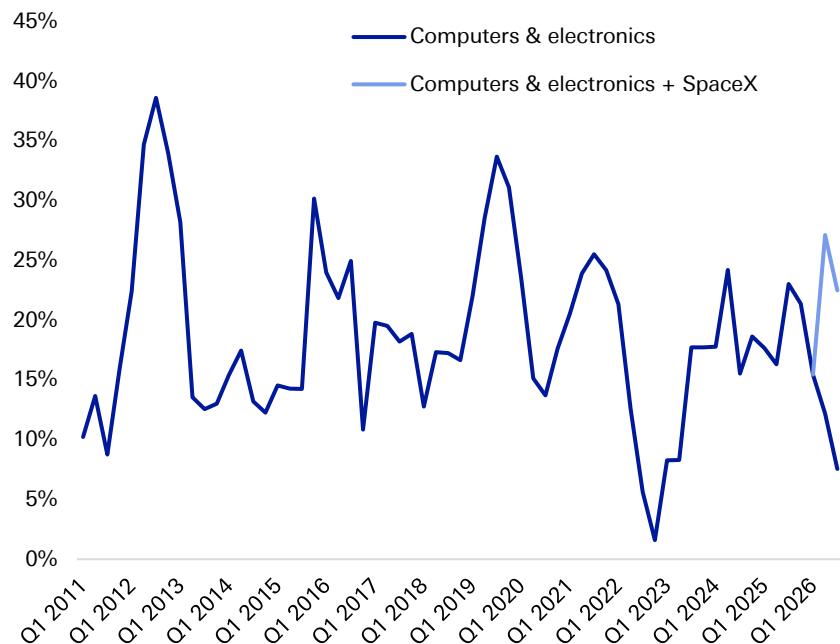
Source: Bloomberg Finance LP, Haver Analytics, Deutsche Bank.

Realisations 2/5

More AI-related businesses are likely to come to the market even after the big AI firms go public. Tech IPOs have been relatively muted despite the boom in financing this decade. Investors' appetite for AI and big capital markets issuances may push more managers to exit tech holdings while the mood lasts. There have been just c \$7bn in pure-play AI IPOs in the US since 2025.

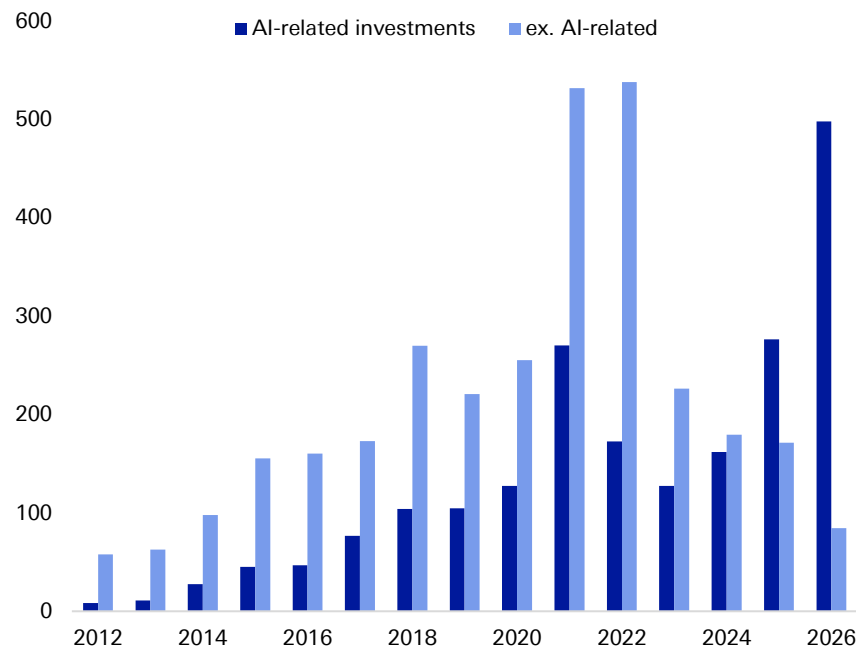


Tech share of total US IPO proceeds, 4q MA



Source: Dealogic, OECD, Deutsche Bank.

Worldwide VC investment in AI (\$bn)

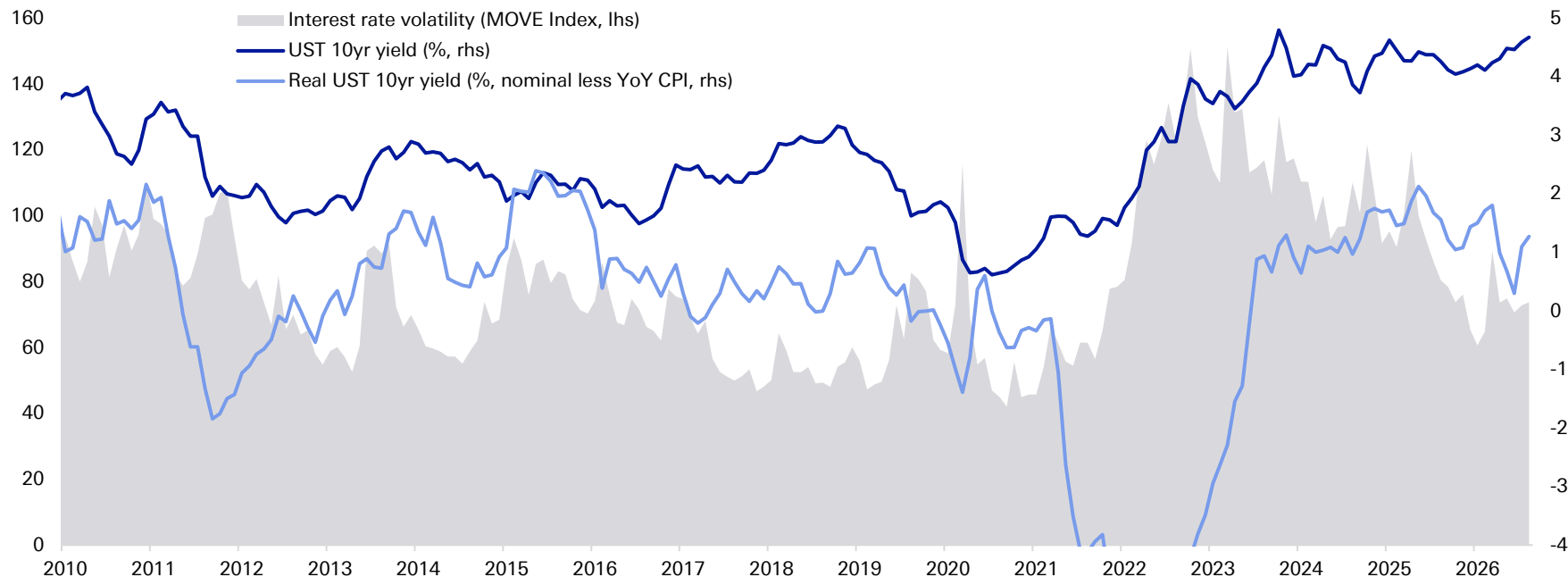


Realisations 3/5

Rising yields are likely to be less of a headwind to capital markets activity than many expect. Meanwhile, both real yields and volatility remain subdued. We have shown previously that dealmaking has been strong even when rates were elevated.



US rates backdrop – monthly averages



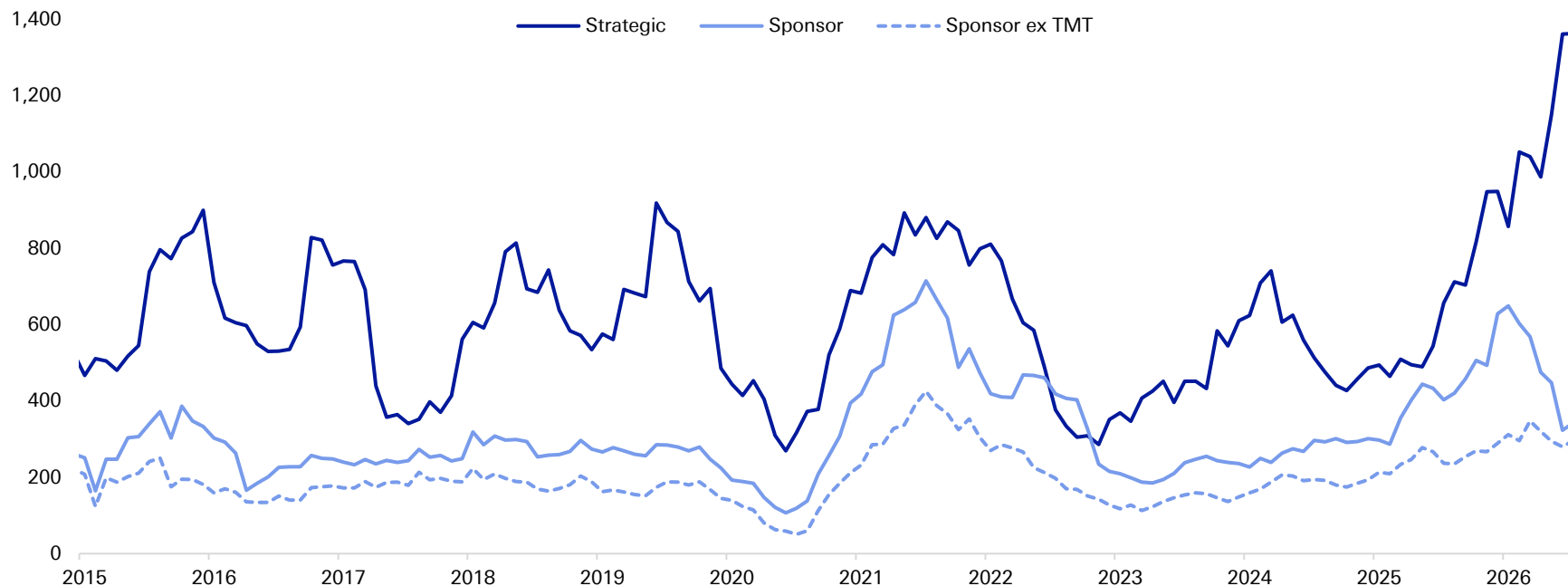
Source: Bloomberg Finance LP, Google Trends, Deutsche Bank.

Realisations 4/5

Rates are also unlikely to grind corporate buyers and thus sale exits to a halt. Meanwhile, sponsor M&A outside TMT has a number of tailwinds.



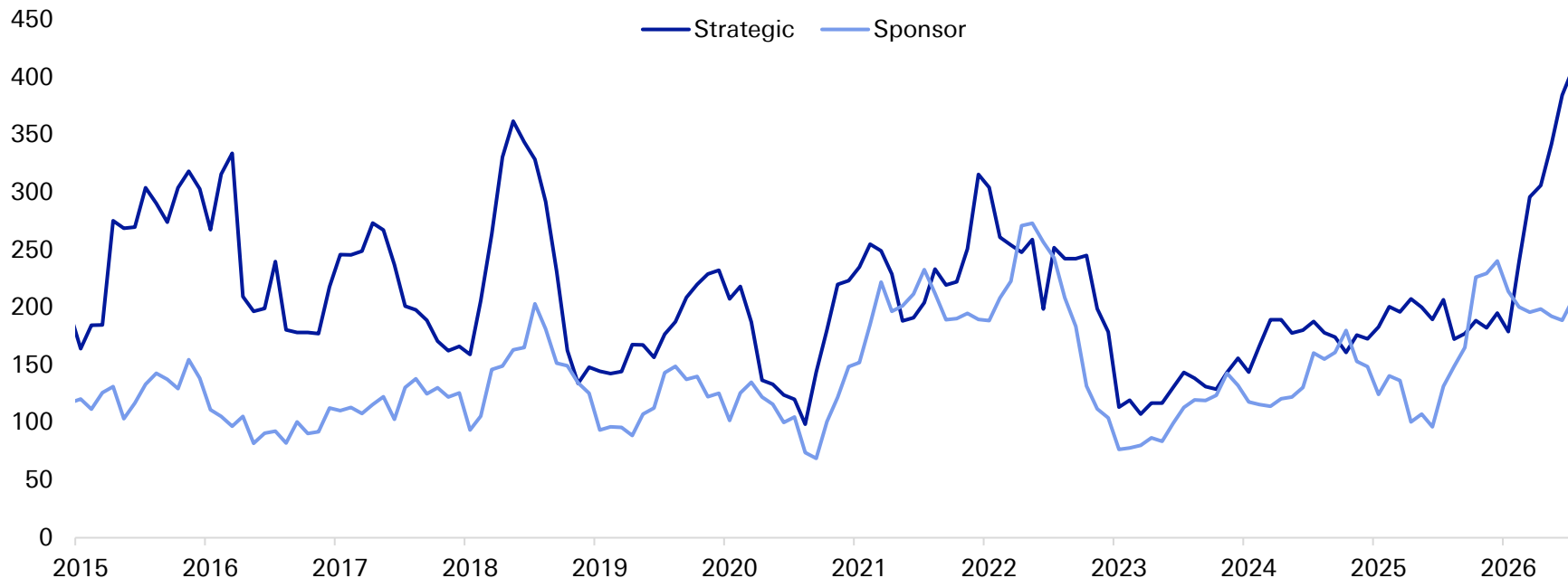
US M&A deal value at announcement (\$bn, 6m sum)



Source: Dealogic, Deutsche Bank. Note: through July 2026.



W Europe M&A deal value at announcement (\$bn, 6m sum)



Source: Dealogic, Deutsche Bank. Note: through July 2026.



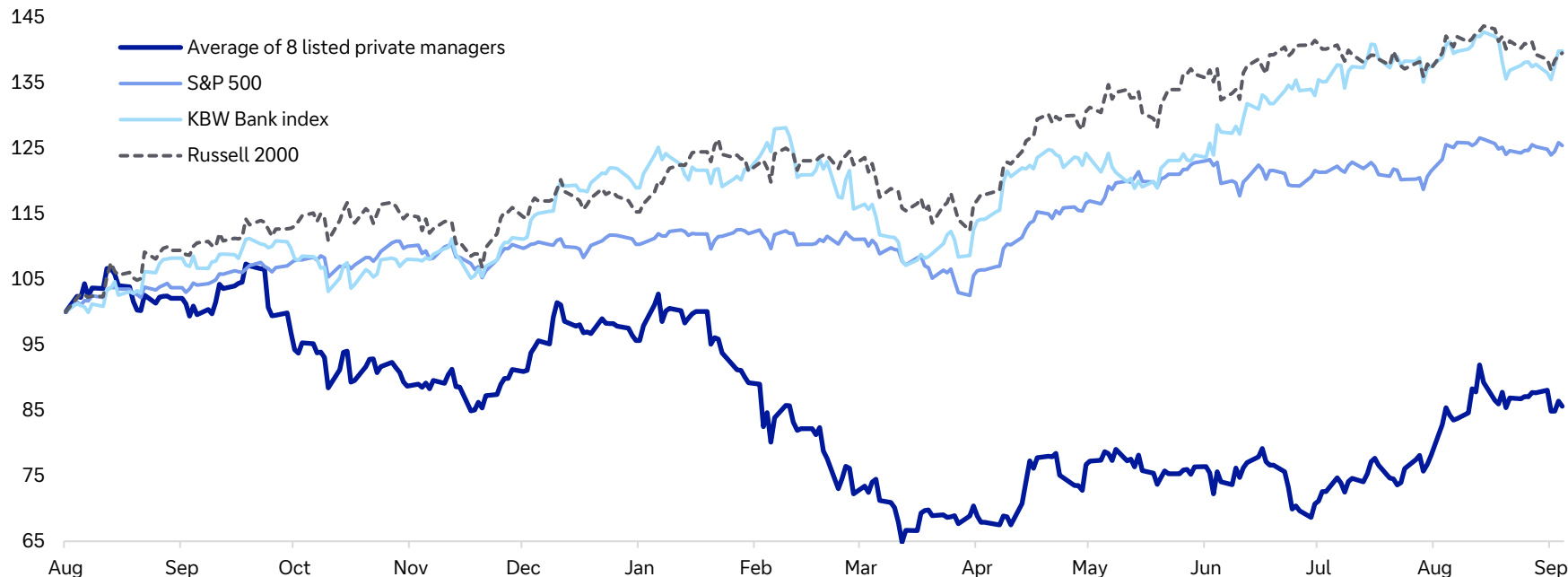
03. Market conditions

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Shares of US-listed private capital managers are up +32% from YTD lows and +25% from June trough as some confidence returns in the asset class. Our index is still down -11% YTD vs +14% for the S&P 500 and 18% for US banks



Private capital managers total return performance relative to broader markets (USD)



Source: Bloomberg Finance LP, Deutsche Bank.

European firms have also rebounded by +23% (from March lows) and +14% (June). The YTD return (-1%) also underwhelms the broader market (+13%) and local banks (+28%)



Private capital managers total return v broader markets in 2025 (EUR)

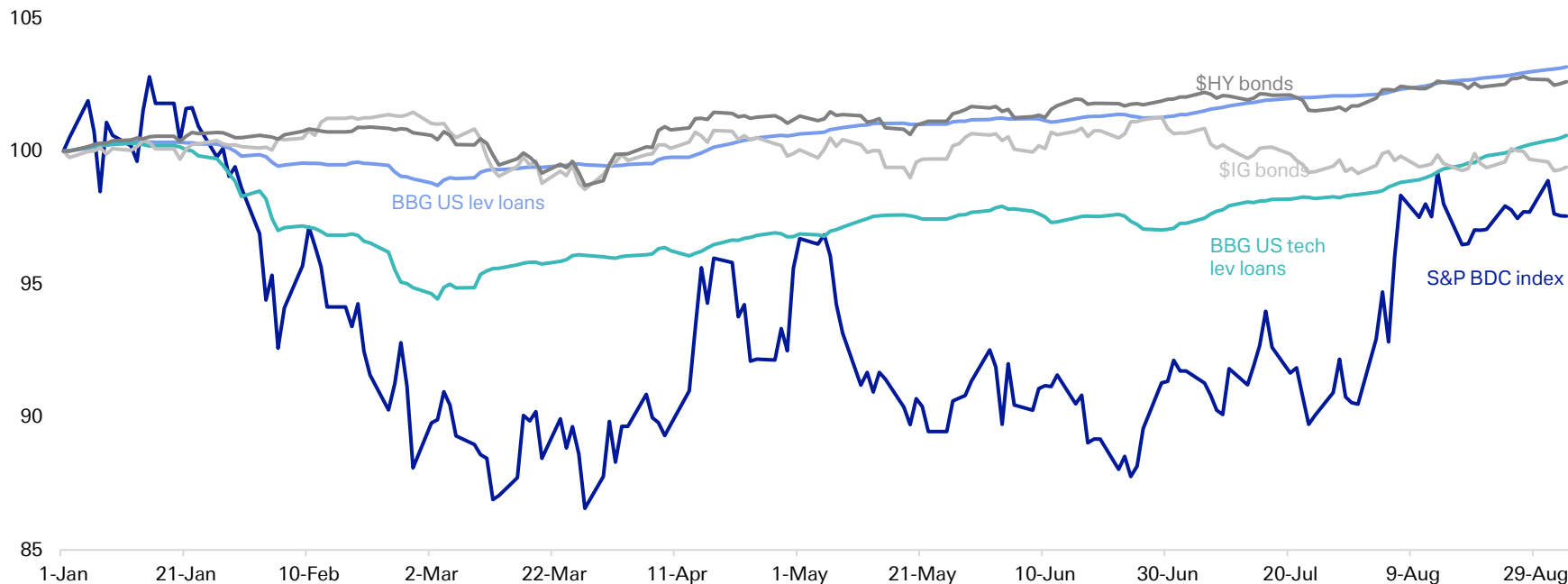


Source: Bloomberg Finance LP, Deutsche Bank.

It's a similar rebound story for BDC returns vs the broader credit markets in recent weeks

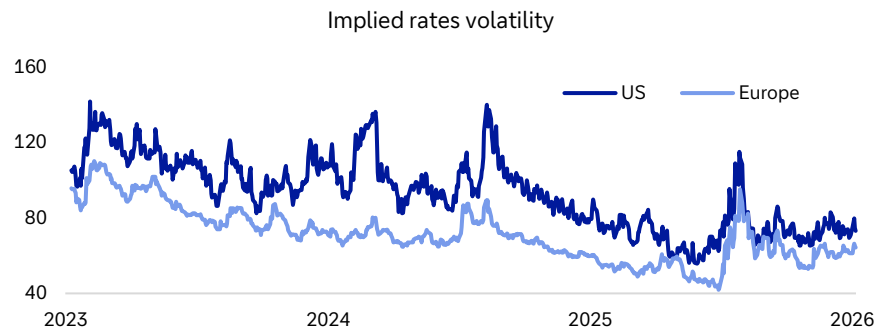
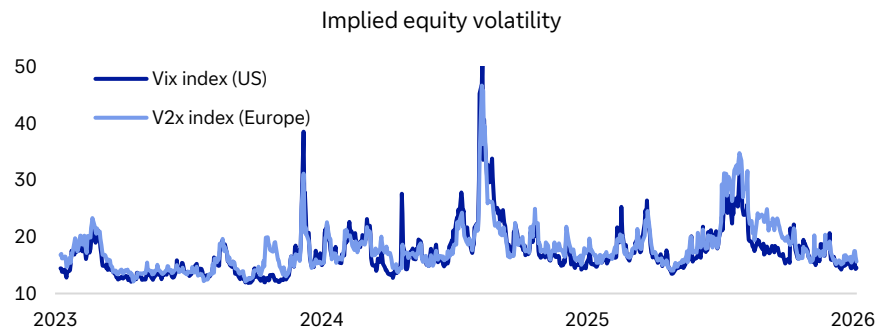
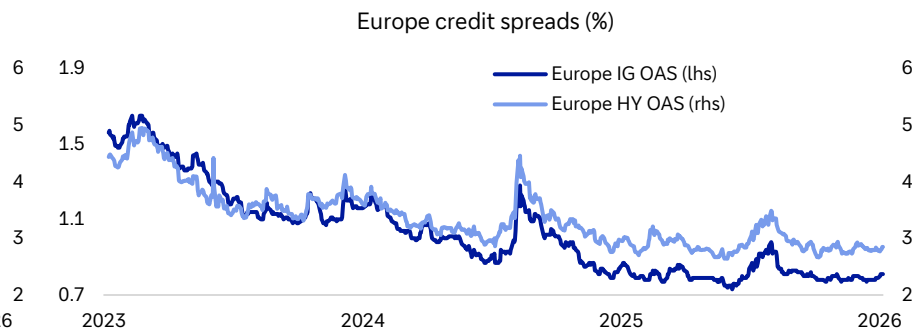
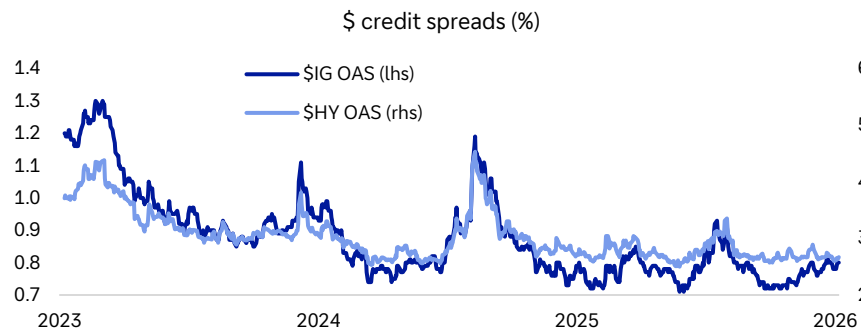


Credit assets YTD total return performance



Source: Bloomberg Finance LP, Deutsche Bank.

Contained volatility, especially in rates amidst rising yields, helps overall risk backdrop. Credit spreads have also been steady.

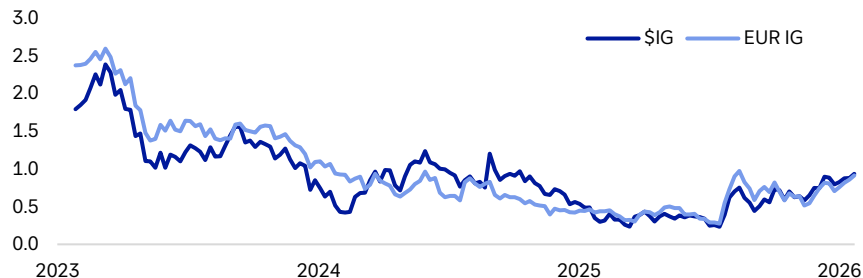


Source: Bloomberg Finance LP, Deutsche Bank.

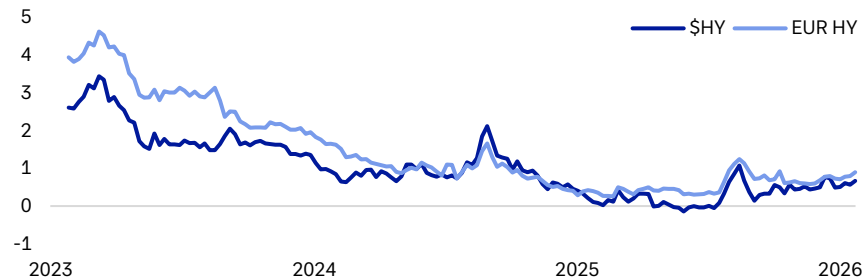
Rising refinancing costs may deter some debt issuance but overall credit conditions remain favourable



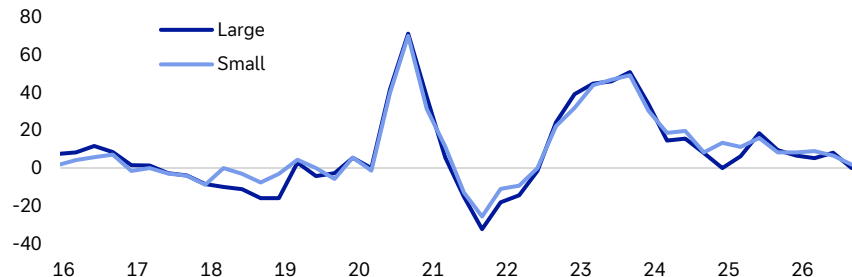
IG bonds - Refinancing proxy in %



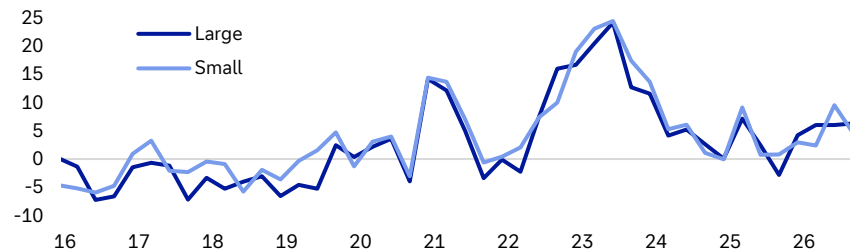
HY bonds - Refinancing proxy in %



Fed SLOOS: banks tightening standards for C&I loans to firms (%)

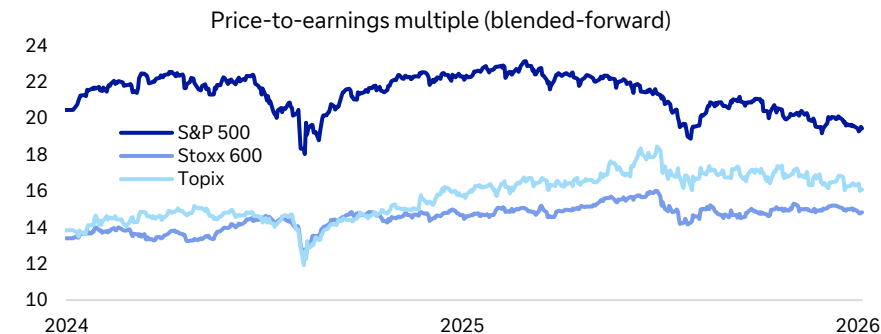
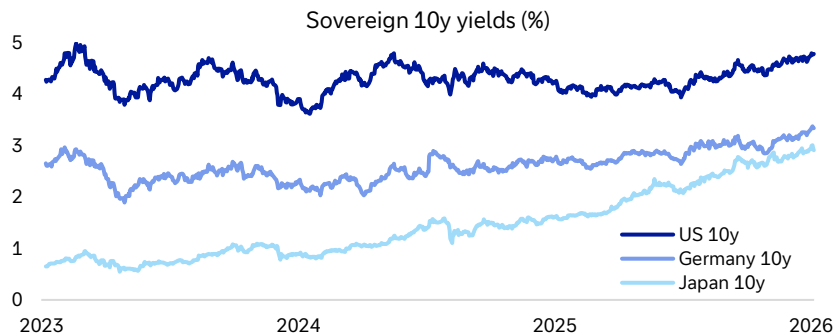
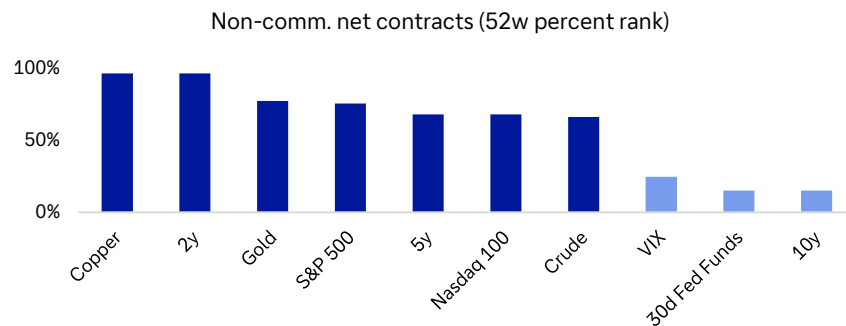
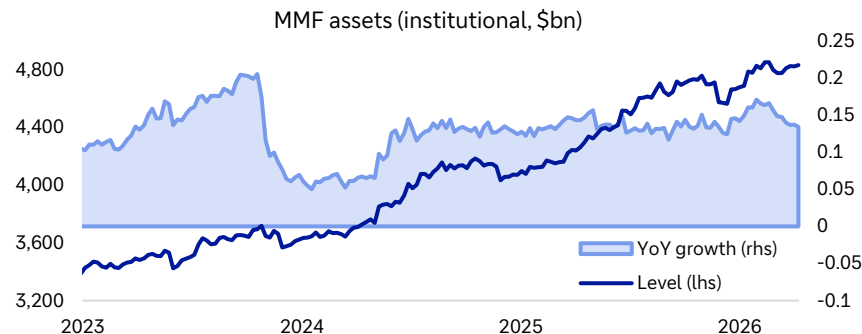


Euro Area bank lending survey: change in credit standards for business loans



Source: Bloomberg Finance LP, Haver Analytics, Deutsche Bank.

Despite rising yields, markets remain in a risk-on mood and there is plenty of cash

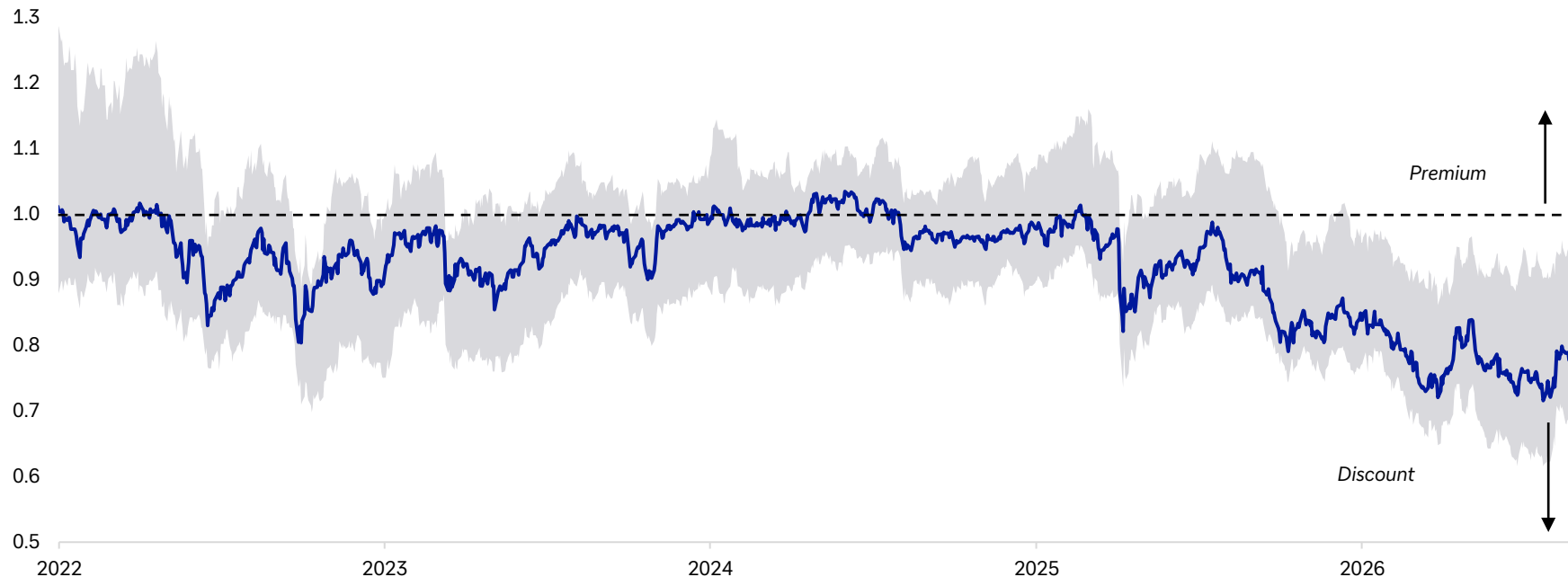


Source: Bloomberg Finance LP, Haver Analytics, Deutsche Bank.

The median BDCs discount to NAV is still wide and now stands at c 22%. Apart from software fears, the focus is on the trajectory of US monetary policy.

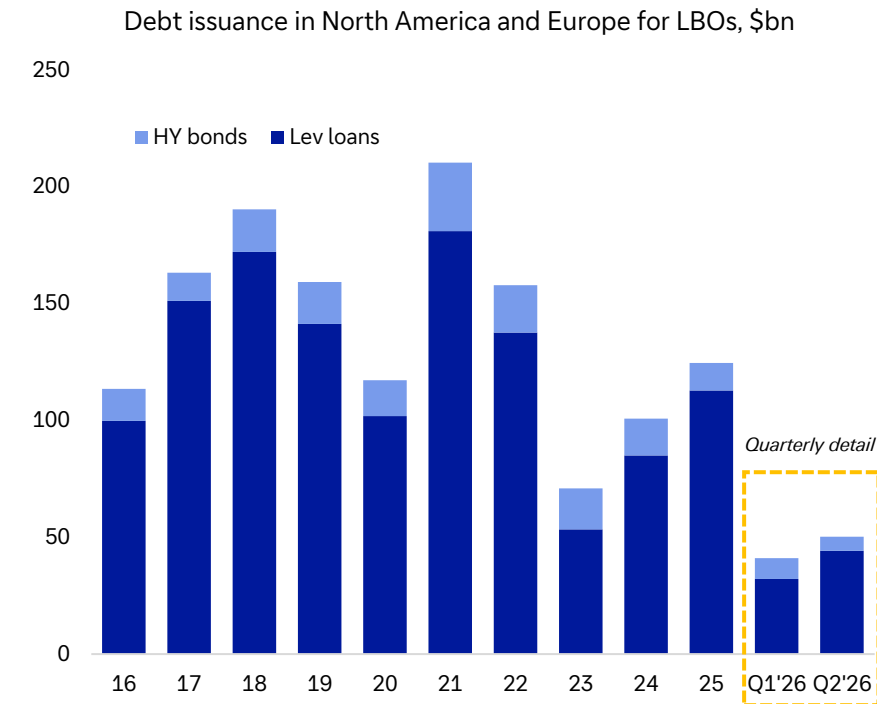


Median price-to-NAV ratio of funds in the S&P BDC index

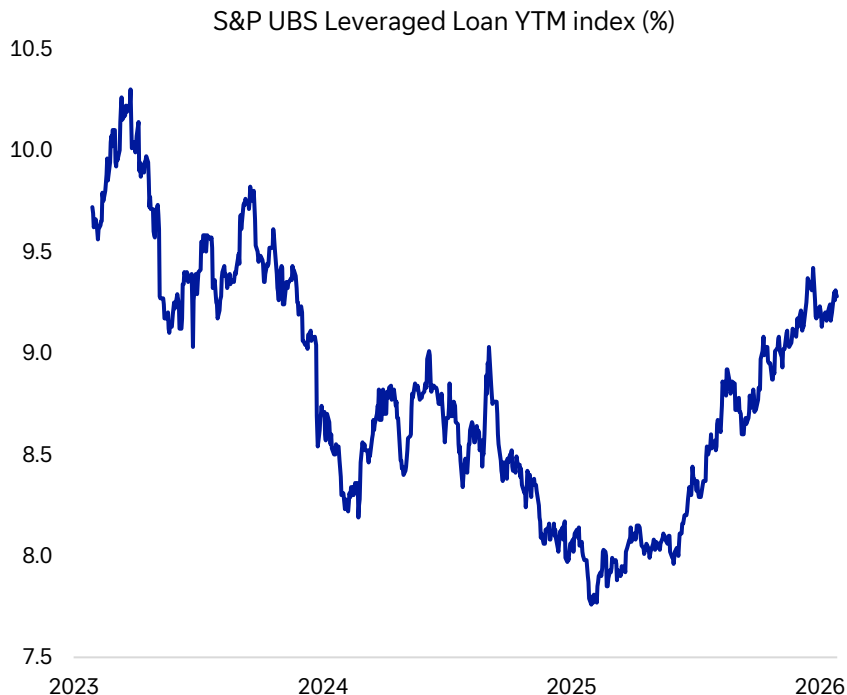


Source: Bloomberg Finance LP, Deutsche Bank.

Debt issuance for LBOs continued to grow in Q2. Rising sovereign rates may dissuade some borrowers but leveraged loan yields remain below recent highs and earnings have been strong.



Source: White & Case, Bloomberg Finance LP, Deutsche Bank.





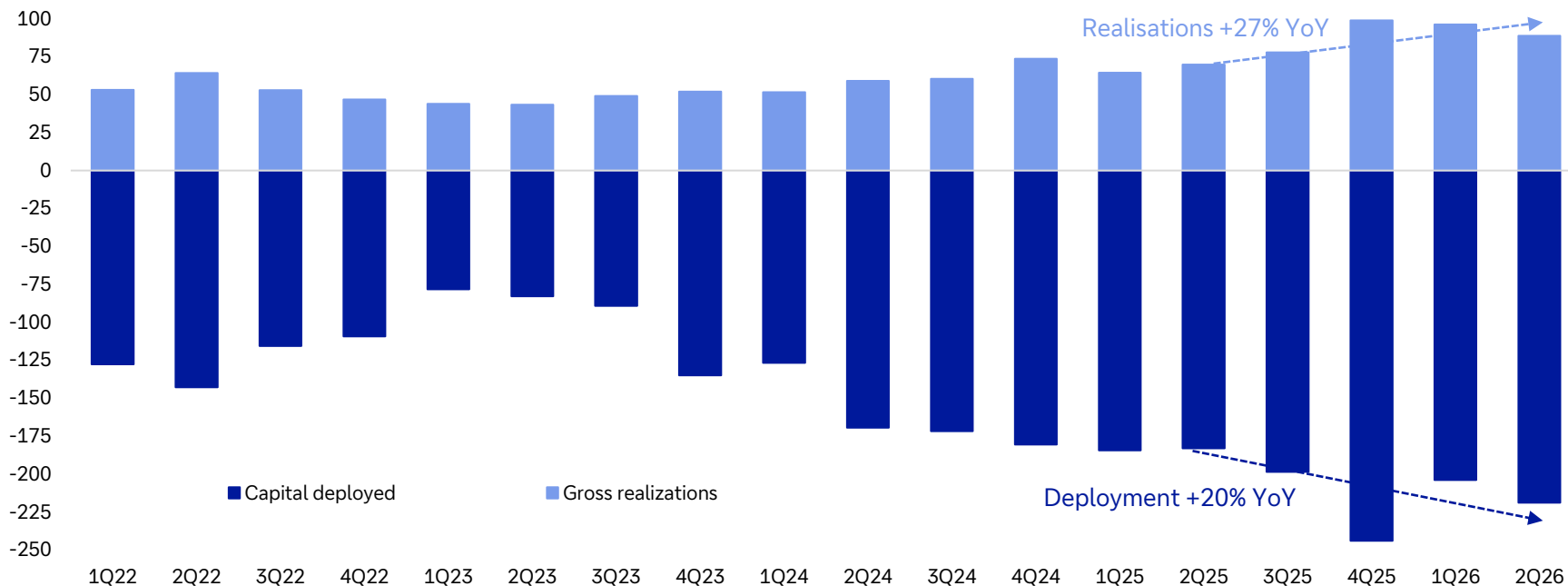
04. Fundraising environment

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Both deployment and realisations are up YoY. Realisations growth will be in focus given uncertainty voiced by managers.



Selected private managers, deployment v realisations (\$bn)

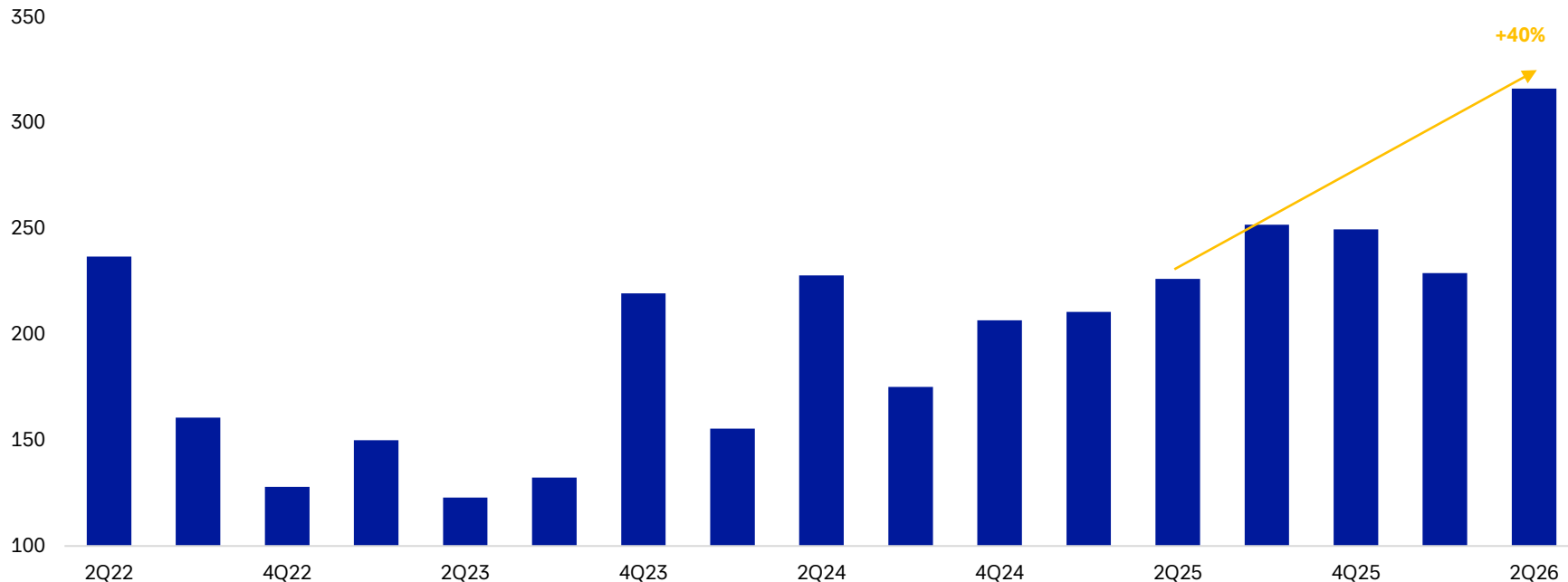


Source: Deutsche Bank company research – [US Brokers, Asset Managers & Exchanges - Deutsche Bank Research](#), Deutsche Bank. * Note: Deutsche Bank company research forecasts.

Aggregate fundraising growth has shot up (in part due to one-off factors) and managers are confident in continued strength.



Fundraising by selected large, listed managers (\$bn)

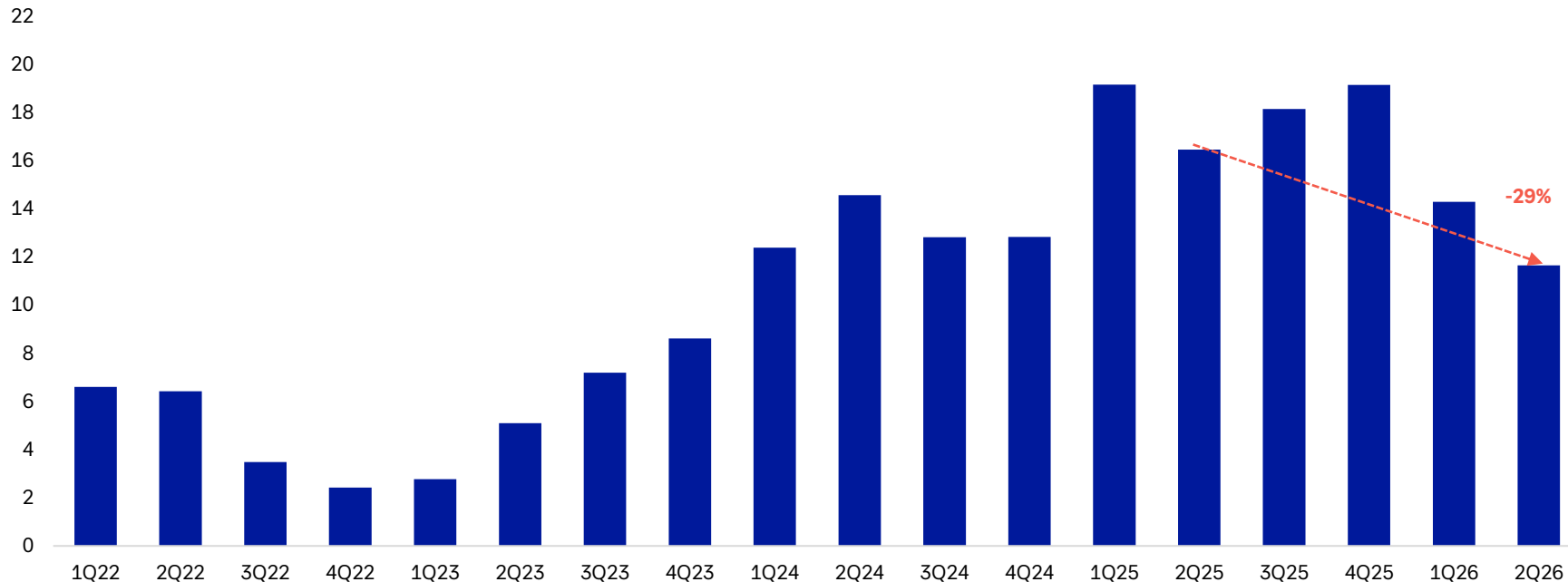


Source: Deutsche Bank company research – [US Brokers, Asset Managers & Exchanges - Deutsche Bank Research](#), Deutsche Bank.

Gross retail fundraising (not taking into account redemptions) continues to come down, now down almost a third YoY.



Retail fundraising by largest listed private capital managers *

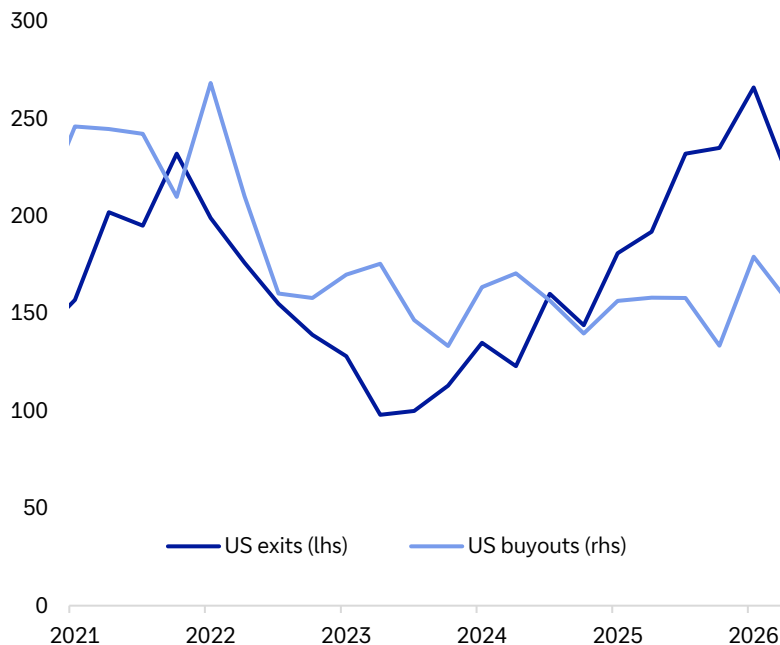


Source: Deutsche Bank company research – [US Brokers, Asset Managers & Exchanges - Deutsche Bank Research](#), Bloomberg Finance LP, Deutsche Bank. * Note: gross, only funds with available Q2 data.

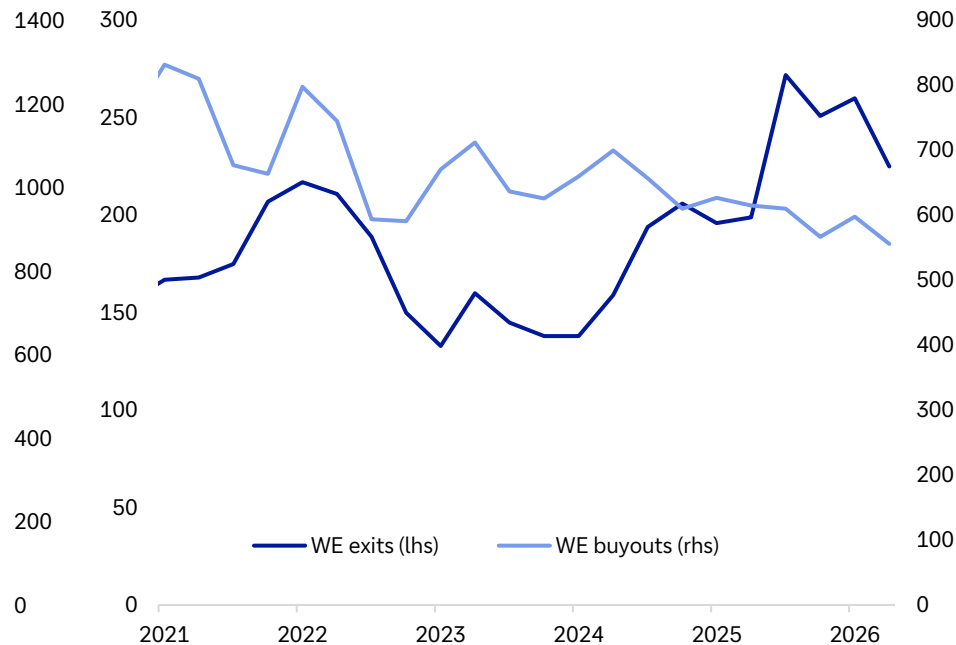
Exit activity remains strong despite some softening relative to recent highs



US PE exits have come down from recent highs



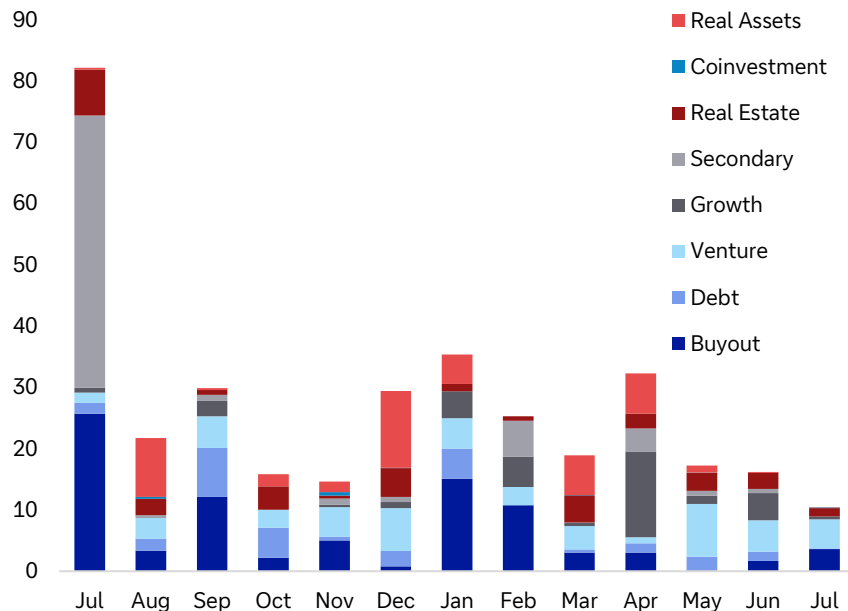
European exits have also slowed but remain high



Source: White & Case, Deutsche Bank.

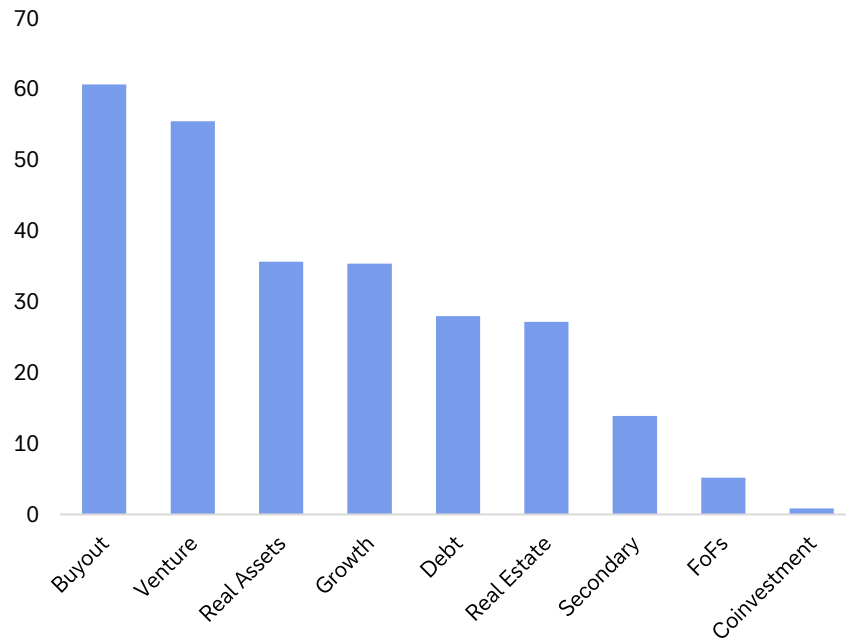


Fundraising funds - target in \$bn by strategy type and launch month, US and Europe



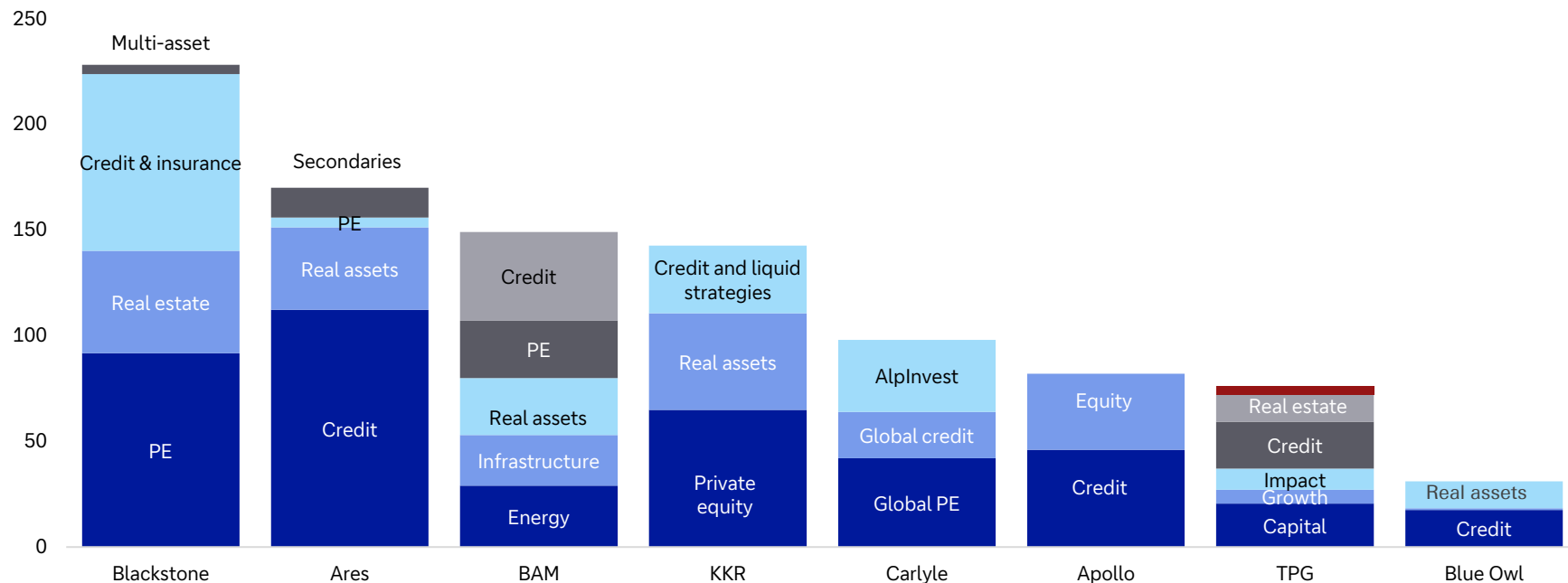
Source: Bloomberg Finance LP, Deutsche Bank.

Fundraising funds in US and Europe, last 12m in \$bn





Dry powder by manager and reported segment, Q2 2026 (\$bn)

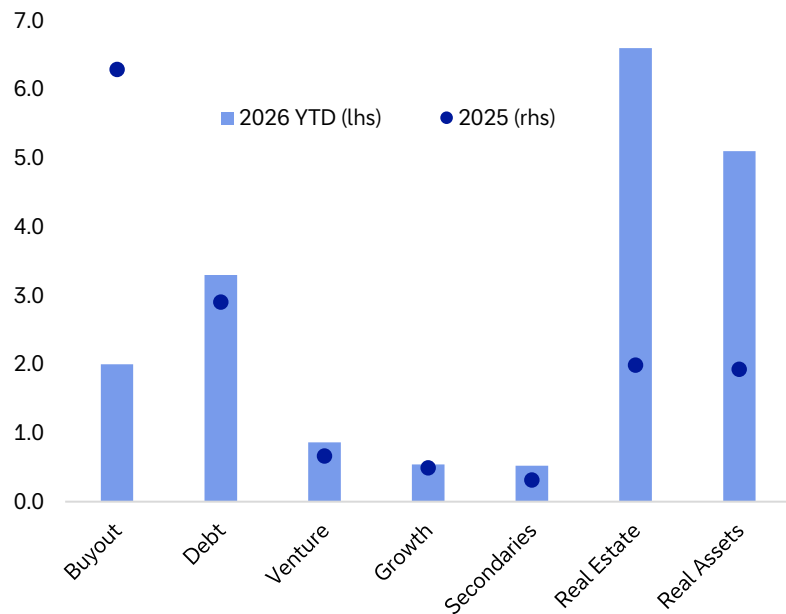


Source: Company reports, Deutsche Bank.

Recent disclosed LP commitments

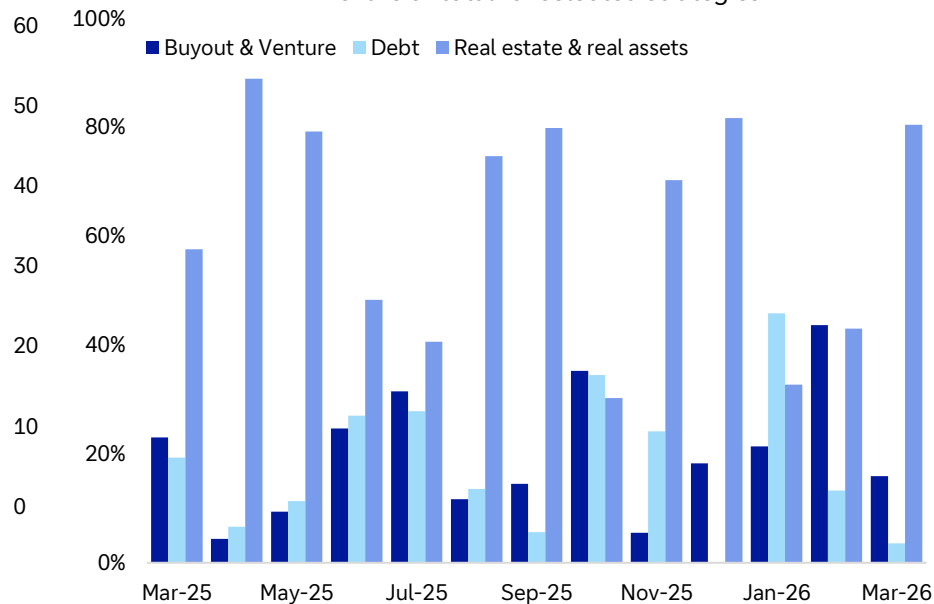


Disclosed LP commitments by strategy in \$bn



Source: Bloomberg Finance LP, Deutsche Bank.

Disclosed LP commitment targets by month and strategy, share of total for selected strategies

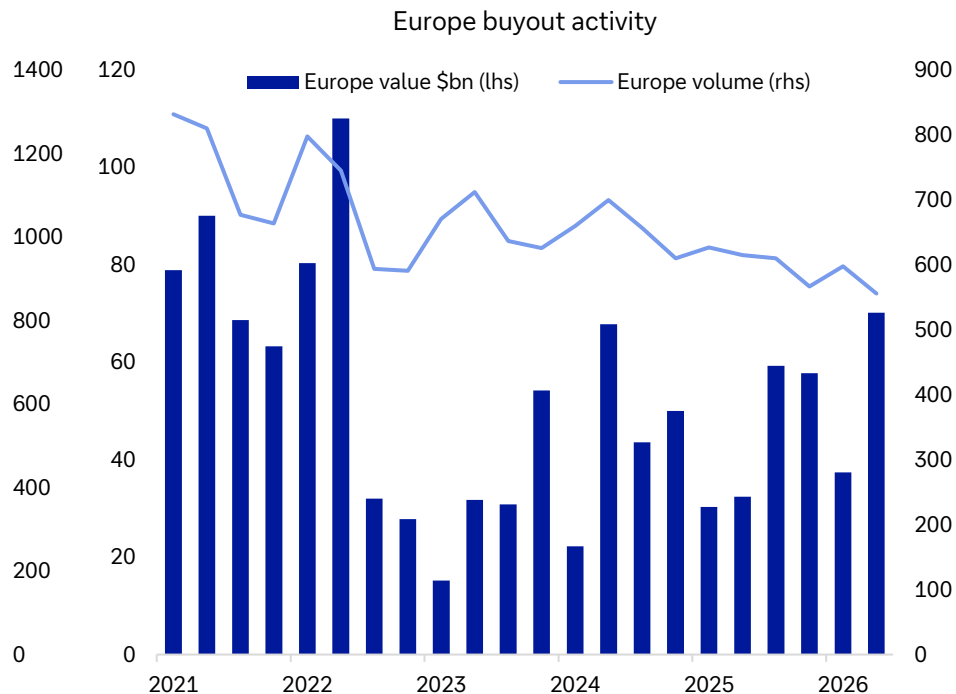
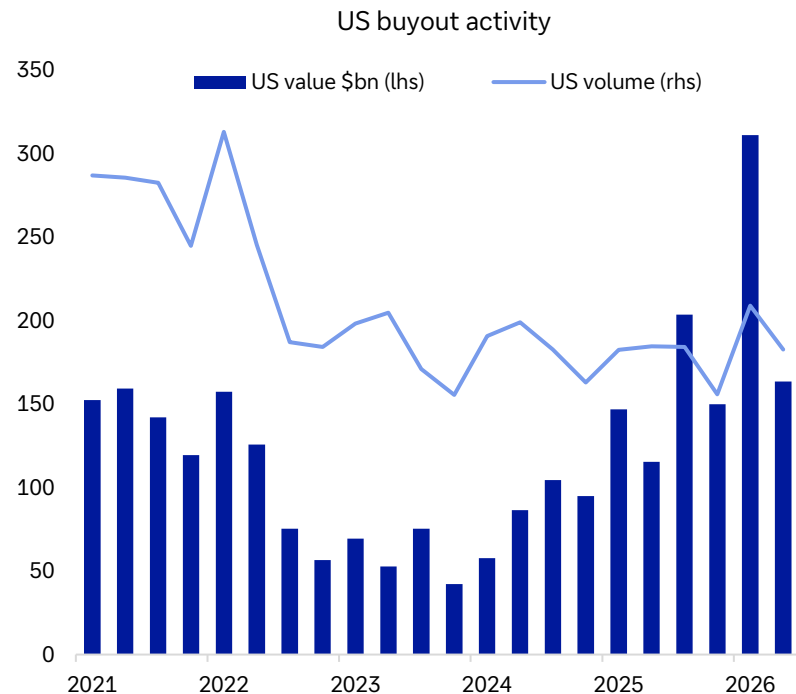




05. Dealmaking activity

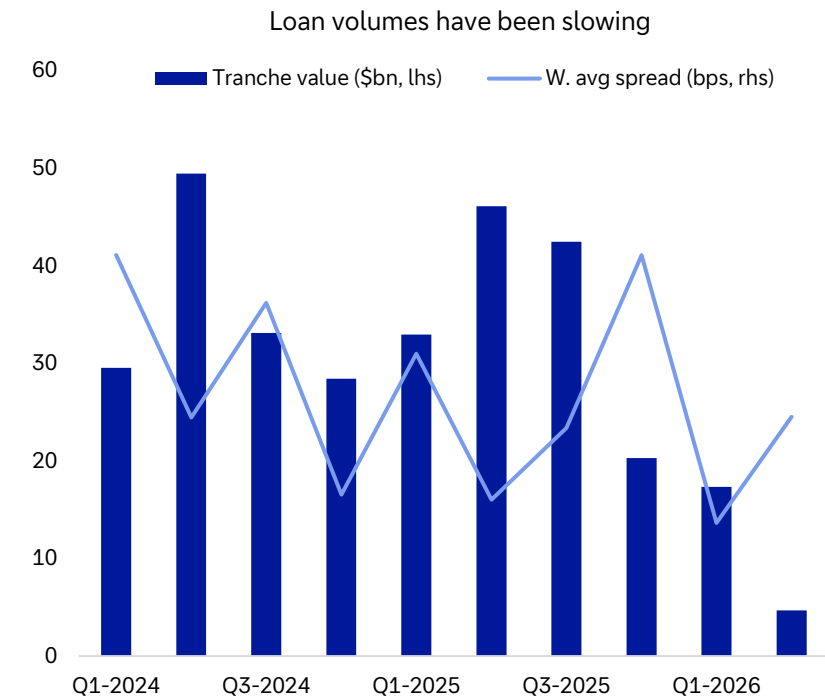
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Dealmaking activity in buyouts

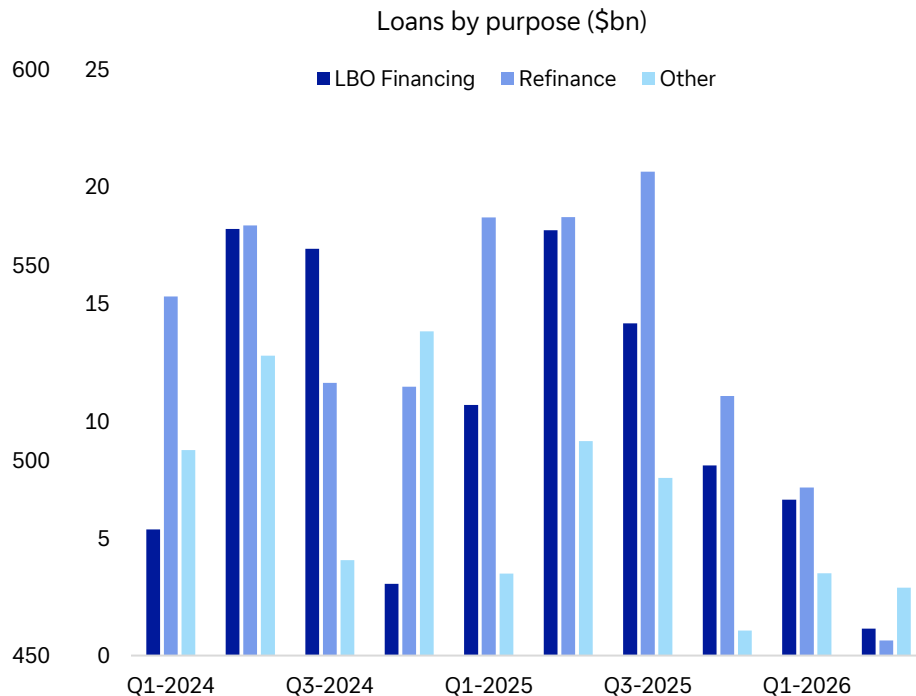


Source: White & Case, Deutsche Bank.

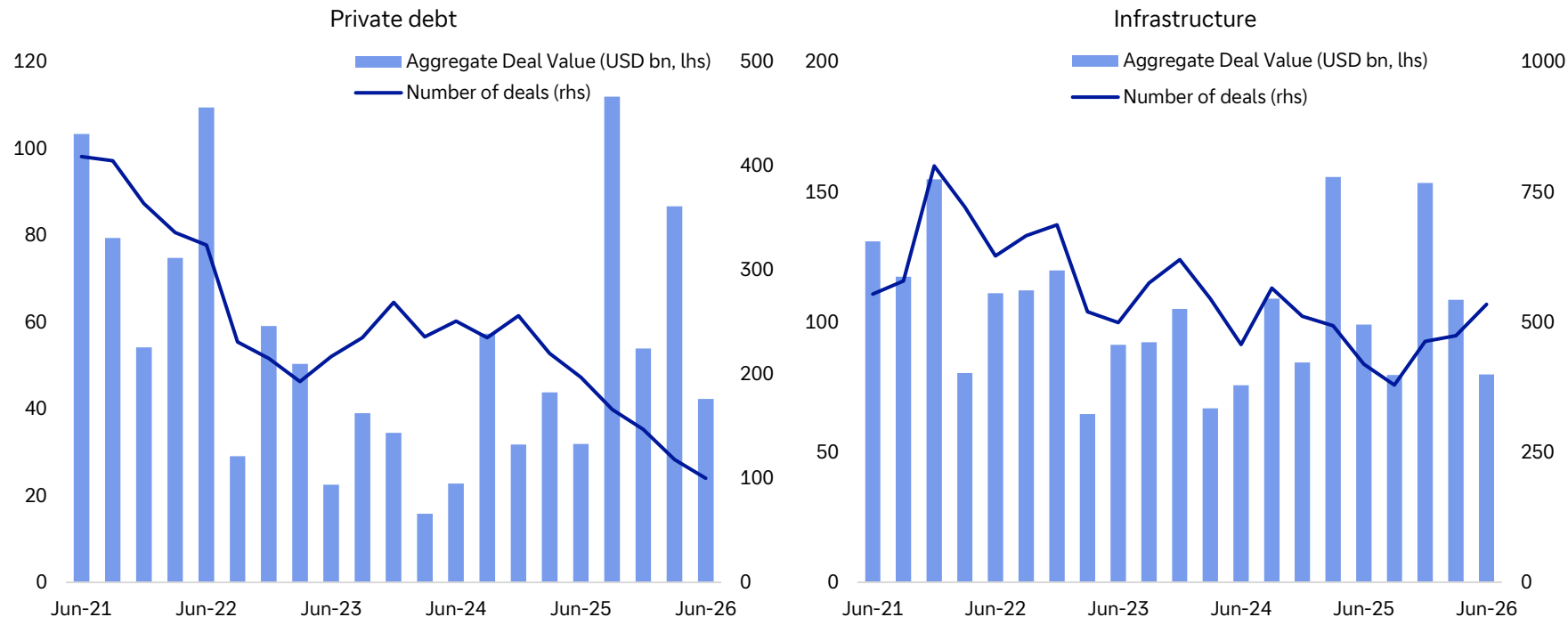
Origination in large private credit loans – loan volumes continued to slow amidst more caution around private credit spilling over from Q1. A recovery in sentiment will support some rebound in volumes.



Source: Bloomberg Finance LP, Deutsche Bank.



Global dealmaking activity in private debt and infrastructure



Source: Preqin, Deutsche Bank.

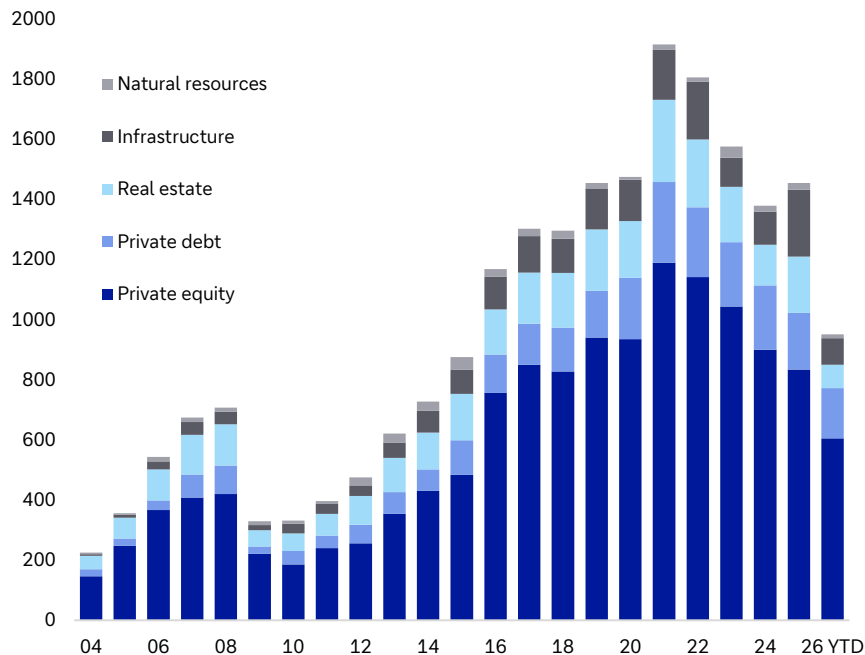


06. Investor allocations

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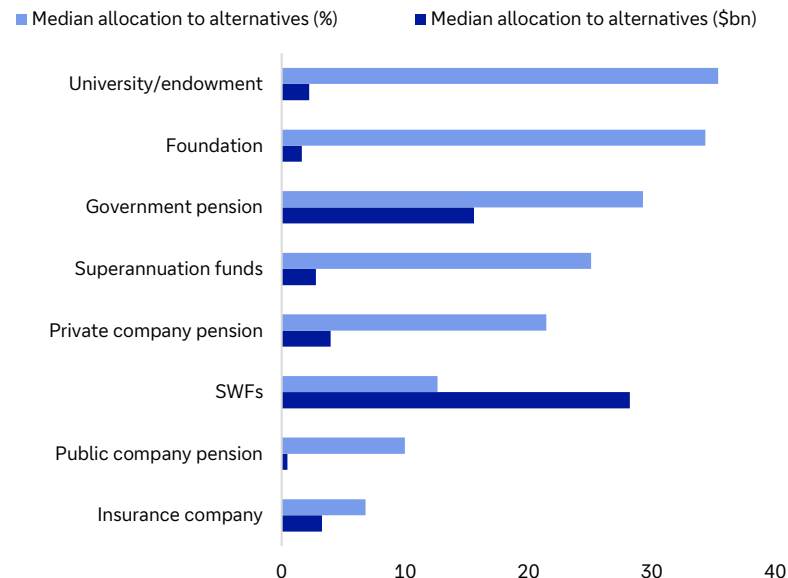


Private capital sum of funds raised (\$bn)



Source: Preqin, Bloomberg Finance LP, Deutsche Bank.

Top 100 LPs in each category, allocations to alternatives



Appendix 1



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Research

The Home Straight....

Jim Reid | Henry Allen | Rajsekhar Bhattacharyya

September 2026

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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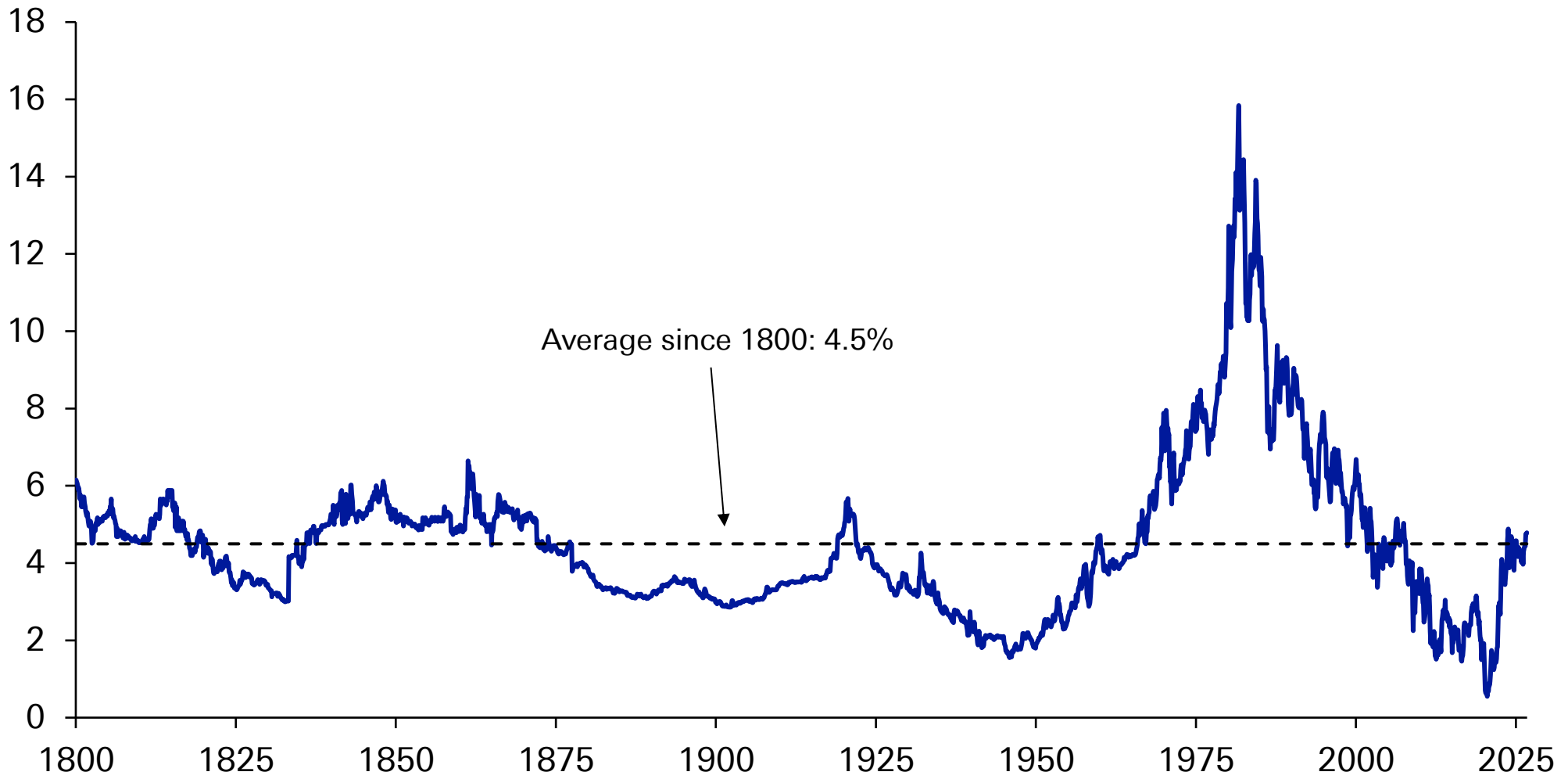


The bond sell-off in perspective...

We would see the recent rise in yields as a continuation of a long normalisation process. 10yr US Treasury yields are now “only” slightly above their long-term average yield since 1800... so they are certainly not extreme....



10-year US treasury yields

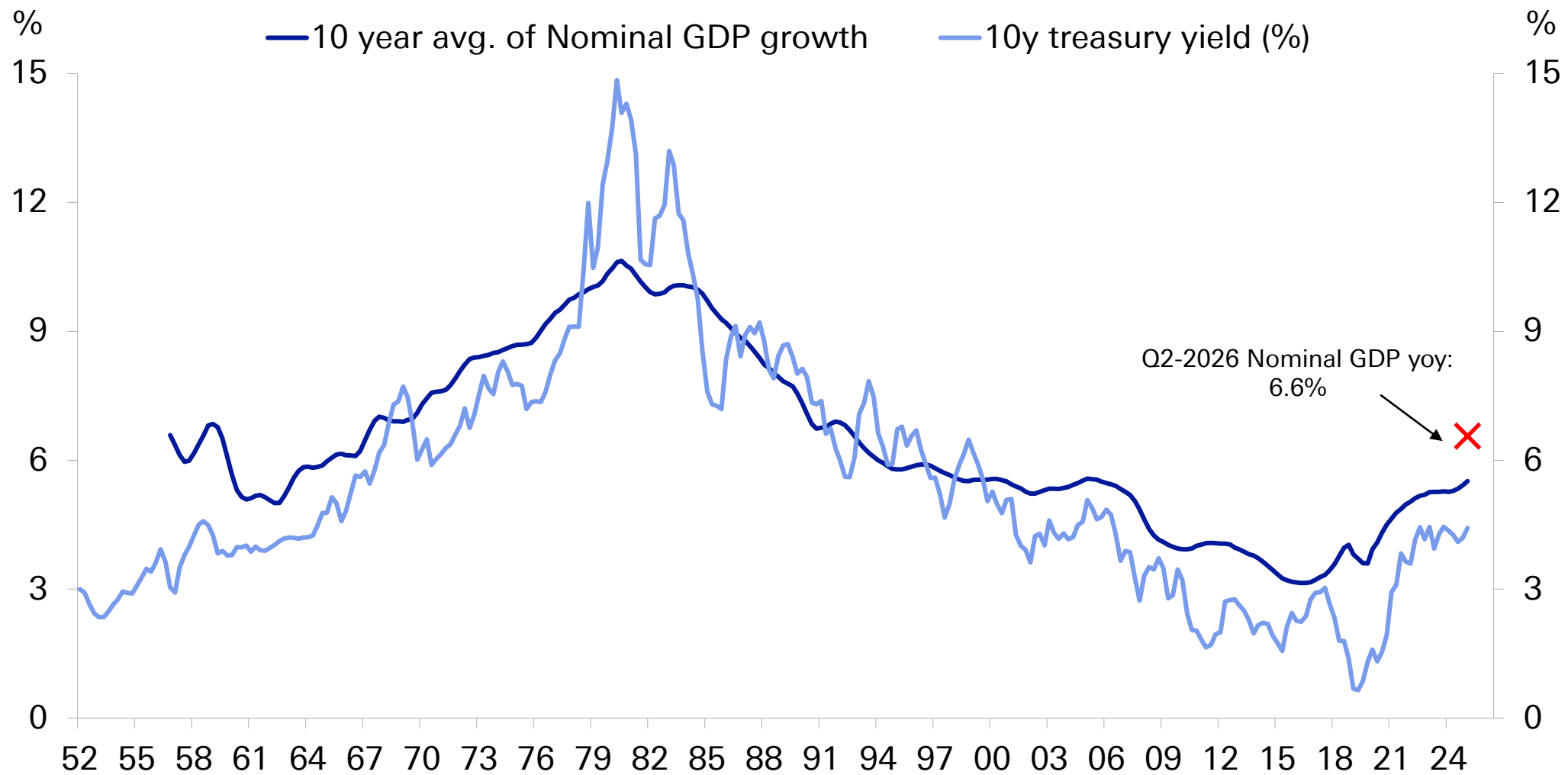


Source: Finaeon, Deutsche Bank



Since the early 1960s, 10 yr UST yields have typically tracked Nominal GDP apart from late 70s/early 80s when they soared above and in the 2010s when they were significantly below. The average gap over this period is around 70bps (NGDP>10yr USTs). 10yr average NGDP is c.5.5% so 4.8% seems broadly fair. However current NGDP is 6.6%. Do we need to factor this in?

10yr average nominal GDP growth and the 10yr Treasury yield

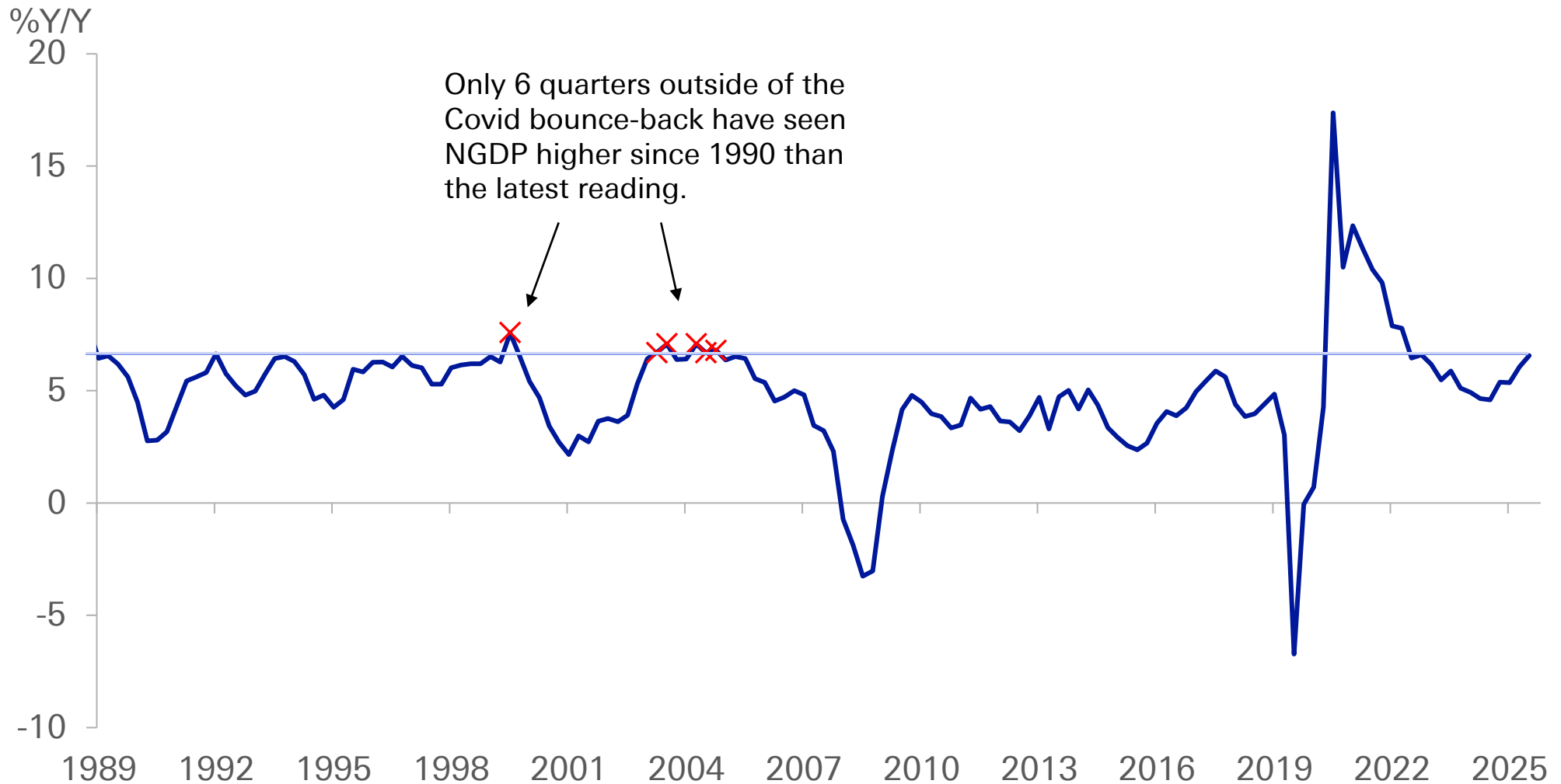


Source: BEA, FRB, Haver, Deutsche Bank

At 6.6%, current YoY US Nominal GDP is at the highest YoY rate since 2005 outside of the Covid bounce-back period. So even if not sustainable at these levels, this level of nominal growth has helped push UST yields higher....



US Nominal GDP growth with red crosses being the only time nominal GDP was higher than the current 6.6% outside of the post COVID rebound

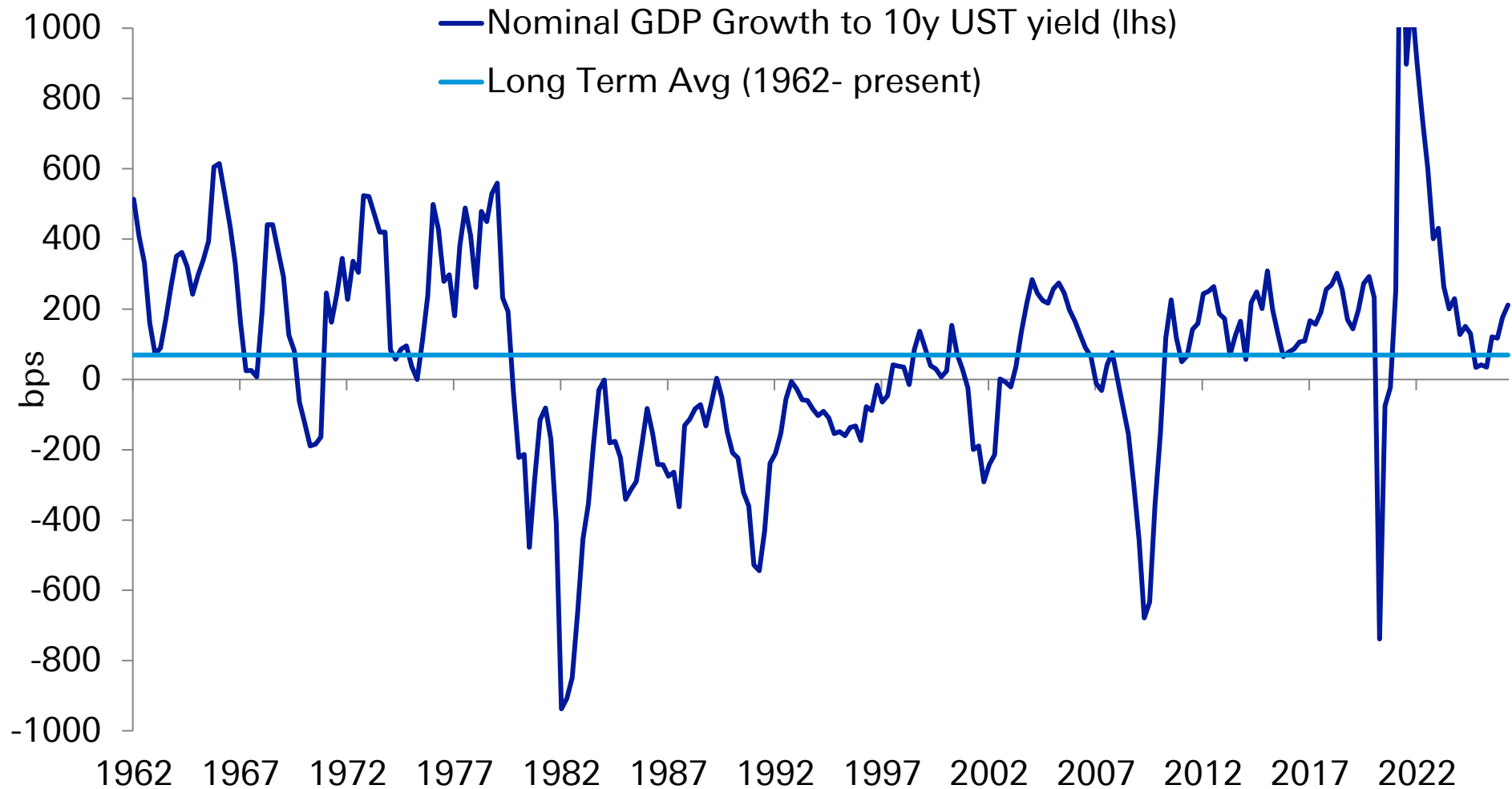


Source : BEA, Haver, Deutsche Bank



With spot Nominal GDP at 6.6% YoY, the current gap to 10yr yields is around 180bps. Clearly nominal GDP may be inflated by energy prices but overall it's hard to say US yields are too high based on this. If nominal GDP settles at 5.5% then 4.8% would be historically average. If we settle at 6% due to AI and inflation, then around 5.3% is reasonable before factoring in fiscal concerns.

US Nominal GDP minus 10yr UST yields

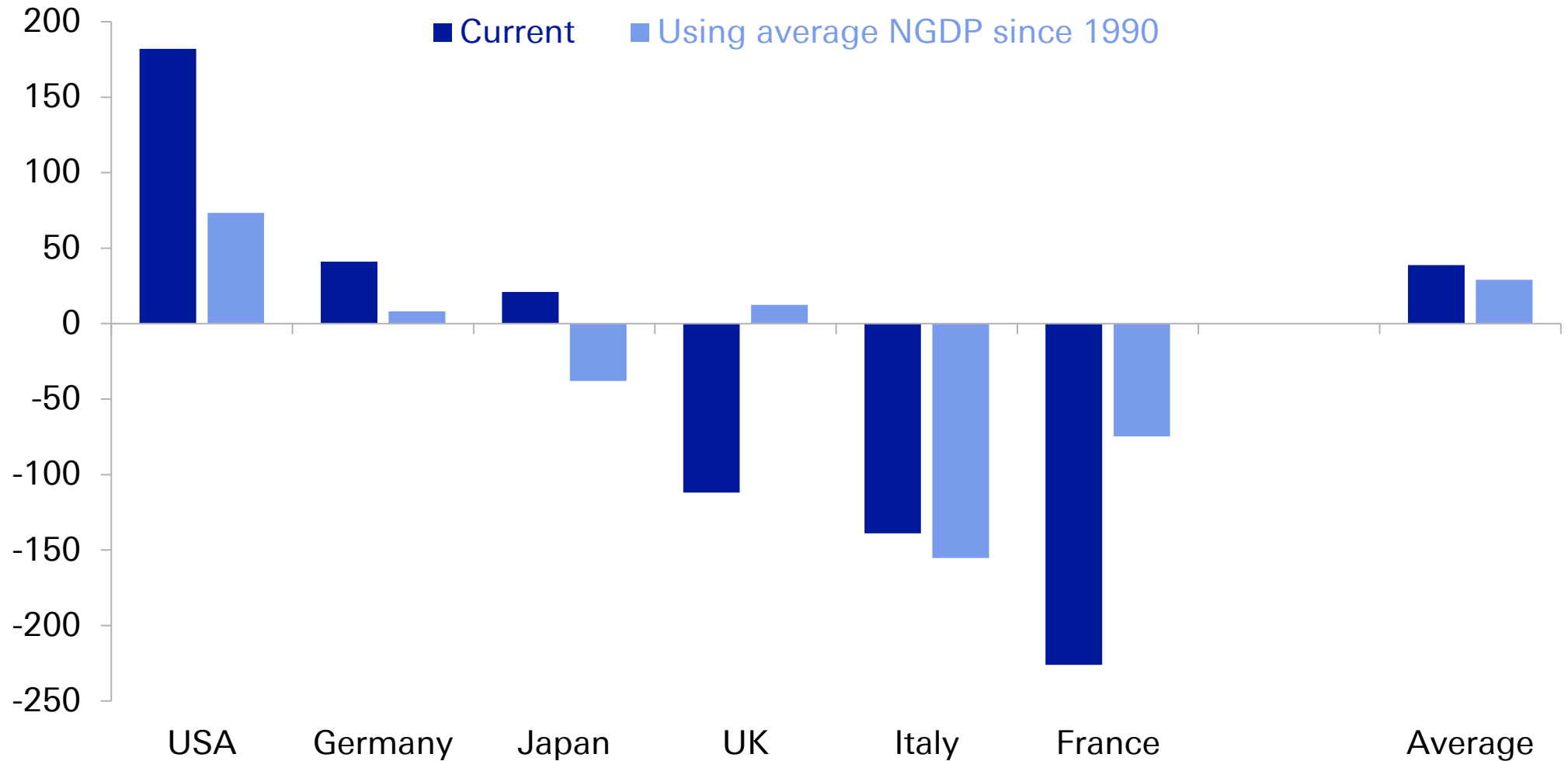


Source: Haver, Bloomberg Finance LP, Deutsche Bank



The US has the largest positive gap between current Nominal GDP and 10yr yields illustrating that growth is currently comfortably above funding costs. France, Italy and the UK are the most negative and have yields comfortably above current nominal growth.

Gap between Nominal GDP growth and 10y yields (bps)

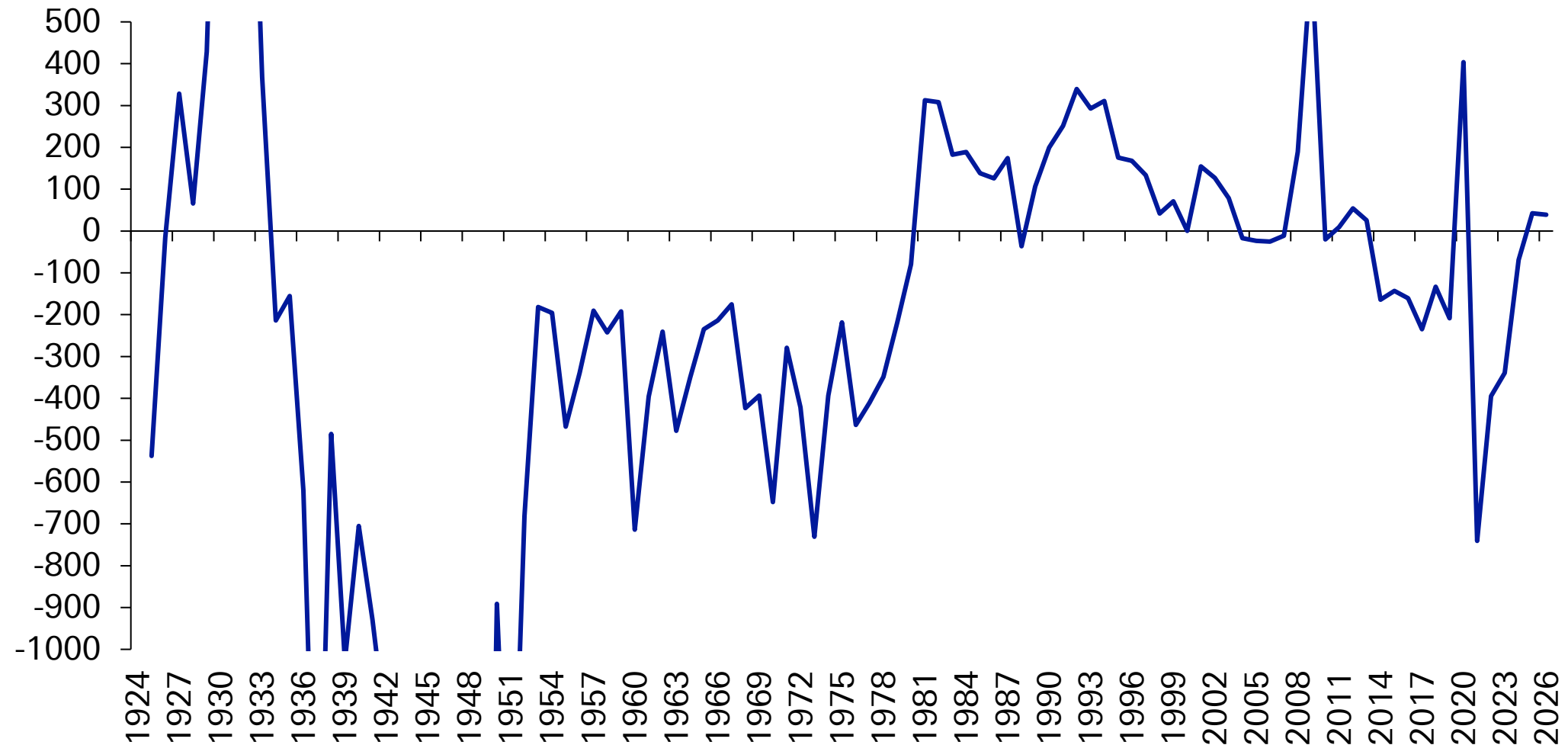


Source: Finaeon, Deutsche Bank

Across DMs, funding costs are now above nominal GDP on average. In the post GFC period of high debt and financial repression, yields were consistently below nominal GDP thus helping the funding situation....



Average difference between Nominal GDP and 10yr Govt. bonds for US, UK, Germany, France, Italy and Japan (in bps)

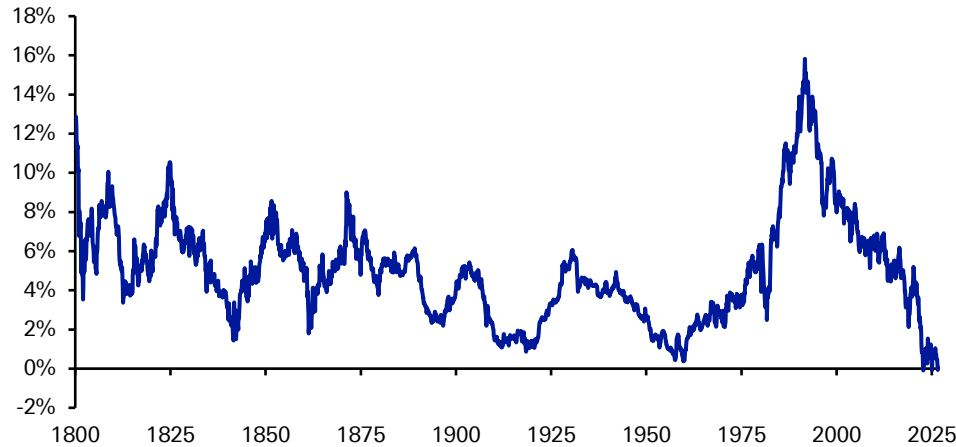


Source: Finaeon, Deutsche Bank

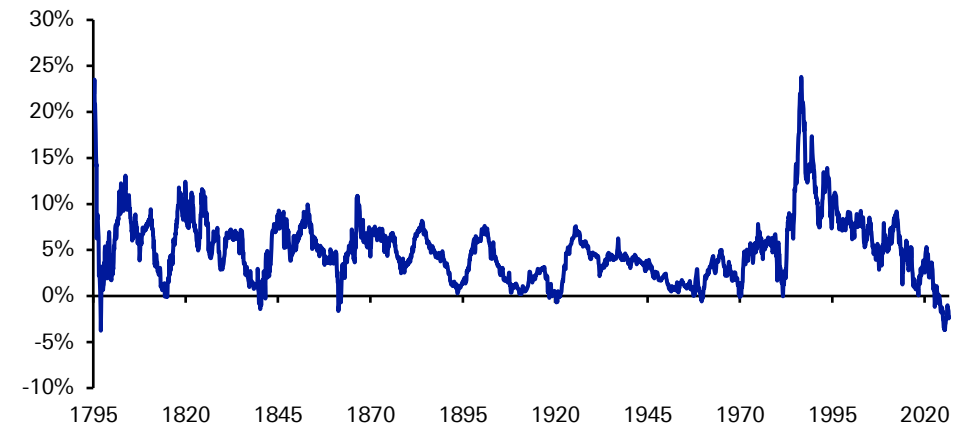


For 10yr Treasury yields, we are still hovering around the worst 5- and 10-year rolling nominal returns on record but it's clear that even with yields again approaching multi-year highs, returns have been stabilising more recently due to the higher coupons...

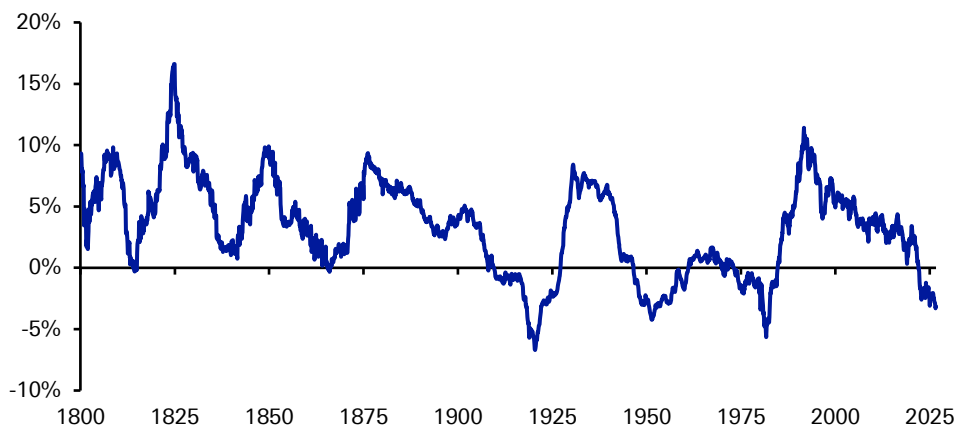
10yr Rolling Nominal Return (Annualised)



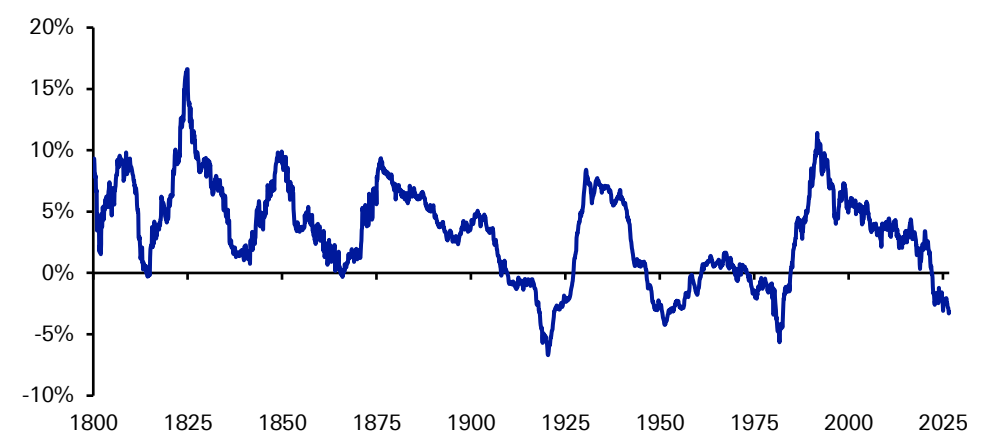
5yr Rolling Nominal Return (Annualised)



10yr Rolling Real Return (Annualised)



5yr Rolling Real Return (Annualised)

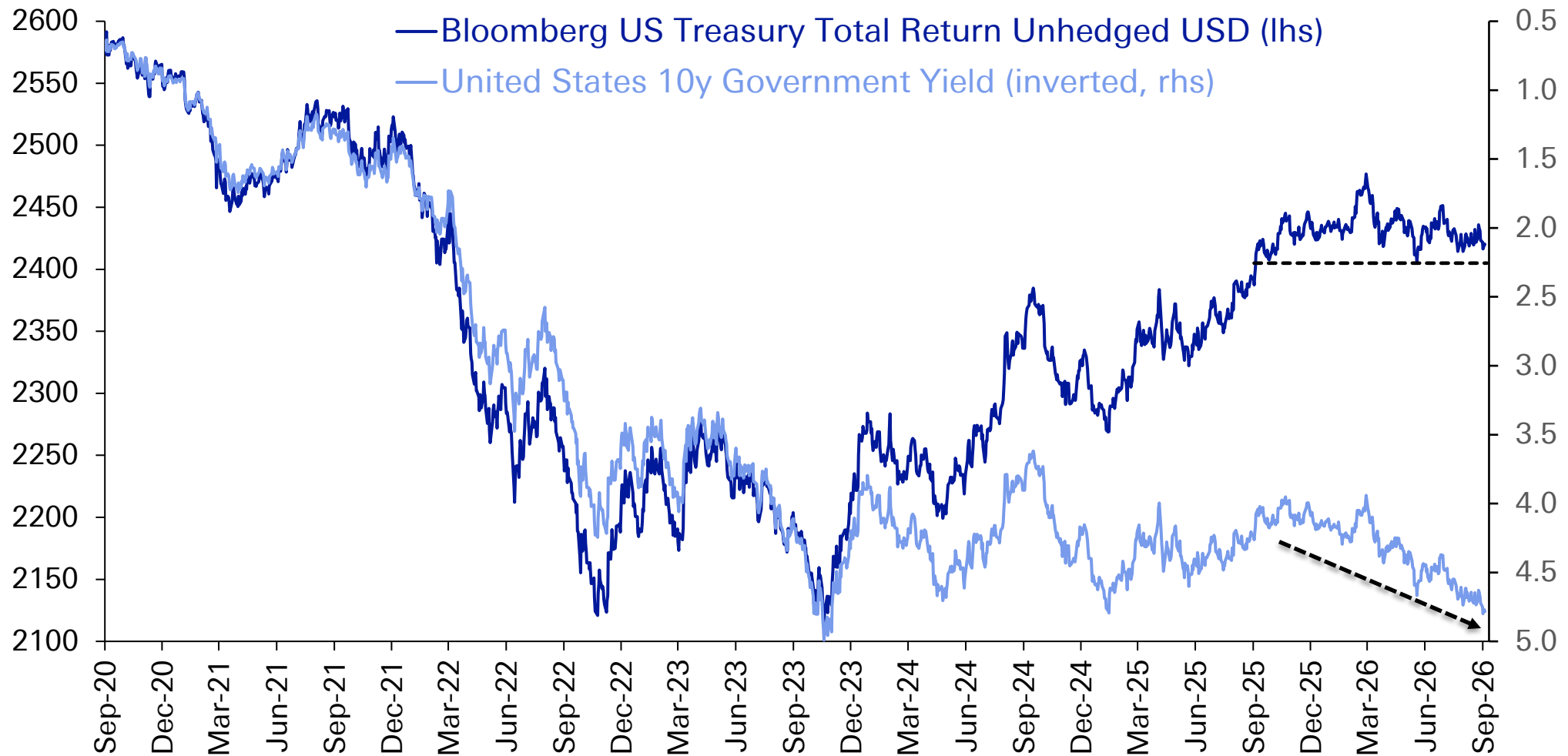


Source: Finaeon, Deutsche Bank



At least we're getting a decent compensatory yield now versus the 2010s and early 2020s. Over the last 12 months the US treasury total return index is higher even with a c.60bp rise in 10yr yields.. Since yields peaked in October 2023, the index has returned nearly 15%

US Treasury Index Returns vs. 10y Govt. Yields. Over the last year returns have been positive even with a notable rise in yields.



Source : Bloomberg Finance LP, Deutsche Bank, Last update: Sep 7

Clearly it's a global phenomenon. For example, 30-year yields are hitting decade plus or multi decade highs in several countries.



30yr Gilt Yields (%)



30yr Treasury Yields (%)



30yr JGB Yields



30yr German Yields (%)

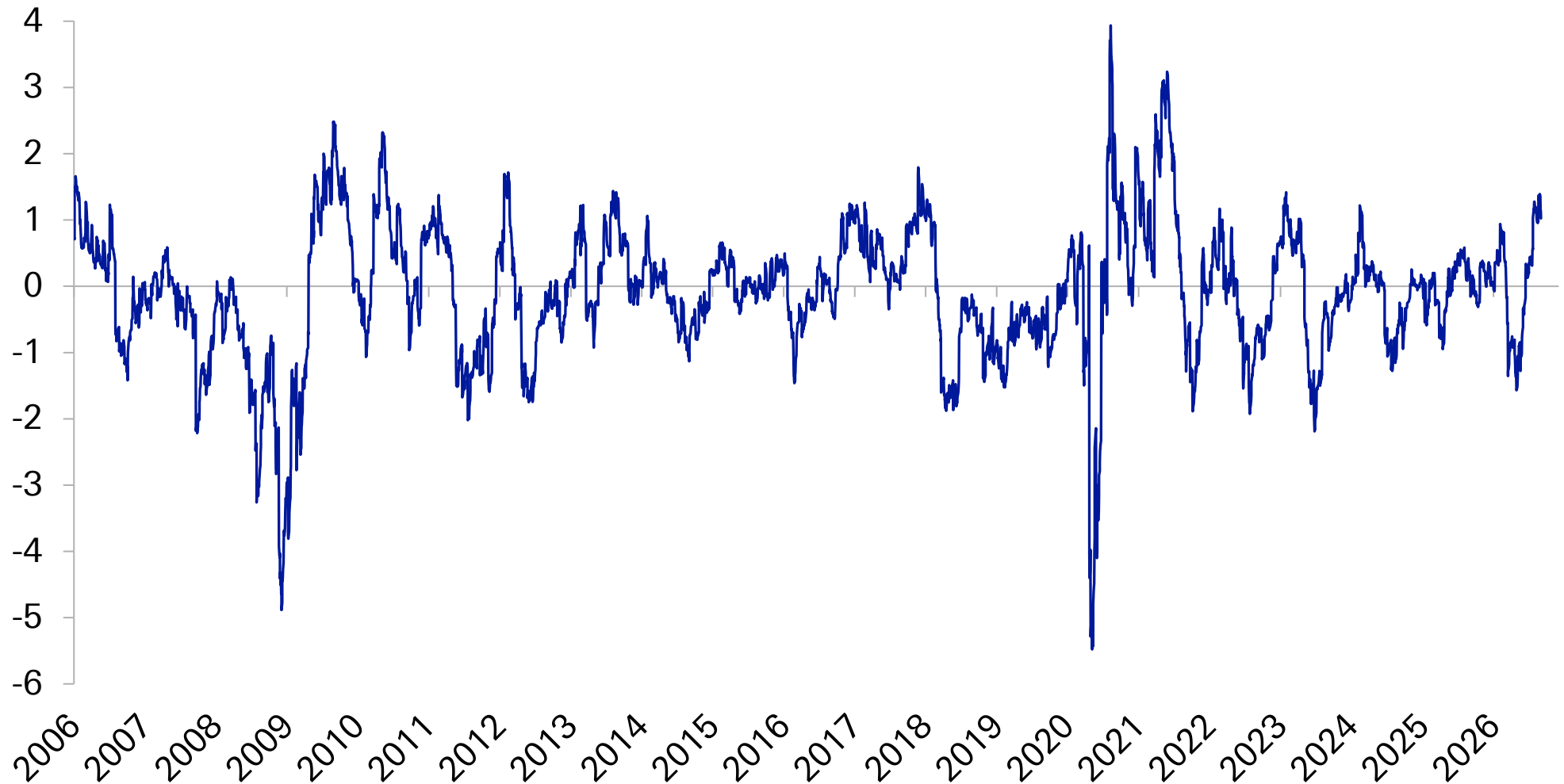


Source: Bloomberg Finance LP, Deutsche Bank, Last update Sep 7

But some of this is justified by the data. For example, European growth surprises are nudging up against 20-year highs outside of the post Covid and GFC bounce-backs.



Deutsche Bank SIREN Surprise Index - Euroarea

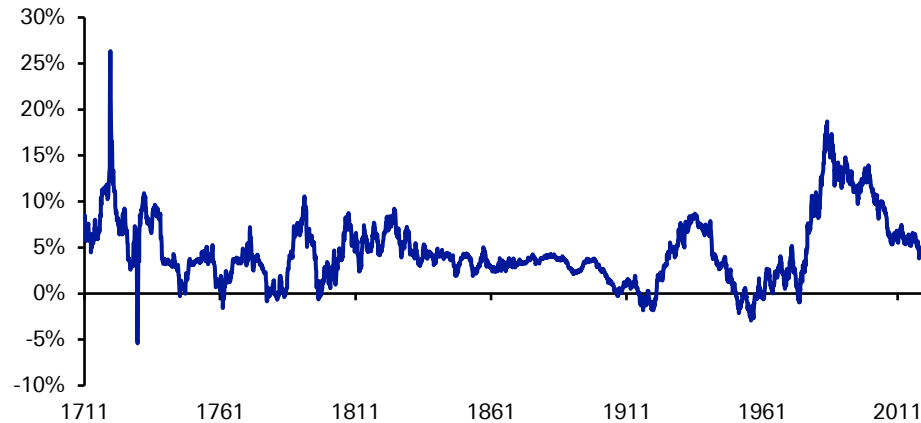


Source: Bloomberg Finance LP, Deutsche Bank; Last update : Sep 7

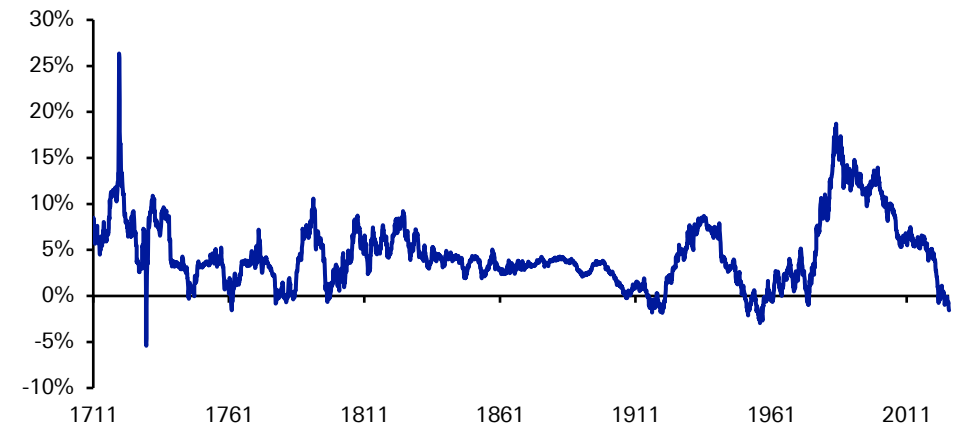
10yr UK gilts have also had a tough time in recent years. Rolling 5 and 10yr total returns are still near the lows..



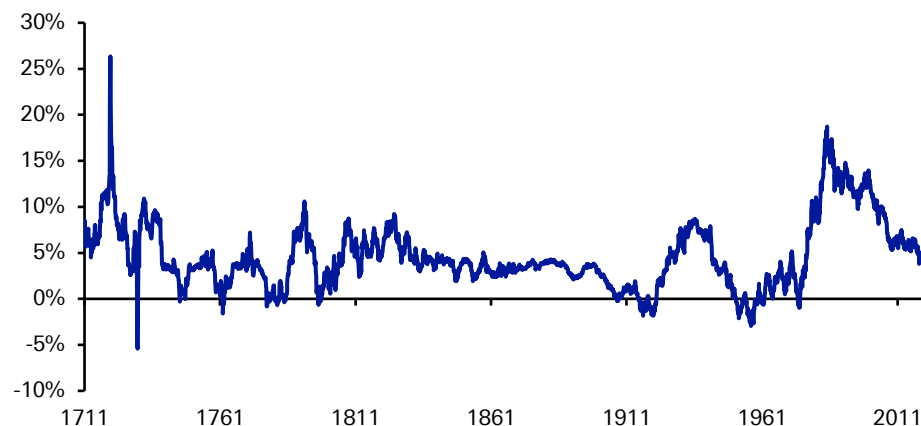
10yr Rolling Nominal Return (Annualised)



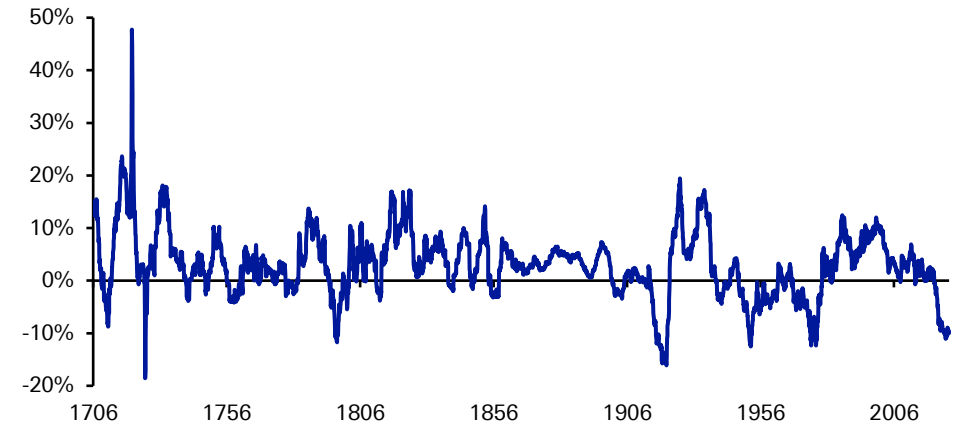
5yr Rolling Nominal Return (Annualised)



10yr Rolling Real Return (Annualised)



5yr Rolling Real Return (Annualised)

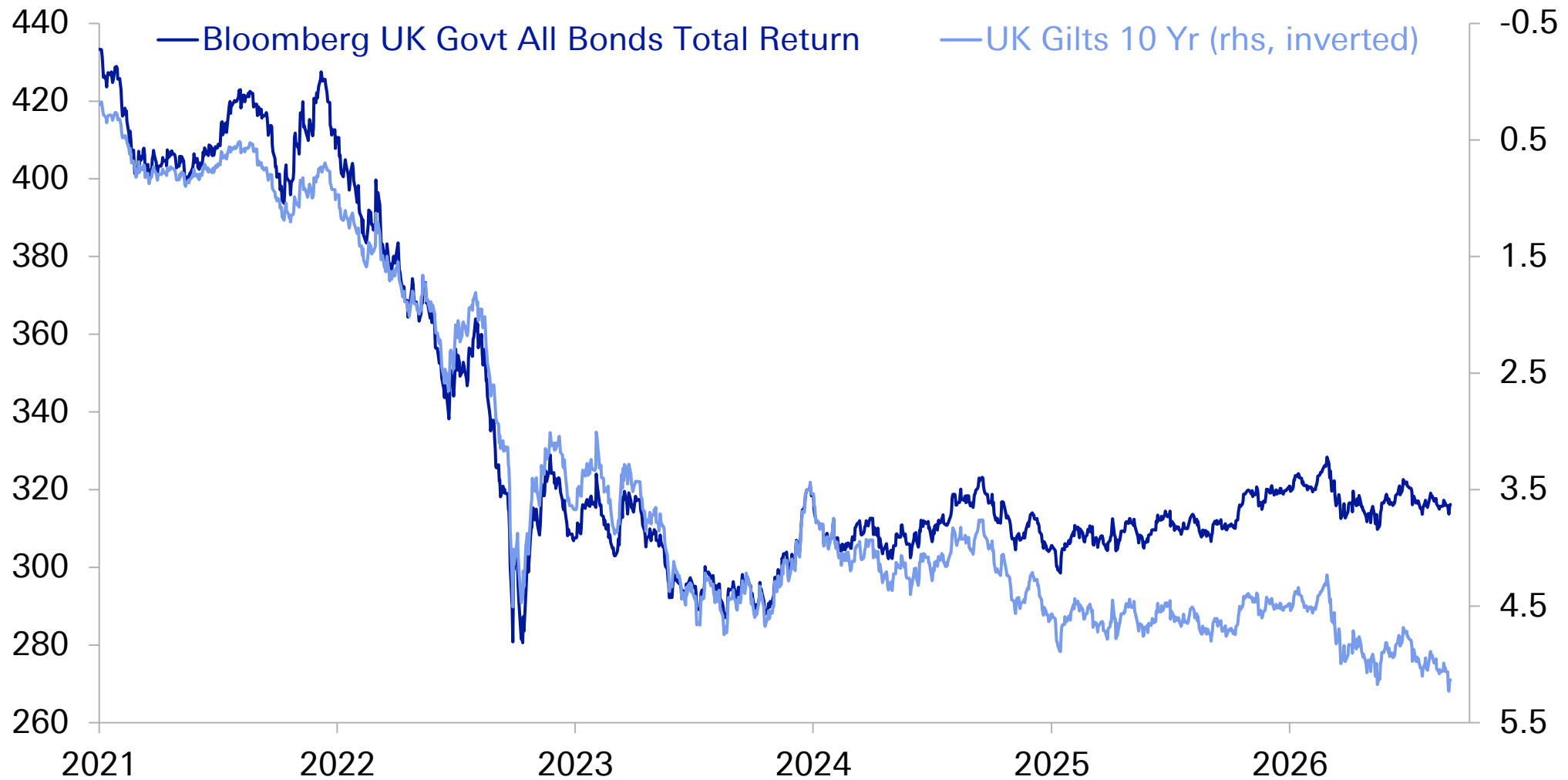


Source: Finaeon, Deutsche Bank

But even with all the negative headlines in the four years since the Truss mini-budget/LDI crisis, Gilt returns have generally crept higher even as yields have returned back to the highs. 10yr yields are currently around 65bps higher than their “crisis” peak in October 2022, yet the total return index is around 12% higher..



UK Gilt Index Returns vs. 10y Govt. Yields.

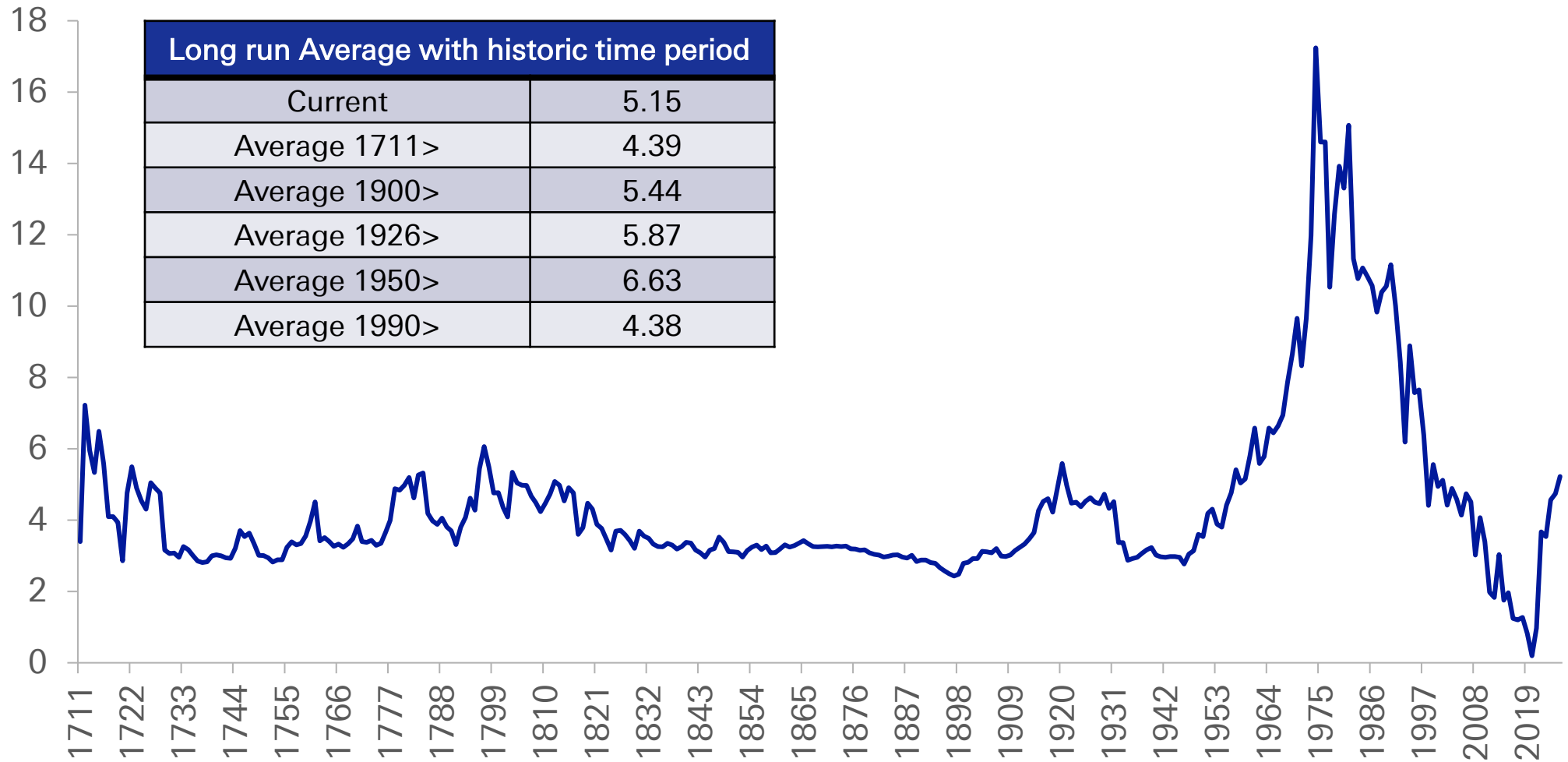


Source : Bloomberg Finance LP, Deutsche Bank, last update Sep 7

What average yield should you use in the UK? Over the long-run (since 1711- and since 1900-) the median is around one percentage point less than the average which is heavily influenced by 1960-1997 where 10yr yields were permanently above 5.5%. Outside of that period the current yield is at the top end of the historical range.



10yr govt yields in UK

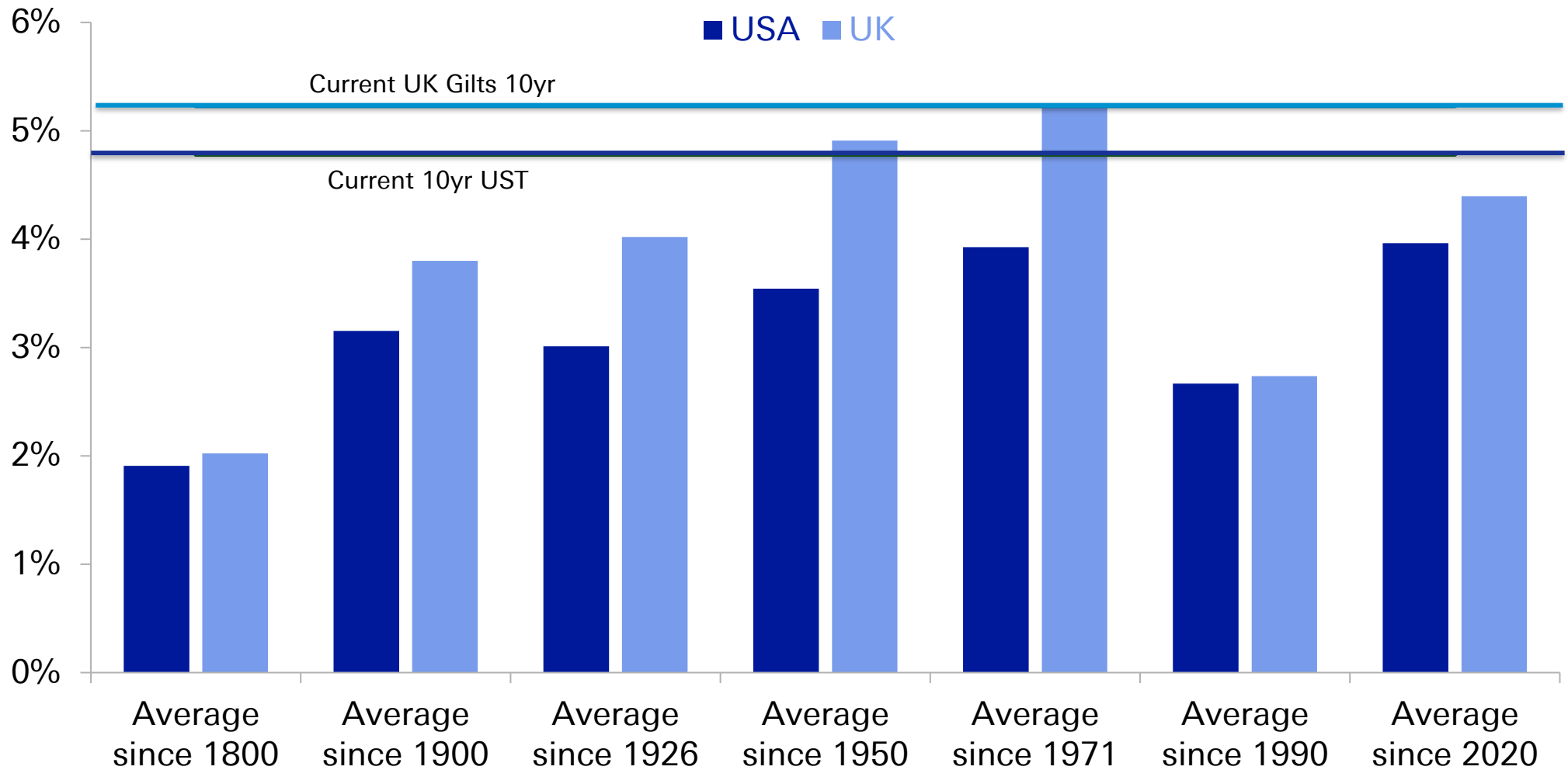


Source: Finaeon, Deutsche Bank

At least US and UK Bond yields are mostly above average inflation seen over virtually all extended periods through history...



Average inflation over various periods in the US and UK and the current 10 year bond yield

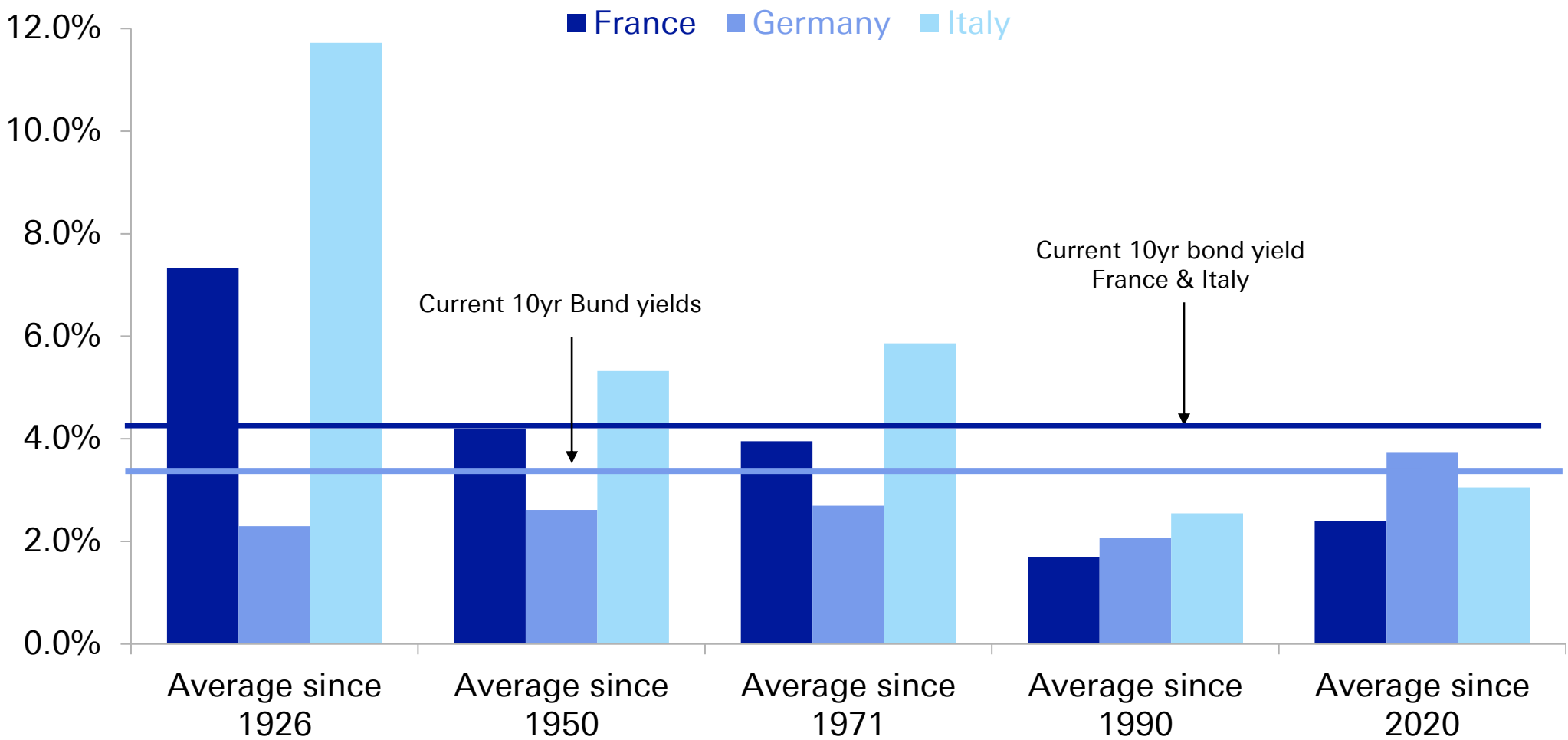


Source: Finaeon, Deutsche Bank

WWII makes this more complicated for European countries, as do lower current yields.



Average inflation over various periods in the France, Germany and Italy and the current 10-year bond yield

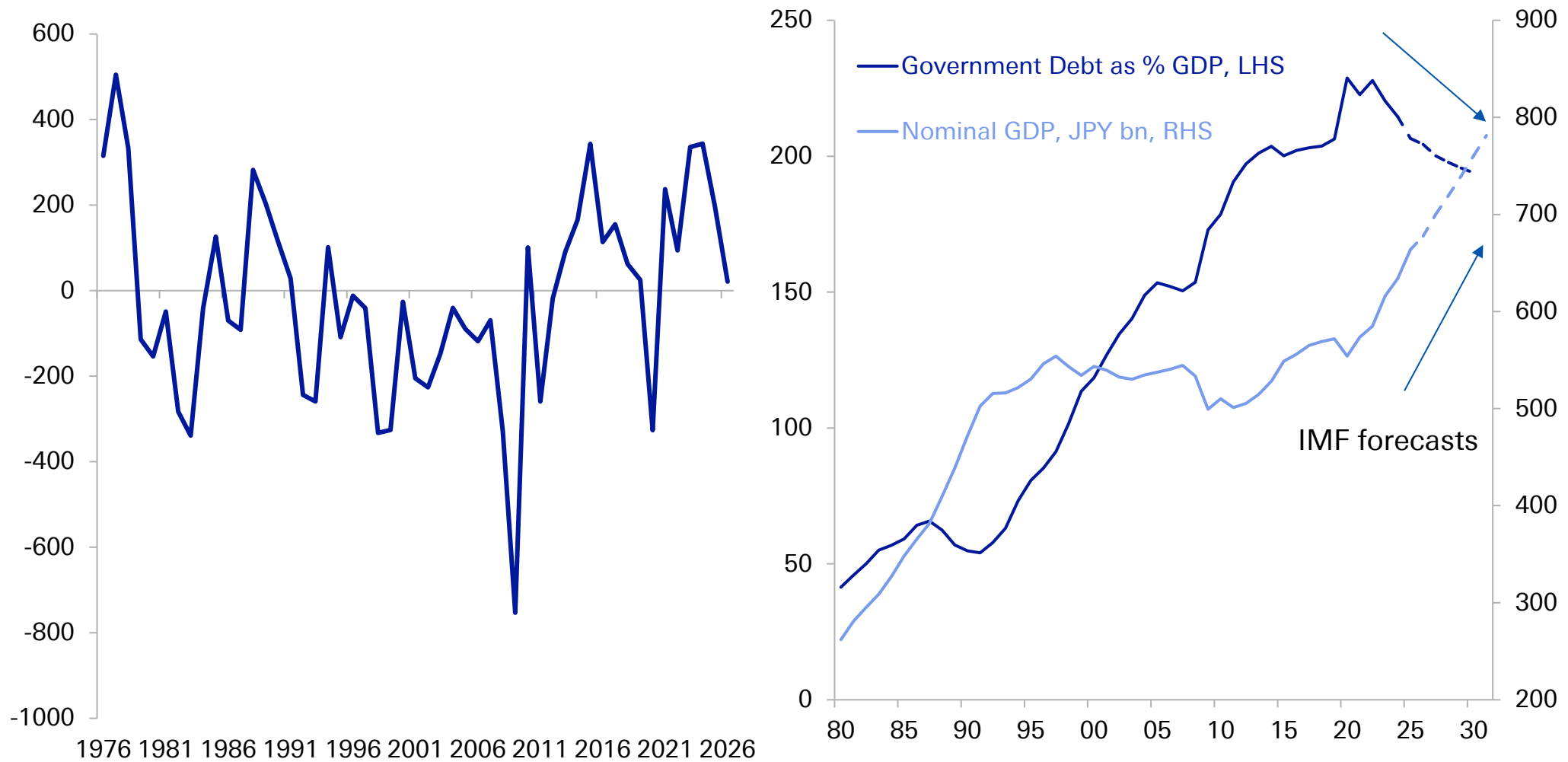


Source: Finaeon, Deutsche Bank

For Japan, rising yields and dipping nominal GDP do threaten what has been a falling debt/GDP since Covid when nominal GDP took off after being broadly flat for a couple of decades. Since 2021, Japanese nominal GDP has averaged over 220bps more than 10yr JGB yields. Today that stands at only around 40bps on a spot basis.



Japan: nominal GDP growth & 10yr JGB spread (left) & nominal GDP vs Govt debt (right)

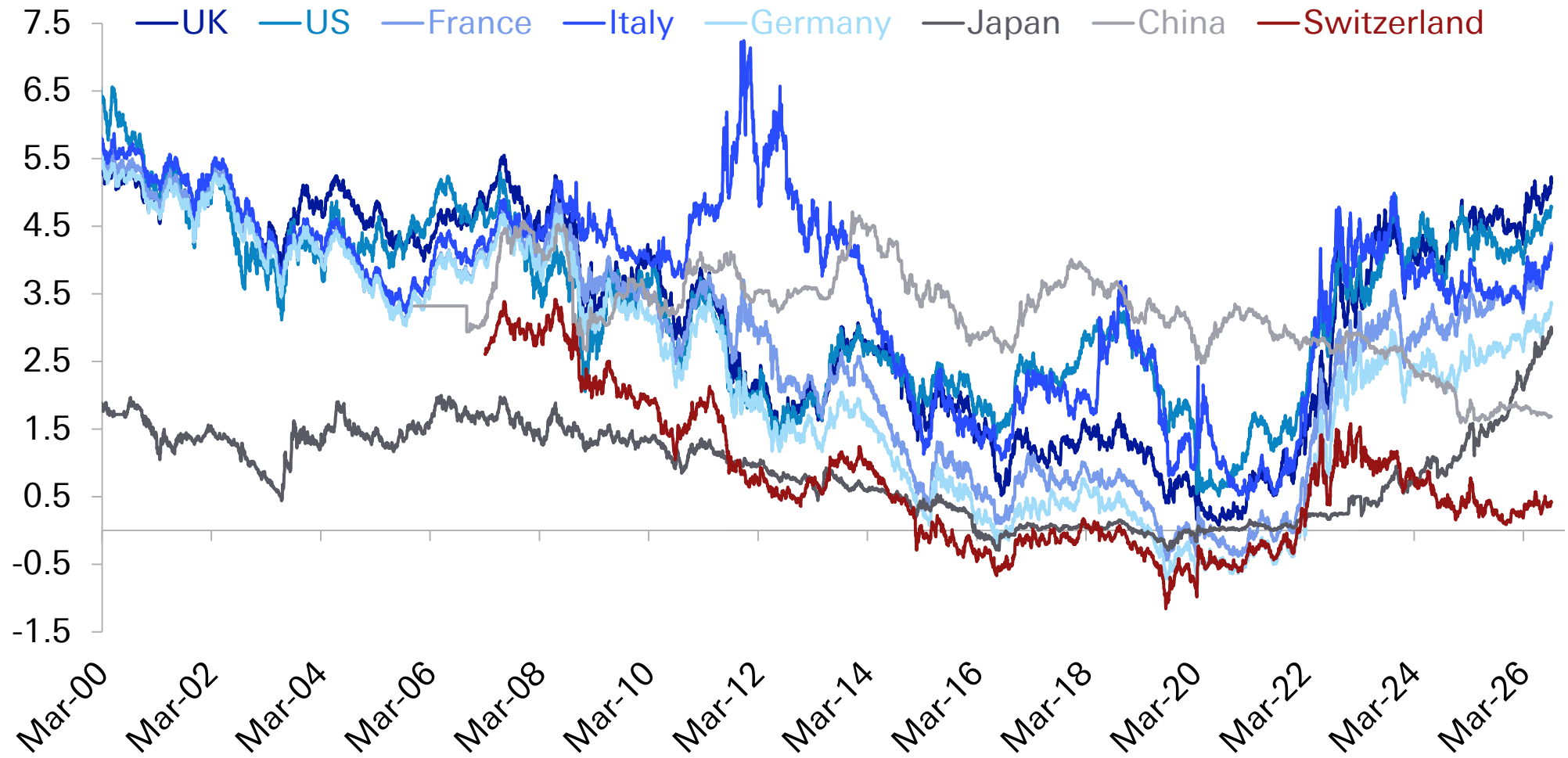


Source: Finaeon , Deutsche Bank Research, Haver Analytics, IMF

Interestingly, while most global yields hit multi-decade highs, Swiss and China yields have been flat or moving lower...



10y Government yields across major economies (%)

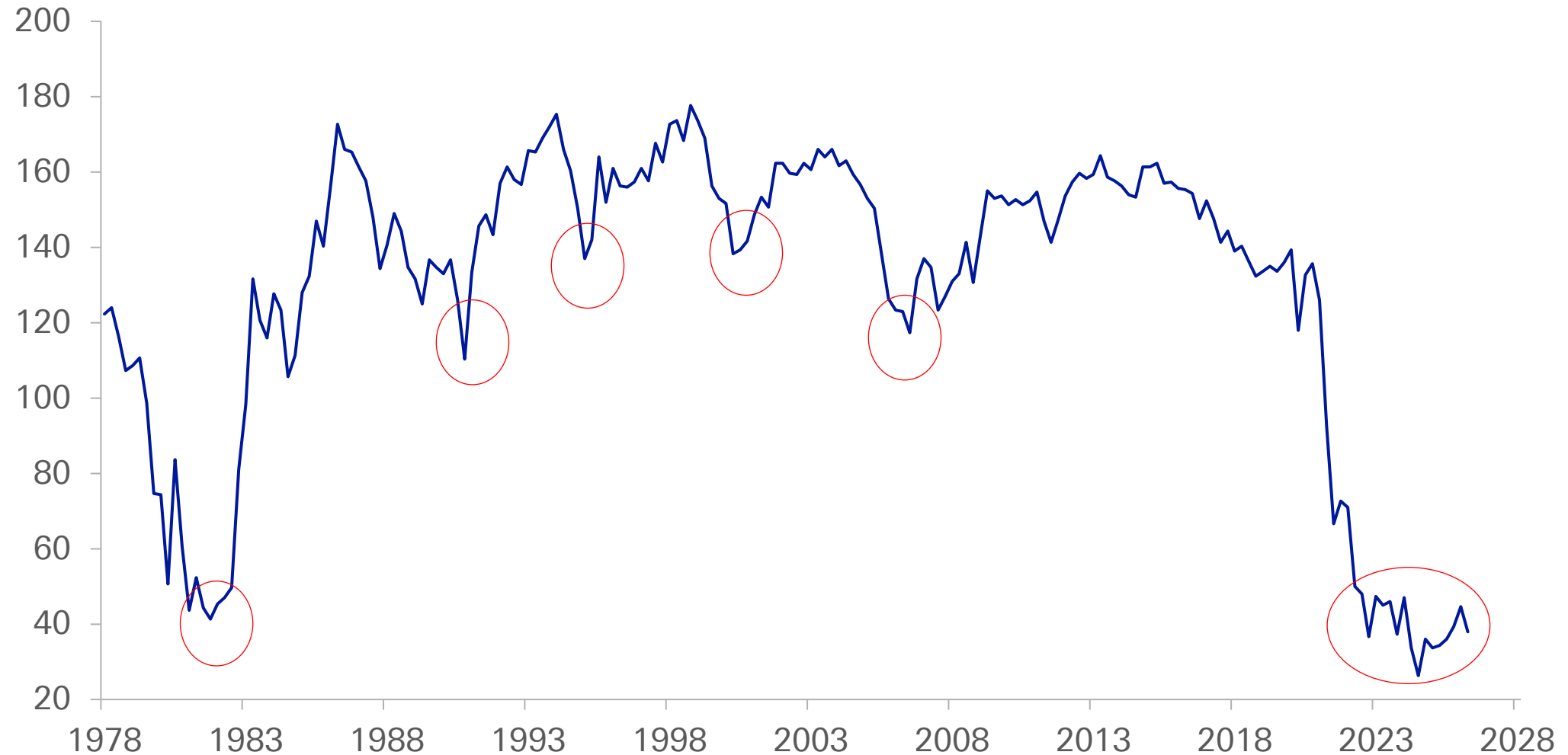


Source : Bloomberg Finance LP, Deutsche Bank, last update Sep 7, Note: Countries are ordered by current yield from high to low

Rising yields make the fiscal arithmetic more challenging but also continue to keep US housing affordability near record lows... Full employment and a lack of activity are the main thing holding prices up



UMICH: Current Conditions for Buying Houses (Index Score): sentiments hit rock bottom for buyers



Source: UMich, Haver, Deutsche Bank

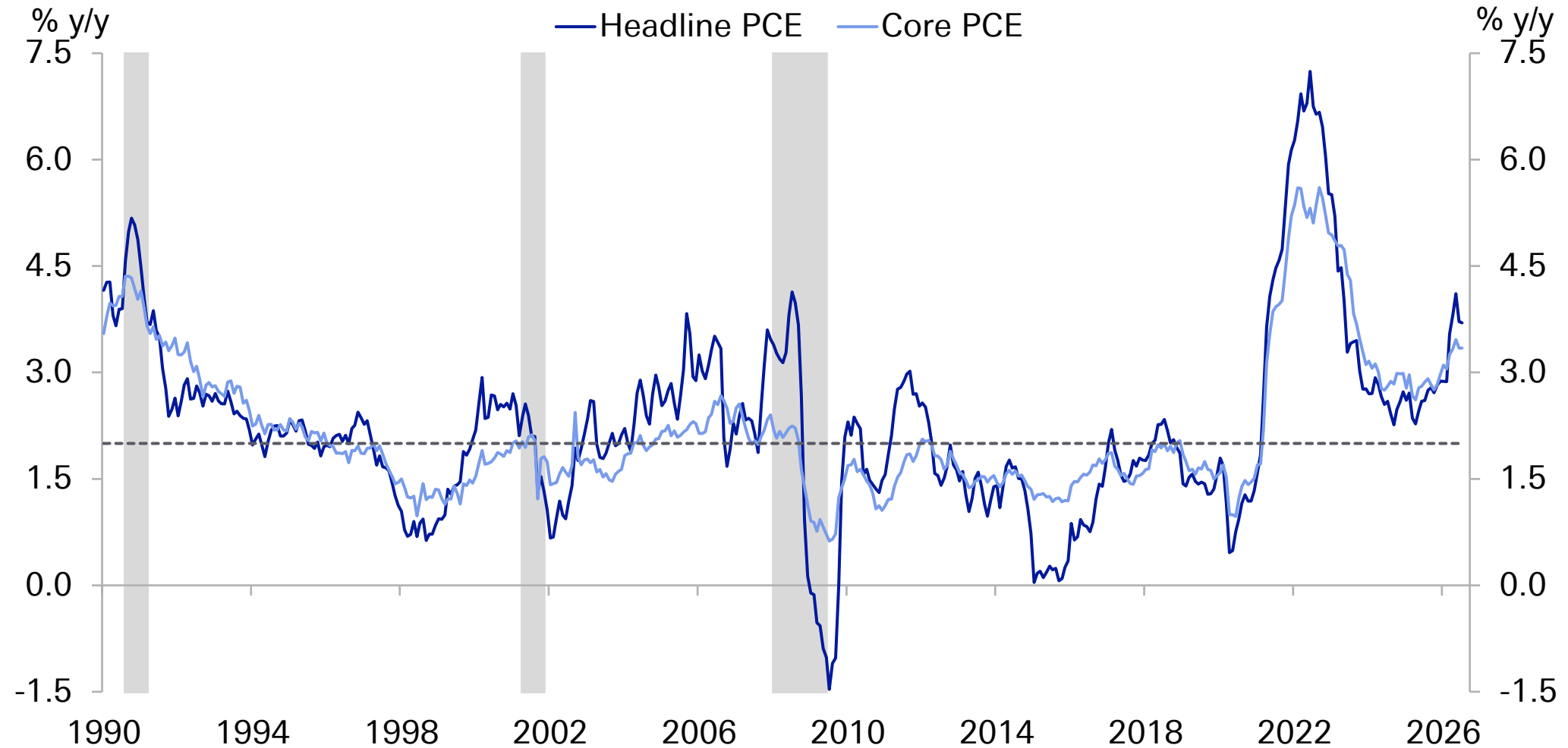


Should the Fed raise rates...?

PCE inflation remains elevated far above the Fed's 2% target. Excluding the pandemic anomaly, core PCE is only 15bps off its 34-year high seen in May



US PCE inflation

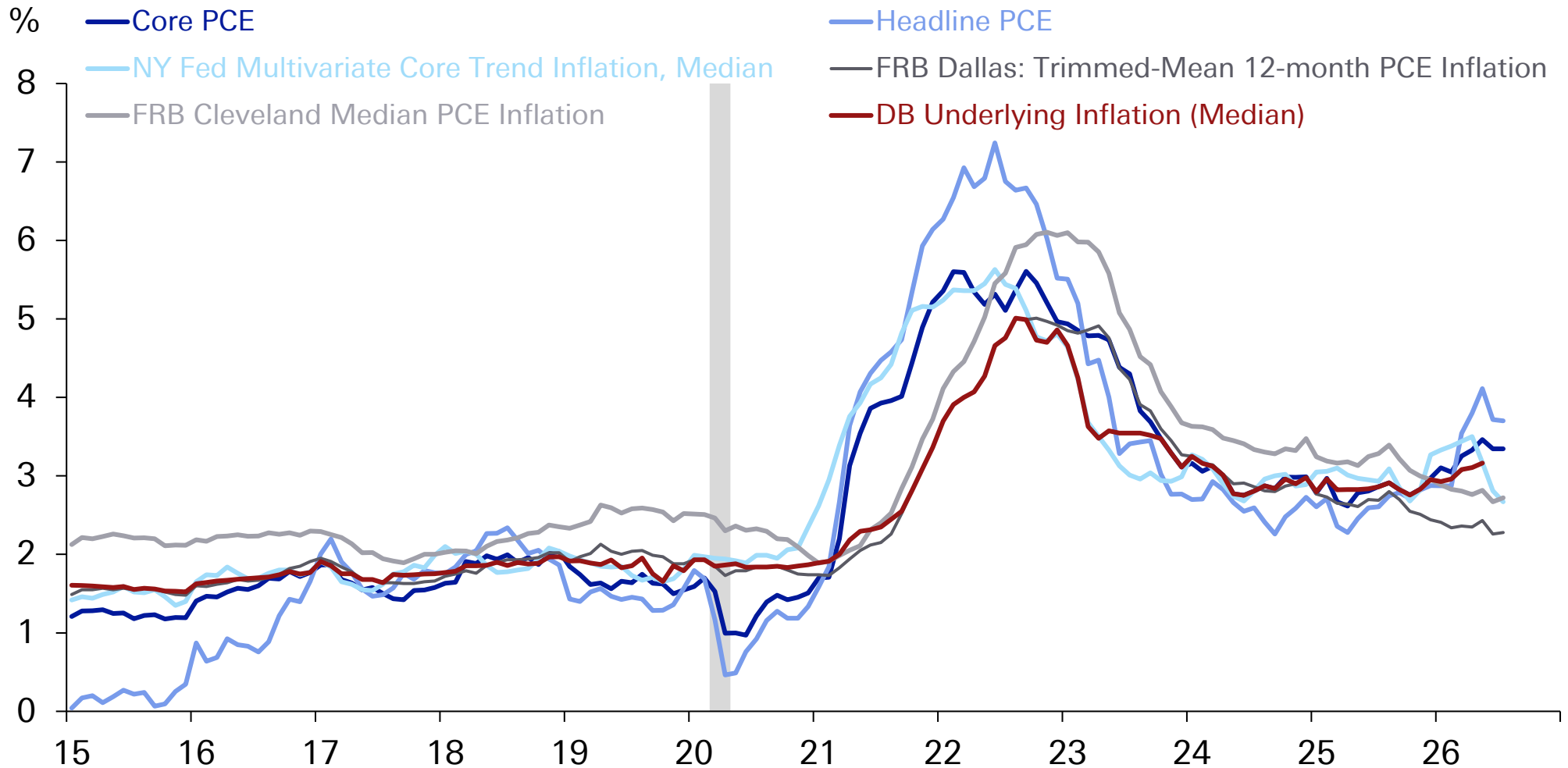


Source: BEA, Haver Analytics, Deutsche Bank

On most measures, trend inflation seems to be sticky at around 3%....



Alternative measures of trend PCE inflation

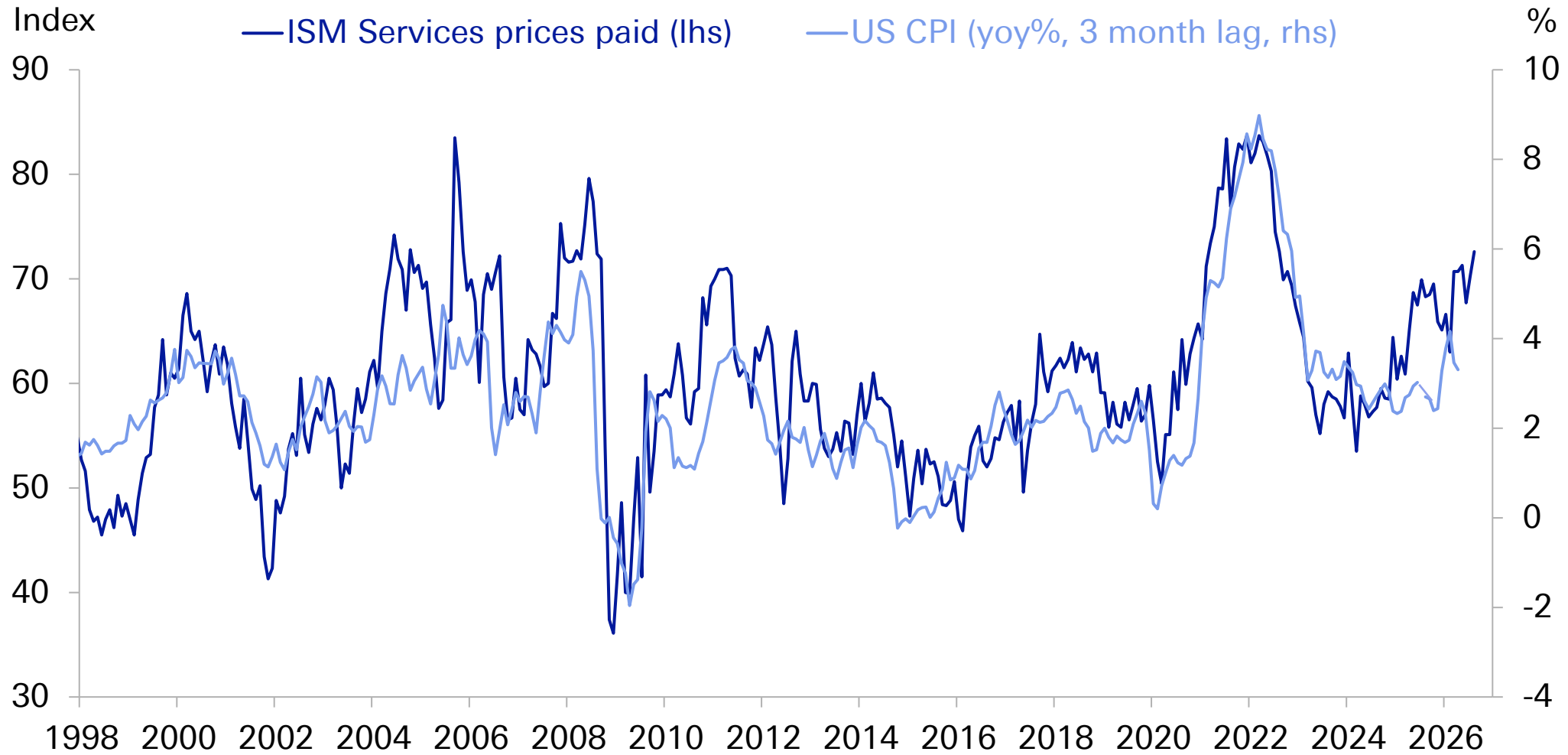


Source: BEA, FRB NY, FRB Dallas, FRB Cleveland, Haver, Deutsche Bank

Meanwhile, the ISM services prices paid component has been tightly correlated to inflation with a 3-month lag in the past... it's at a 4-year high and was consistent with CPI above 5% in the last inflation wave... While the correlation shouldn't be taken too literally, it highlights that price pressures remain in the pipeline...



US CPI Inflation and ISM Services Prices Paid

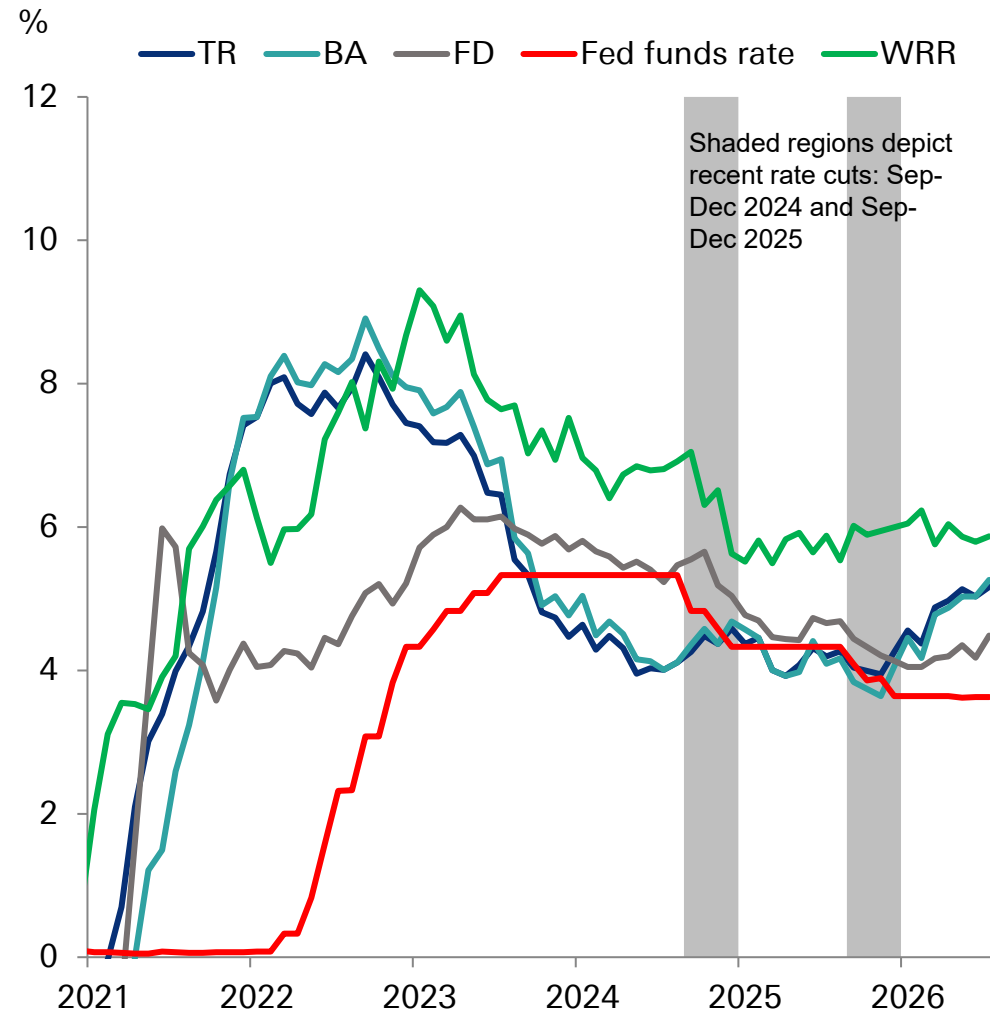


Source: ISM , BLS, Haver Analytics, Deutsche Bank

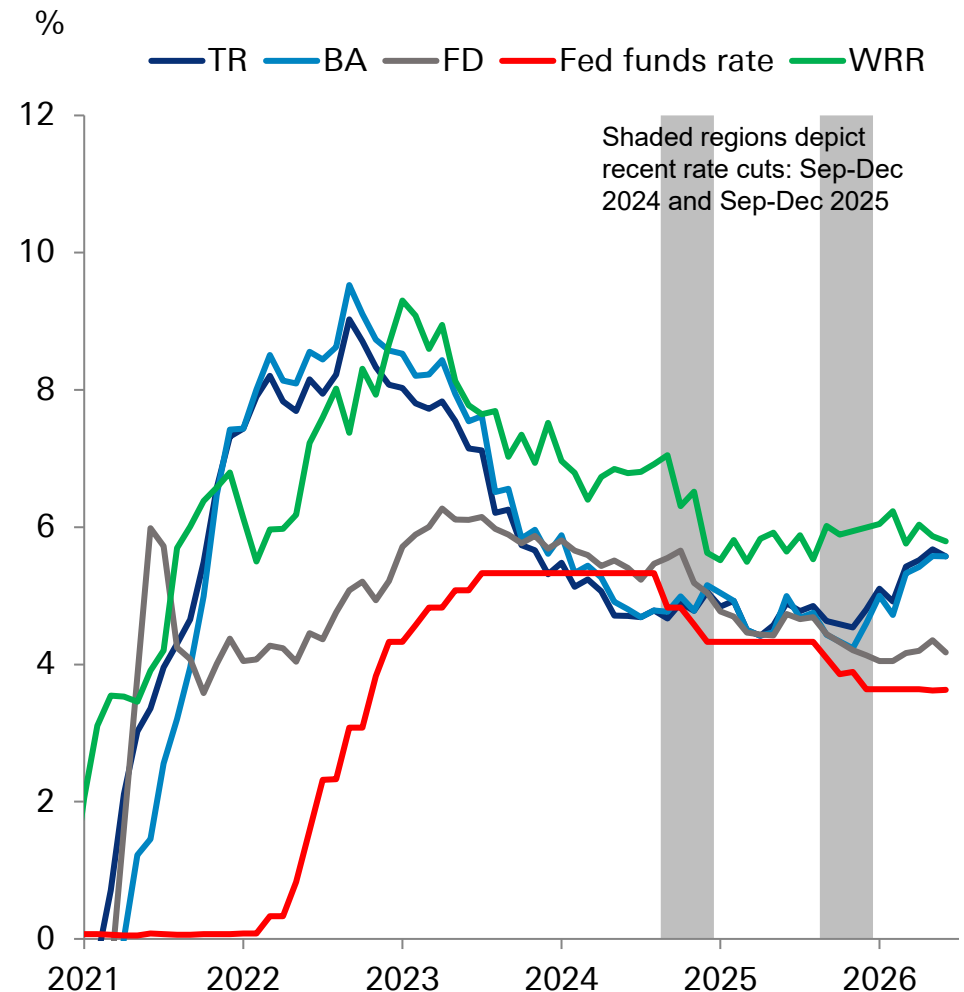
The policy rate has moved comfortably below policy rules after last year's "insurance" cuts



Fed funds rate versus rules w/ Fed r-star



Fed funds rate versus rules w/ DB r-star

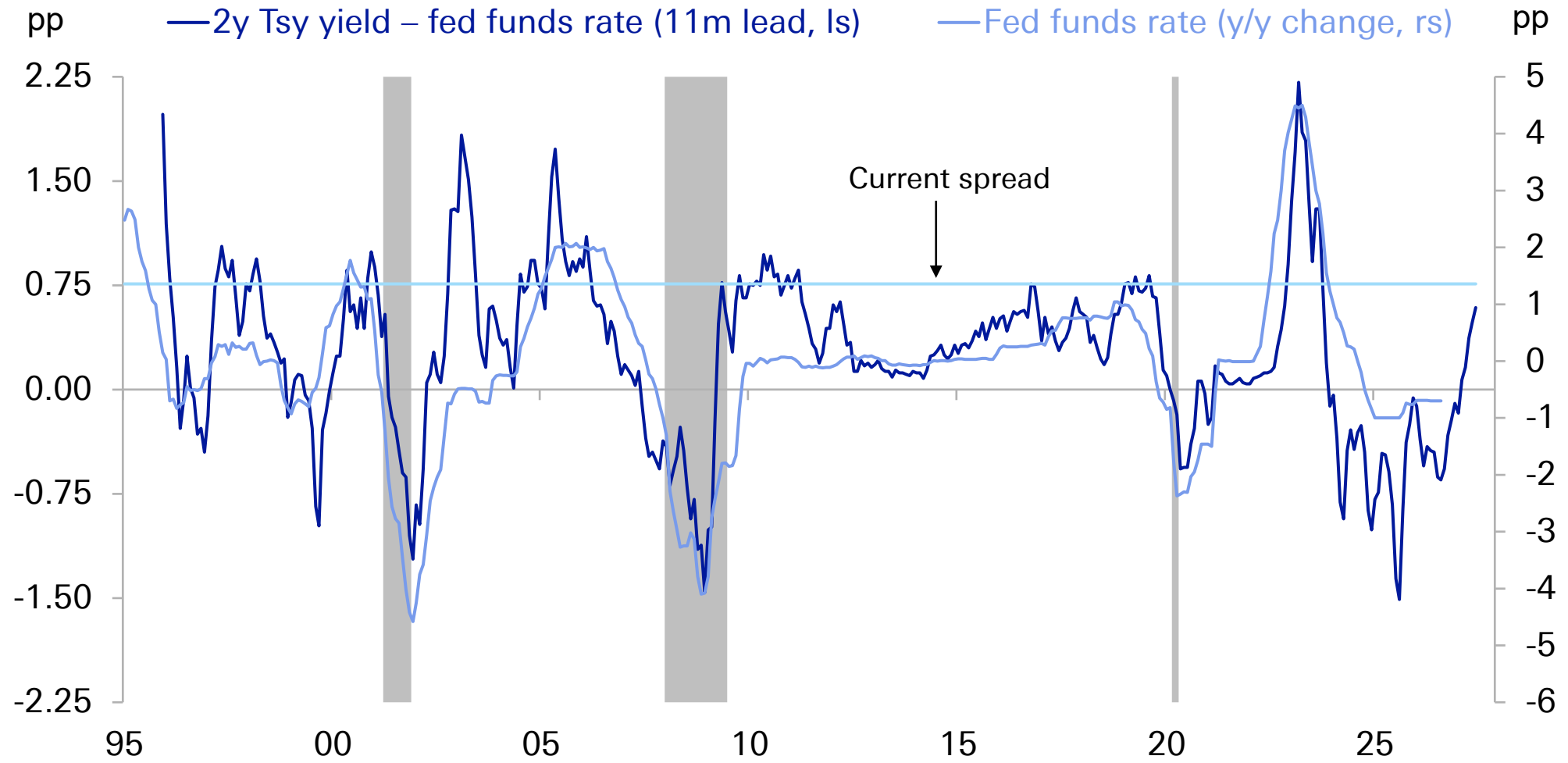


Source: FRB, BLS, CBO, Haver Analytics, Deutsche Bank

The wide gap between 2y yields and fed funds rate also predicts rate hikes



2y yield – fed funds rate and future rate changes

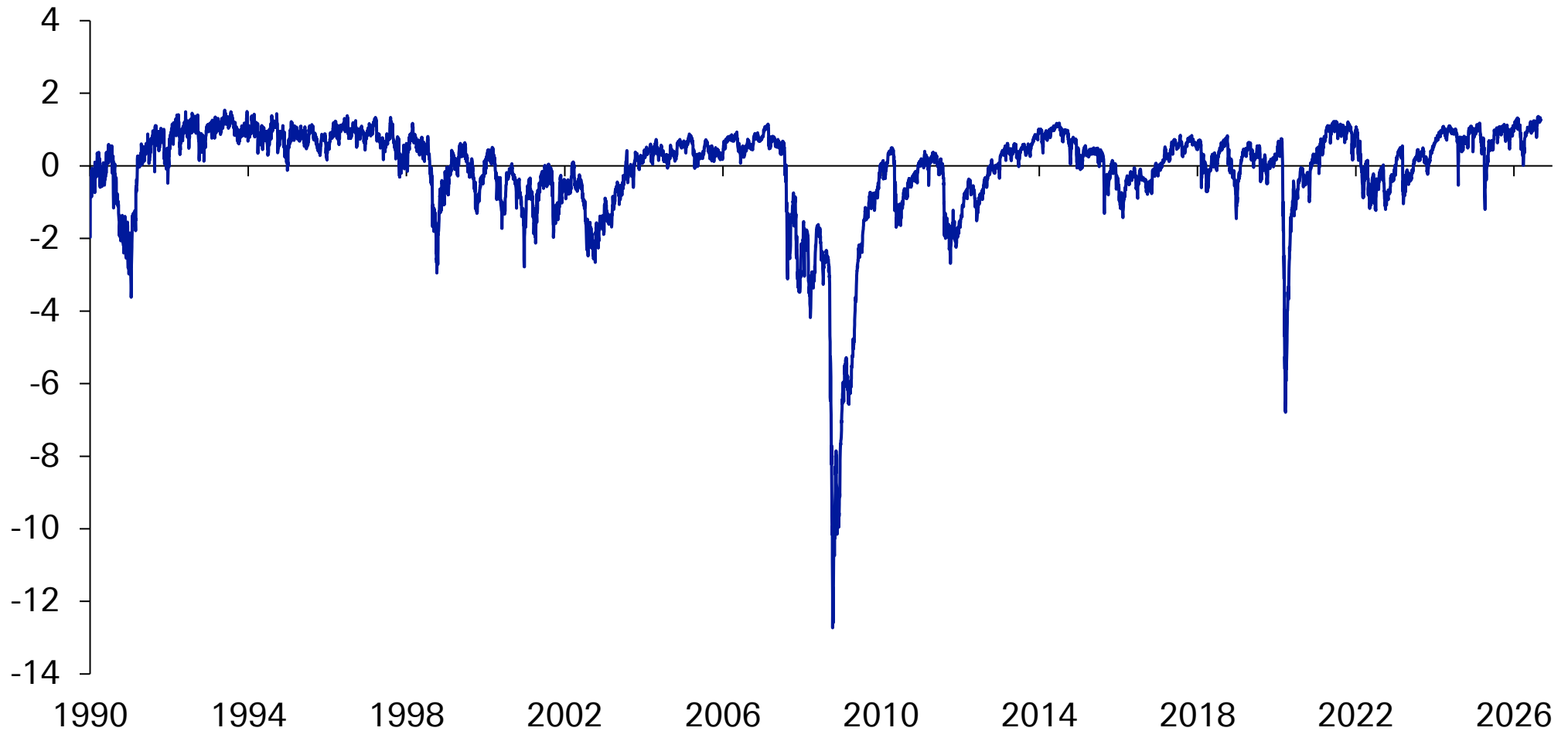


Source: FRB, Haver Analytics, Deutsche Bank

Financial conditions are broadly bumping up against the loosest they've been since 1990...



Bloomberg United States FCI

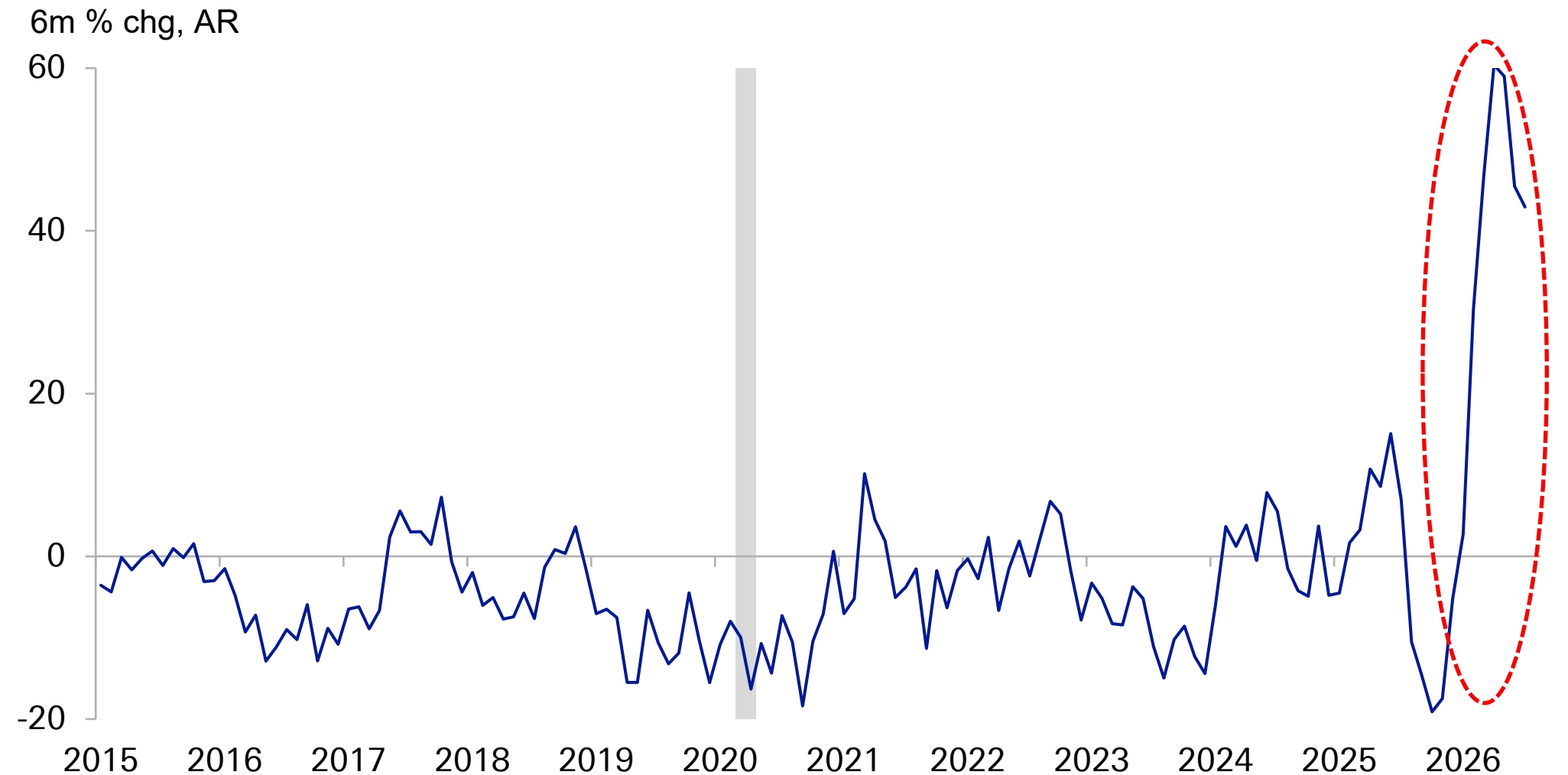


Source: Bloomberg Finance LP, Deutsche Bank; Last update : Sep 7; Note: Bloomberg U.S. Financial Conditions Index tracks the overall level of financial stress in the U.S. money, bond, and equity markets to help assess the availability and cost of credit. A positive value indicates accommodative financial conditions, while a negative value indicates tighter financial conditions relative to pre-crisis norms.

AI is an inflationary force today



PCE: software and accessories inflation

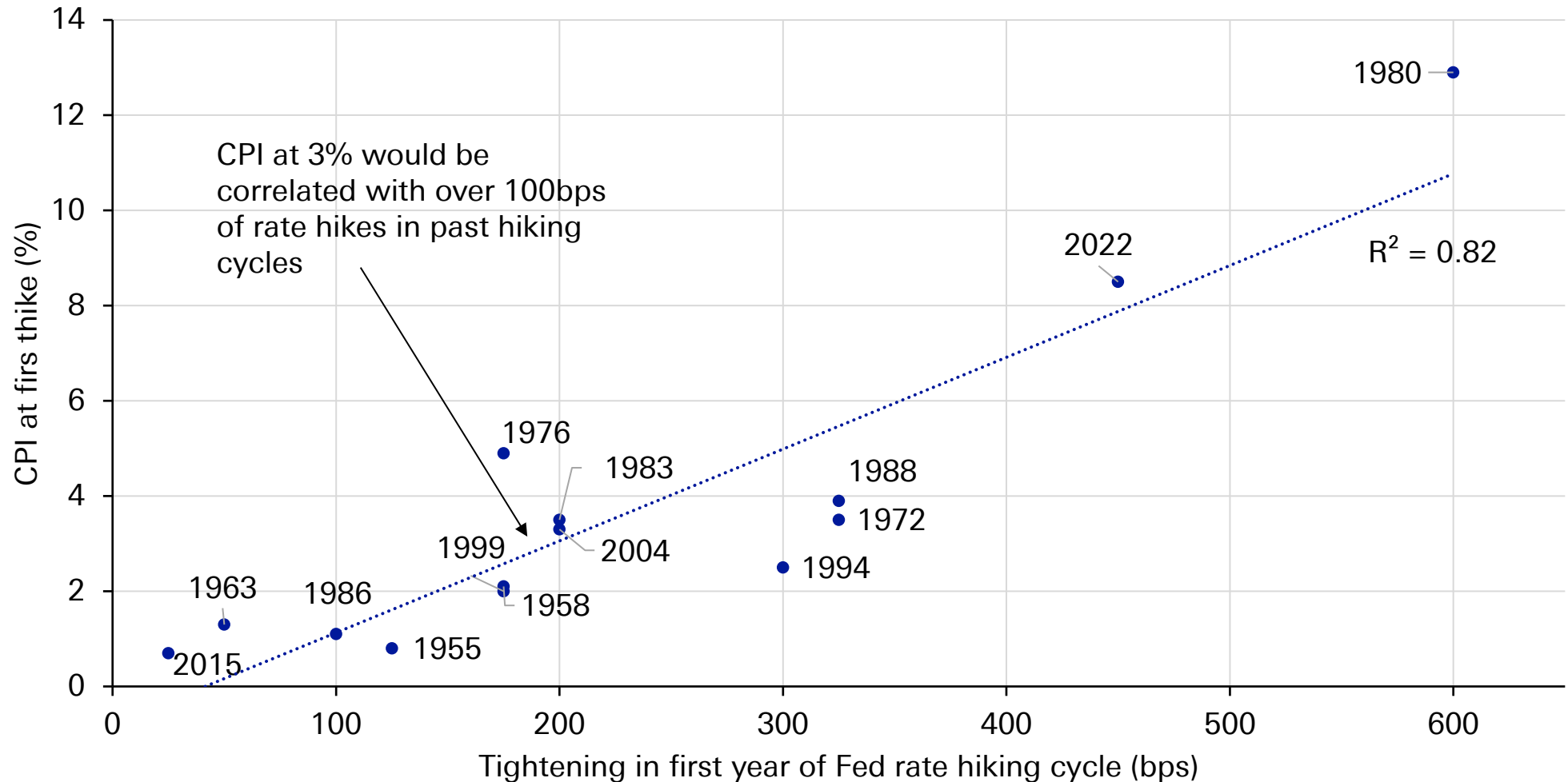


Source: BEA, Haver Analytics, Deutsche Bank



Historically, there's been a tight correlation between CPI inflation when the Fed starts hiking and how much they hike in the first year... with CPI at 3.4% in July, that would correlate to well over 100bps today... far above all expectations...

CPI and the amount of Fed tightening in the first year of a hiking cycle

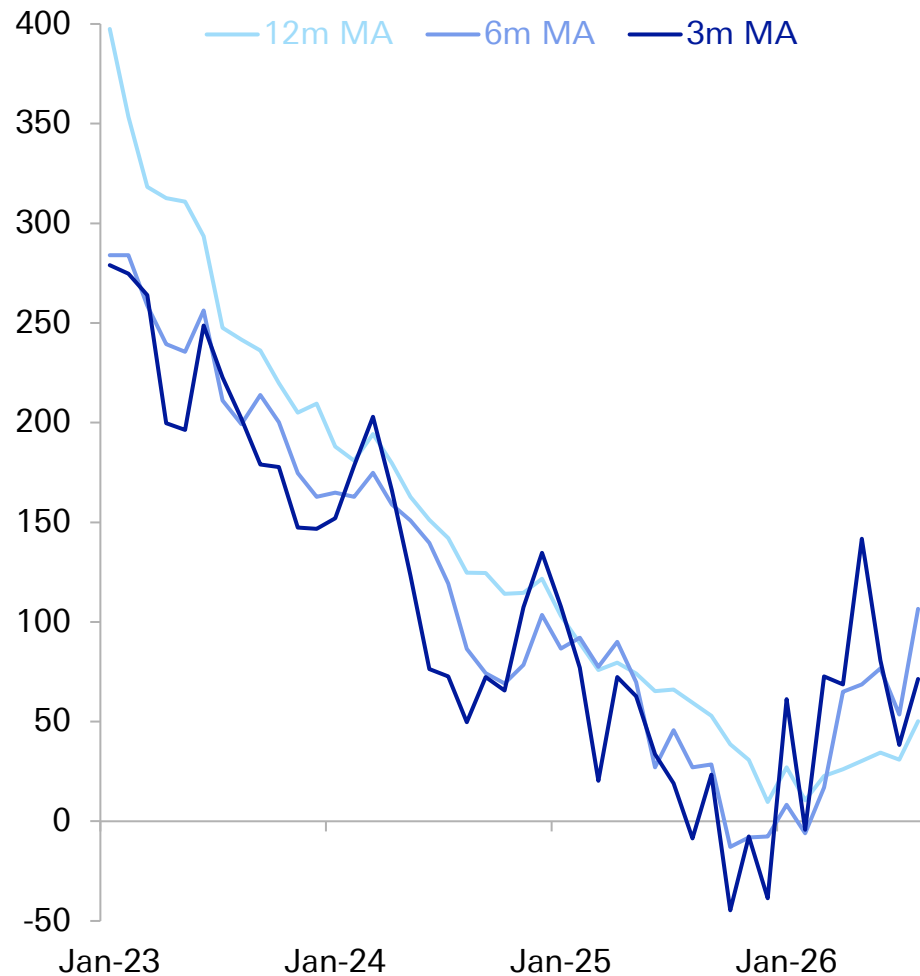


Source: Bloomberg Finance LP, Haver Analytics, Deutsche Bank

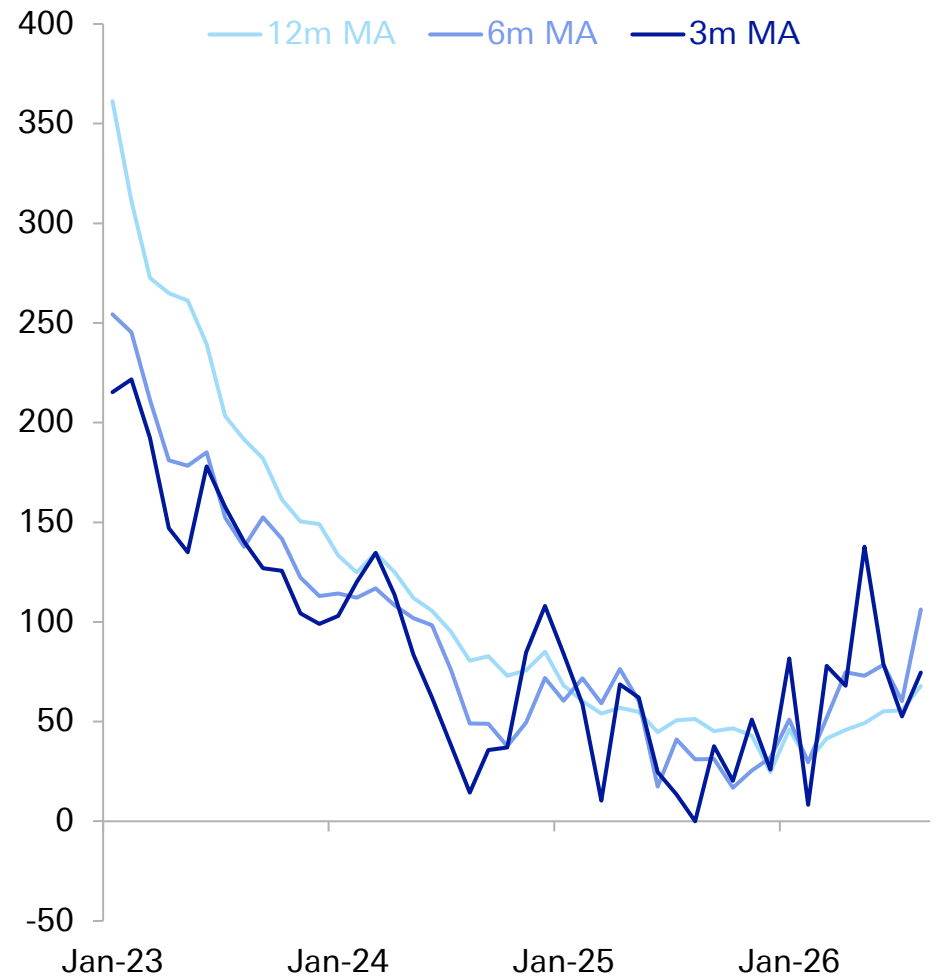
Payrolls did slow in 2025 which encouraged the insurance cuts but although volatile, there are clear signs that they have trended up in 2026.



Smoothed monthly gains in nonfarm headline payrolls



Smoothed monthly gains in private nonfarm headline payrolls

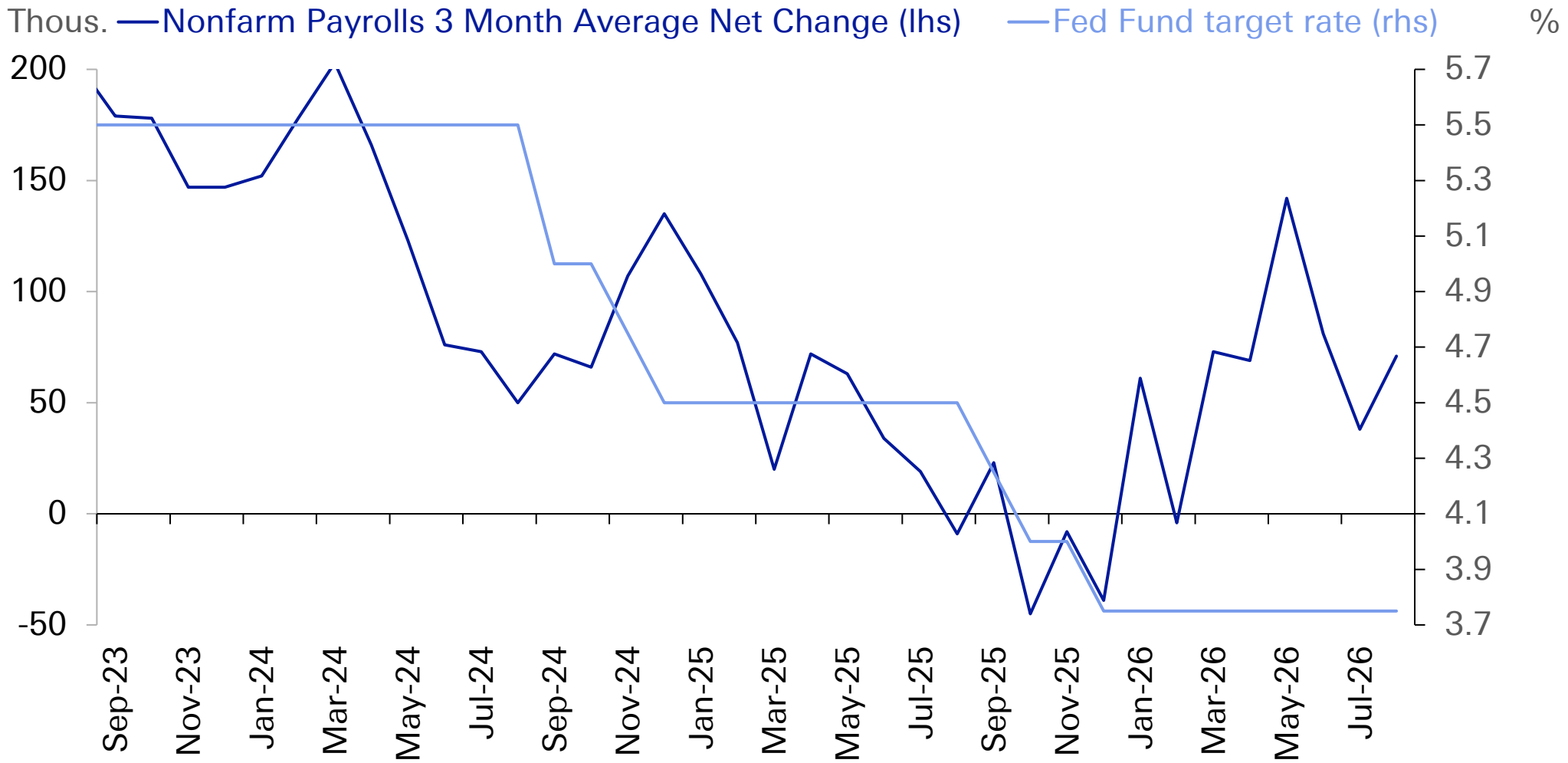


Source: BLS, Haver, Deutsche Bank

Indeed, with payrolls turning higher, those insurance cuts look ripe to be reversed...



Nonfarm payroll vs. Fed funds



Source: Bloomberg Finance LP, Deutsche Bank

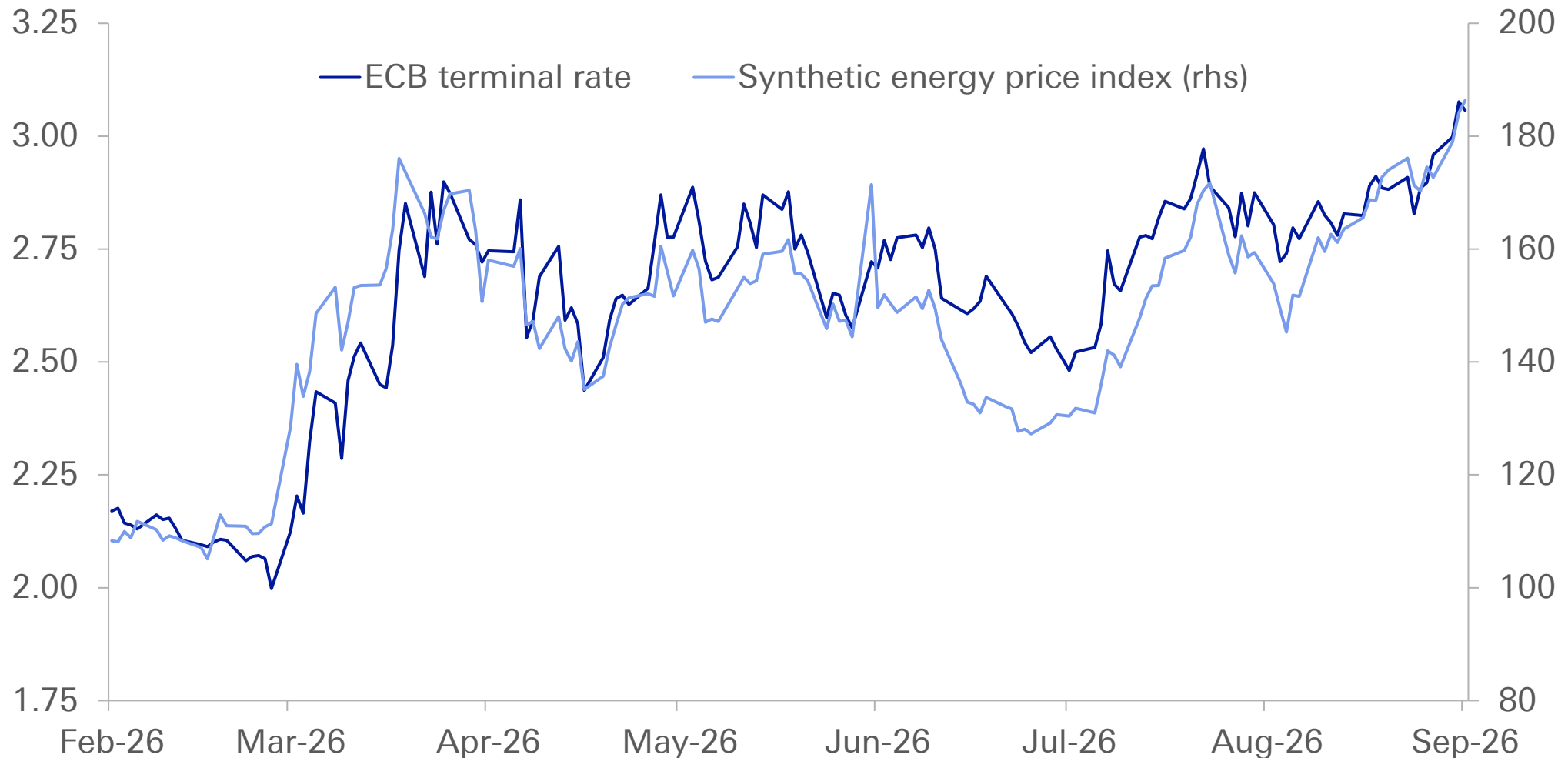


Should the ECB raise rates...?

ECB rate expectations (75bps more hikes) have tracked a synthetic energy price index.



ECB rate expectations & synthetic energy prices

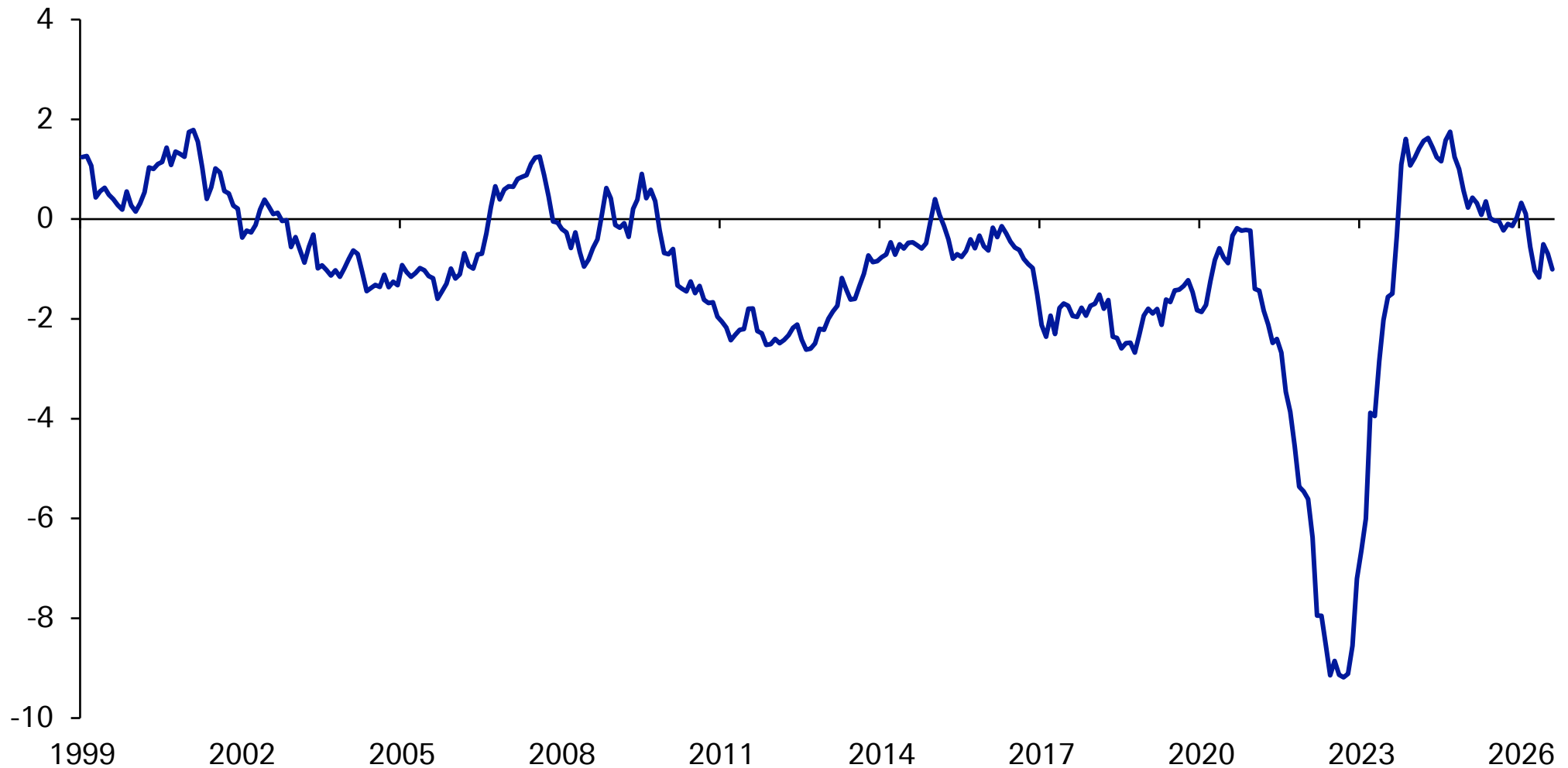


Source: Bloomberg Finance LP, Deutsche Bank. The index is an equal weighting of spot and 12M forward oil and gas prices

In February before the Iran conflict, the real adjusted ECB's deposit rate was still positive. But the energy shock pushed it back into negative territory again...



Real ECB Deposit Rate (the Deposit Rate minus year-on-year HICP inflation)

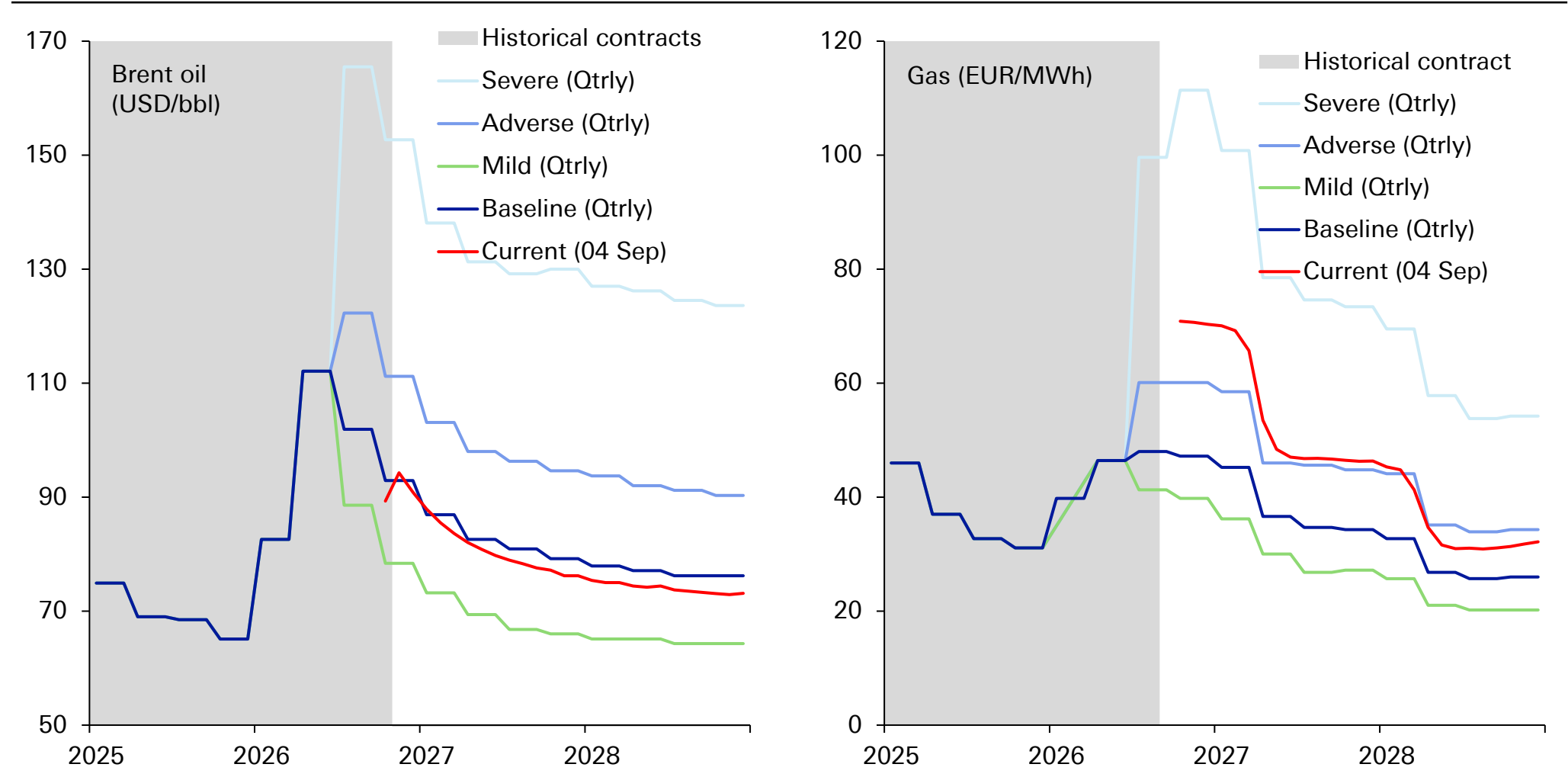


Source: Haver Analytics, Deutsche Bank



Brent is currently in-line with the ECB base case scenario from June but gas is now just above the adverse one. You can see our economics team's preview of the ECB meeting [here](#). They expect a hike this week and another to 2.75% in December. Note they have also upgraded '26 and '27 GDP to 0.8% and 1.2% (up 0.3pp and 0.1pp) respectively.

Oil prices (left) and Gas prices (right) vs. ECB baseline and alternative scenario



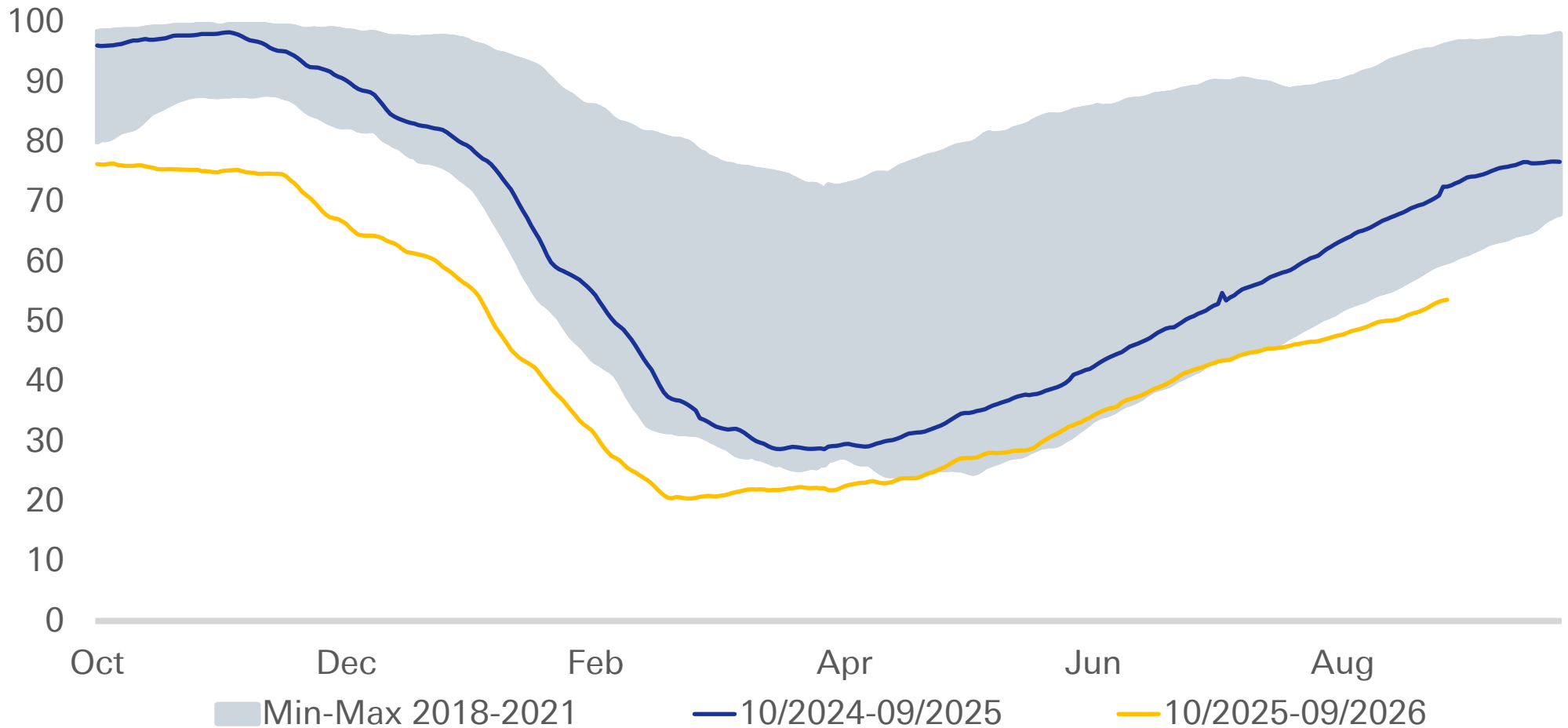
Source: ECB, Bloomberg Finance LP, Deutsche Bank Research

Gas storage levels haven't been an issue since the early days of the Russia-Ukraine conflict but we're starting to see low storage levels again ahead of the winter heating season. See Eric Heymann's blog [here](#) for more on this... Maybe El Nino will help prevent a cold winter!!



Gas Storage level in Germany

Gas storage level in Germany, %

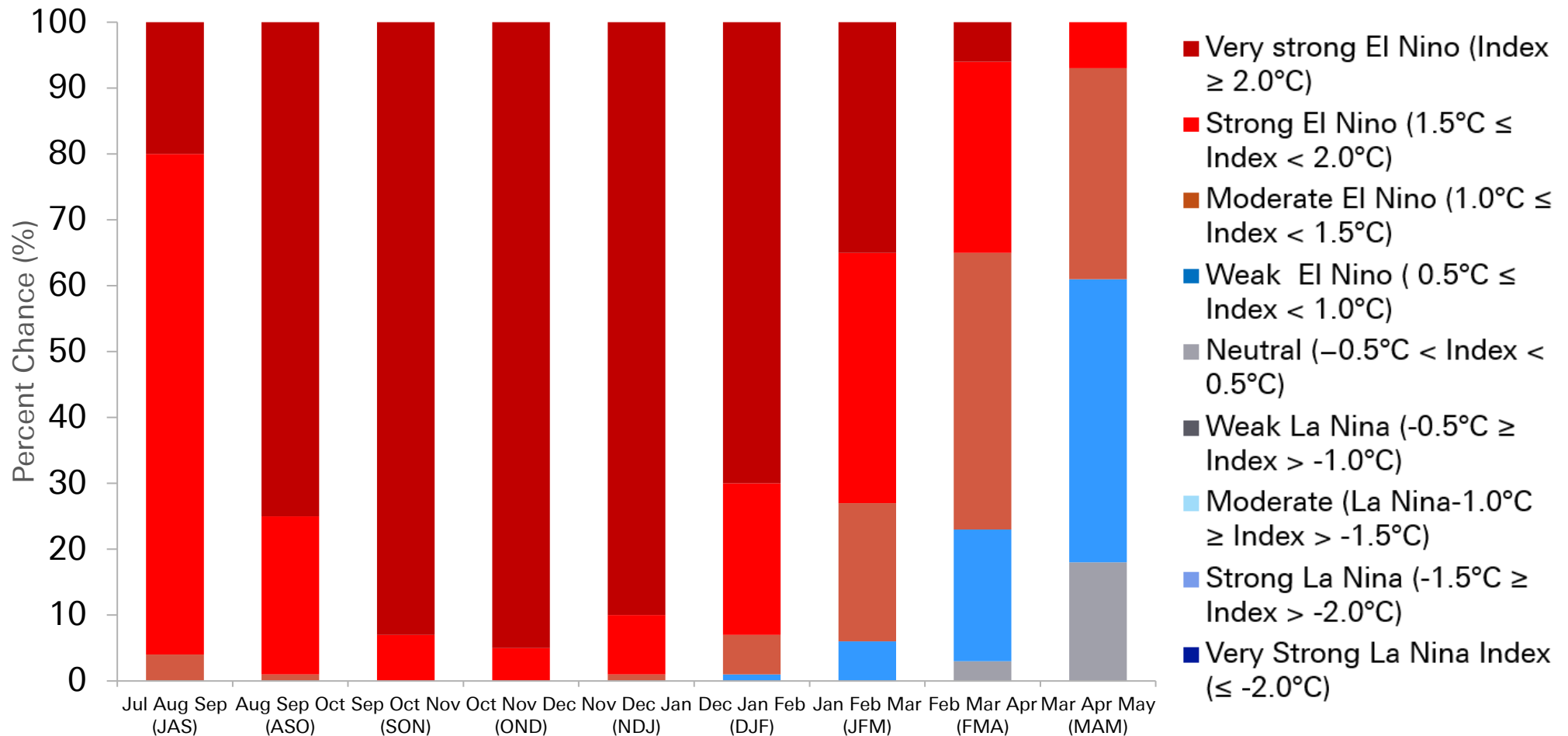


Source: Federal Network Agency, Deutsche Bank Research

Note that by the end of the year we could be seeing an El Nino unlike any we've seen before. Darkest red indicates the highest heat probabilities. This may have inflation and macro implications...



Official NOAA CPC ENSO Strength Probabilities (Issued August 2026)



Source: NOAA, Deutsche Bank ; Note: The bars show the chance of El Niño, ENSO-Neutral, and La Niña provided for 9 upcoming, overlapping 3-month seasons (each letter represents a month).

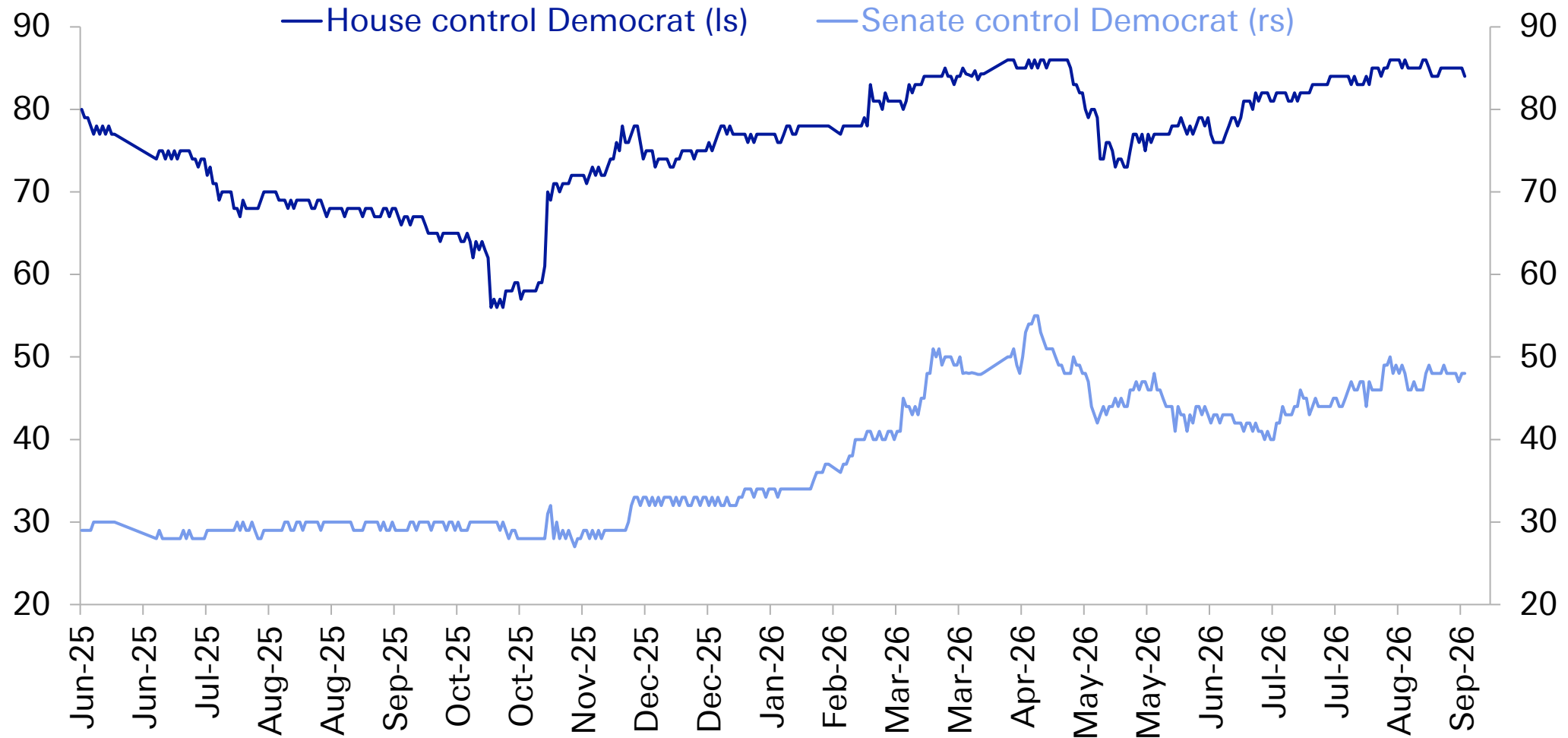


Midterms are fast coming into view...

While the House race has long been in favour of the Democrats, the Senate was firmly leaning GOP in prediction markets until this year and especially since the Iran conflict lifted inflation...



Predicted 2026 Mid-term election outcome

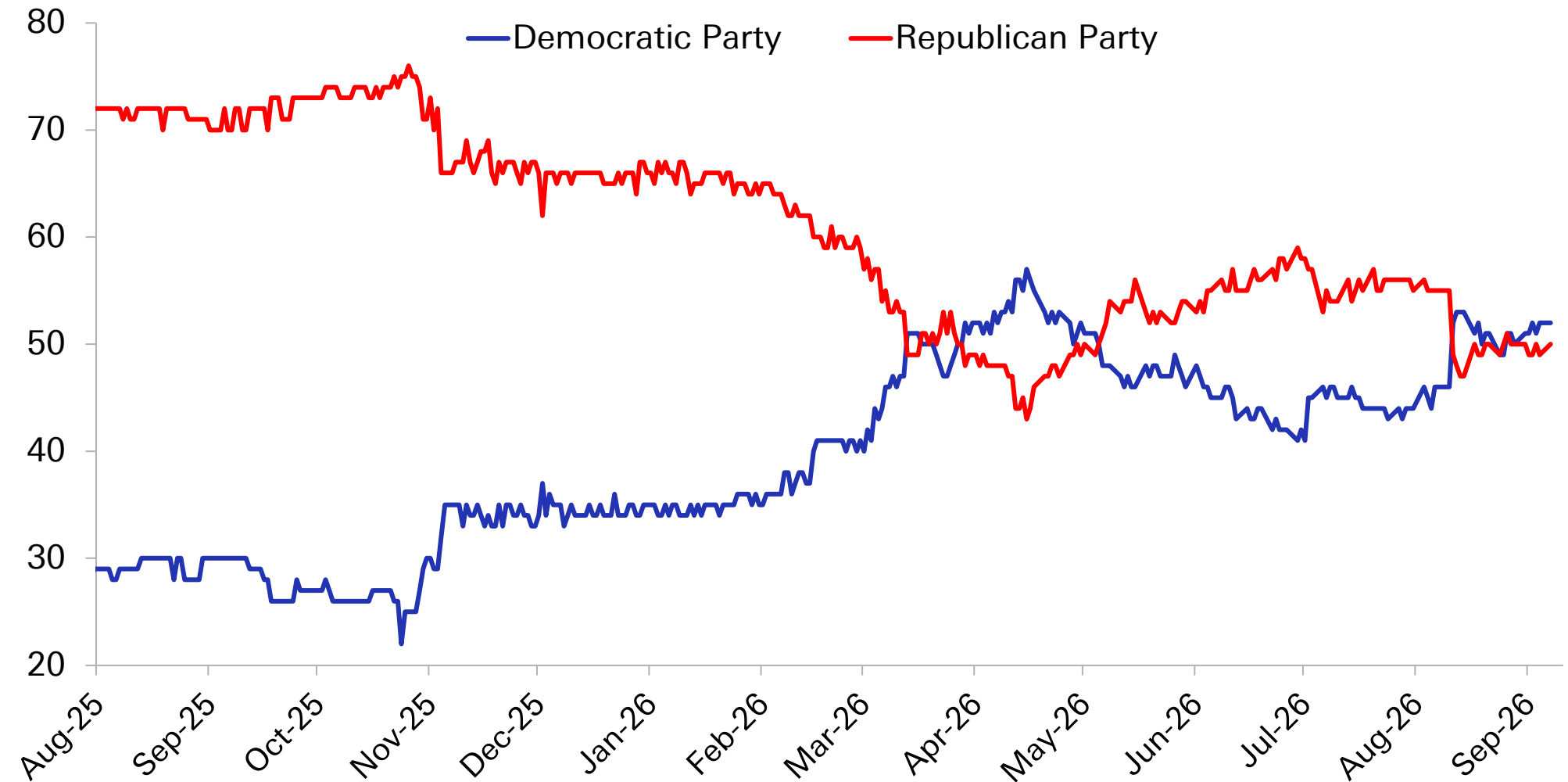


Source : Bloomberg Finance LP, Deutsche Bank, Polymarket, updated through Sep7

Since around the time the Iran strikes began, the Democrats have caught up in the race to retake control of the Senate... that would have a big impact on appointments like Fed Governors, cabinet members and Supreme Court justices, which all require Senate approval... Surely the Senate matters to Trump...



US Polymarket Party control Senate after 2026 Midterm election

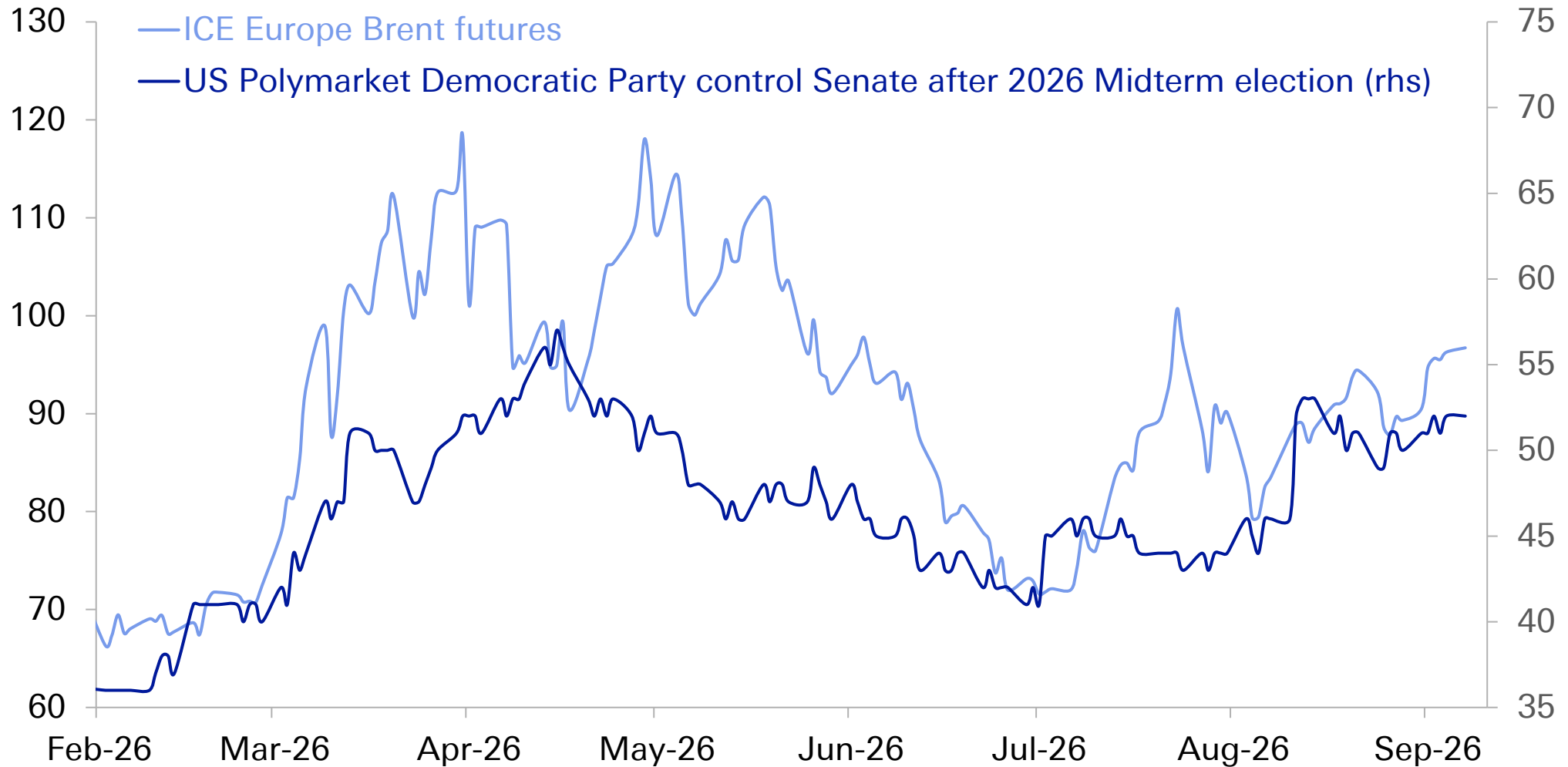


Source : Bloomberg Finance LP, Deutsche Bank, updated through Sep 7

There is a decent correlation between Brent and the probability of Senate control



Probability of Democrat control of the Senate after mid-terms and Brent price

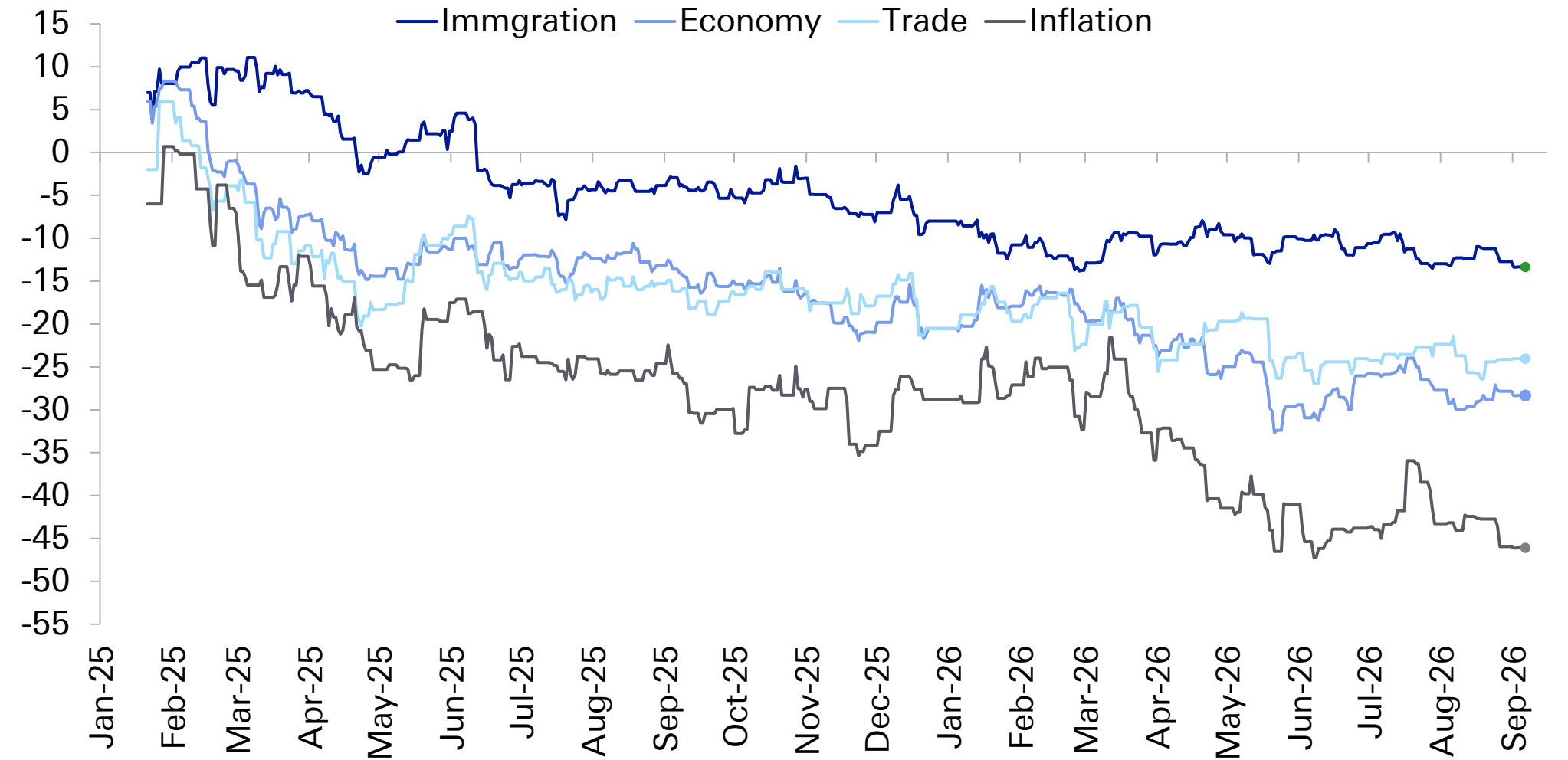


Source : Bloomberg Finance LP, Deutsche Bank, updated through Sep 7

Inflation is where Trump's approval rating seems most challenged and fell further after the Iran conflict began.



An updated polling average of Donald Trump's net approval rating on immigration, the economy, trade, and inflation, accounting for each poll's quality, recency, sample size, and partisan lean



Source: Nate Silver, Deutsche Bank, Last update : Sep 7



Mid-term years often see a weaker start to H2 for the S&P 500 and a fall into October as summer fades. After mid-terms the market tends to be strong regardless... 2026 initially followed the script but markets have been climbing since early August...

Average S&P 500 performance (first trading close on or after July 1 = 100)

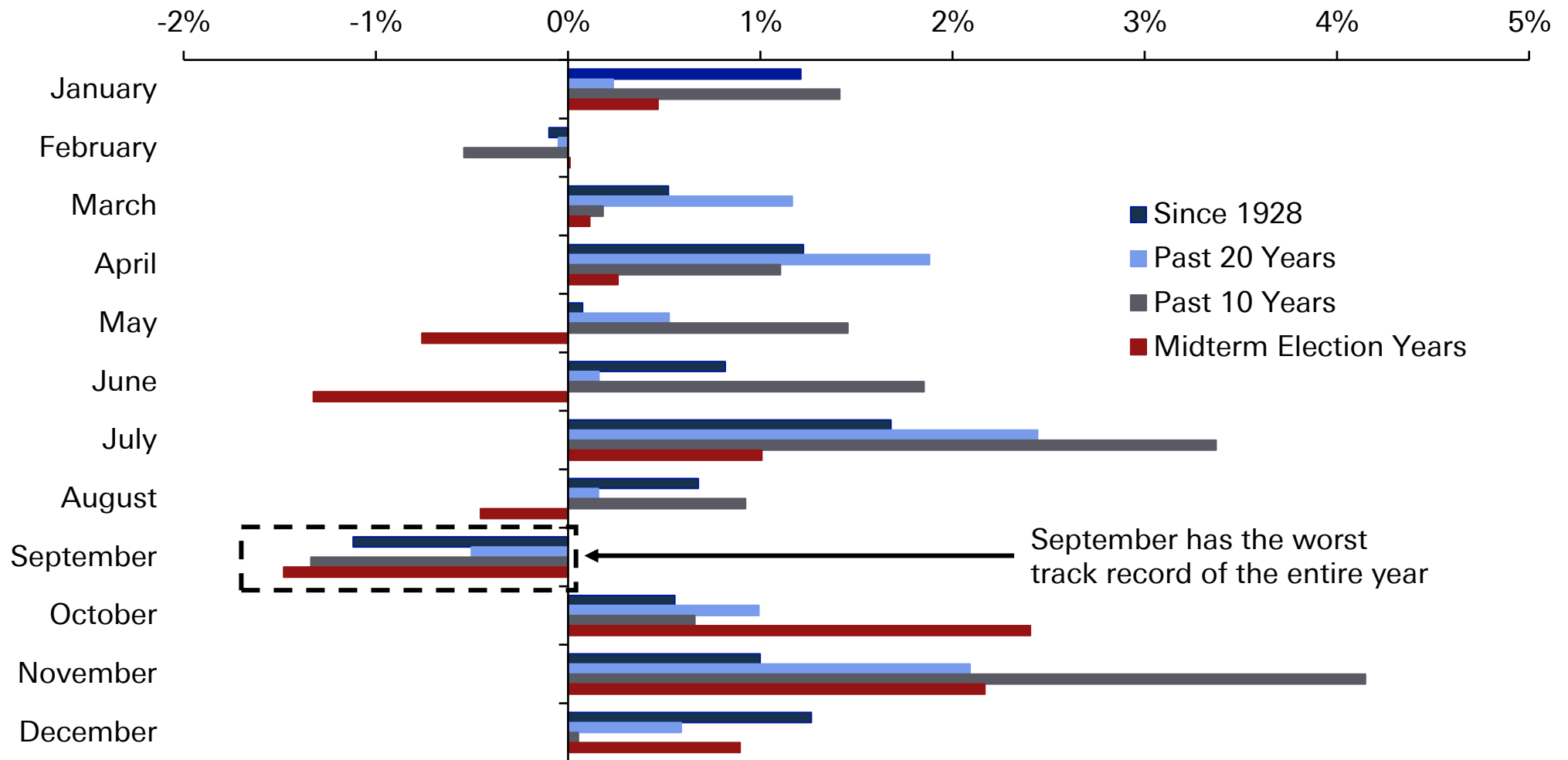


Source: Bloomberg Finance LP, Deutsche Bank

Throughout history and in recent years, the S&P 500 has consistently seen its worst month of the year in September... Midterm years are tougher but note that just before and after mid-terms, the market fully recovers so is this enough to justify lightening up?



Monthly Performance of the S&P 500 (1928-2025)

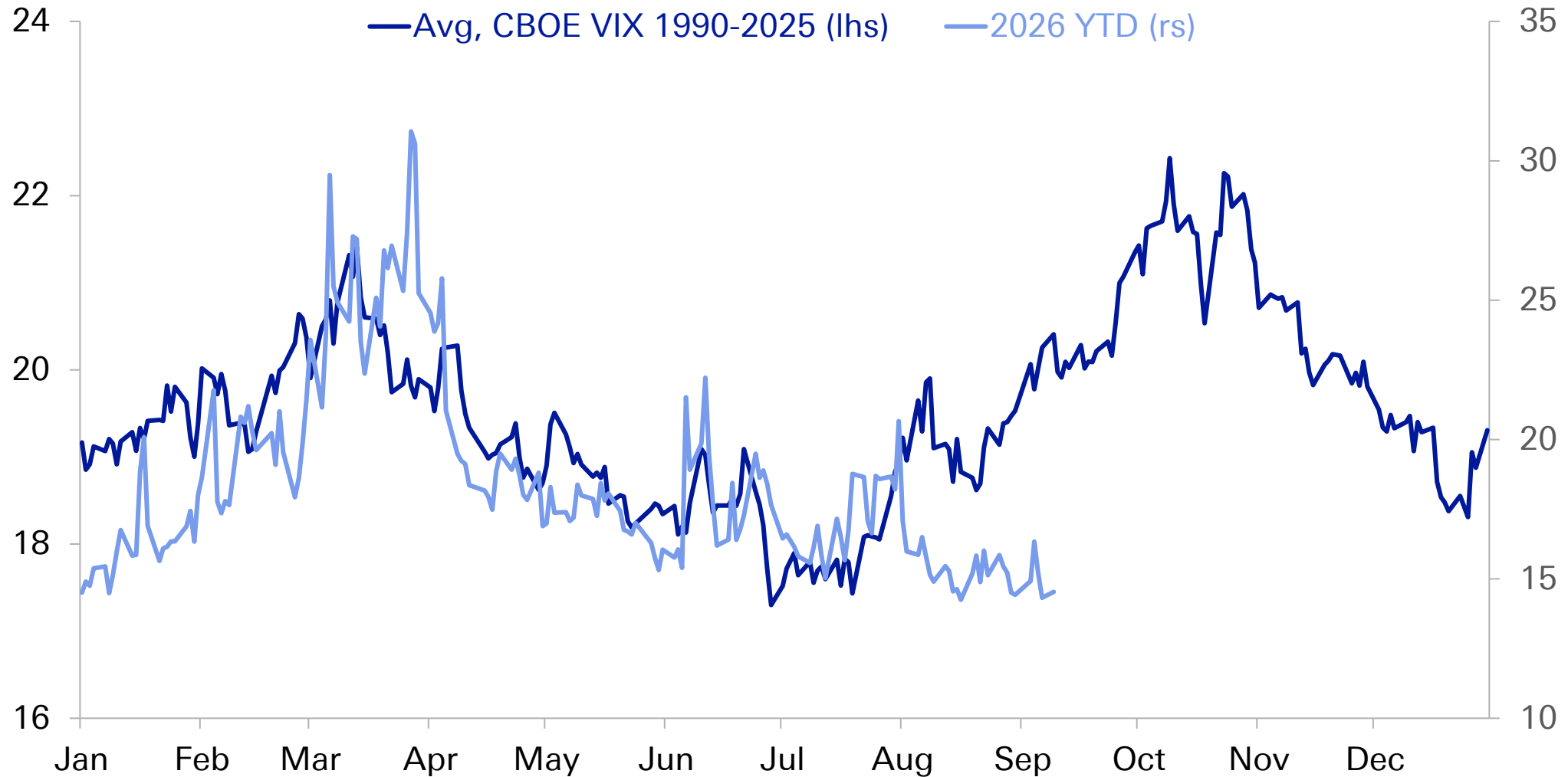


Source: Bloomberg Finance LP, Deutsche Bank

Even though there are a few ongoing concerns (e.g. Iran, oil, rising yields, mid-terms etc.), so far 2026 is resisting the seasonal pattern in terms of volatility...



Seasonality of the VIX index



Source: Bloomberg Finance LP, Deutsche Bank, Last update: Sep 7



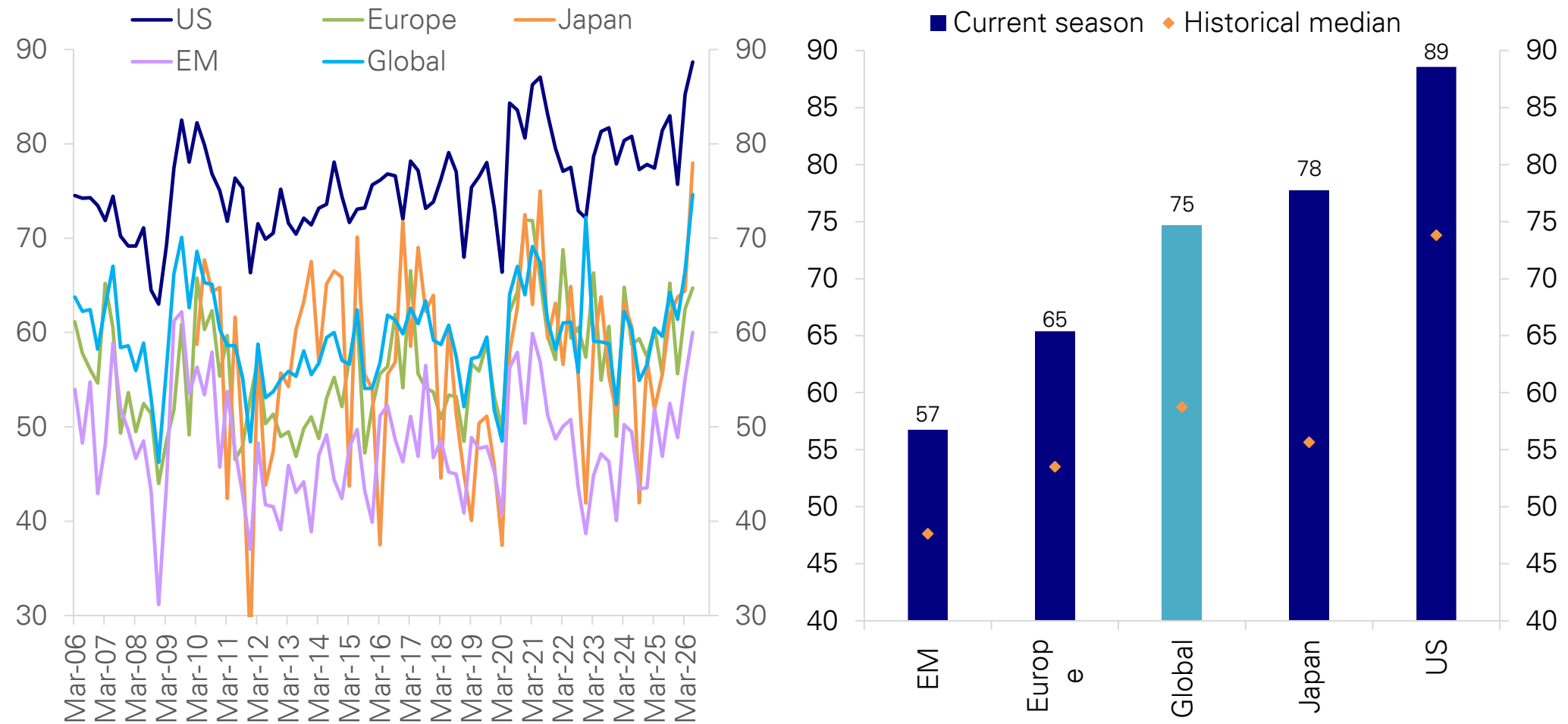
Global earnings may never have been stronger...

Please see Parag Thatte and Binky Chadha's excellent piece reviewing global earnings [here](#)



The breadth of companies beating expectations hit record peaks in Q2 in the US and Japan and remained well above historical averages in Europe and EM

Proportion of companies beating on earnings (%)



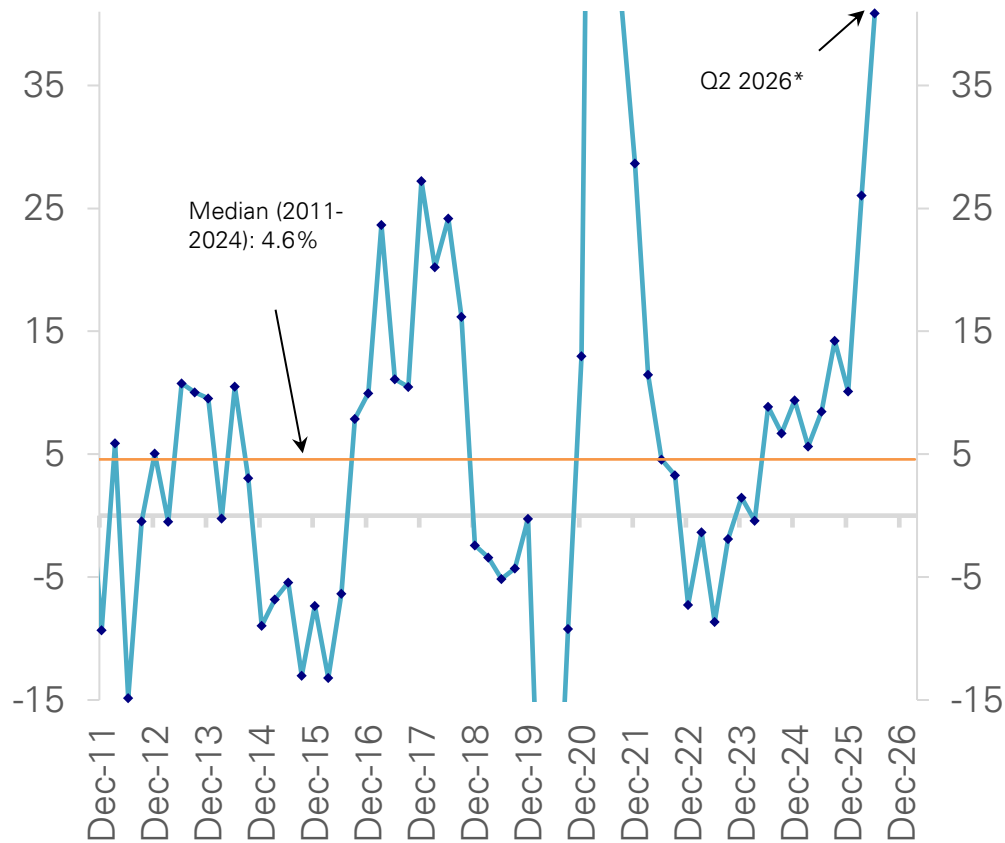
Source: Bloomberg Finance LP, Deutsche Bank Asset Allocation

And we haven't beaten low expectations. In Q2 we saw the strongest non-pandemic YoY global earnings expansion on record, driven by synchronised cyclical and commodity tailwinds.



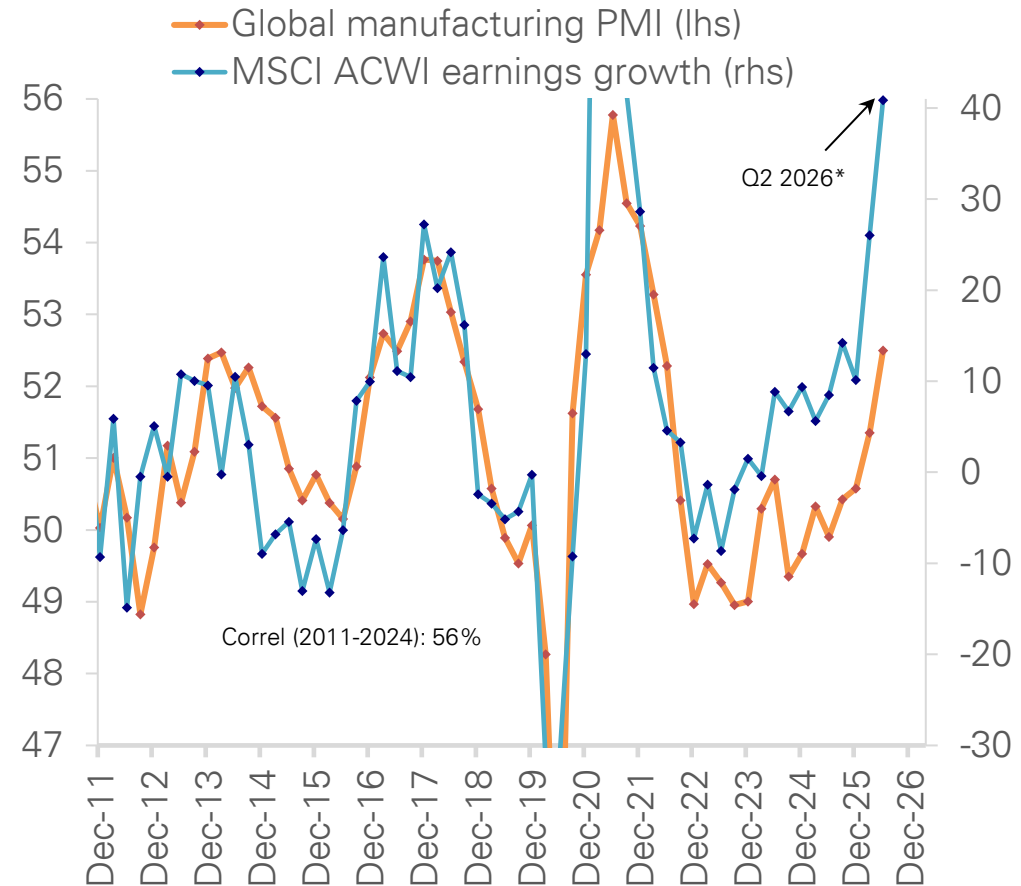
Global YoY Earnings growth

MSCI ACWI aggregate earnings growth (% yoy)



*Q2 is blended, assumes companies yet to report beat at current beat rate; growth in USD

Global earnings growth vs manufacturing PMI



*Q2 is blended, assumes companies yet to report beat at current beat rate

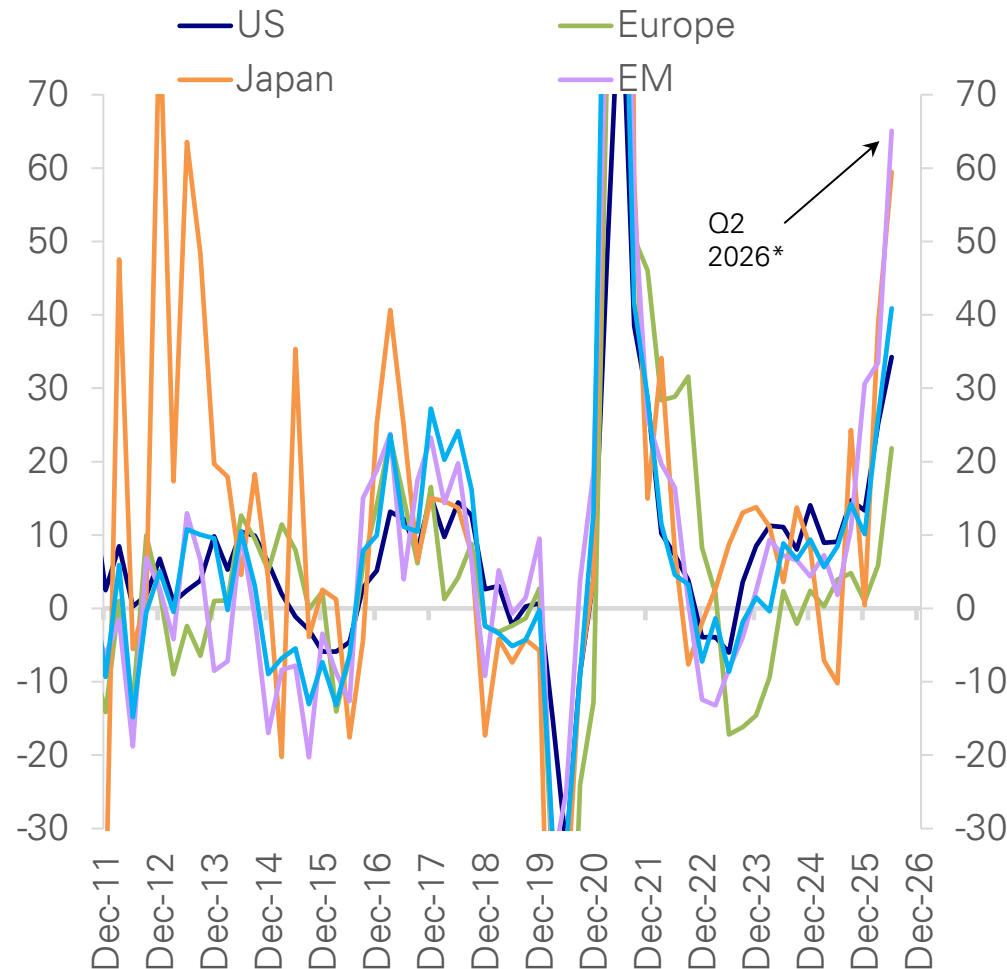
Source: Bloomberg Finance LP, Deutsche Bank Asset Allocation

Earnings growth has accelerated across all regions

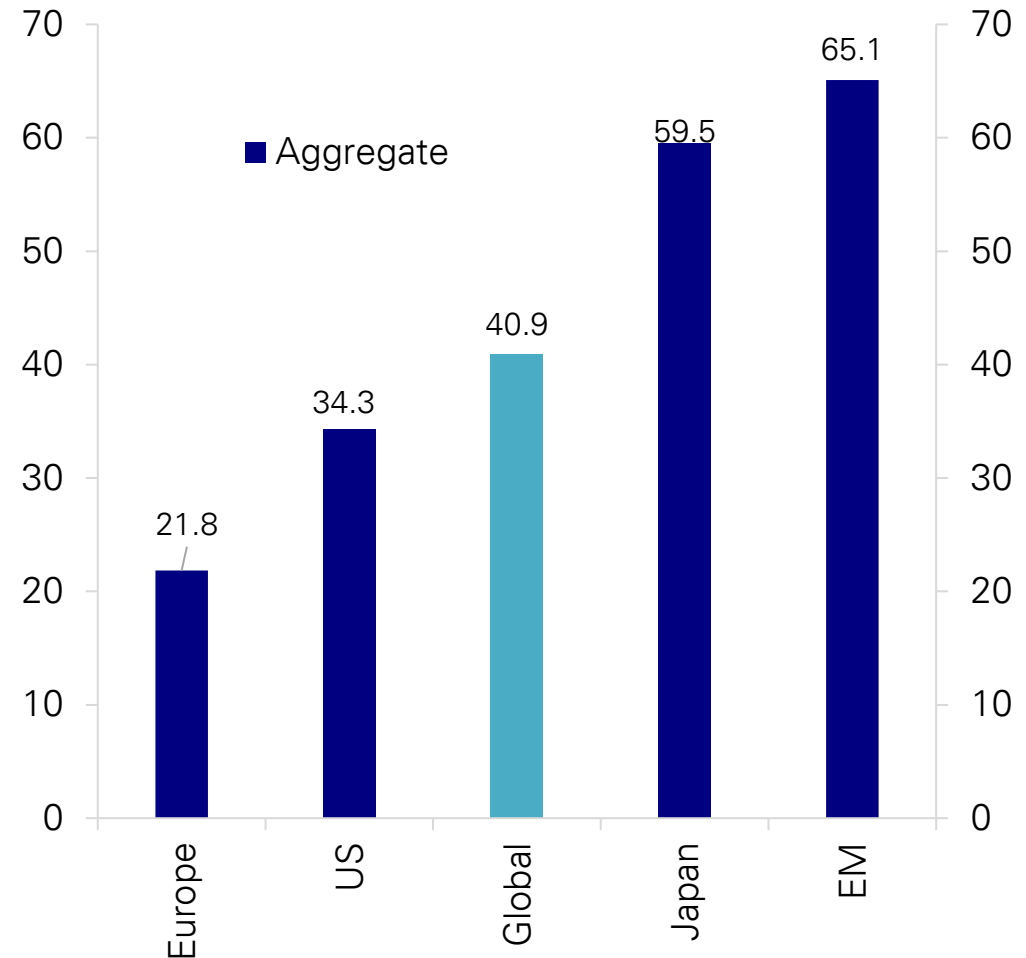


Quarterly aggregate earnings growth (% yoy)

Q2 2026 blended earnings growth (% yoy)



*Q2 is blended, assumes companies yet to report beat at current beat rate



*Q2 is blended, assumes companies yet to report beat at current beat rate

Source: Bloomberg Finance LP, Deutsche Bank Asset Allocation

It's been a blockbuster Q2 for the US...

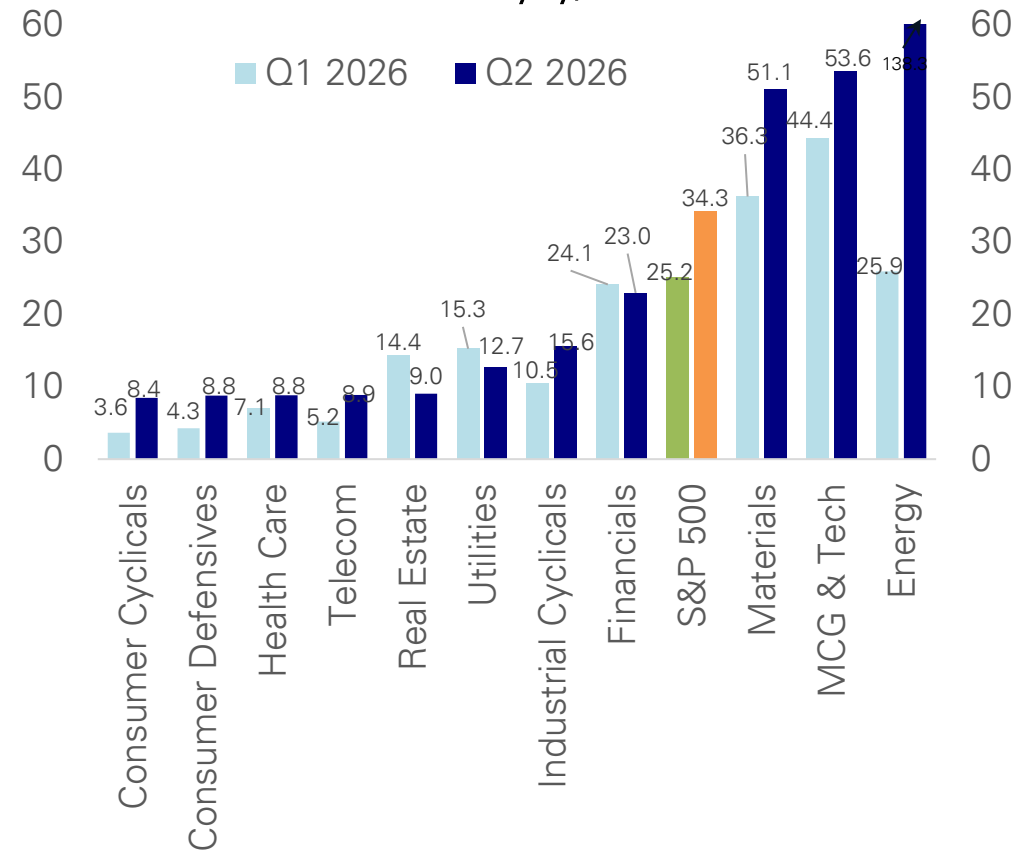


S&P 500 YoY earnings growth with sectors performance...



* Excluding 2018 tax cut

Earnings growth by S&P 500 sector group (% yoy)



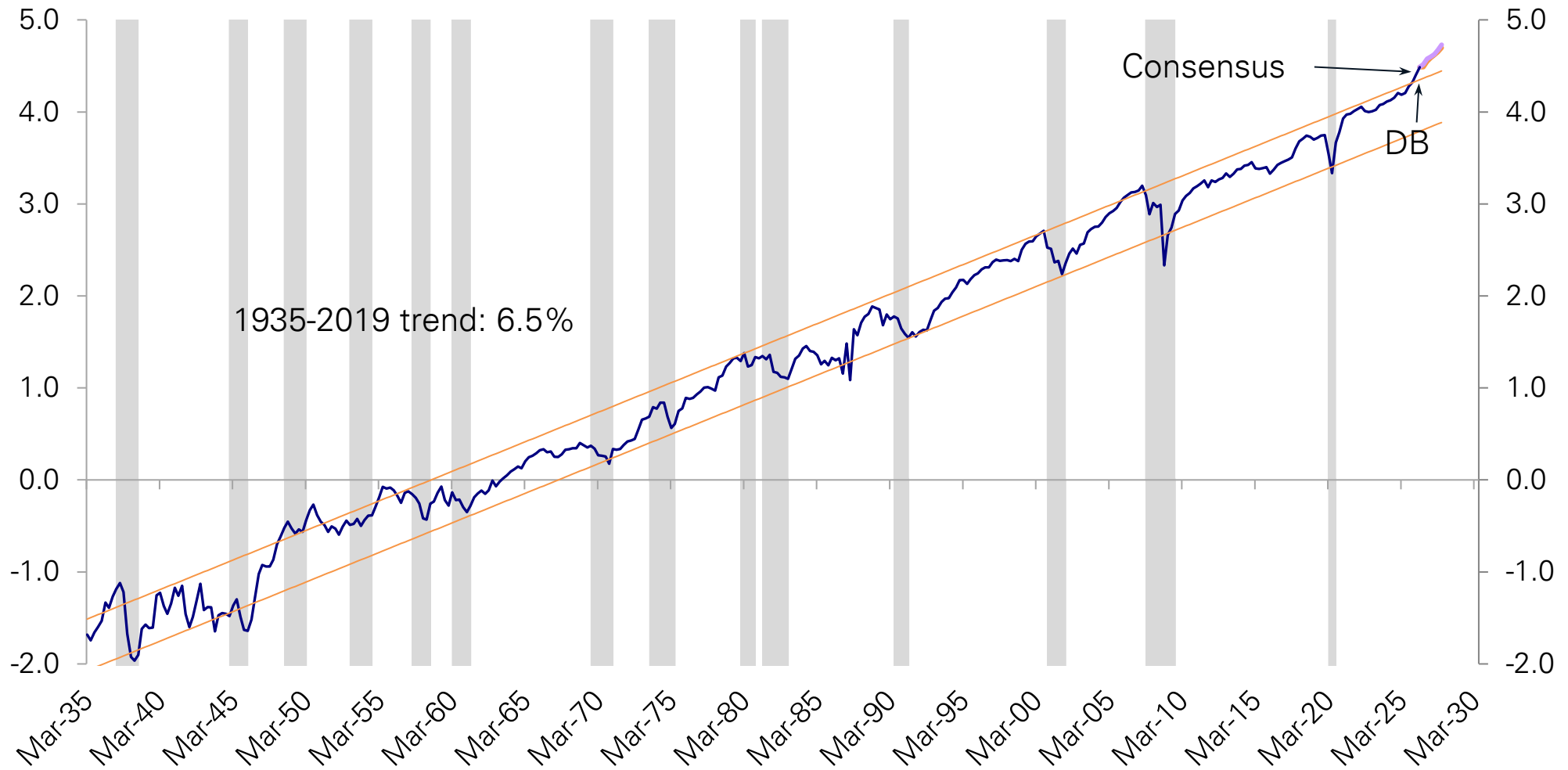
* Blended assumes companies that that report beat at the current beat rate

Source: Bloomberg Finance LP, Deutsche Bank Asset Allocation



US earnings are breaking out of a 90-year channel.... On the upside which is incredible when you consider the high growth periods of the past.... The debate will rage on as to whether this is a sustainable new plateau or an artificial period of AI related elevated earnings...

S&P 500 quarterly EPS seasonally adjusted (log)

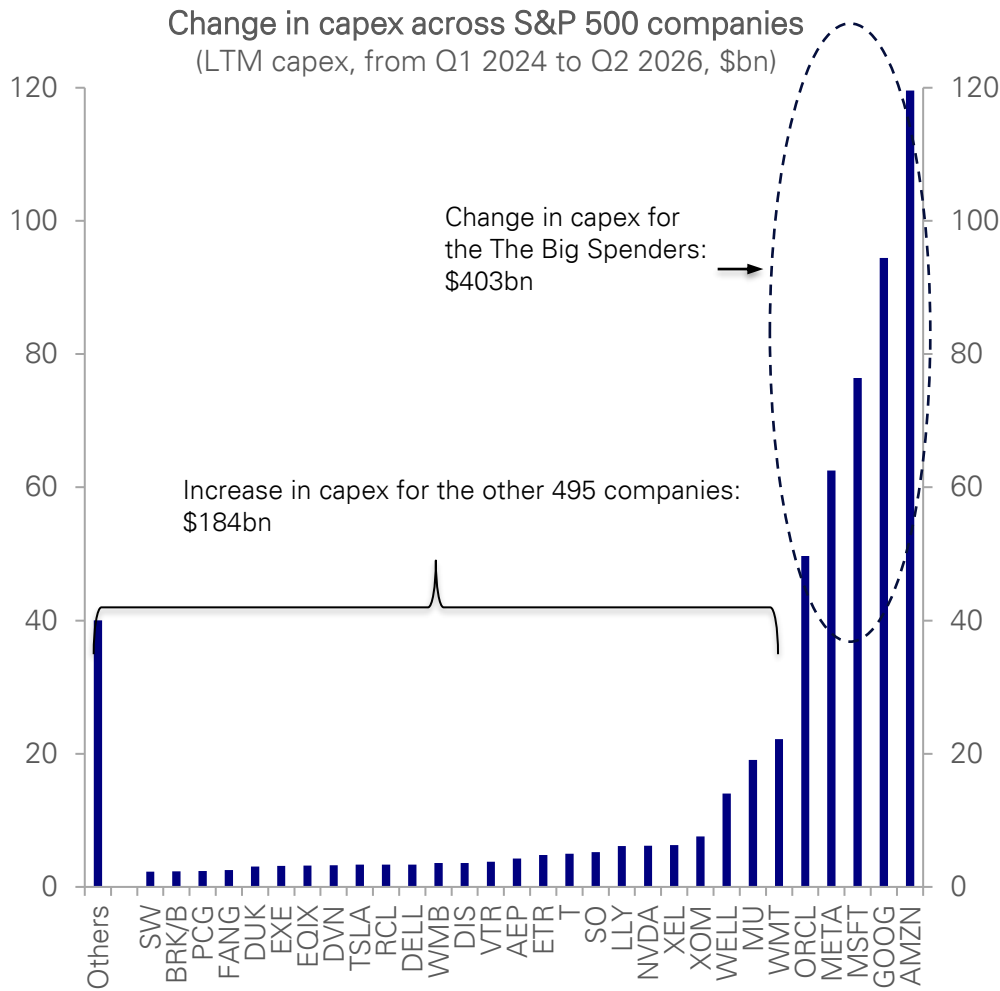
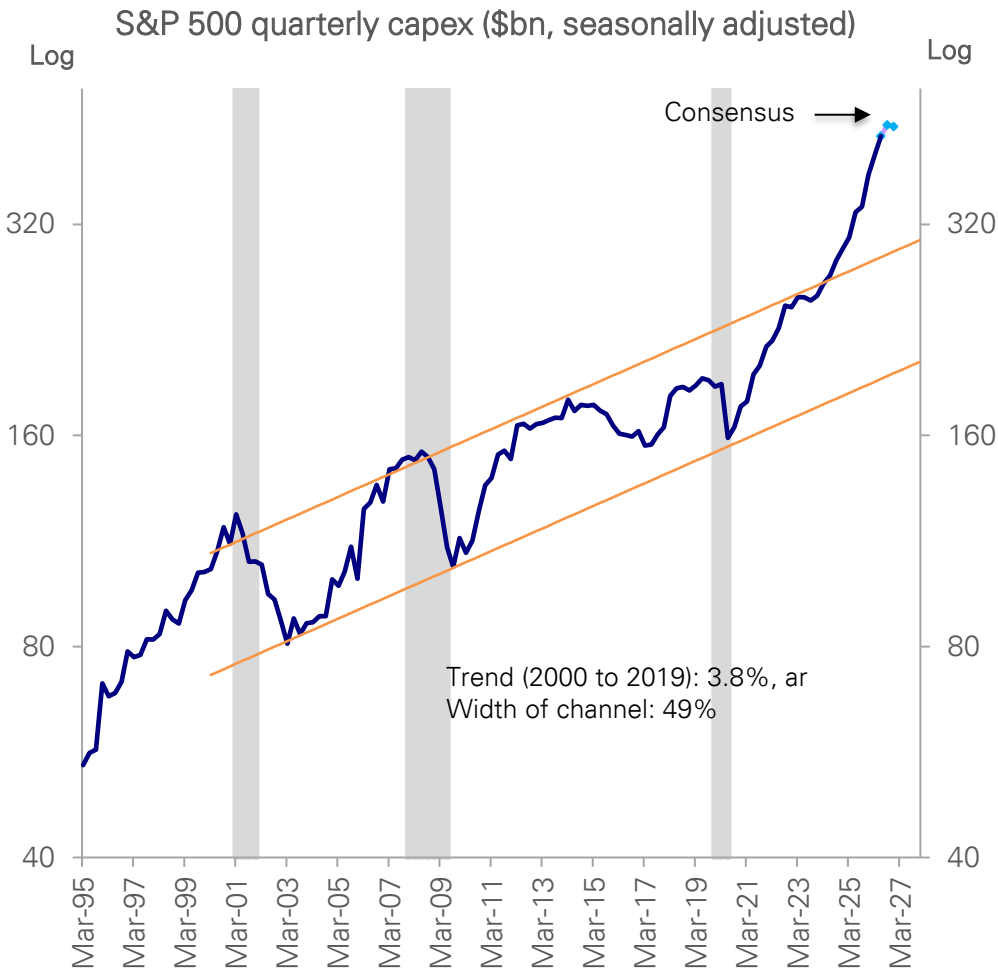


Source: Bloomberg Finance LP, Deutsche Bank Asset Allocation

We are clearly in an AI Capex boom. The top 5 capex spenders are spending more than double that of the remaining 495 in the S&P 500



S&P 500 Capex spending...

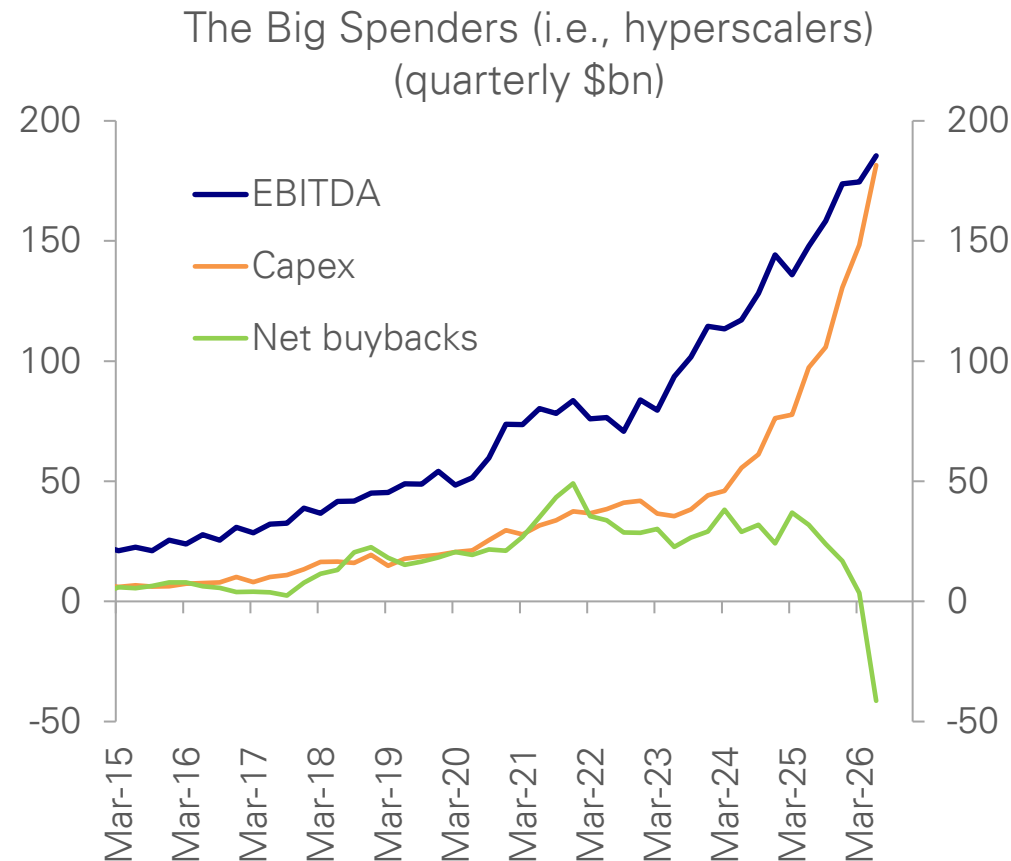


Source: Bloomberg Finance LP, Deutsche Bank Asset Allocation

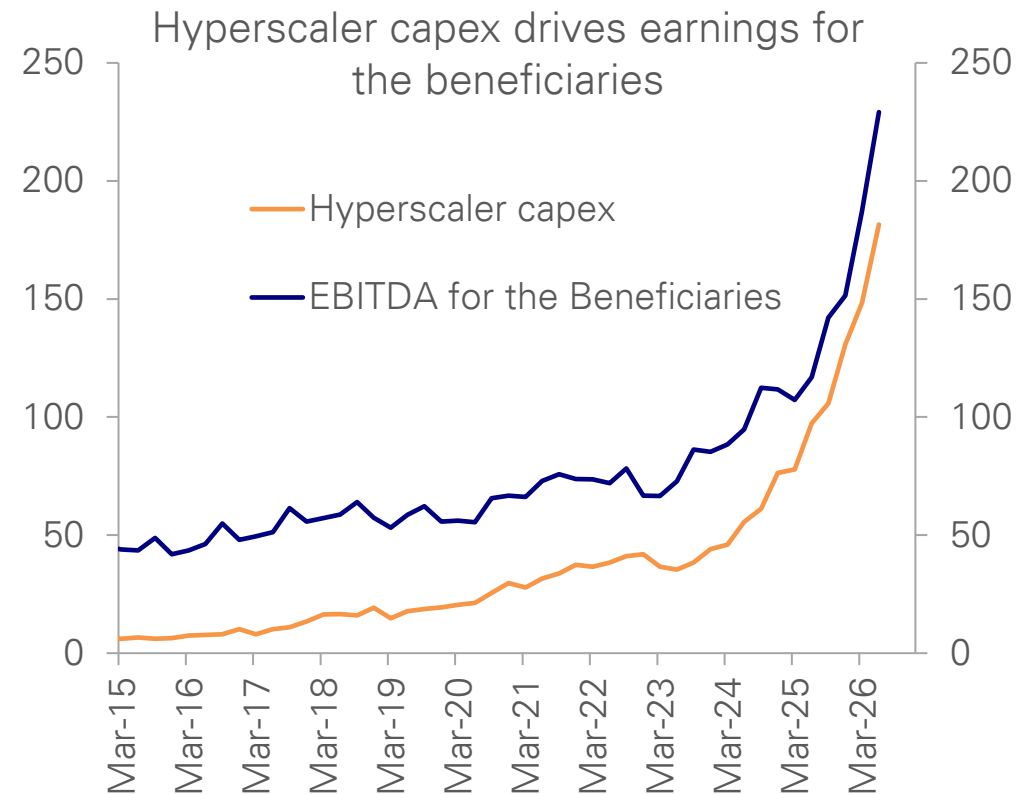
Buy-backs have turned into issuance for the hyperscalers but their spending is massively helping the AI food chain...



US Hyperscaler Capex, earnings and buybacks



Hyperscalers: AMZN, GOOG, MSFT, META and ORCL



Hyperscalers AMZN, GOOG, MSFT, META and ORCL

Beneficiaries: Semis, Tech H/W & Equip., Construction & Engg., Electrical Equip., Electric Utilities, Independent Power Producers and Data Center REITs

Source: Bloomberg Finance LP, Deutsche Bank Asset Allocation



The AI train rolls on...

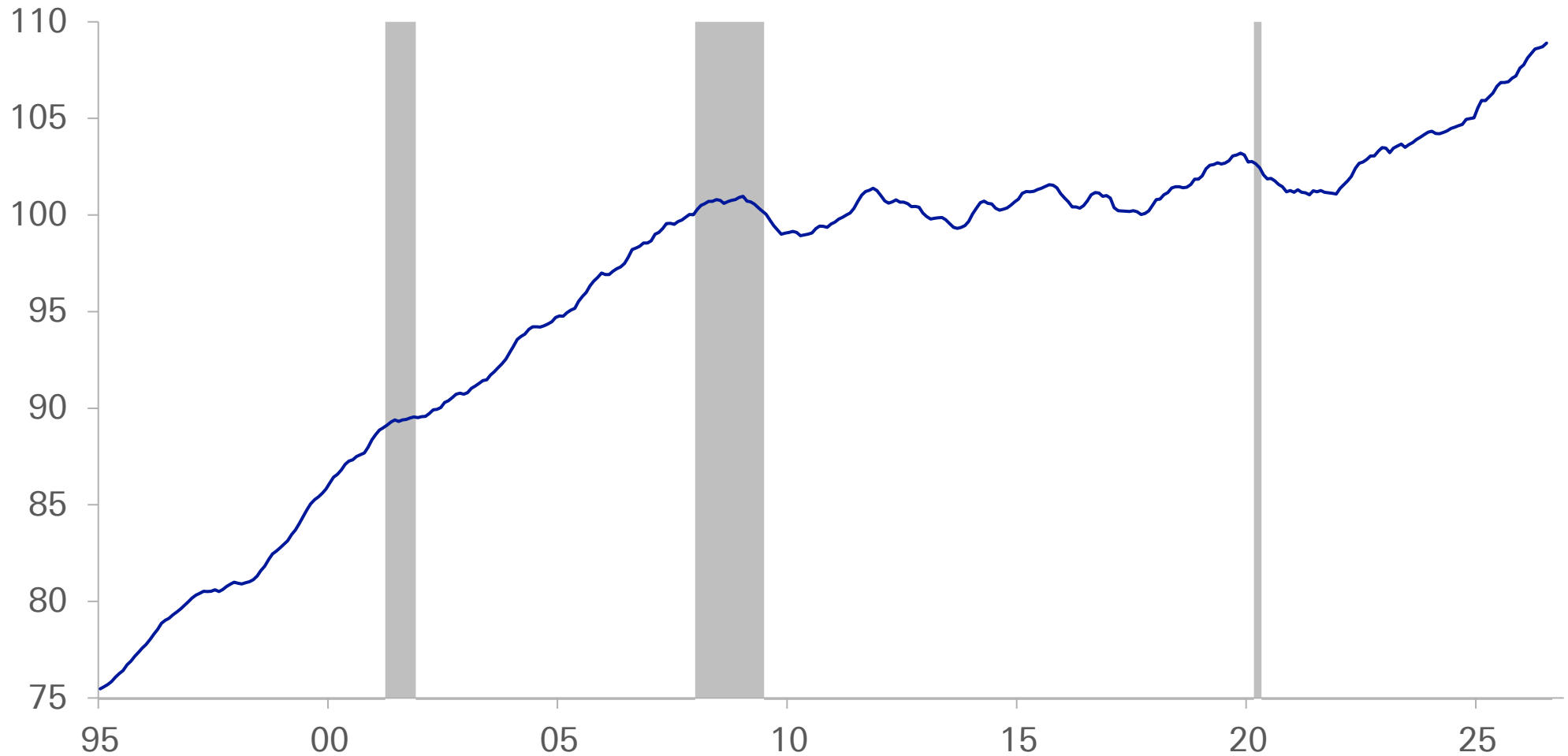


100% 90% 80% 70% 60% 50% 40% 30% 20% 10% 0%

This exponential growth continues to drive demand for energy after a decade of stagnant demand...



IP: Electric Power Generation, Transmission and Distribution , 24-month Moving Average

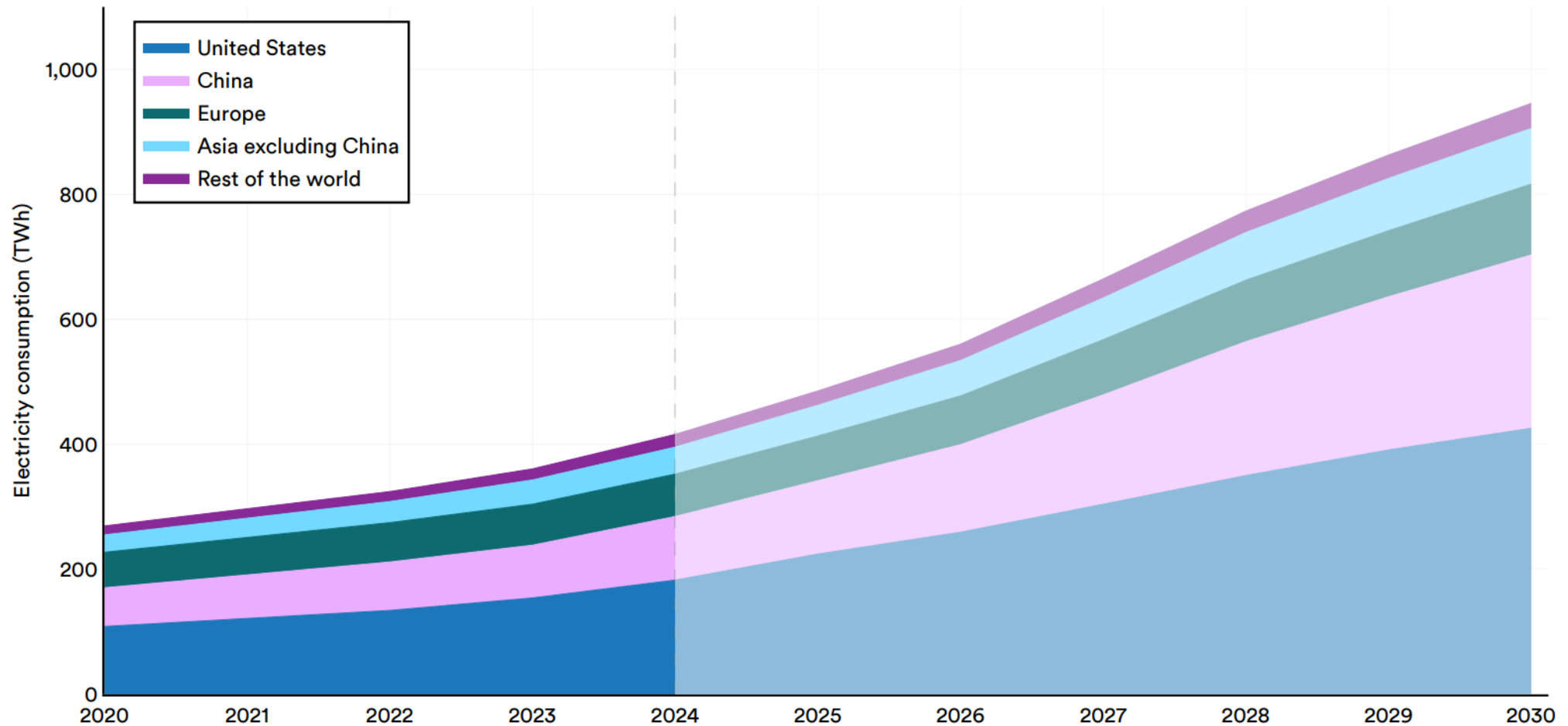


Source: FRB, Haver, Deutsche Bank

There is little sign that energy demand is going to wane...



Data center electricity consumption by region, 2020–30



Source: International Energy Agency (2025), Energy and AI, IEA, Paris | Chart: 2026 AI Index report, Deutsche Bank



Token costs are collapsing again amid cheaper compute, more efficient systems and pricing pressure from cheaper open-weight models. Lower prices and more sophisticated uses are fuelling adoption, even if the economics of some of the expensive frontier models may be tested

Silicon Data LLM Token Expenditure Index

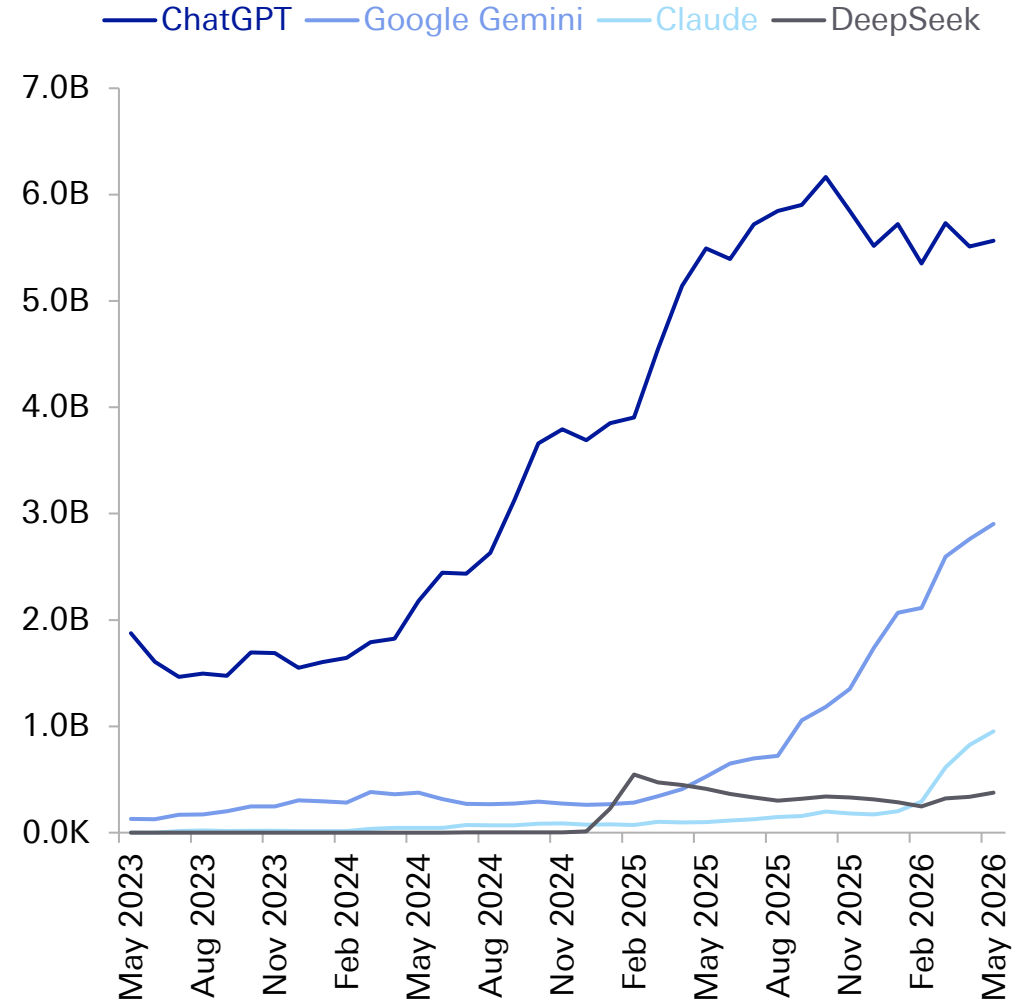
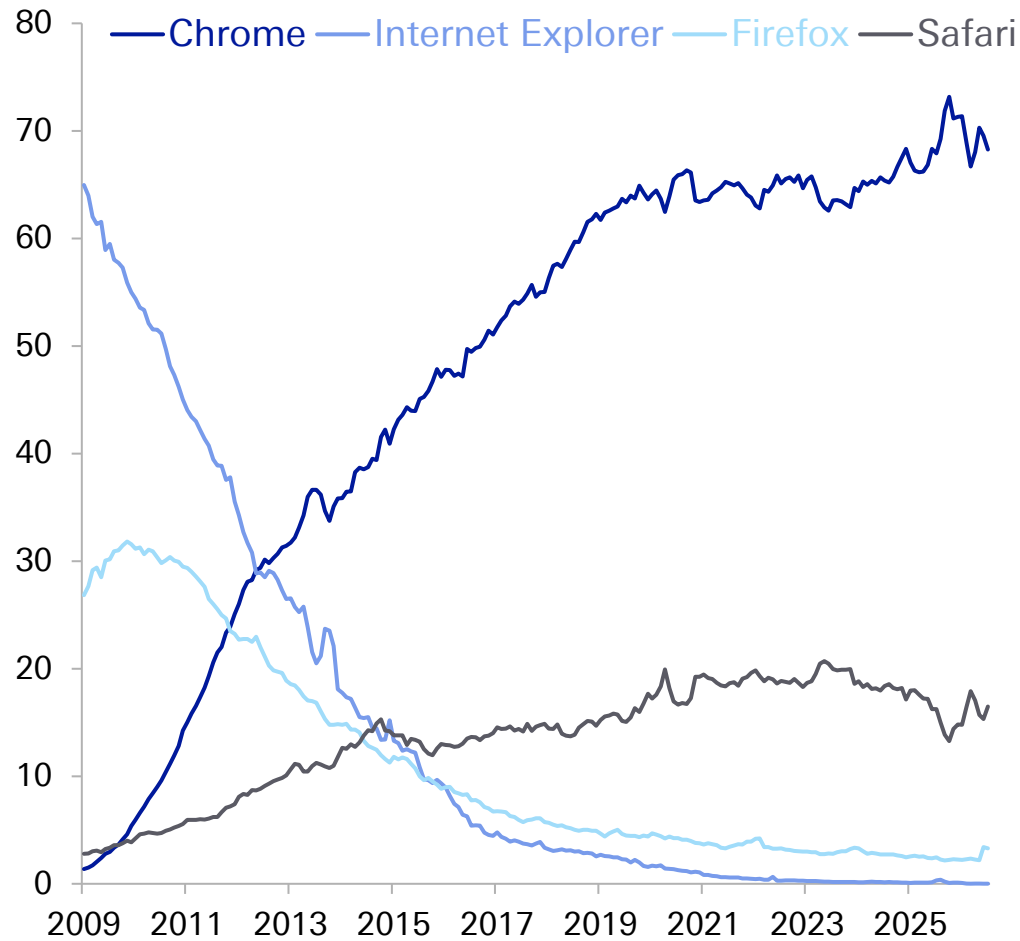


Source: Bloomberg Finance LP, Deutsche Bank, Last update Sep 7

Remember the early winners don't always end up on top....



Browser market shares (%) (2009 to today) (right) and Top four generative AI platforms: monthly visits worldwide (desktop and mobile) (left)

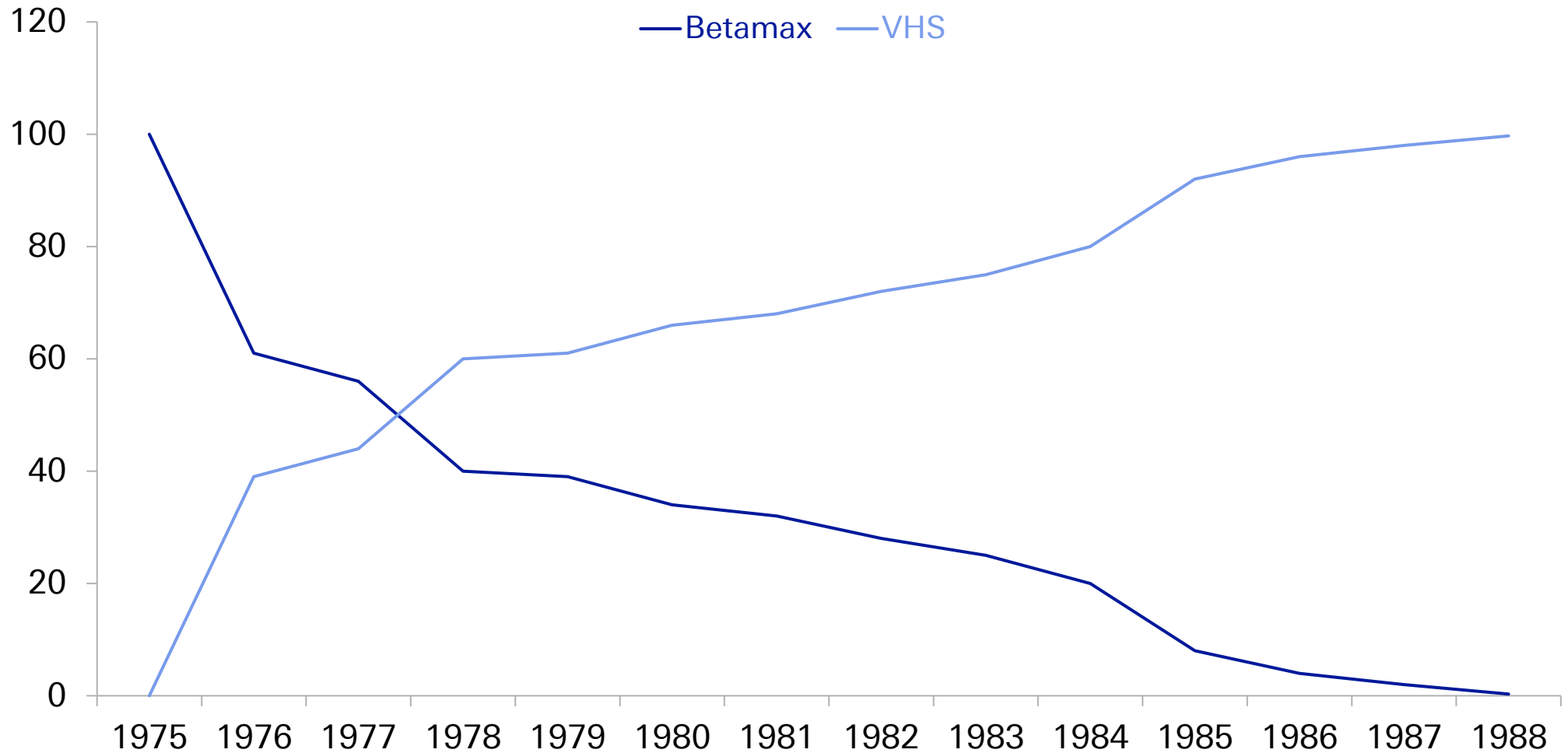


Source: Similarweb, Deutsche Bank

My favourite historical example of why winning early—and even having the better product— isn't always an advantage...



Betamax vs VHS market shares as proportion of annual production

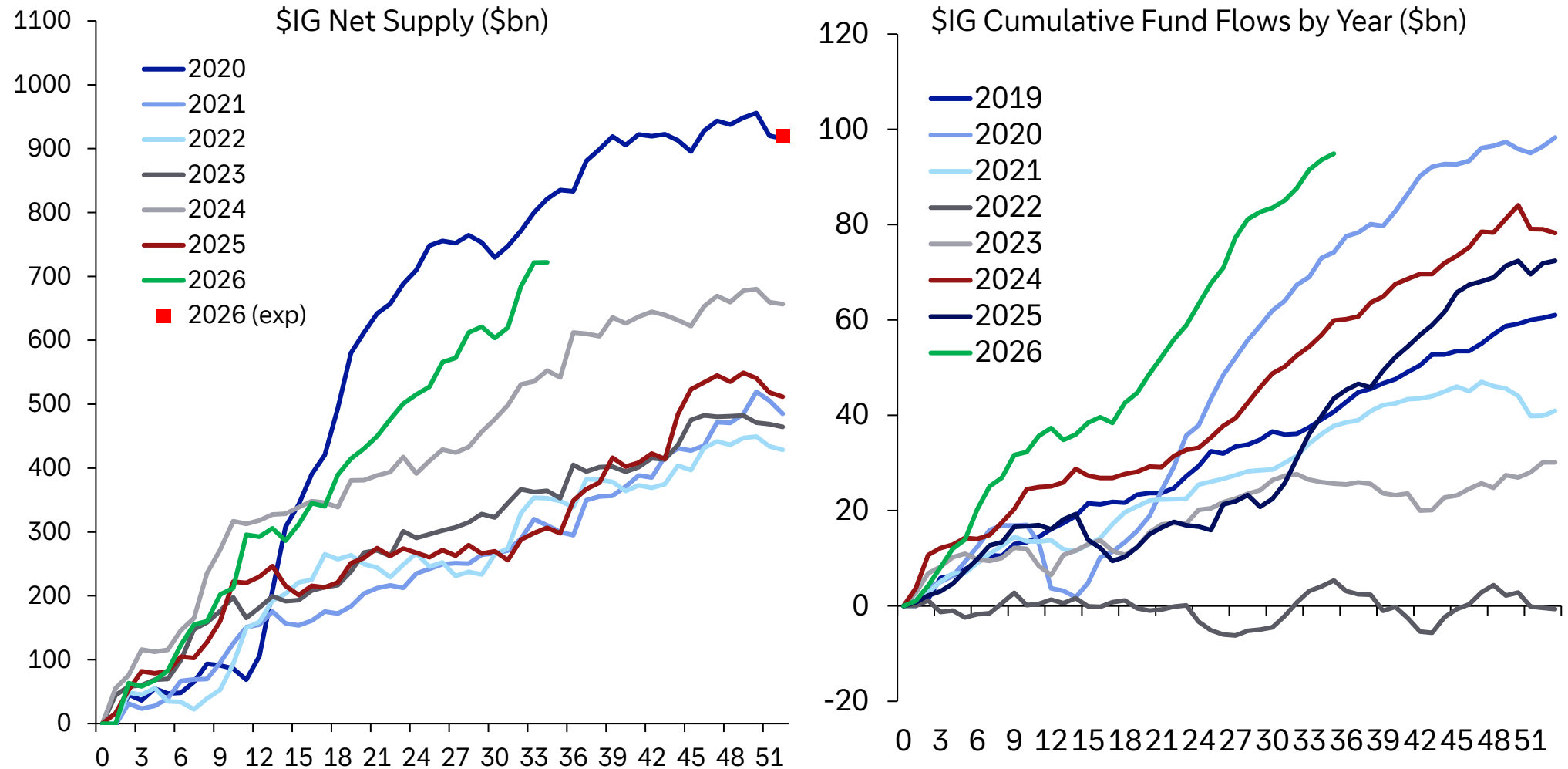


Source: Strategic Maneuvering and Mass-Market Dynamics (Cusumano, Mylonadis, Rosenbloom, 1992), Deutsche Bank Research

Technology—especially hyperscaler issuance—is helping put 2026 on course to surpass 2000 as the record year for US IG corporate net supply. However demand is also very high.



US IG Corporate Net Supply is likely to be the highest on record in 2026, but demand is also likely to be at a record...

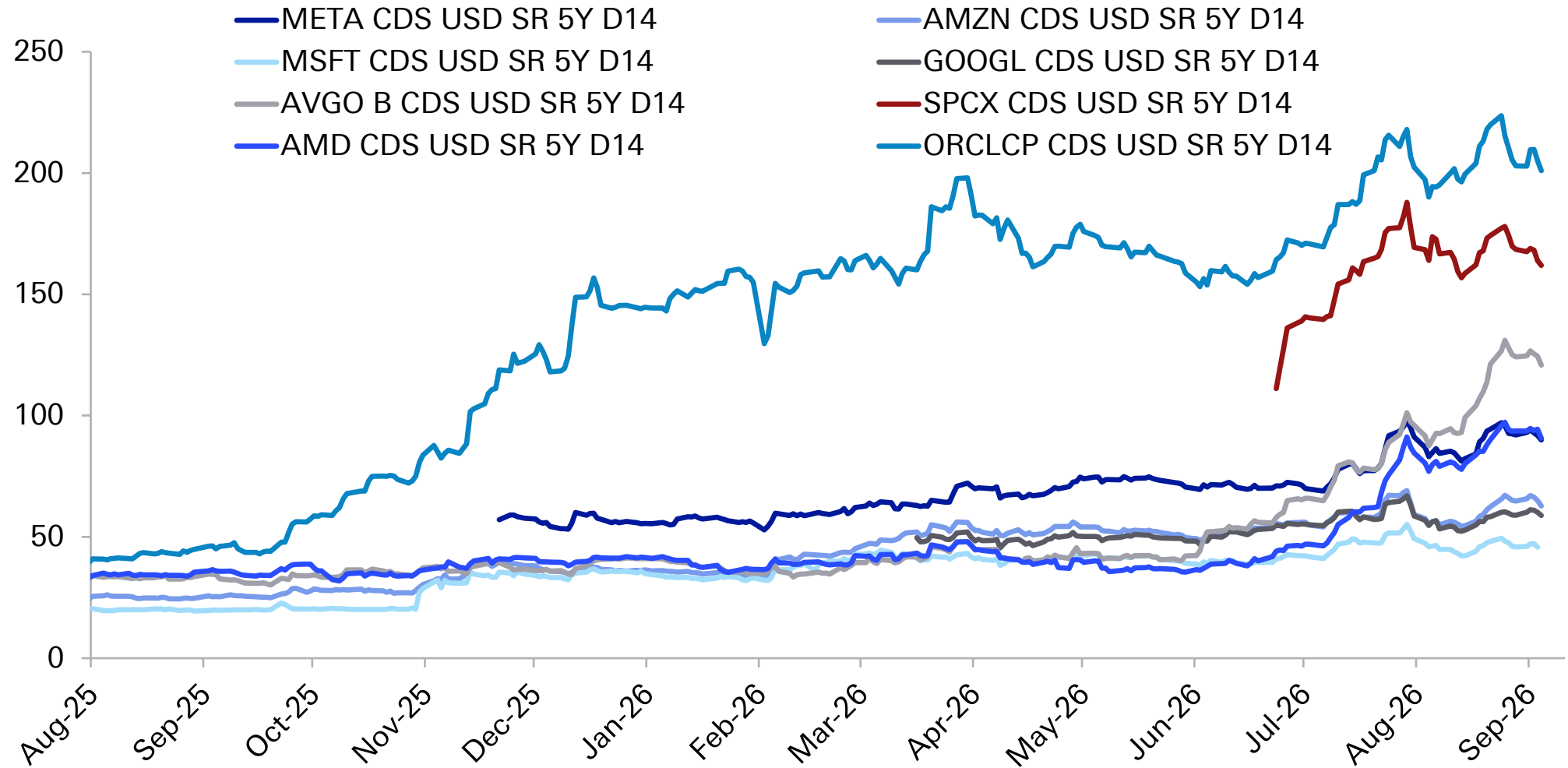


Source : EPFR, Bloomberg Finance LP, Deutsche Bank,

Hyperscaler (and related) CDS is climbing partly due to supply and increasing Capex but also as a hedge for all things AI-related so you have to be slightly careful in overinterpreting the moves...



Hyperscaler CDS

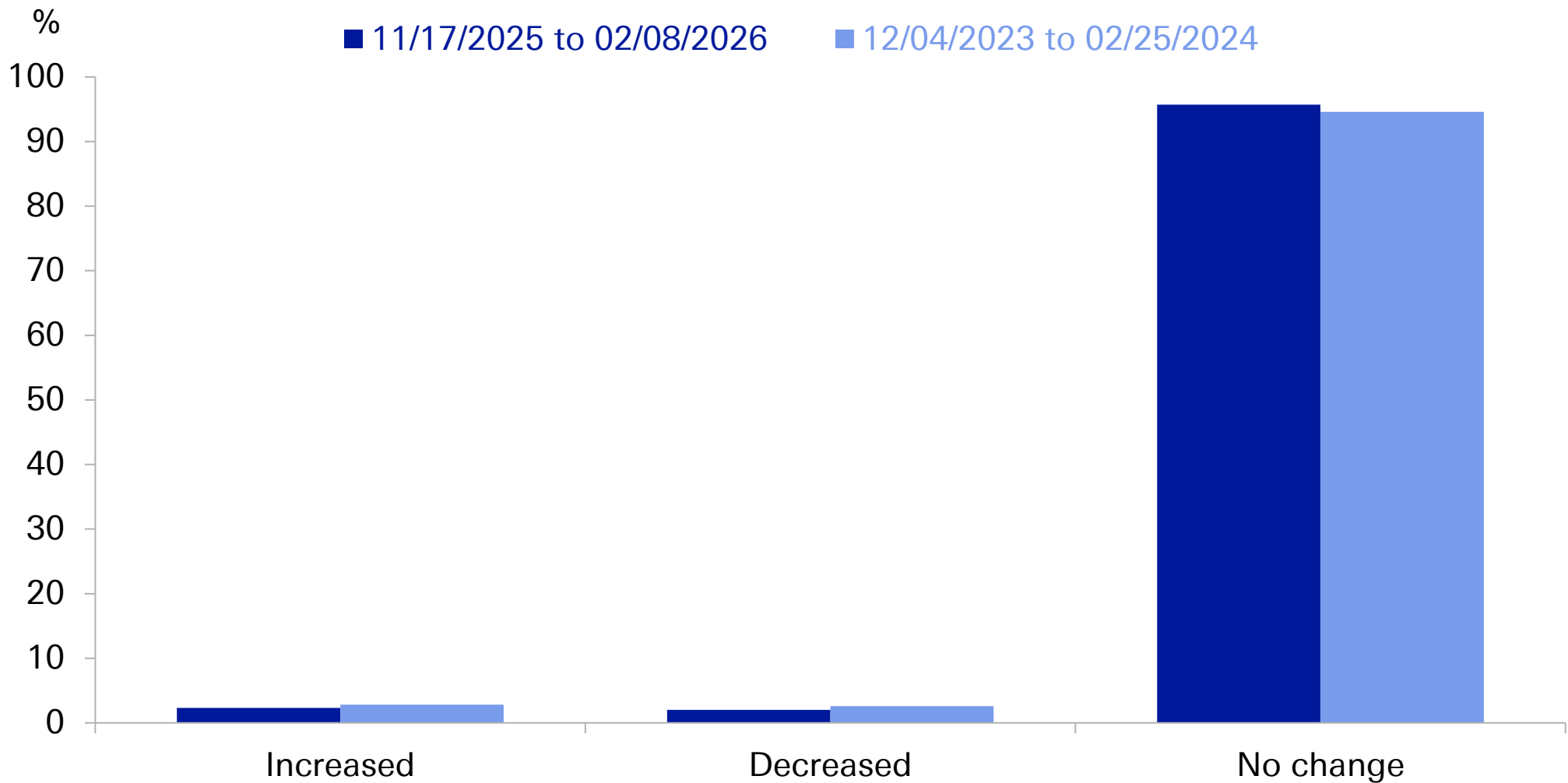


Source: Bloomberg Finance LP, Deutsche Bank, Last update Sep 7

While AI adoption grows, a significant percentage of enterprises have not altered their workforce size as a direct result. We're still a long way from a revolution...



In last 6 months, how did use of AI affect business employment?

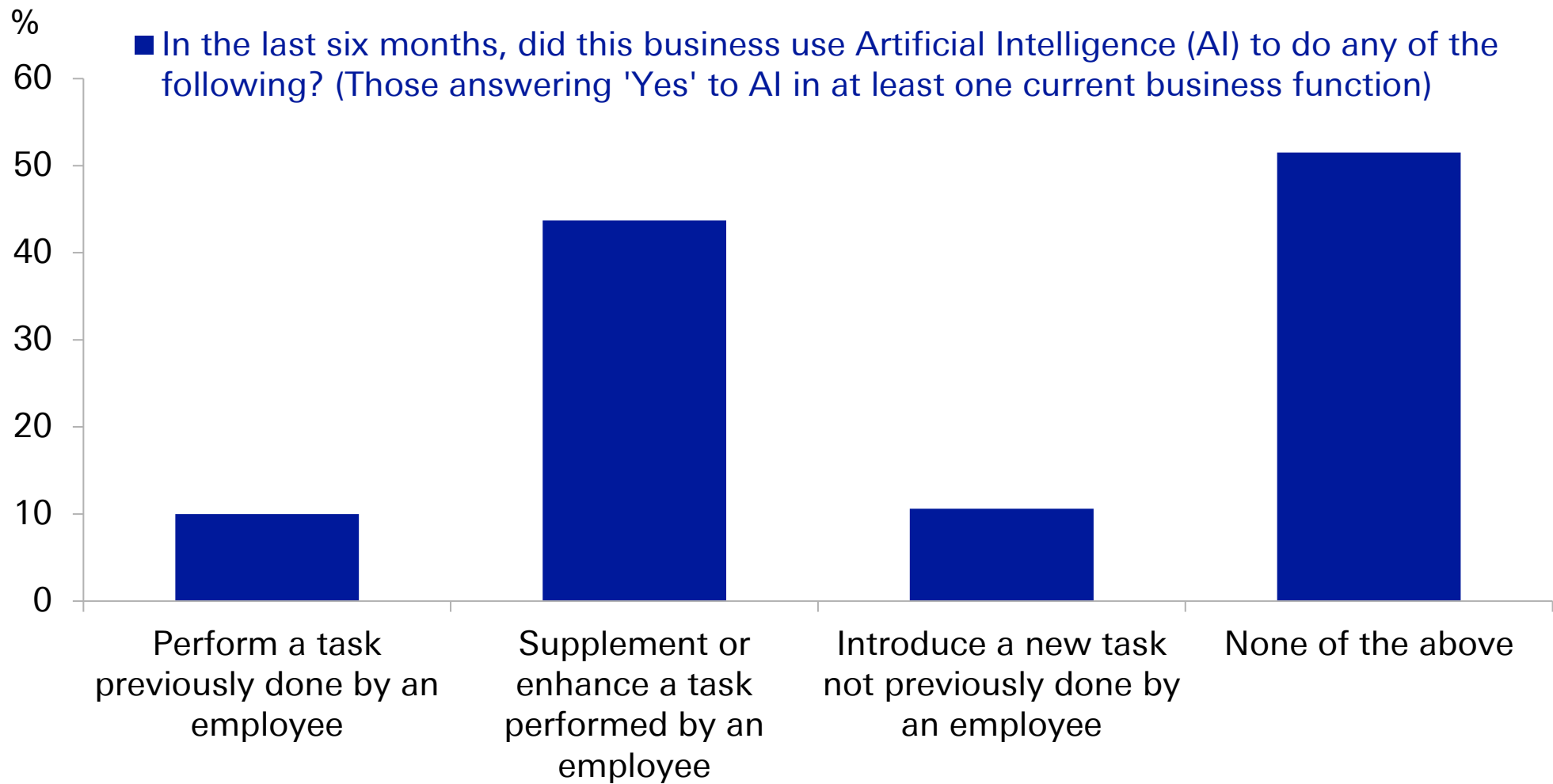


Source: BTOS, Census, Deutsche Bank

Companies adopting AI currently utilise it mainly to augment and enhance their employees' existing roles.



In the last 6 months, did this business use AI to do any of the following?

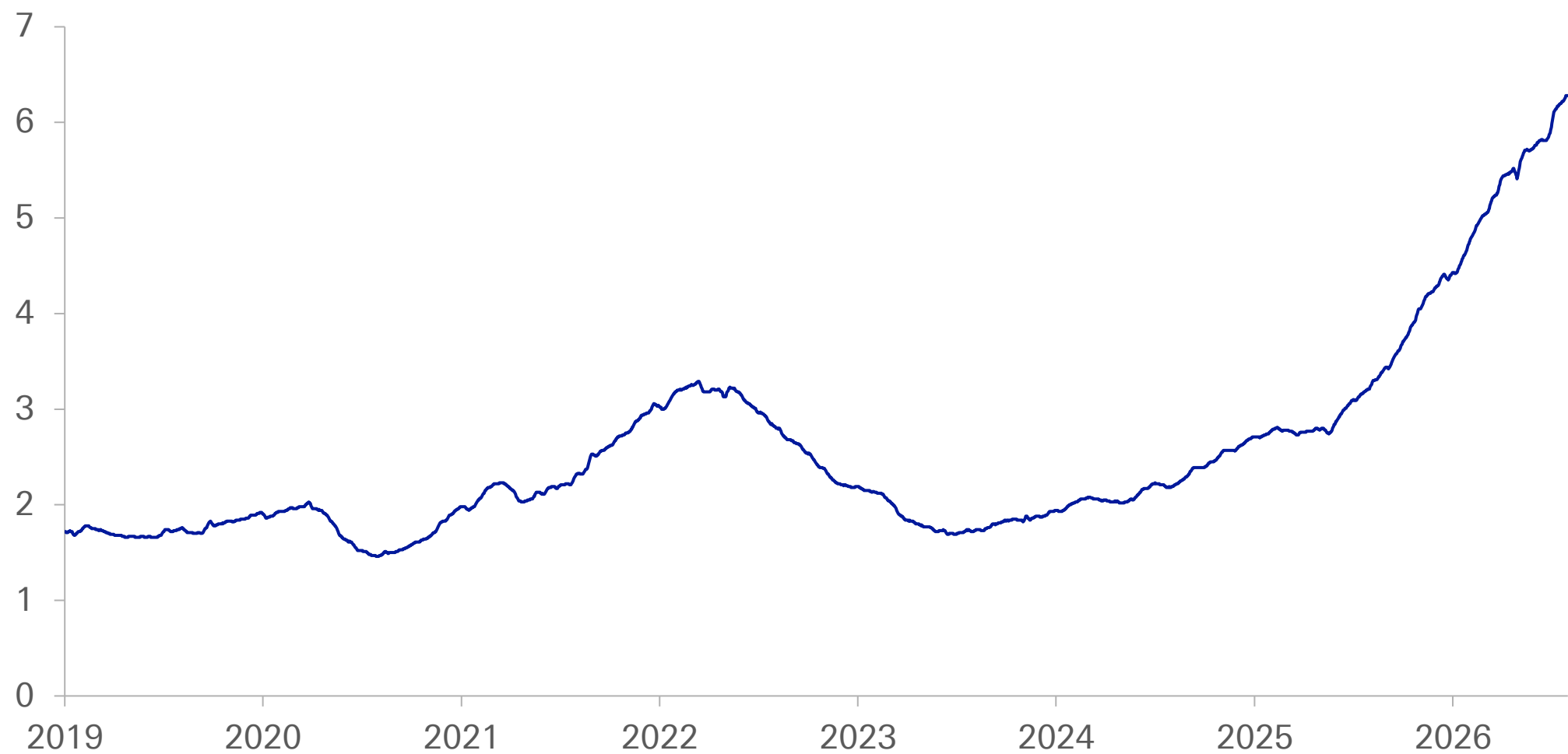


Source: BTOS, Census, Deutsche Bank

AI-related job postings have increased notably year-over-year in the US



Indeed AI Tracker: Share (%) of job postings on Indeed containing Artificial Intelligence (AI) terms in United States

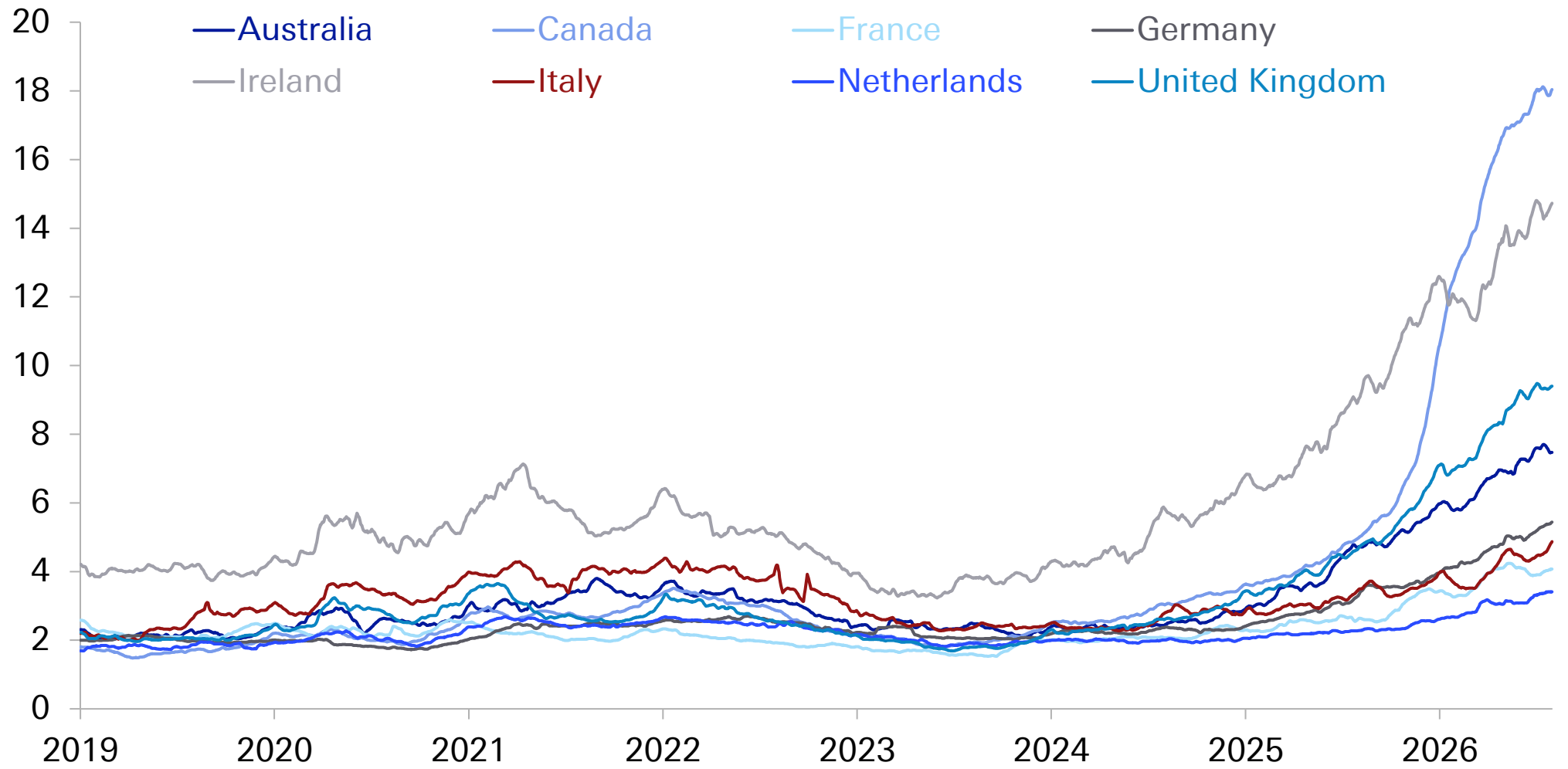


Source: Indeed Hiring Lab, Deutsche Bank

The share of job postings referencing AI is rising across countries. Is AI actually creating job openings rather than seeing them disappear?



Indeed AI Tracker: Share (%) of job postings on Indeed containing Artificial Intelligence (AI) terms

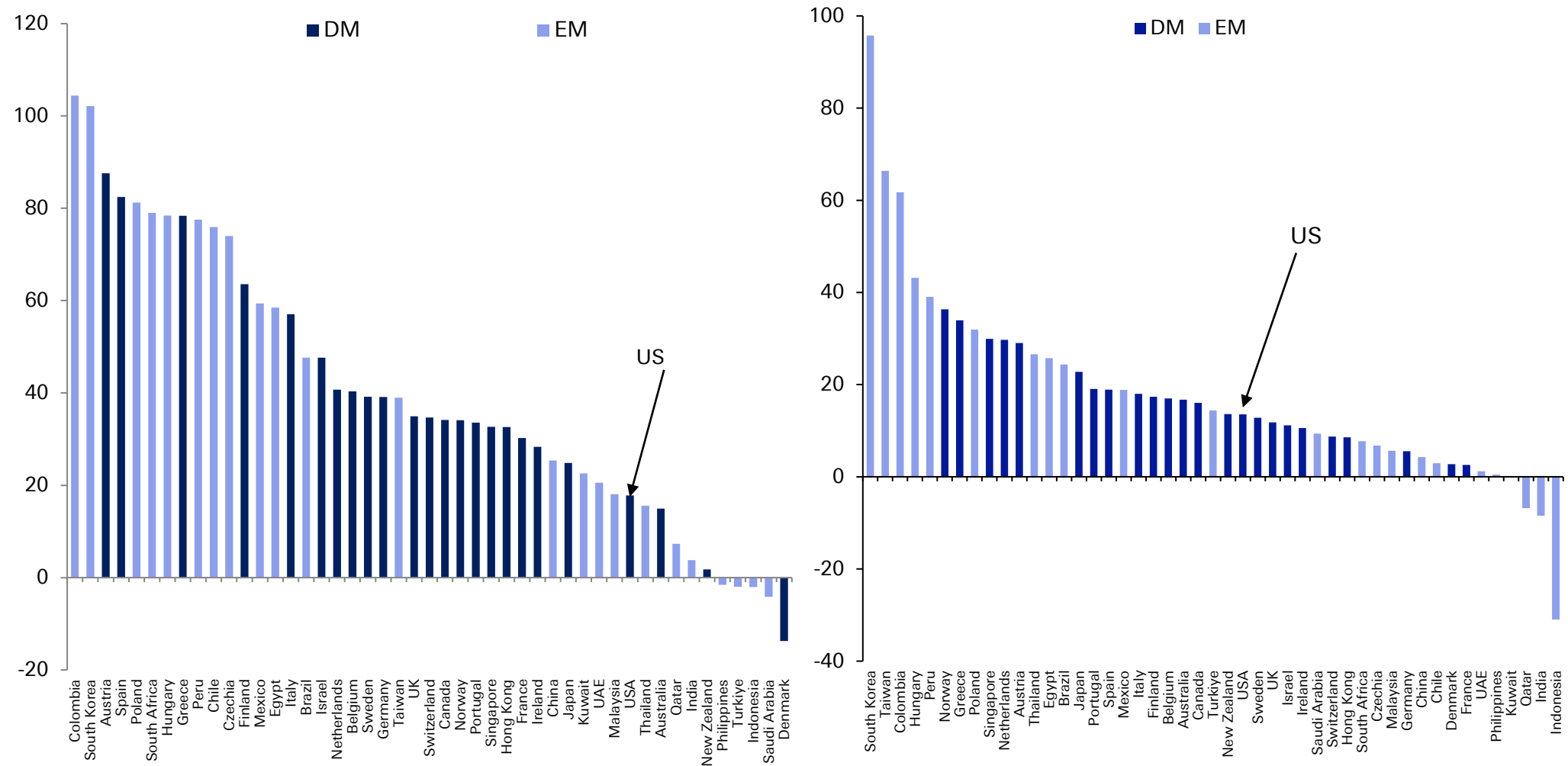


Source: Indeed Hiring Lab, Deutsche Bank

Global equity leadership has changed over the last two years as the AI cycle matures. The US has had two decent years but is in the middle of the global pack in 2026, after being around the bottom quartile in 2025...



Bloomberg large/mid-cap country stock market indices: 2025 total return in USD (%) (left) and 2026 YTD (right)

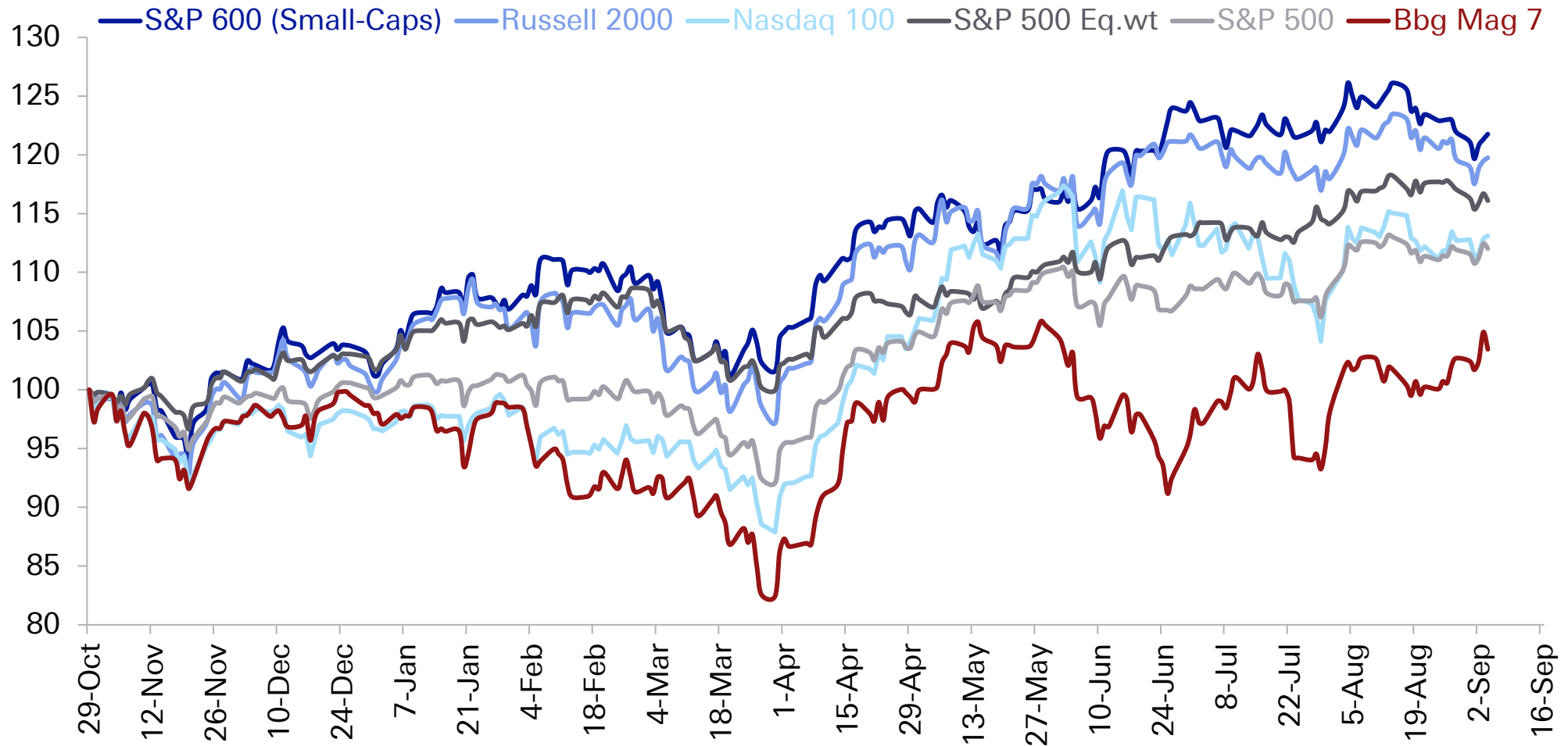


Source: Bloomberg Finance LP, Deutsche Bank, Note: 2026 YTD is updated through Sep 7

Since the end of last October, the Mag-7 have notably underperformed and are up less than 5%. However, other parts of the US market have performed strongly.



Rotation away from large cap tech. US Index performance (Oct 29, 2025=100)

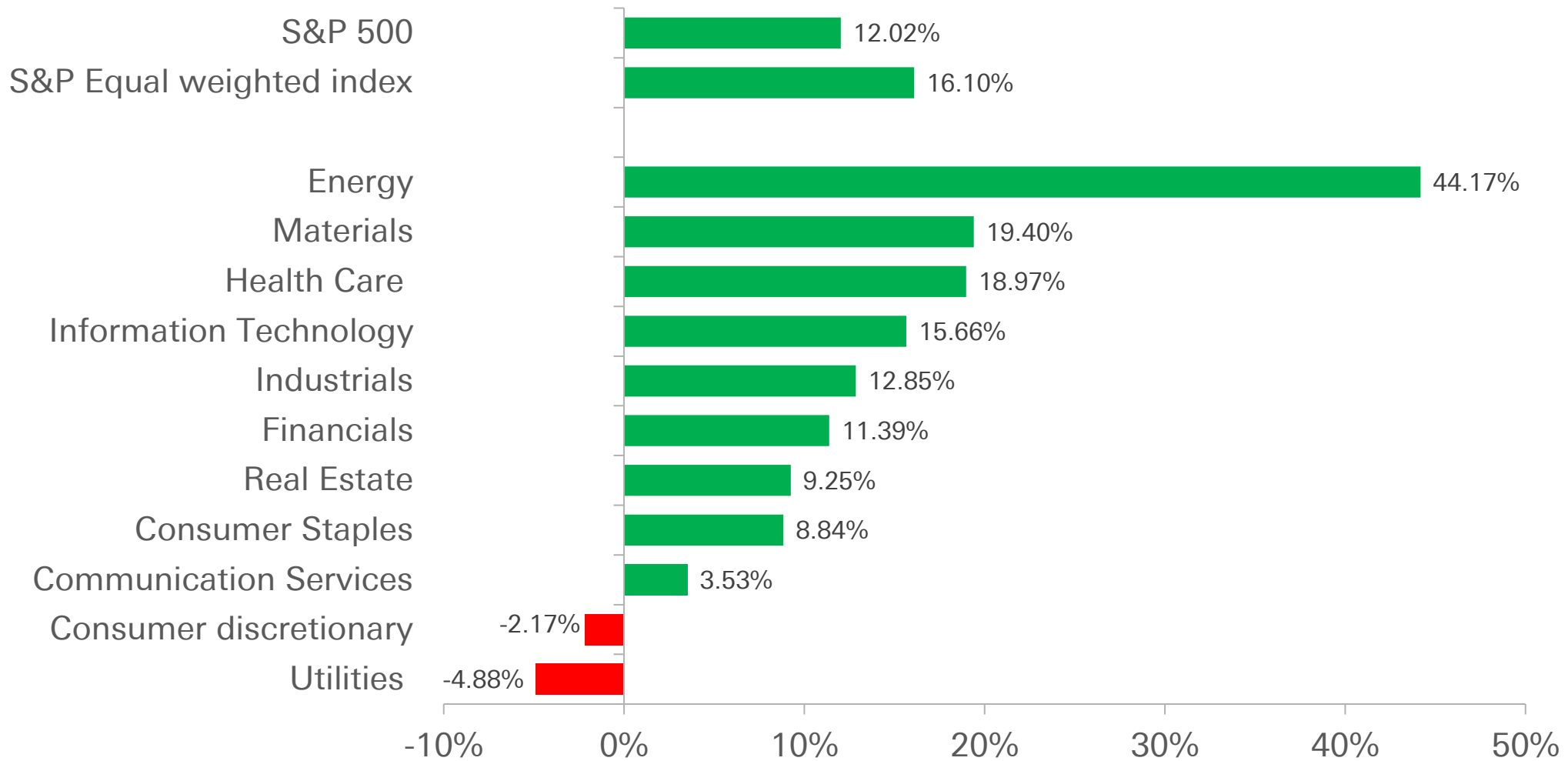


Source: Bloomberg Finance LP, Deutsche Bank., Last update Sep 7

Even with the Mag-7 underperforming, tech has performed in line with the market since tech started to see winners and losers emerge at the end of October last year. But tech has been outpaced by Energy, Materials and Health Care. The equal weight S&P 500 has outperformed.



S&P Sectors return since October 29, 2025

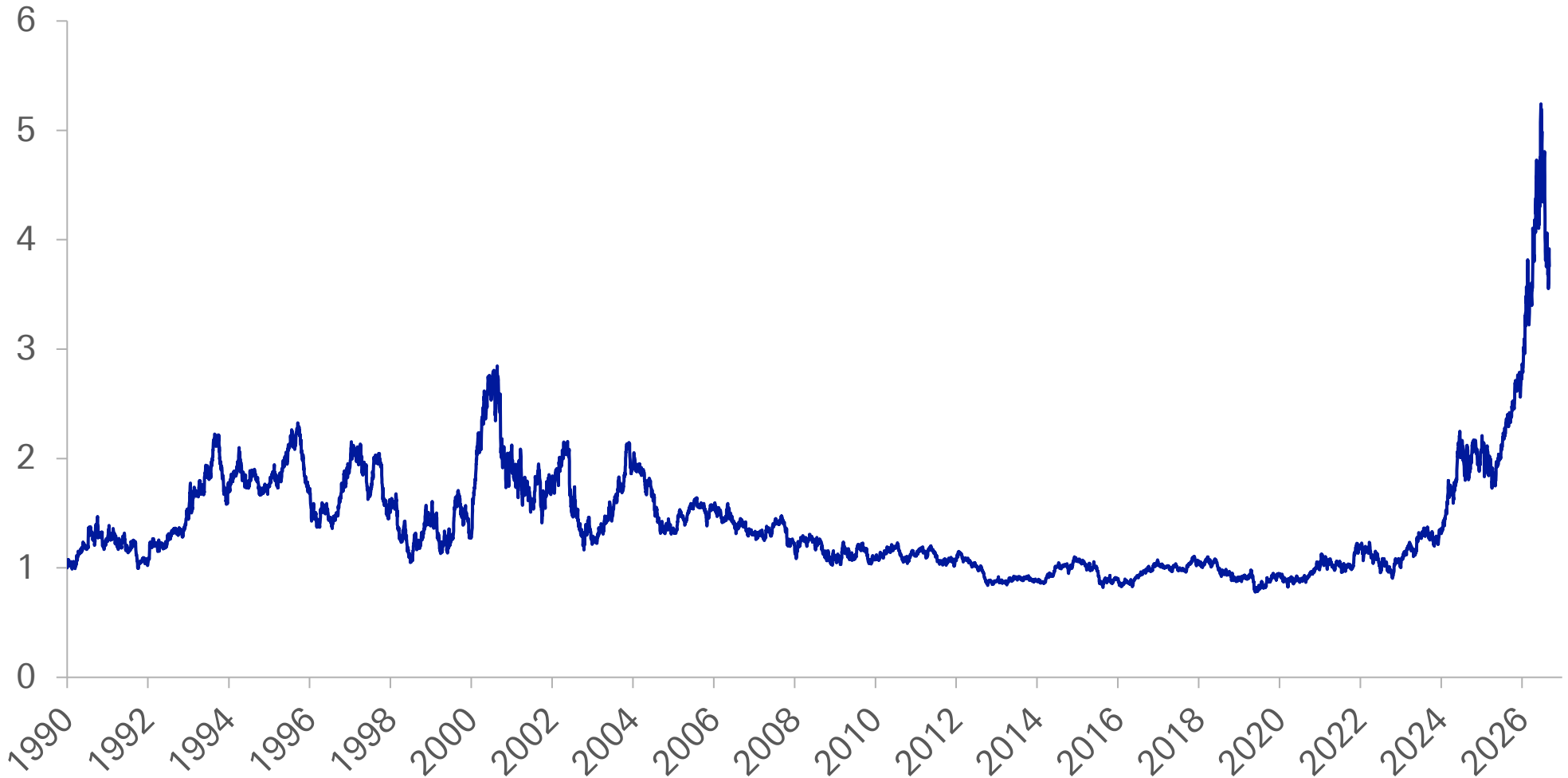


Source : Bloomberg Finance LP, Deutsche Bank., Last update : Sep 7

The tech winners and laggards theme can best be summed up in this chart... both are sizable sub-components within the tech index... Since the end of June software and services have made a comeback relative to semis but the major underperformance over the past 18 months is clear ...



S&P 500 semiconductors & equipment relative to software & services, total returns ratio, since 1990 (=1)



Source: Bloomberg Finance LP, Deutsche Bank, Last update: Sep 7.

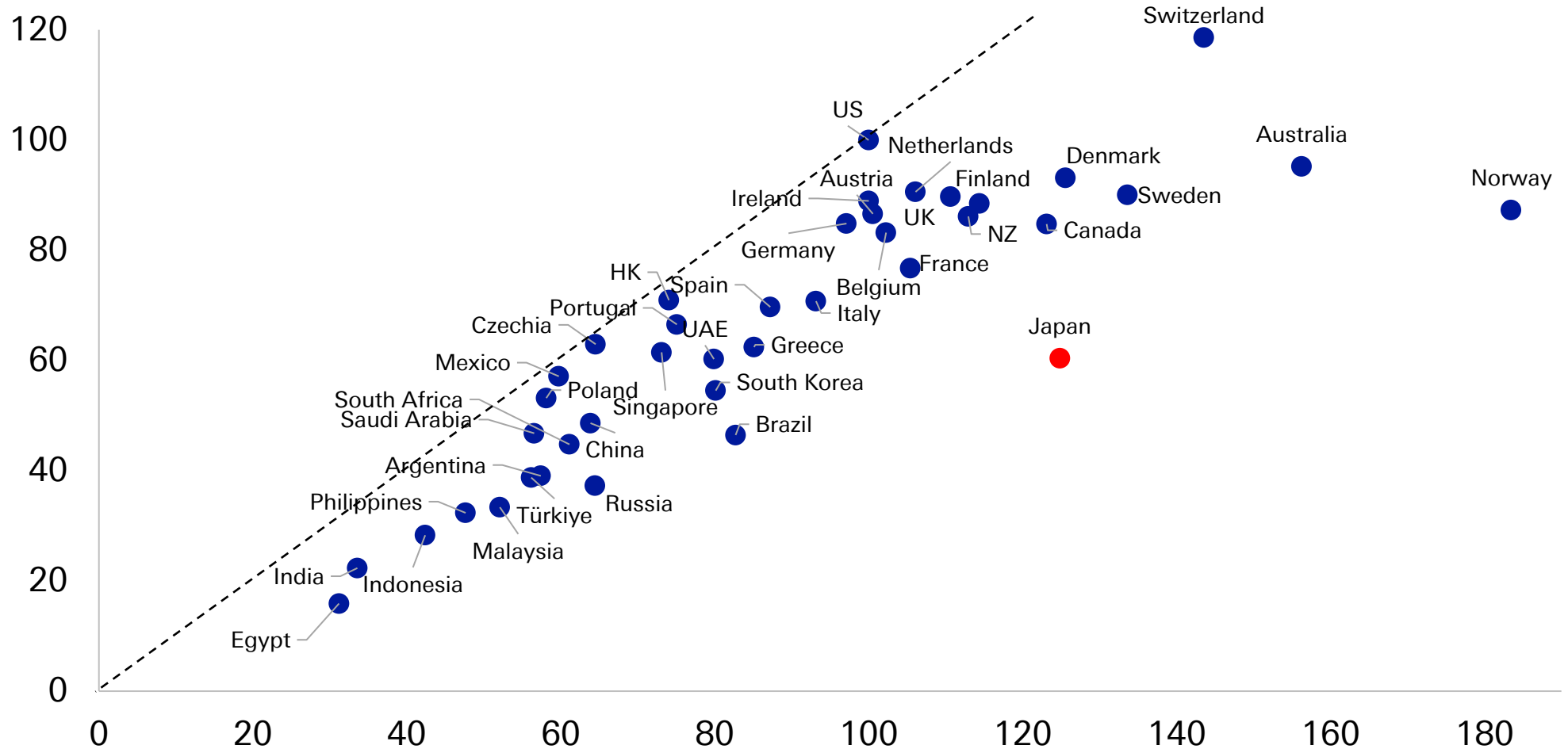


Japan is worth a watch...

Japan has cheapened up extraordinarily since the early 2010s....



Relative Price Levels as Implied by PPP (US = 100), 2012 (x-axis) and 2025 (y-axis)

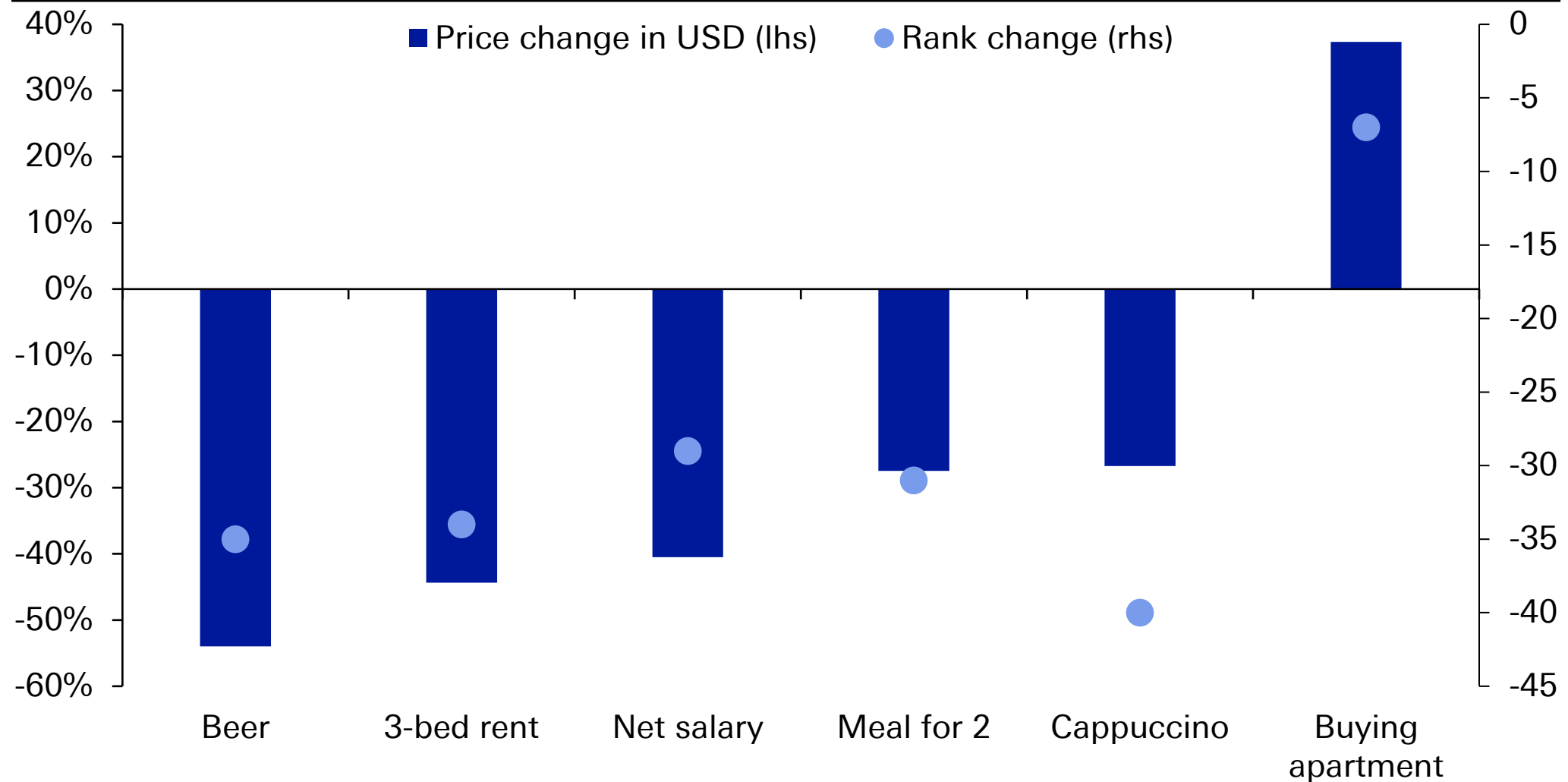


Source : IMF, WEO, Haver Analytics, Bloomberg Finance LP, Deutsche Bank calculations; Note: The PPP conversion rates as published by the IMF have been adjusted with actual foreign exchange rates (as on 31 December each year) to derive the implied price levels. Also note that some of the above numbers may not match those published last year exactly due to data revisions.

Is Tokyo the cheapest city in the world relative to its economic prowess? Maybe see you there for next year's holiday... its very cheap... See Mapping the World's Prices 2026 [here](#) for more evidence



Changes in prices of goods and services in Tokyo relative to 2012 in USDs



Source : www.numbeo.com, Deutsche Bank

The Yen is bouncing off 40-year lows...



USDJPY Spot Exchange Rate - Price of 1 USD in JPY

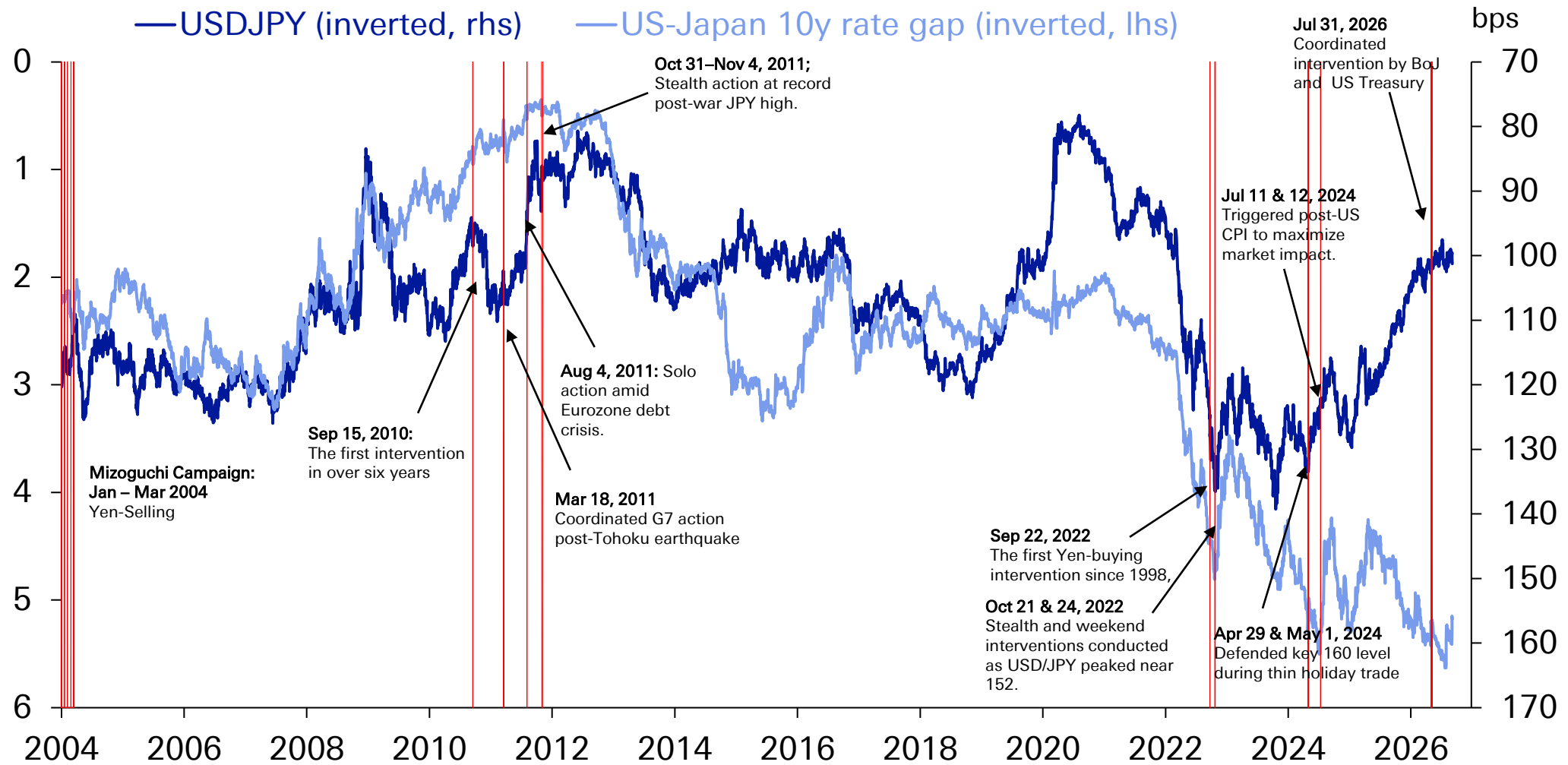


Source: Bloomberg Finance, Deutsche Bank; Last update: Sep 7

Since early 2025, the Yen has decoupled from its traditional anchor: interest rate differentials, compressing yield gaps have failed to spark JPY strength... The BoJ is the most likely source of a sustainable rally



USD/JPY and 10y spread with BoJ intervention episodes annotated (inverted axis)

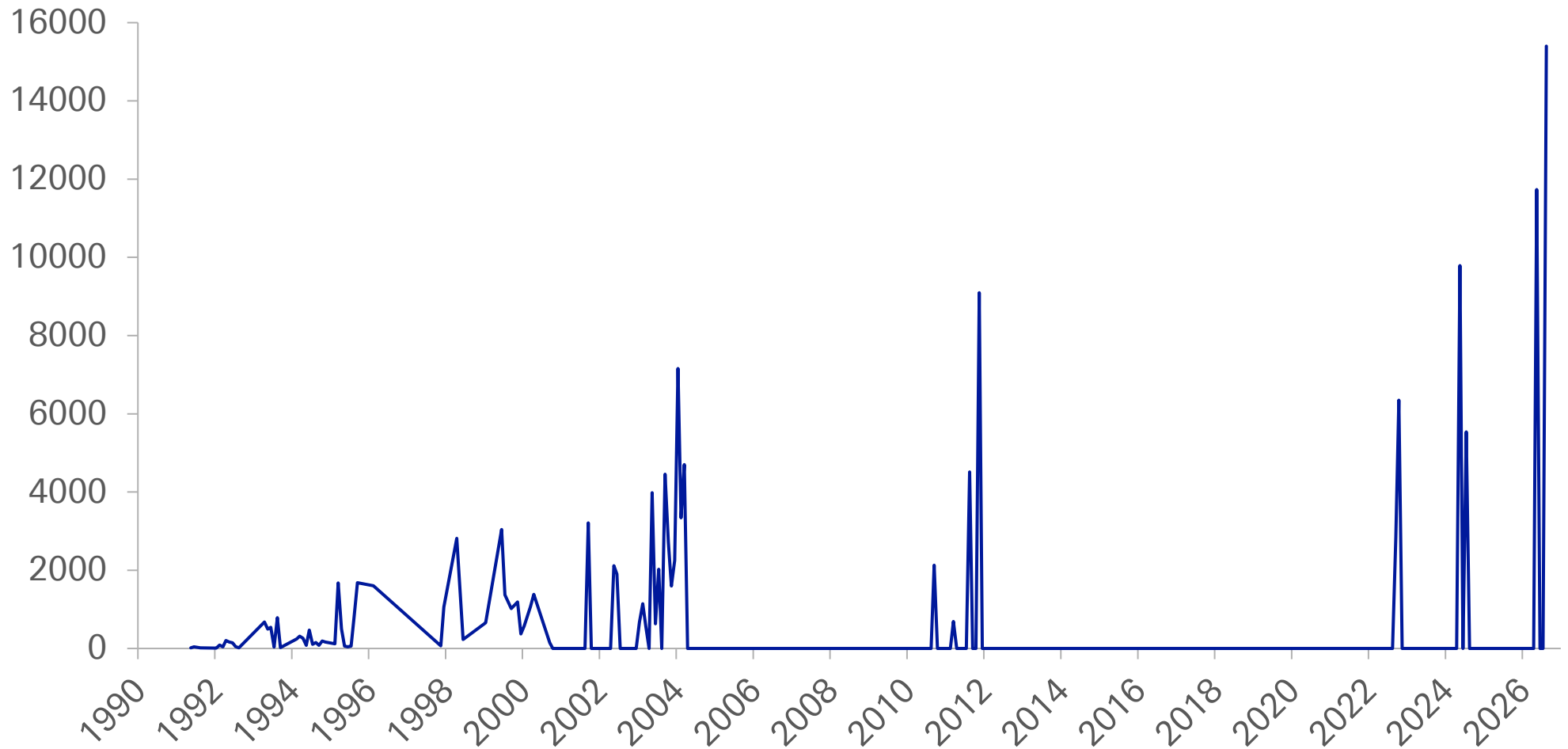


Source: Bloomberg Finance LP, Deutsche Bank, Last update Sep 7

Japan spent ¥15.4tn on market intervention from April to August—the most since 1991



BOJ Japan Foreign Exchange Intervention Operations - Total Amount (JPY bil.)



Source: Bloomberg Finance LP, Deutsche Bank, Last update: Sep 7

Appendix 1

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The Drone Frontier: Geopolitics in a New Era of Conflict

September 07, 2026

Executive Summary

- Policy incentives and spending hikes are driving the expansion of the global military drone market, with a projected CAGR of 8% through 2030. This looks set to unlock new investor opportunities across the wider drone ecosystem. Multi-billion-dollar investment pledges in the UK, US, Japan, India, and the EU are among those that reflect this rapid build-out.
- The surge in global spending will likely benefit established manufacturers in markets including the US, UK, and Türkiye. Ukrainian companies are well-positioned to capture post-war market share, as their footprint expands across Europe, the Gulf, and Asia-Pacific. Japan and the Philippines are among Asian markets interested in Ukrainian capabilities.
- The rapid development of drone technology – particularly hardware – risks leaving governments with obsolete stockpiles. For investors, this makes the sector harder to assess than the headline growth numbers suggest. Companies able to upgrade quickly across many drone models, or those specialising in software that can be rapidly upgraded, may be better positioned than those tied to a single platform.
- Organised crime, terrorist, and insurgent groups are increasingly adopting the technology, creating localised security risks that could drive further spending and prompt new restrictions on commercial drone use.
- National security concerns will likely drive efforts to cut critical dependencies, strengthen supply chains, support domestic manufacturers and secure an edge in AI. The US, EU, India, and Japan are among markets that will seek to reduce reliance on China for critical minerals and key components, in our view, but Beijing's dominance makes this a multi-year process requiring sustained investment across the value chain.

Authors

Miha Hribernik

Chief Geopolitical Strategist, Asia



Context

On 2 September, Ukraine announced it would soon finalise a bilateral drone production and development deal with Germany. This followed the signing of drone production and export agreements with Denmark, the Netherlands, and Estonia on the sidelines of the NATO Summit in July. The deals are among the latest developments marking Kyiv's emergence as one of the global leaders in drone technology. The agreements also come against the backdrop of a worldwide surge in investment in the sector. This trend looks set to continue for years, as governments absorb the lessons of the conflicts in Ukraine and the Middle East.

This report assesses the key geopolitical implications of the accelerating global race for drone dominance.

Disrupting Defence: The Rise of Cost-Effective Drone Power

The proliferation of inexpensive drones is reshaping military doctrine, defence spending, and procurement worldwide. While military drones date back decades, the technology has reached a number of major milestones and 'firsts' in the last several years, reflecting the accelerating pace of development, production and deployment (see timeline). The wars in Ukraine and Iran in particular have demonstrated the limitations of traditional costly military assets in the face of numerous, increasingly capable and AI-enabled, and often disposable drones.

Figure 1: Selected key milestones in military drone production and use, 2020-2026



Source : Deutsche Bank Research

Drones are unlikely to fully displace other military assets, including the need for 'boots on the ground' to seize and hold territory, but the next several years will likely see a global push toward adoption and integration with conventional military forces on an unprecedented scale.



Ukraine: From Necessity to Supremacy

Ukrainian companies could capture a sizeable share of the expanding market after the war. The need to counter Russia's manpower advantages has driven an innovation model linking frontline demand with fast domestic production. This has established Ukraine as a global leader in unmanned operations and high-volume production, particularly of attack drones. This includes utilising AI for navigation, targeting, and autonomous "last mile" attacks.

Kyiv reportedly [doubled](#) its drone procurement y/y during H1 and [plans](#) to eventually use robots for 100% of frontline logistical needs. According to President Zelenskyy, Ukraine now produces [10 million](#) drones per year (likely [more](#) than all of NATO combined, although Ukrainian figures probably include a large proportion of small tactical drones), and the country could scale production to 20 million units if its international partners provide sufficient funding. Nevertheless, once active combat ends and stockpiling needs are met, production could slow given how quickly drone capabilities change (see below).

Ukraine's manufacturing footprint is likely to extend further beyond its borders.

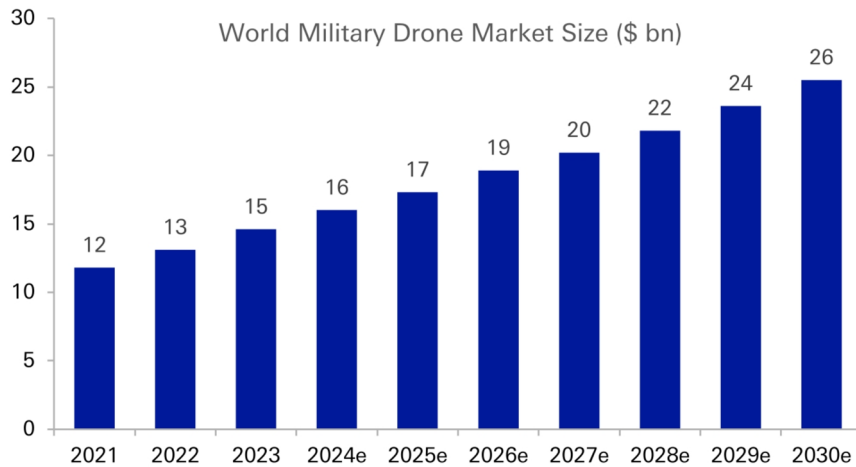
It is already expanding to facilities in countries including [the UK](#), [Denmark](#), [Norway](#), and [Germany](#), fostering technological diffusion and the emergence of a Europe-wide supply chain. Ukraine is also pursuing joint production with countries in the Middle East, Europe and the Caucasus, and has struck 10-year cooperation agreements with Saudi Arabia, the UAE and Qatar. More recently, [the WSJ](#) reported on 21 July that a new joint venture will produce Ukrainian sea-based attack drones in the US.

A Worldwide Surge in Spending

Policy incentives and spending hikes are set to fuel an expansion of drone and counter-drone arsenals in the coming years. The global military drone market is expected to grow at a CAGR of 8% through 2030, with growth in some regions – such as the Asia Pacific and Europe, the latter with an expected CAGR of 13% – projected to be even higher, according to [Global Market Insights](#). The forecasts signal robust, sustained expansion as global defence forces accelerate their transition to unmanned platforms. **This trajectory should create opportunities** for investors and companies in the wider ecosystem – from drone manufacturers to component, technology and software suppliers, as well as contractors providing support and maintenance.



Figure 2: The global drone market is expected to grow by a CAGR of 8% by 2030



Source : Global Market Insights, Deutsche Bank

Recent key developments include record procurement and spending requests in the US – already the world’s largest military drone market. Other notable initiatives include a historic shift toward a Hybrid Navy in the UK, the creation of a “Drone Wall” on the EU’s eastern flank, a mass purchase of commercial drones for military use in South Korea, and a record US\$2 billion investment push in India, which aims to bolster domestic drone manufacturing and reduce reliance on foreign suppliers (see table).

These investment pledges – set against the backdrop of rising global defence spending, which reached US\$2.6 trillion in 2025 – are **set to boost demand among existing manufacturers beyond Ukraine, including established players in the US and the UK**. This trend is reflected in **Türkiye’s** recent export boom. Türkiye is the world’s largest military drone supplier, accounting for some **65%** of armed-drone export sales. The country saw its defence exports triple over the past five years, and reportedly aims to double them in the next two.

The Asia-Pacific will likely continue to emerge as a major market. Geopolitical tensions in the Taiwan Strait and the South China Sea could see more investment in [surface and undersea drone capabilities](#). Ukrainian defence companies also [reportedly](#) aim to expand manufacturing and technological cooperation into the region, with Japan, the Philippines and other regional economies among potential markets. Regional companies are also seeking Ukrainian expertise, as reflected by a recent Japanese acquisition of two Ukrainian drone startups.



Figure 3: Selected recent developments, including planned or proposed spending increases

United States	The US Department of War has requested a 24,000% increase in FY2027 funding for its Defense Autonomous Weapons Group, which oversees mass-produced drones with AI capabilities. The Pentagon has placed an order for 30,000 one-way attack drones (the largest such order to date), and expects to receive more than 300,000 by early 2028.
United Kingdom	Aims to invest GBP5 billion (US\$6.7 billion) in advanced uncrewed systems, including a Hybrid Navy which integrates autonomous vessels and undersea drones. Plans to boost Army lethality ten-fold by investing in capabilities including AI and autonomous systems.
European Union	The creation of a new “ Drone Wall ” or “ Drone Shield ” will entail a multi-billion euro spending commitment.
France	Plans to expand its stock of explosive drones by up to 400% by 2030.
Australia	Intends to more than double funding for anti-drone capabilities over the next decade.
Germany	Announced plans to invest around EUR16 billion (US\$18.2 billion) in unmanned reconnaissance, attack and defence capabilities by the end of the decade.
China	Continues to invest in new drone capabilities amid rising defence spending. Recent developments include the reported construction of the world’s largest unmanned submersibles and the demonstration of a drone swarm operations system in March 2026.
India	Preparing to place a record US\$2 billion order for domestically-built military drones, reportedly as part of a plan to reduce dependence on foreign suppliers.
Japan	Tokyo allocated JPY 277 billion (US\$1.7 billion) for unmanned system in this year’s defence budget, more than double the figure from 2025.
South Korea	Intends to train 500,000 military drone operators and procure 11,000 commercial drones for military training in 2026.

Source : Deutsche Bank Research, external sources (see links in table).

Tech Turnover: The Obsolescence Dilemma

The scramble for drones risks leaving governments with stockpiles of obsolete equipment. The technological ecosystem around drones, and counter-drone systems, is not yet mature. Battlefield pressure forces rapid innovation, sometimes within weeks, risking large stockpiles becoming quickly **obsolete**. The rapid growth of the industry, and the proliferation of new companies entering the sector, also raises questions about the risks of an eventual industry consolidation.

For investors, this makes the sector harder to assess than the headline growth numbers suggest. The better-positioned companies may be those that can upgrade quickly across many drone models – through software, flexible or modular manufacturing and battlefield feedback – rather than those tied to a single drone platform. With hardware vulnerable to obsolescence, software could prove to be a material differentiating element, and several manufacturers are prioritising software platforms as key to upgrading their systems.

Rogue Drones: The Proliferation to Non-State Groups

Drone technology is a ‘great equaliser’ for non-state groups that could fuel a rise in global security risks. Organised crime, terrorist, and insurgent groups have expanded their drone capabilities in recent years. Groups like Hezbollah, Hamas, guerrillas in **Colombia**, and **Mexican** drug cartels have already adapted cheap commercial drones for attacks and demonstrated an ability to **absorb** the lessons of the war in Ukraine.

The proliferation of drone technology means that defences must cover a wide range of potential targets – from critical infrastructure to public events and government buildings – while being updated to close gaps that small, adaptive groups can exploit. Although there is little evidence to suggest that these groups have used AI-



enabled drones at scale, this could become an additional medium-term risk. **The need to counter evolving threats from non-state actors could be another driver of increased spending and may prompt more [restrictions](#) on commercial drone use.**

Strategic Autonomy: Diversifying Drone Supply Chains

National security concerns will likely drive efforts to reduce dependencies, strengthen supply chains, support domestic manufacturers, and secure an edge in AI. China continues to dominate drone supply chains, including for processed critical minerals used in key drone components such as motors and actuators, batteries and infrared / optical components (such as lenses). Major markets, including the US, remain highly dependent on China-linked supply. The table below highlights three key areas. We anticipate continued efforts to diversify critical mineral and component supply chains amid accelerating geopolitical competition. This process will likely dovetail with policies intended to reduce dependence on China for processed critical minerals (see [Geopolitical Outlook 2026: At the Mid-Point](#)).

Figure 4: From Mine to Mission: China's Supply Chain Footprint

Key raw material	China share	Processed material	Drone component	Drone function affected
Rare-earth compounds and metals Rare-earth inputs used in permanent-magnet supply chains	High China supplied 71% of US rare-earth compounds and metals imports in 2021–24; USGS identifies magnets as the leading global end use of rare earths.	Permanent magnets	Motors and actuators	Propulsion, control and payload lift
Natural graphite	High US net import reliance for natural graphite is 100%; China supplied 46% of US natural-graphite imports in 2021–24 and 55% of US graphite battery-anode material imports in the first eight months of 2025.	Battery anode material	Lithium-ion battery cells / packs	Battery capacity and electric-power availability
Gallium and germanium	Very high for gallium; high for germanium China produces close to 90% of global gallium and around 60% of global germanium.	Gallium- and germanium-containing semiconductor and optical materials	Radio-frequency communications, power electronics and infrared/optical components	Signal transmission, infrared/optical sensing and power-electronics efficiency

Source : Deutsche Bank Research, USGS. Note: This table lists three key touchpoints relevant for drones but is not meant to be exhaustive.

However, diversifying sources of commodities and specialised components will likely take years and require significant capital expenditure across the entire value chain. The drone supply chain remains highly globalised, and a single unit can contain hundreds of specialised components sourced from dozens of manufacturers across multiple countries. While derisking could ultimately benefit suppliers/producers in the US, Japan, [South Korea](#) and partner states, it is likely to progress unevenly. Given the scale of the task, eliminating dependencies in military drones will likely continue to take priority over dual use (civilian/military) drones.

Supply chain diversification is a multi-pronged process. Recent developments include efforts by US drone makers to [build](#) domestic supply chains amid new restrictions on China, including the drone [tariffs](#) announced on 13 August. Meanwhile, Taiwanese drone exports to the EU [reportedly](#) increased 40-fold in 2025, while [Japanese and Taiwanese](#) companies are purportedly seeking closer cooperation. Efforts to reduce dependencies include plans to advance technological solutions such as rare-earth-free electric motors and more sustainable navigation and power systems for drones. Some countries, such as [the UK](#), have moved to restrict the use of China-made components on national security grounds.



Expanding export controls could impact diversification timelines. During 2026, China has introduced new restrictions curbing access to critical minerals, rare earths, advanced electronics and other dual-use components. On 24 April, China banned dual-use item exports to seven European companies, [purportedly](#) to counter EU-Ukraine-Taiwan security cooperation. On 22 June, Beijing added ten US defence-linked entities – including drone companies – to its export control list restricting access to dual-use items. This was followed in subsequent weeks by additional curbs on Japanese drone and advanced-technology firms. On 5 August, China tightened drone-related exports to the US in response to Washington's recent tech and economic curbs.



Appendix 1

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Capitalism and the dollar: the AI race, tokenization, and the dollar's future

September 03, 2026

A massive race for capital

The US is sprinting in its reach for capital. Private markets, debt, and public equities have been setting records, and the US is enjoying unprecedented amounts of foreign equity inflows. In this paper, we discuss how the US is leveraging the power of capitalism – supercharged by blockchain – to try and win the AI race. But the US reach for capital is not only about AI spending; it also reflects growing fiscal deficits and the need for foreign private funding. The US is increasing its reliance on private capital to offset public sector vulnerabilities. Meanwhile, China is running a very different race – given large trade surpluses, domestic excess savings, and a "low-cost" AI approach, China is raising less capital, with much less from abroad.

America embraces asset tokenization

The US is using blockchain technology to support US dollar and capital market dominance. We wrote last year about the crucial role of stablecoins. We are now at the next stage: asset tokenization – converting ownership rights to financial assets into digital tokens recorded on blockchain. Asset tokenization is not futuristic – it is happening now at the heart of US markets. Securities depositories and exchanges are building the infrastructure to tokenize US assets, with regulatory oversight.

TINA meets Tokenization: the future of the dollar

TINA already plays a key role attracting capital to the US. Tokenization could add to this by making US assets more attractive and accessible. The collateral velocity of tokenized assets could be a superpower. And if US assets are trading 24/7 with instant settlement, this could impact global behaviour, particularly for retail. But lowering the hurdles for inbound capital also exposes the US to capital going out. The dollar is becoming riskier, more equity sensitive, and leveraged to the AI race. If AI proves uneconomical, or the US starts to fall behind, the dollar is very exposed.

Open markets meet Open technology: the future of capitalism

The US is taking an "open markets-closed tech" approach to the AI race, while China is taking an "open tech-closed markets" one. The US is widening the doors for capital but keeping technology closed to preserve corporate pricing power. Meanwhile, China is pushing out low-cost models with open-weights. This is not just a contest between AI models, but economic models. If China's approach undercuts US profits, this could threaten the US model of shareholder capitalism itself. China's challenge is not only about technology, but the central maxim of US power: that free and open capital markets always deliver the winning innovation.

Authors

Mallika Sachdeva
Strategist



Part 1: A massive reach for capital in the US

The US is sprinting at record speed in its reach for capital. Huge sums of money are being raised to fund the buildout of AI infrastructure and development of frontier AI models, with US companies expected to spend around \$800bn in AI capex this year.

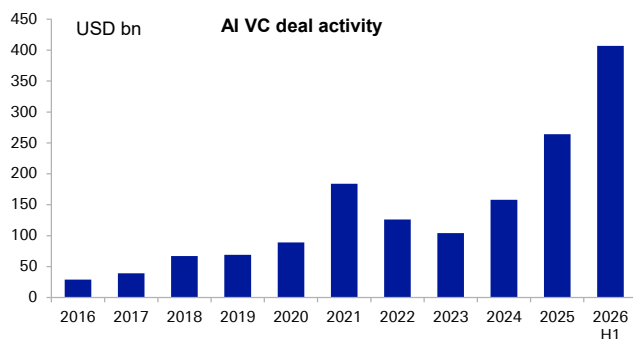
We argue in this paper that the US is leveraging the full power of capitalism to fund capital investments seen as crucial to winning the AI race

The US is raising enormous amounts of private capital

We simplify the key trends in the US' capital-raising landscape into a few observations. First, AI labs are turning to private markets to raise large sums of equity capital. Second, hyperscalers are turning to investment grade debt markets to issue meaningful amounts of debt for the first time. And third, public equity markets are back in focus for capital funding with some of the world's biggest companies returning to issue stock, and major IPOs in the pipeline.

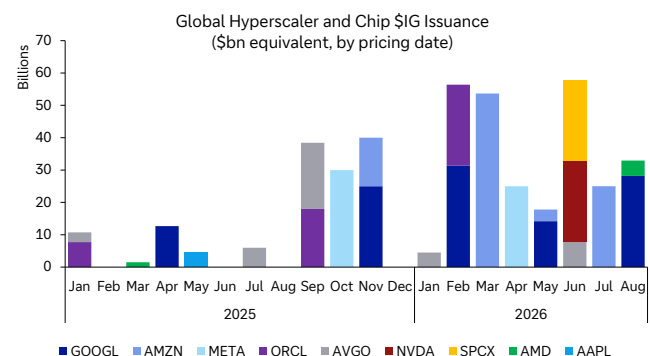
- The AI venture capital industry has raised over \$400bn this year alone, with over 90% of the biggest deal volumes in the US ([Pitchbook](#)). The two largest AI labs in the US have raised a combined \$217bn this year – valuing each company near \$1tn. We note that xAI had raised \$20bn earlier in the year, before being acquired by SpaceX in February. On an annualized basis, AI VC capital is running at three times last year's pace.
- In debt capital markets, the shift has also been staggering. While free cash flows remain an important source of funding for hyperscalers, the capital intensity of the infrastructure buildout is outrunning cash flows, forcing companies to turn to debt. Hyperscalers like Google, Meta, Oracle and Amazon raised an average of just USD25bn per year between them over 2020-24. In 2026 YTD, they have raised around ten times that amount through IG credit markets. Microsoft has yet to turn to debt markets.
- In public equity markets, Google also had its first primary equity raise since its 2004 IPO in June 2026, raising \$85bn. Returning to issue is a big reversal from years of buying back stock. This is also setting up to be a historic year for IPOs. SpaceX achieved the largest total market valuation of a newly listed company in history near \$2tn in its June 2026 IPO. The largest US AI labs are expected to come to the IPO market soon which could change the AI boom as we have written about [here](#).

Figure 1: US AI companies are attracting record amounts of private capital



Source : Deutsche Bank Research, Pitchbook

Figure 2: US hyperscalers have turned to public credit markets to issue huge amounts of debt



Source : Deutsche Bank Research, Bloomberg Finance LP



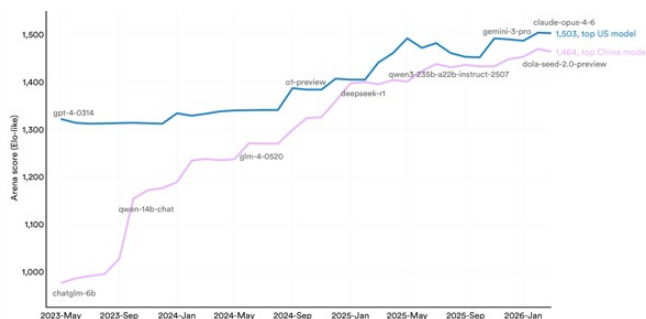
China is running a different race

On many metrics, Chinese models are closing the gap to the US in terms of performance and capabilities (Figure 3). While the US remains ahead in frontier development, Chinese models account for more than half of US company token usage on Open Router (Yahoo Finance).

But from a capital perspective, China appears to be running a different race - raising much less capital, less from private sources, and less from abroad. While US companies are expected to spend around \$800bn in AI capex this year, that number is closer to \$100bn in China. While the biggest US AI labs have raised nearly \$250bn in private funding rounds this year, Chinese labs have raised far less. The largest capital raise came from DeepSeek which raised \$7.4bn earlier in the year, and may be returning for a further \$8bn. China's National AI fund has participated in a number of capital raisings. Stanford University's [recent AI report](#) made a similar observation: noting that in 2025, US private AI investment was 23 times the \$12.4bn invested in China, although Chinese numbers are likely understated given government guided funds.

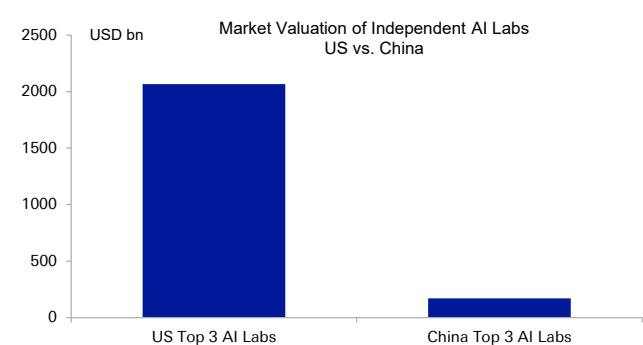
The stark difference in the approach to capital raising is also reflected in markets. With US AI labs raising far more capital for ownership stakes, the market valuations of US AI companies compared with their Chinese peers are many orders of magnitude apart (Figure 4). Furthermore, while Chinese hyperscalers like Tencent, Baidu and Alibaba have been tapping debt markets, they are doing so in smaller size, with a no major uptick compared with earlier years. They are, however, increasingly funding in dim sum bonds, suggestive of a desire to de-dollarize funding. See more on the Rise of the RMB Financing Ecosystem [here](#).

Figure 3: The gap in performance between US and Chinese models has narrowed a lot



Source : Deutsche Bank, Stanford University HAI

Figure 4: US AI labs have much larger implied market valuations than their Chinese peers



Source : Deutsche Bank, Bloomberg Finance LP, Various news sources; Note: Valuations are based on market cap for publicly listed companies and announced or intended funding rounds for private companies.

The US will likely want all the foreign capital it can get

The dramatic reach for capital in the US is not only reflective of an American economic model that prioritizes open capital markets, it also reflects the US' growing need for foreign financing given the US' large twin deficit position. The US is running a 6%+ fiscal deficit combined with a near 4% of GDP current account deficit. In the absence of fiscal savings, incremental private sector capex will require funding from abroad. At a time when private sector financing requirements for AI capex are rising, it is not a surprise the US is widening its reach for capital from abroad.

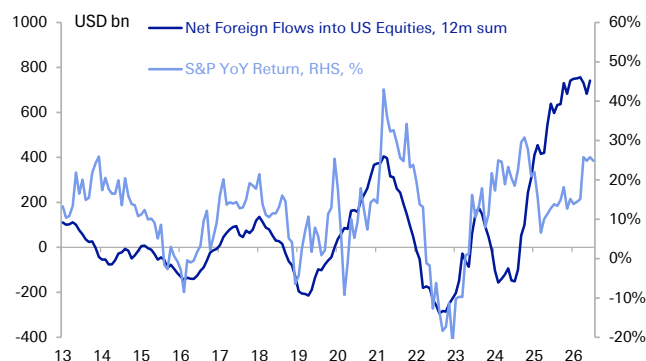


The US is in fact enjoying an unprecedented amount of equity capital inflow. In Q2 2026 alone, the US received over USD400bn of foreign equity capital, well above any previous quarter. The amount of capital coming into US equities is also well above what US equity performance alone would imply (Figure 5), and now well above debt flows (Figure 6) – the historical mainstay of US capital account financing.

We have written in the [past](#) about the big rotation underway in US financing. Geopolitical ruptures in long-standing alliances in defence and trade are hurting long-term appetite for USD government debt, particularly by the foreign official sector. But at the same time, technology is pulling enormous equity capital in from other types of foreign investors like retail. The gap between net equity flows to the US (increasing) and debt flows (falling) has never been wider, and the shift from official to private funding sources is becoming more noticeable. With the US on the cusp of a potentially significant IPO cycle, we find it notable that US companies are picking up on the funding shift. SpaceX reportedly initially earmarked 30% of its IPO for retail investors, more than three times as much as a traditional IPO offering.

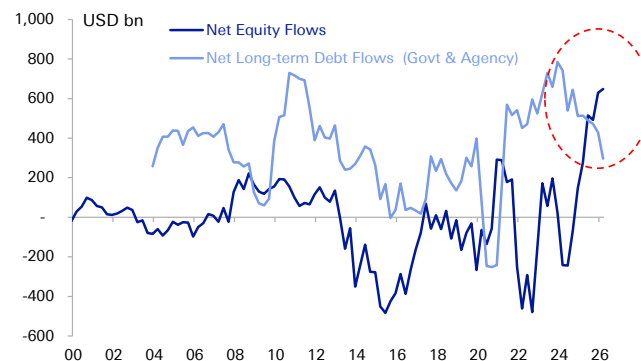
As we show in this paper, the US will likely remain very focused on attracting foreign private capital to fund both its AI ambitions and its deficits.

Figure 5: The US is attracting a record amount of foreign equity capital - more than market returns alone would suggest



Source : Deutsche Bank, Bloomberg Finance LP, Haver Analytics

Figure 6: Equity flows have replaced debt flows as the main source of foreign capital financing for the US



Source : Deutsche Bank, Haver Analytics

Where blockchain technology comes in

The US is taking a very capital-intensive approach to the AI race, raising record amounts of capital, and attracting huge amounts of it from overseas. Foreign capital not only helps support the dollar by providing deficit funding, it also improves the cost of financing for US companies. It is natural for the US to leverage technology that supports these capital flow trends.

We have [written](#) at length about how the US is using the power of blockchain technology and a new form of private money – stablecoins – to democratize access to the dollar. **We are now at the second stage of evolution, where the power of blockchain technology is being used to democratize access to US capital markets. This is called real world asset (RWA) tokenization.** While stablecoins aim put the US dollar "on chain" as a digital money token, asset market tokenization could look to put the entirety of US asset markets and their underlying financial



architecture on chain.

The two forces of tokenized money and tokenized assets are themselves highly symbiotic. The existence of tokenized assets should widen the appeal of holding tokenized money like stablecoins, that can then be more easily deployed for returns and yields "on chain", without needing to be moved off-chain back into fiat currency to invest in traditional assets. Both stablecoins and tokenized bank deposits could become important means of payment for tokenized assets, with a meaningful impact on the settlement process. We will explore the nuances of this in a forthcoming FX Special Report on Programmable Money.

From a macro perspective, we argue that US tokenization ambitions are deeply linked to supporting capital inflows into the US. Our earlier research on [stablecoins](#) discussed how they could support the diversification of dollar debt to private sector holders. Not only did the GENIUS Act introduce 1:1 reserve backing requirements for dollar stablecoin issuance, that helps power demand for front-end US T-bills; more importantly, if stablecoins help keep the world paying in dollars, they help retain the role of the dollar as a private sector savings instrument, which could help demand for assets like longer-dated Treasuries. If stablecoins are now combined with a wider universe of tokenized assets, this could encourage demand for many more US assets. This should help what we have termed a "geopolitics-to-technology funding swap". The US is moving away from attracting official, long-term, geopolitically motivated capital flows, towards private sector, shorter-term, technology motivated capital.

America's full-hearted embrace of asset tokenization – compared with China's turn away from it – can be understood in the context of both the AI race and the relative external funding position of both states. If the US is taking a capital-focused approach to the AI race, with a starting point of twin deficits, the US will naturally benefit from lowering the hurdles for foreign and private capital as much as possible. That is indeed what it is doing.

Part 2: America embraces Asset Tokenization

We begin with a definition for asset tokenization, borrowing from the [DTCC](#), the custodian for close to \$115 trillion in US assets and an organization that finds itself at forefront of this development. **Asset tokenization is the "process of converting ownership rights to a financial asset – such as stocks, bonds or real estate – into a digital token recorded on a blockchain. These tokens represent the same legal rights as the original asset but can be transferred, tracked and managed digitally."** They note that tokenization introduces new capabilities, from fractional ownership, faster cross-border transfers, transparent records and smart contracts. In a tokenized world, instead of waiting days for a U.S. Treasury bond to settle through multiple intermediaries, a tokenized version could settle in minutes. Also see "Asset Tokenization 101" from the DB Research Institute [here](#).

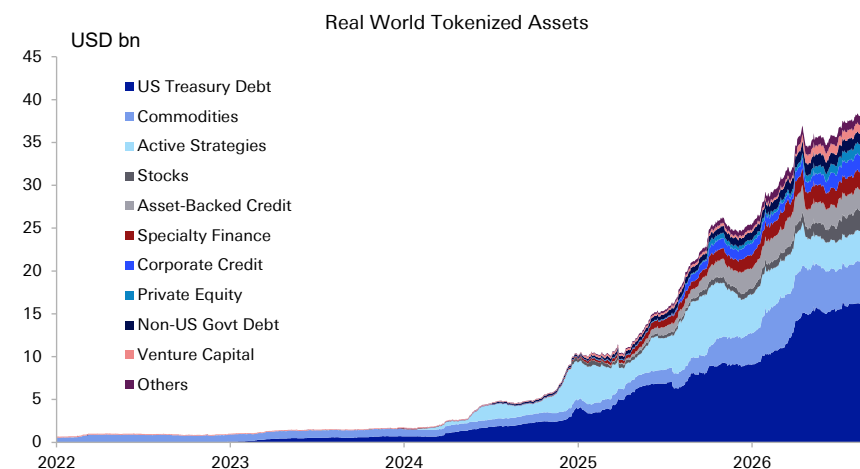
Only around \$40bn of real-world assets have been tokenized [to date](#), but the size of the tokenized asset universe is expected to grow much further and faster from here given adoption by traditional finance that is just getting underway.

In Part 1 of this paper we explored the *macro* motivations for America's embrace of tokenization: to reduce the hurdles for foreign capital to flow into the US and fund its AI ambitions in an increasingly competitive and existential AI race. In this



section, we look at some of the more micro motivations for tokenization, and how they come together.

Figure 7: Real world tokenized assets have been growing, but are still a fraction of the over USD100tn US asset universe - leaving big headroom



Source : Deutsche Bank, RWA.xyz

Some further motivating drivers for asset tokenization

The underlying technology that enables tokenization is distributed ledger technology (DLT) which allows the ownership rights to anything from money, to a stock, to a painting, to be represented as a digital token on a decentralized network like blockchain. But new technology alone cannot drive real world adoption and change, without the financial and economic incentives to use it.

So, why is DLT being used to tokenize real-world assets today and how have incentives aligned across economic agents – from retail investors, asset managers, corporates, financial firms, securities depositories and exchanges – to motivate this adoption? Beyond the macro, state-level motivation for attracting capital to fund the AI race and US deficits, we argue that private agents in the financial system are also likely to be deeply invested for their own reasons.

First, the revolution in overseas retail trading demonstrates the case for widening access to markets. Retail trading has been spurred on in recent years by a number of developments: from Covid lockdowns, the rise of zero-commission platforms, social media, fintech brokerages, improved access to information and charting, and even tax incentives to increase financialization of household savings (i.e., Japan's NISA program). Korea is arguably the poster child of the retail revolution, and offers a potential template for what might become possible at a global scale if technology is used to lower hurdles for entry. Korea represents 2% of global nominal GDP, but accounted for 10% of the \$740bn of foreign equity inflows to the US last year, a lot of which were powered by retail investors, or the "seohakaemi" (overseas stock ants). Access arguably played a big role. Korea began allowing for the fractional trading of foreign stocks by certain Korean brokerages as early as 2019, even earlier than for domestic stocks. Korean brokerages also signed up with a number of US alternative trading systems that offered daytime trading of US stocks in Korean hours. The combination of a highly digitized, connected population with lower hurdles for access was a dramatic flow of Korean capital into US equities. To us, the case of Korea demonstrates what is possible if US markets were to become easily



accessible 24/7 to everyone.

Second, corporates and institutions still face a number of inefficiencies around cash and capital in traditional off-chain markets. We focus on three examples here: trapped cash for settlement, inefficient collateral management, and manual corporate actions. While securities markets in the US have successfully moved from T+2 to T+1, this gap still involves the lock-up of capital to act as a backstop for trade settlement. If securities and payments were to settle atomically and instantly on blockchain, this could free up a huge amount of capital. Collateral markets are also still relatively inflexible today. Today, repo markets allow financial entities to borrow cash in exchange for lending collateral like US Treasuries. Repos are structured around overnight to longer-term financing and are constrained by end-of-day settlement cycles and operational windows. Digital assets held on a blockchain could arguably be posted instantly for much shorter periods, creating intraday repo possibilities and increasing the velocity of collateral. As a final example, NASDAQ highlights that processing corporate actions today alone "carries costs of more than \$58 billion across the industry, driven by fragmented systems, manual workflows, and duplicative record-keeping." The programmability offered by digital assets would help address this as corporate actions could be programmed into smart contracts attached to the asset. In sum, there are clear incentives for agents – from corporate issuers to institutional asset managers – to embrace a tokenized asset environment.

Third, the tokenization train had already left the station but was at risk of being mediated via third-party offerings and fragmenting market liquidity. In the traditional space, we note that fund managers like Blackrock and Templeton had already begun issuing some of their funds like BUIDL and FBOXX on public blockchains. This involved making the funds accessible as digital tokens, but importantly the underlying assets themselves remained off-chain in traditional financial instruments. More importantly, the presence of blockchain technology has allowed for entry and innovation by new private players that have begun offering US assets as digital tokens on blockchain rails. New tokenized equity frameworks offer *representations* of real US stocks and ETFs, backed 1:1 by underlying traditional equities, held in regulated custody, and issued as on-chain tokens on many different blockchains. They are available to non-US clients in 100+ countries, and allow fractional purchases with as little as \$1 in select US stocks and ETFs, some of which are available to trade 24/7. They do *not*, however, confer shareholder rights like voting or dividends in cash – instead, they offer price exposure to the underlying asset in an always-on format, via a *third-party* token. If tokenization was already underway, the incentive for traditional asset managers and exchanges to be part of the process was becoming more evident – to avoid losing funds and liquidity to third parties. Rather than simply tokenizing *claims* on US securities, developments underway are now building the infrastructure to allow underlying US securities *themselves* to be issued, traded, and held as tokens.

As we discuss in the next section, asset tokenization is not a futuristic or frontier concept – it is live and is now happening at the heart of traditional markets, led by core players in the US financial ecosystem.

What is happening?

In a nutshell: tokenization is going mainstream, with adoption by core, traditional financial players. Securities depositories and exchanges under the aegis of regulatory guidance are building the market infrastructure to tokenize the US asset universe, with participation from asset managers and banks.



Regulatory efforts arguably kicked off in force with the launch of [Project Crypto](#) by the SEC in July 2025, alongside the [White House's Digital Asset Report](#) that called for laying the groundwork for the US to become the deepest and most liquid digital asset market in the world. One of the first tasks was to develop a [taxonomy](#) for crypto assets, where tokenized securities were importantly defined as *remaining "securities"*. In January 2026, Project Crypto evolved into a [joint inter-agency framework](#) between the SEC and the CFTC that would allow for "coordination, coherence, and a unified approach to the federal oversight of crypto asset markets". Crucially, it is actions from regulatory agencies in the US that are driving progress, independent of pending efforts in Congress to pass the CLARITY Act.

The big green light for mainstream tokenization came in December 2025 when the SEC issued the DTC with a ["no-action" letter](#) which legitimized the idea that the same security could trade on two rails. DTC could therefore tokenize assets in its custody from US stocks, ETFs and bonds, and the digital version of these assets would have "the same entitlements, investor protections and ownership rights" as the traditional asset. This was followed up by a [ruling](#) in January 2026 which noted that the "format in which a security is issued or the methods by which holders are recorded (e.g., on-chain vs. off-chain) does not affect application of the federal securities laws". Assets can be held traditionally or in tokenized form (as a digital twin) and they will be treated the same, and will refer to the same security. In other words, **both traditional and tokenized architectures can now coexist, run side-by-side, and be interoperable, with the trust of DTC custody to back them**. While the passage of the CLARITY Act is still pending, the above SEC notifications and the commitment of Project Crypto to bringing US capital markets on chain means that the regulatory umbrella for tokenization has been sufficiently opened. This has been enabling of developments underway this year.

The progress at DTC remains crucial to track. DTC is the main central securities depository in the US – it is the venue where trades settle and clear, and the custodian where financial firms hold their assets. DTC has around \$115 trillion of US assets under its custody. Tokenization at DTC has now begun. In July 2026, the DTC converted the first asset into tokens that were used for real world trades. 40 different financial firms participated. Amongst many examples, the SPDR S&P500 ETF Trust (SPY) – the world's first ETF – was tokenized. In another test, JP Morgan successfully posted tokenized assets to satisfy a margin requirement at the CME. The full set of tests that day can be found [here](#). DTC intends to launch its tokenization service by October 2026, but full-scale tokenization will be a long process. The DTCC CEO has argued it could proceed over many years with markets "crawling, then walking, then running" and the pace to be set by market demand to hold assets as tokens. DTC is beginning by tokenizing the Russell 1000, ETFs, and US Treasuries before widening it out. Market [estimates](#) currently range from \$2tn to \$30tn for the size of the tokenized asset universe by the 2030s, but these estimates may well be revised higher as the infrastructure becomes operational.

US exchanges – NYSE and [NASDAQ](#) – are shifting to accommodate "always-on" trading. In current financial architecture, exchanges are the *trading* layer, where buyers and sellers meet and prices are discovered, while DTC is the *post-trade* infrastructure layer for settlement and custody, maintaining a central record of securities ownership after trades occur. The biggest development at US exchanges is that they are set to move towards round-the-clock trading. NASDAQ has also announced an intention to offer trading 23/5 trading by December 2025, while NYSE is aiming for 22 hours by later this year. As noted earlier, synthetic equities



have already been trading outside market hours on permissionless blockchains and alternative trading systems, so increasing exchange hours will allow that liquidity to come back on exchange, and prevent tokenized equities from becoming off-exchange products that fragment liquidity.

Exchanges are also building new platforms and tools to enable a tokenized trading experience. The NYSE's [new digital platform](#) – being built in conjunction with Securitize – aims to not just offer 24/7 operations, but also instant settlement, fractional share trading, and stablecoin based funding. NASDAQ has also announced the launch of an "[equity token design](#)" that will allow public companies to leverage tokenization to program shareholder engagement from corporate actions to proxy voting. This will address the current condition of third-party digital tokens that offer investors economic exposure but without the underlying shareholder rights and corporate actions. NASDAQ aims to have full tokenization capabilities by 2027.

The first mover advantage

Tokenization is going mainstream in the US and this is happening now: as we have described above, the biggest exchanges and securities depositories are making significant upgrades to allow US asset markets to be traded and held on chain. While tokenization of the US asset universe could proceed over many years, the ship has left the harbour.

With the US moving to provide rapid regulatory clarity and support for the development of tokenized securities, and building the settlement infrastructure to support it, it stands to benefit from a first mover advantage. Not only might the US be setting the standards for what on-chain financial trading and settlement looks like, it could benefit from future network effects. Just as the US benefits from agency over networks like SWIFT and USD correspondent banking, it may exercise them over the future on-chain infrastructure. **If the majority of on-chain private money (stablecoins) is dollarized, and the majority of on-chain assets are US listed, this could reinforce USD payments and capital markets dominance.** And if access to capital is increasingly concentrated in the US, global companies too may be increasingly interested in listing and issuing in the US to access them. As a potential harbinger, it is perhaps notable that SK Hynix made the largest foreign debut in US markets with its \$26.5bn ADR this summer, favouring US markets for their capital raising.

We look at some of the broader implications for markets from the interplay between the AI race and asset tokenization in Part 3.

See next page



Part 3: The future of the dollar, imbalances, and capitalism

In this final section, we look at how the AI race and approach to tokenization could impact the dollar, and how it might shape the future of global imbalances and capitalism itself.

Tokenization meets TINA

US equities already account for more than 50% of global equity market capitalization, and US Treasuries for more than half of OECD sovereign debt. For many investors, this gives US assets "There is No Alternative" status, popularly known as TINA: investors have no alternative but to hold assets which are such a large part of the global market portfolio. The position of US companies in the AI race thus far has further contributed to TINA, with investors compelled to have exposure to technology which could change the world. The ultimate outcome of the AI race will no doubt play an enormous role in whether the US' share of the global market portfolio grows or shrinks: it is not just how earnings and valuation multiples evolve, but also how terminal values of the corporate sector could change, and how debt sustainability could be impacted.

TINA is already playing a big role in attracting capital to the US, but tokenization could add to this: by making US assets more *attractive* and more *accessible*.

First, on attractiveness. The collateral implications of holding US assets on chain could be very attractive for investors. Fed Governor Cook noted in a [speech](#) that the most significant benefit of tokenization would be implications for capital and liquidity management for institutions. A tokenized US asset becomes more liquid and deployable, and capital and cash thus becomes more productive. In simple terms, you are more likely to hold a tokenized US equity or bond instead of cash, if you can easily repo your tokenized asset on chain when you need liquidity. If the value of US assets is going up, the cost of capital for US firms is going down. **The improved collateral velocity of tokenized assets could be another superpower for US assets, and another arrow in the quiver of US capital attractiveness.**

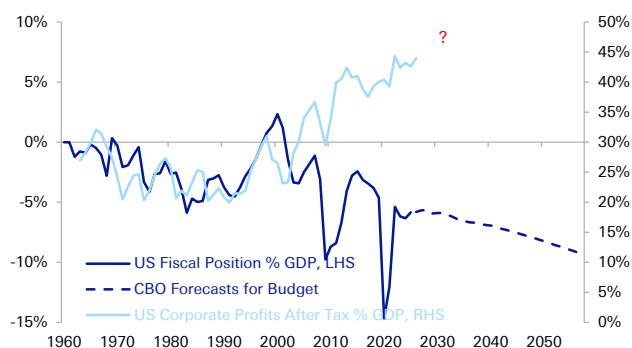
Second, on accessibility. We had argued that stablecoins could give anyone with an internet connection access to the dollar on chain. Asset tokenization could do the same for US assets. **In a world in which US equities and bonds are trading 24/7, with instant settlement and much lower costs and frictions, this could have dramatic knock-on effects on global behaviour, particularly for retail investors that could find access much easier.** We find it notable that [NASDAQ](#) openly acknowledged that "the investor base of public equities has the potential to expand as markets remain open across multiple time zones, making it increasingly important to have modern tools for public issuers to engage with investors worldwide." Tokenization need not also be limited to equities and bonds. One can imagine a world in which a person with even \$1 in a place 10,000 miles away could own a tiny fraction of a tokenized Times Square on chain.

From an FX perspective, if tokenization succeeds in making US assets more attractive and accessible, this could help attract private capital flows. It is perhaps not surprising that the US is leveraging the power of technology – both AI and blockchain – to attract more foreign private capital at a time when US geopolitical leadership could be under pressure (see important examples [here](#), [here](#), and [here](#)) and crucially when US fiscal deficits have been widening. Both increase the need for alternative sources of private sector funding. **The US is arguably going "all-in" on private capital to offset public sector vulnerabilities.**



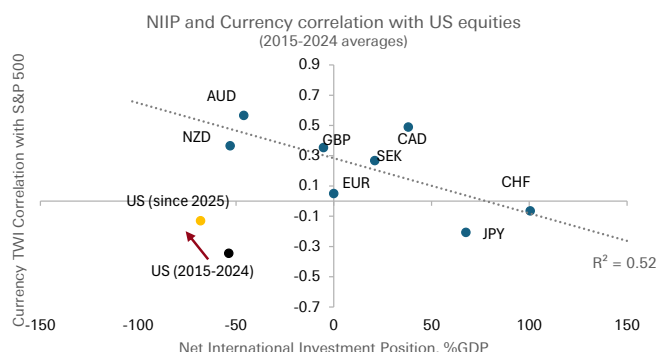
But lowering the hurdles for inbound capital, also exposes the US to capital going out. Tokenization would meaningfully increase capital mobility, in and out of US assets. The shift in US funding towards private sector, potentially shorter-term flows into equities – motivated largely by technology returns – will make the dollar riskier, more equity sensitive, and leveraged to the AI race. We have covered in earlier research how the dollar's historical properties of diversifying equity risk could be waning, with the dollar likely to become more positively correlated to risk. If for instance, AI proves uneconomical or the US starts to fall behind in the AI race, the dollar could be very exposed.

Figure 8: The US is increasing reliance on private capital - motivated by technology leadership - at a time of growing public sector vulnerability



Source : Deutsche Bank, Federal Reserve Bank of St Louis, CBO

Figure 9: The dollar is becoming a weaker diversifier for equity risk



Source : Deutsche Bank, Haver Analytics, Bloomberg Finance LP

Europe is on board, but China is taking a very different approach

How does the highly enabling regulatory environment towards tokenization in the US compare with the other major economic blocs, namely Europe and China?

Europe has been receptive to the opportunity presented by tokenization.

Europe's DLT Pilot Regime introduced a regulatory sandbox for institutions to test the trading and settlement of tokenized securities on DLT as early as March 2023, even earlier than the US. ECB Governor [Lagarde](#) has noted that tokenization offers the prospect for Europe to "tackle the fragmentation that has held us back for decades." Note that Europe has over 30 central securities depositories across the union. An integrated market for tokenized assets could allow Europe to "leapfrog the fragmentation of existing legacy systems" (ECB). Europe has also embraced the role of tokenized securities as collateral, with DLT assets held in European CSD's eligible as collateral at [Euroclear](#) from March 2026. This puts digital assets on the same footing as traditional ones in Europe.

Where Europe sits apart from the US is perhaps in: the scale of its crypto ambitions, an explicit commitment to retaining a two-tier monetary system with central bank money as the final settlement asset, and the size of the AI economy itself. Europe is alive to the opportunities presented by tokenization for market unification – an important step to increasing Europe's capital market attractiveness, but the US arguably has more clearly stated ambitions to become "the crypto capital of the world". Moreover, Europe appears committed to keeping central bank money as the final settlement instrument between financial institutions even in a tokenized world, with initiatives like Pontes and Appia



underlying this. In the US, it remains to be seen whether private stablecoins become the primary instrument of settlement, disintermediating central bank money, or whether the existing system of bank deposits and reserves is tokenized. Ultimately, the biggest divide between Europe and the US is in the AI economy itself – Europe has a much smaller footprint in data centres and independent AI labs, and thus may not be where the appetite to raise or deploy capital – even in a tokenized asset world – is most compelling.

Meanwhile, China is taking a very different approach altogether; while the US is embracing stablecoins and RWA tokenization, China is tightly controlling – if not restricting – them. In February 2026, the PBOC and CSRC along with eight other national organization issued [a notice](#) that updated China's 2021 ban on crypto-related activities. The notice defined RWA tokenization and tightened regulation around it – noting that such activity or intermediation of it was prohibited in China, except with regulatory approval on specific financial infrastructure ([Xinhua](#)). The notice also banned entities from issuing stablecoins pegged to RMB abroad, without regulatory approval. We note that Chinese regulators prevented Ant Group and JD.Com from proceeding with plans for yuan stablecoins in Hong Kong in October 2025. Instead, in January this year, PBOC began to allow payment of interest on e-CNY, retail CBDC deposits suggesting relative encouragement for controlled forms of central bank digital money. We look further at some of the drivers of this very divergent approach between the US and China.

Open markets versus open technology

The US is pursuing a hyper-capitalist approach in the AI race: throwing open the doors for private capital to fund large capex, and using the power of asset tokenization to supercharge it. We call this the open markets approach. Under the US model of shareholder capitalism, US companies prioritize profit maximization, keep AI model weights closed to allow companies to control pricing via subscriptions, and offer the carrot of return participation to shareholders to attract in global capital. The US then uses those huge sums of capital to make large capital investments to innovate and stay ahead of the competition. Importantly, the open markets approach is paired with a closed technology one.

China is taking a very different approach in the AI race, pushing out lower-cost models with open-weights into the world. We call this the open technology approach, and it is paired with a closed markets one. Chinese companies are not only achieving similar model results, and charging clients less for tokens, they are offering their models with open-weights. Open weights allows users to download and customize models on their own servers, promoting global proliferation and usage among corporates. See more from the DB Research Institute on this [here](#). China may be counting on the same penetration of its AI models around the world as it has achieved with manufactured goods. One can perhaps start to look at China's approach to AI as an extension of China's broader involution approach to manufacturing where state-based alignment combined with fierce domestic competition achieves low pricing, improving quality, rising market share and ultimately industrial dominance. That China is now trying this in the technology space, where the US has thus far remained a leader, bears close watching. The US' open markets are meeting China's open weights.

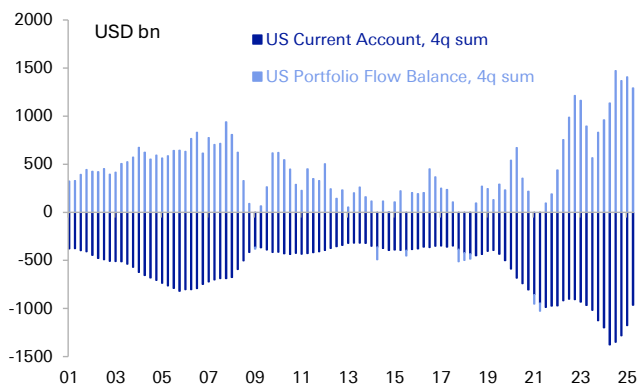
In sum, the US is taking an "open markets-closed technology" approach to the AI race, while China is taking an "open technology-closed markets" approach. This divergence is not just a reflection of different economic models, but underlying global imbalances, and very different needs for global funding.



A test for global imbalances and capitalism itself

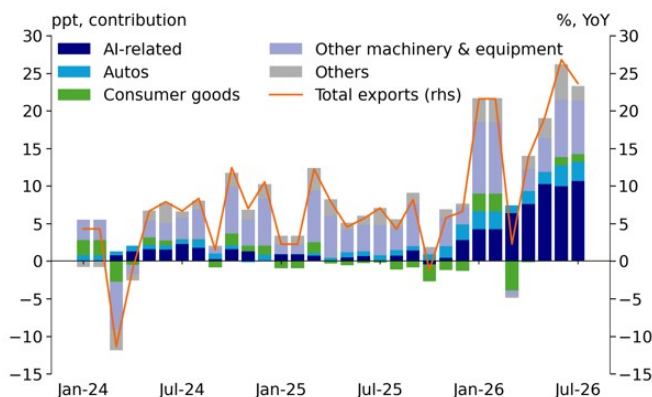
The US' open embrace of tokenization, compared to China's more closed approach to foreign capital is a reflection of underlying imbalances. The US' current account deficit combined with a fiscal deficit means the US requires foreign financing to fund incremental private capital expenditure, in the absence of domestic fiscal savings. In comparison, China is running a large and growing current account surplus, reflective of excess savings. China's trade surplus is in fact growing – with the AI cycle boosting AI-related exports. China does not need to turn to foreign capital because it can fund capital expenditures from abundant current account savings. This affords China the ability to maintain a more managed capital account, and explains the relatively restrictive approach to the high-speed blockchain rails that the US is so actively building.

Figure 10: The US is dependent on foreign capital flows to fund its deficits



Source : Deutsche Bank, Haver Analytics

Figure 11: Chinese exports are growing even faster as a result of AI



Source : Deutsche Bank, CEIC
Note: See "China Macro: Trade Imbalance and Rates Divergence" (31 Aug 2026)

The AI race could further reshape the state of global imbalances. As one potential path, imbalances could become more extreme. If China leverages AI to make industry even more efficient, adopting it for production and industrial automation, this could boost China's manufacturing dominance and good-driven surplus. One can imagine an alternative scenario in which the US current account improves, narrowing global imbalances. We have written in earlier research how AI revenue streams could help the US current account by around 1% over the coming decade via higher services exports. But this is crucially dependent on US AI labs retaining pricing power with private subscribers, and monetizing an international subscriber base. But China's low-cost, high-proliferation approach could cap the excess profits of US AI companies, and their ability to sell intellectual property for profit. How things evolve may come down to whether US and Chinese models become complements that coexist and serve different market segments, or whether they become substitutes in a winner-takes-all outcome. There is a lot at stake.

Ultimately, the AI race between US and China is more than a battle between AI models. It is a battle between economic models. The US is doubling down on capitalism: operating on the premise that increasing capital market openness will pull in the funding for large AI capital expenditures; in turn, this will help the US remain dominant in AI technology, allow corporates to maintain pricing power, and for the US to pull ahead in the race. Meanwhile, China does not need to leverage open markets to attract capital flows. China's industrial dominance and record



trade surpluses mean it has ample and cheap sources of domestic funding. This allows China to keep a more closed market approach that need not prioritize profits firsthand, and allows it to compete with low-cost, open-weight AI offerings. If China's approach starts to undercut US profits, this could threaten the US model of shareholder capitalism, which has become critical not only to attracting funding for AI, but for growing US deficits. China's challenge is not only about technology, but the central maxim of US power: that free and open capital markets always deliver the winning innovation. The AI race is a big test for capitalism itself.



Appendix 1

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The good, the bad and the ugly

September 03, 2026

- As the summer holidays come to an end, we take a look at the three questions that are likely to occupy our time in the months ahead: Is the euro area economy proving resilient to the energy shock? Will the ECB stop at 2.50% or continue hiking until policy is restrictive? And will public finances and elections prove a destabilising combination?
- The European economy is proving remarkably resilient, with some elements of positive demand shock offsetting the negative energy supply shock. Expectations are improving, but some caution is still needed: the energy shock is not over and the AI trade deficit is widening.
- With respect to the ECB, a rapid normalisation of energy prices may be needed to stop the hiking cycle at 2.50%. That said, the inflation outlook is not yet consistent with an outright restrictive stance either.
- Budget season is about to begin, debt dynamics are vulnerable and elections imply political risk and market volatility. The degree of tightening of financial conditions will be a function of political choices, the resilience (or not) of growth and the restrictiveness (or not) of ECB policy.

Q1: Is the euro area economy proving resilient to the energy shock?

Strictly speaking, as a net energy importer the euro area should be highly exposed to the negative energy supply shock triggered by the Iran war. In any case, HICP inflation jumped to 3.0% in Q2, 1.2pp higher than expected at the start of the year. The household real income shock alone should have knocked more than 0.5pp off GDP.

And yet euro area GDP was not weak in Q2, it was surprisingly strong. At 0.44% qoq, GDP growth exceeded expectations. We don't have detailed components yet, but on the basis of available data household consumption grew despite the inflation shock.

It is possible that elevated household savings buffered the real income shock on the assumption that it would be fleeting. There was also a targeted and temporary easing of fiscal policy (lower energy taxes), but this was not substantial and only moderated the rise in inflation slightly. That said, the euro area fiscal stance was already easy for 2026. Not nearly as easy as 2022, and skewed heavily towards Germany. But the stance is easy, not tight.

Most encouraging are the signals of rising investment spending: the BLS points to demand for loans for fixed asset investment, the flow of long-term loans to corporates has been a little volatile but hints at investment spending, and the rise in German industrial orders from elsewhere in the euro area has historically been consistent with a euro area investment cycle too.

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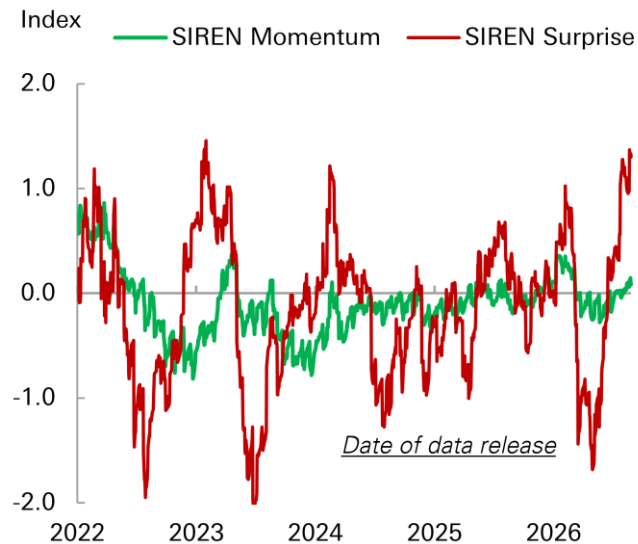
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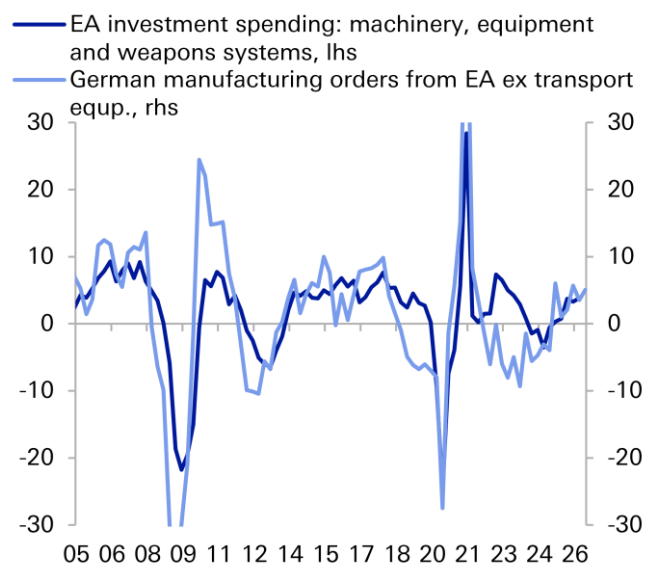


Figure 1: Euro area activity data showing strong surprises and building momentum



Source: Bloomberg Finance LP, Deutsche Bank Research

Figure 2: Encouraging signals of a capex cycle building



Source: Eurostat, German Federal Statistics Office, Haver Analytics LP, Deutsche Bank Research

We need to be careful. We might be seeing the lagged transmission of the earlier ECB rate cuts. Private sector balance sheets are healthy, which would aid transmission. Maybe structural demand from AI investment and defence spending is offsetting higher energy costs and geopolitical uncertainty.

In short, maybe what Europe is facing is a positive demand story, not just a negative supply story. Having quickly dropped to 48.8 in April, the composite PMI rose to 52.1 in August and might have been higher were it not for heat-related headwinds in services.

That said, there are reasons to be cautious about growth.

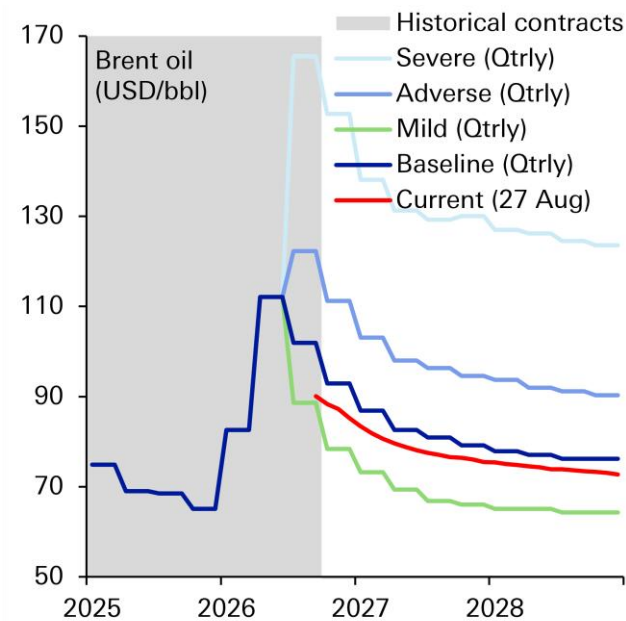
The energy shock is not over. Gas prices have reached a new high for the year, raising the prospect of an electricity-based production cost shock to add to the existing oil-based transport cost shock.

The energy shock might also be masked by temporary factors like stock-building. The shock in the Middle East went beyond energy into chemicals and metals. There was a threat to global supply chains. Fear of disruption might have triggered stock-building, which might explain why manufacturing has held up.

There is no evidence of domestic inventory building. But Europe might have benefited from temporary stock-building in its export partners. This might help explain export resilience.

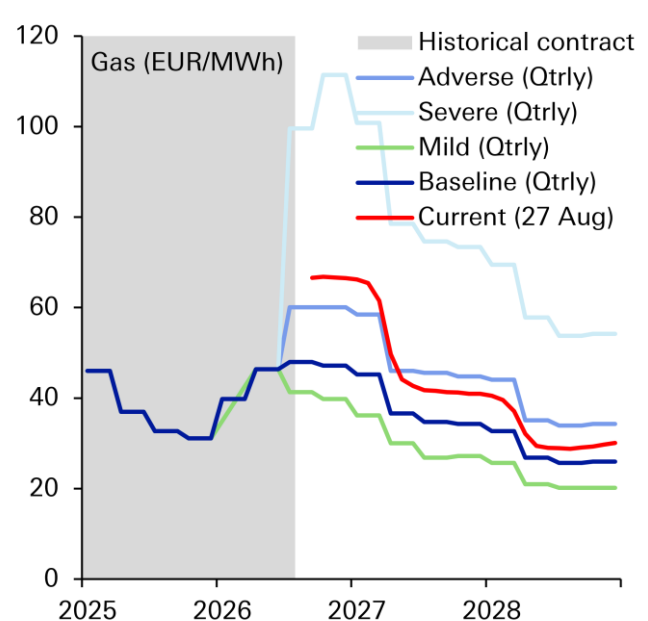


Figure 3: Oil prices vs. ECB baseline and alternative scenarios



Source: Bloomberg Finance LP, Deutsche Bank Research

Figure 4: Gas prices vs ECB baseline and alternative scenarios



Source: Bloomberg Finance LP, Deutsche Bank Research

AI investment spending should keep rising. But the AI spending boom might be less beneficial for GDP than it appears, however, as Europe's AI-related trade deficit is widening as both AI-related exports weaken and AI-related imports grow.

Monetary policy is tightening too, which will restrain growth with a lag (see our latest review of upstream and downstream financial conditions [here](#)). Concerns about sovereign debt sustainability could generate additional growth-sapping risk premia.

Finally, competitiveness remains weak and the Second China Shock and increasingly direct competition with China could weaken European production and employment.

The bottom line is the economy has been remarkably resilient to the energy shock so far and our euro area GDP forecasts are probably too low at 0.5% in 2026 and 1.1% in 2027 (our German team have raised their 2026 forecast from 0.5% to closer to 1.0%).

Q2: Will the ECB stop at 2.50% or continue hiking until policy is restrictive?

Since early March we have been expecting the ECB to hike twice (Jun and Sep), raising the policy rate to 2.50%, the upper end of the range of neutral. This felt like good risk management. It signals the ECB commitment to price stability on the one hand without overly burdening economic growth in the context of a negative supply shock on the other.

If growth was weakening amid a supply shock, inflation expectations were comprehensively anchored and data showed no risk of energy price pass-



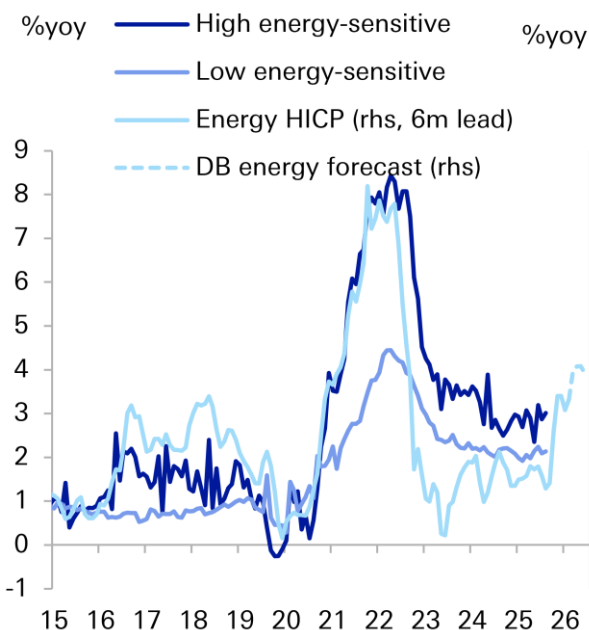
through, it would be easier to call into question the ECB's decision to hike rather than "look through" this energy supply shock.

In the round, the data, the ECB messaging (even this week Isabel Schnabel said the market "very well" understands the ECB reaction function, repeating what President Lagarde said in the July press conference) and the lack of resolution in the Middle East and still elevated energy prices mean a hike to 2.50% in September seems all but certain. Indeed, this is the impression from the July ECB Accounts. The question is, will the ECB stop there?

There are two scenarios in which the ECB hikes to 2.75%.

In one scenario, the ECB concludes that the upper end of the range of neutral is 2.75% rather than 2.50%. The persistent strength of credit growth in recent months despite higher lending rates and despite banks saying that lending standards are tightening might reveal that neutral is higher than expected. The scale of fiscal easing, even if led by Germany, could push r^* higher. If this is right, the hurdle to raising policy rates to 2.75% might be lower than assumed and might not need strong or clear evidence of indirect or second-round inflation.

Figure 5: Maybe it's too soon to expect indirect energy effect to show up



Source: Eurostat, Haver Analytics LP, Deutsche Bank Research

Figure 6: The credit impulse is consistent with a robust domestic economy



Source: ECB, Eurostat, Haver Analytics LP, Deutsche Bank Research

The argument is not one-sided. The strength of credit growth might reflect the lagged transmission of the earlier cuts rather than a higher neutral rate. In any case, our composite indicator of restrictiveness, based on a range of data that might reveal neutral, says that policy was still mildly restrictive in mid-2026 when the ECB started to hike. If that's right, it might be more difficult to argue 2.75% is still within the range of neutral.

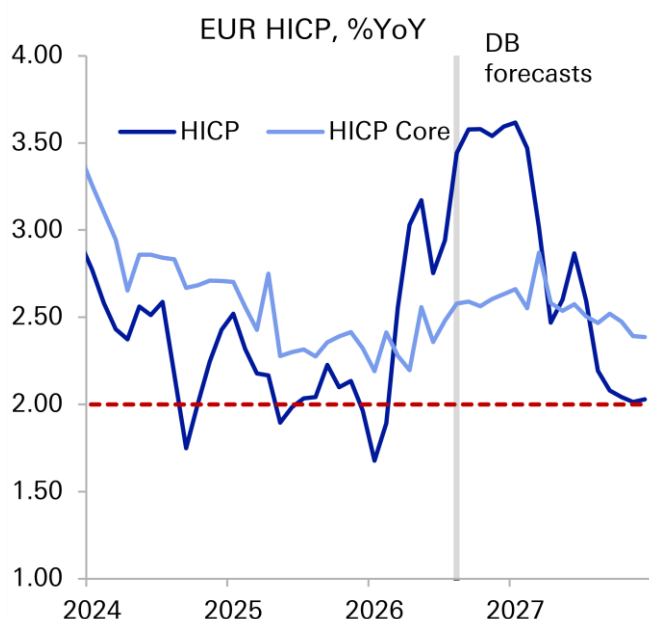


The other scenario for getting to 2.75% (or higher) is the ECB decides that a restrictive policy stance is appropriate. In the ECB's current framework, this means moving from a "measured" tightening to a "forceful" tightening. This requires the ECB to believe that the inflation shock is moving from a 'large but not too persistent' shock to a 'more significant and sustained overshoot of the target'.

There are different routes to believing in a more significant and sustained overshoot of the inflation target.

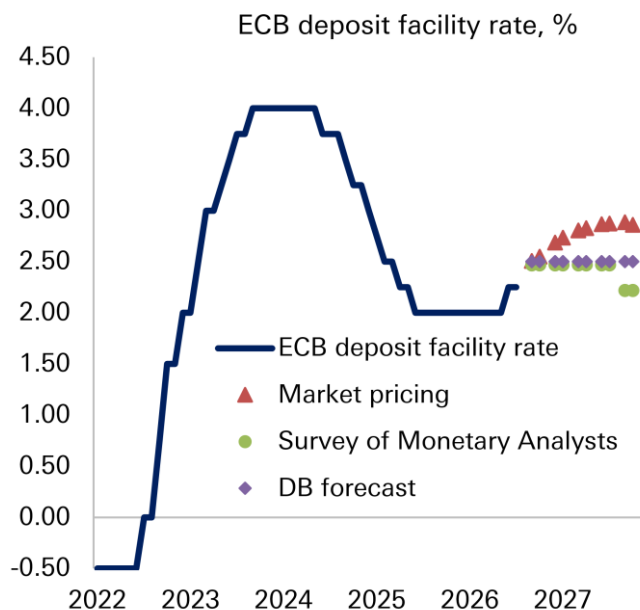
One would be if energy prices remain at elevated levels for longer than expected. Oil prices are in line with baseline, but gas prices are more consistent with the adverse scenario (at least in the near term). Given the lagged transmission of rising gas prices into inflation, this could add persistence to the headline HICP shock, lengthening the period during which energy inflation could spillover into core.

Figure 7: Headline and core HICP: DB forecasts



Source: Eurostat, Haver Analytics LP, Deutsche Bank Research

Figure 8: ECB policy rates: Market priced hikes to 2.75-3.00%



Source: ECB, Bloomberg Finance LP, Deutsche Bank Research

Another way would be if larger than expected indirect effects and in particular second-round effects materialised. So far, there is no compelling evidence of indirect effects and no evidence of second-round effects. However, both would only materialise with a lag. For example, it was six months before high energy-sensitive core HICP started rising in the 2022 shock.

A third way would be if stronger growth added persistence risk to inflation. The recent resilience of growth would add to this risk if it lasts. That said, if AI lifts productivity, the additional growth might not be as inflationary.

How the ECB staff forecasts change in September will be very important to the outlook for monetary policy. In June, the baseline forecast had core HICP



averaging 2.5% in 2027 and staying at least 30bp above target through the year. Importantly, however, headline HICP was expected to be back down to the 2% target from Q3-27 and stay there through to the end of 2028.

It is important to remember the baseline staff forecast in June was conditioned on policy rates rising to 2.75%. But despite sticky core inflation, it is difficult to imagine monetary policy tightening much further if headline inflation is sustainably returning to target within a year.

As such, the near-term path for policy could depend critically on whether the Governing Council believes core inflation will converge down to headline in the medium term or headline converges up to core. The former is less hawkish, the latter is more hawkish.

Which of these two outcomes seems more or less likely to the Governing Council will depend on how it judges the risk of inflation persistence. This is a function of pricing power, bargaining power and purchasing power. Our view is there is some risk of persistence, but not as much as 2022 and not enough to warrant an outright restrictive policy stance. Not yet, at least.

Our 2.50% baseline might be too optimistic, but 3.00% feels too restrictive. 2.75% is a grey zone: a non-negligible possibility unless the situation in the Middle East is resolved quickly and energy prices normalise rapidly. Our work on policy rules shows that the ECB 'baseline' scenario macro forecasts are consistent with 2.75%. The 'milder' scenario forecasts are consistent with 2.50%.

There may not be any new policy signals from the ECB at the Jackson Hole symposium. Unlike in the past, there is no global policymaker panel. Isabel Schnabel is the ECB member on the official agenda – speaking about tokenized finance. However, markets will be on the watch for press interviews by ECB attendees.

Q3: Will public finances and elections prove a destabilising combination?

The environment around European public finances is becoming more challenging for a number of reasons. Government bond markets are under pressure globally, driven currently by the US. The 'i minus g' gap at the heart of debt dynamics is becoming less favourable as the benefits of the low rates regime of the last decade expire, amplified by rising inflation risk, tighter monetary policy, rising term premia and generally low rates of economic growth.

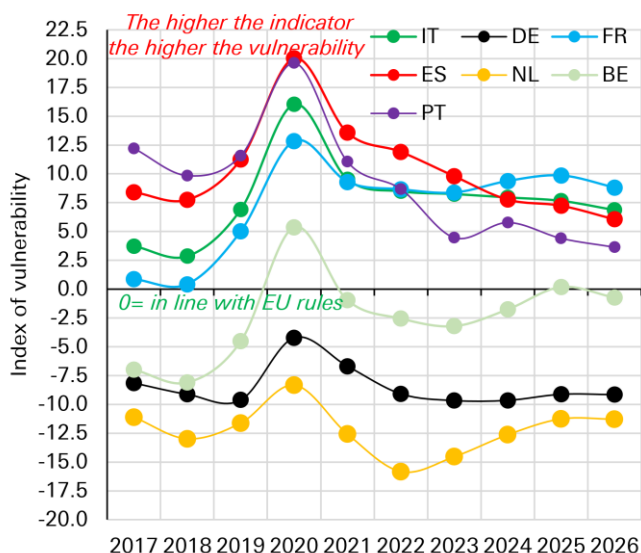
At the same time, public debt to GDP ratios have not returned to pre-pandemic levels, fiscal vulnerability is elevated even when accounting for improved external balances, and European governments are facing numerous spending demands, from defence and energy to climate change and technology.

The combination of more vulnerable public finances and elections in several significant member states over the next 12 months creates the potential for tighter financial conditions. National elections lie ahead in Italy, Spain, Greece, Finland, Estonia, Poland and, perhaps most consequentially, in France. Financial conditions are well-behaved so far (the ECB's state-of-the-art Macro-Finance FCI has hardly tightened). The degree of tightening of financial conditions going



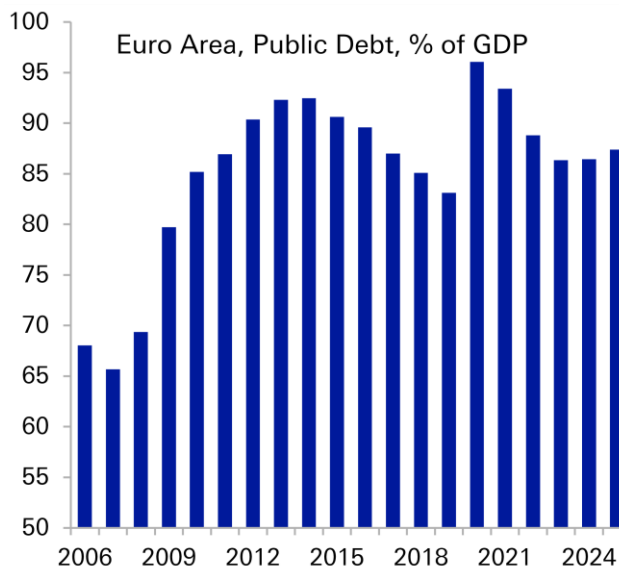
forward will be a function of the fiscal policy choices made by governments, the resilience (or not) of growth and the restrictiveness (or not) of ECB policy.

Figure 9: DB fiscal vulnerability indicator incl. net external position – France most vulnerable



Source: European Commission, Deutsche Bank Research

Figure 10: Public debt remains high and above pre-pandemic levels



Source: European Commission, Deutsche Bank Research

With regard to France, the opinion polls say that Rassemblement National (RN) has a credible path to winning the Presidency with Marine Le Pen. If that happens, there is a good chance an early legislative election will be called too.

OAT markets have been on edge since President Macron's ill-judged legislative election in 2024 resulted in a deeply divided parliament with little capacity to drive strong fiscal consolidation. On the basis of what is known at the moment, there is a lack of a credible consolidation trajectory in the RN fiscal plans. Even under moderately optimistic assumptions about consolidation, France's debt ratio will likely remain broadly flat in the range of 120-125% of GDP over the next decade. A flat trajectory means vulnerability to any new negative surprises on growth, funding costs and/or deficits – and as we have said before, the deficit is under pressure from inflation (indexed spending), legacy policy decisions (stalling the pension reform), defence, ageing, etc.

Much will depend on the course of French fiscal policy, particularly RN's fiscal policy. This could be revealed in three distinct stages over the year ahead.

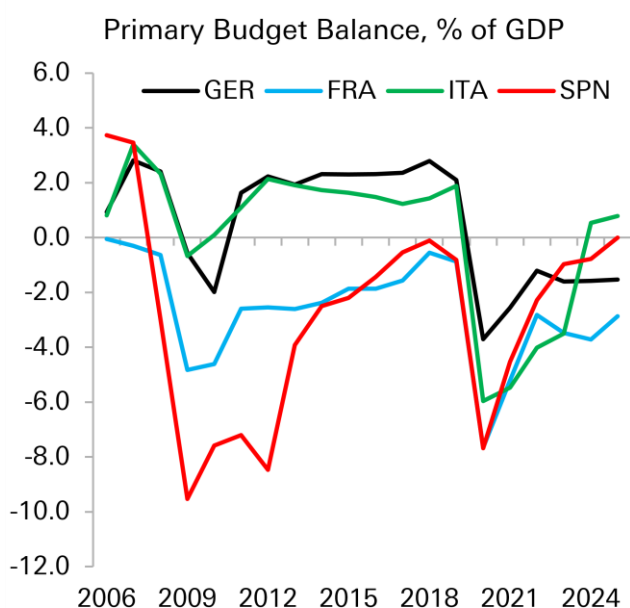
- **The 2027 budget.** The incumbent Lecornu government needs to have a budget agreed by year-end. A deeply divided parliament and approaching election will make strong consolidation difficult. Importantly, however, RN is signaling that it won't oppose a budget. The alternative ("special law") would mean a less stable financial position in 2027. RN would prefer to have some breathing space if it is elected.
- **The election campaign.** This will reveal more about RN's fiscal plans. The limited details available so far imply a wider deficit. However, RN's fiscal



identity is unclear and RN President Bardella's admiration for Italy's market-friendly fiscal stance under PM Meloni suggests an understanding of the benefits of a stable bond market. The extent of the RN's intention to challenge Brussels will be important too. This could influence market confidence in the availability of European backstops (ECB TPI, ESM, etc).

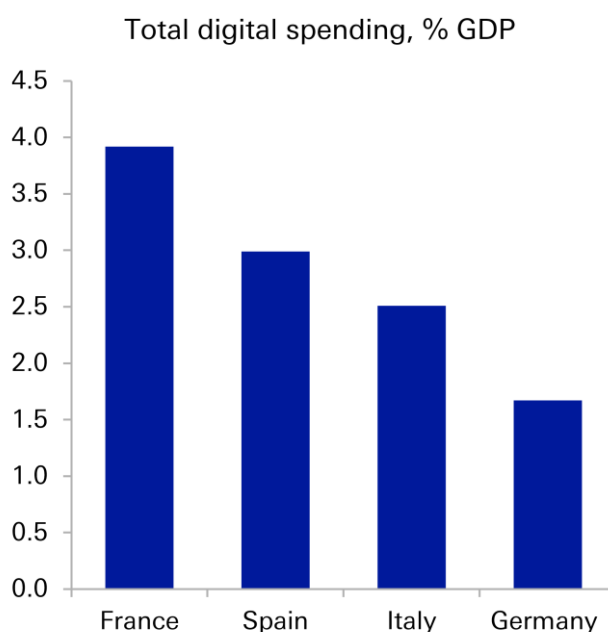
- **Post-election decisions.** Assuming an RN President would dissolve parliament in the spring to trigger legislative elections by the summer, the first task for the new parliament would be the preparation of the 2028 budget. This assumes a budget for 2027 has already been passed. If France is operating under a special law, a 2027 budget would take immediate precedence. These would reveal RN's fiscal plans and their consistency with EU rules.

Figure 11: France has the largest primary deficit among the Big4...



Source: European Commission, Deutsche Bank Research

Figure 12: ...but France might have better growth prospects, helped by AI (digital investment spending)



Source: Eurostat, Deutsche Bank Research

Counterbalances. France will be the focal point for the market. As we have argued (here, here and here), the key theme for investors is not outright regime change, but a constant, volatility-inducing push and pull between RN's political objectives – shifting in places – and the fiscal, legal, and market constraints that would ultimately shape its capacity to govern. The push for political change will generate friction.

There are several buffers that could help keep the country off the more challenging paths.

- **The market is already pricing weak fundamentals.** French yields have been gradually adjusting to these weak fundamentals over the last couple



of years. For example, the current yield already anticipated further credit downgrades.

- **Robust banking.** French banks are well-capitalised and unlike the crisis-hit periphery a decade and a half ago hold relatively little direct exposure to OATs. As a global safe asset, OATs are widely held abroad.
- **Backstops.** Unlike the sovereign debt crisis 15 years ago, there are backstops in place. For example, the ESM can lend to crisis-hit member states and the ECB can intervene in the market for eligible member states with TPI. If foreign investors offload OATs in scale, the French banks have some capacity. However, as bank exposure increases, the potential for a more destabilising sovereign/banking doom-loop would increase.
- **Economic growth.** There are several reasons why French economic growth could outperform its peers and help stabilise debt. One is AI. Among the larger member states, France is spending the most on digital investment and was the only EU member state to launch a notable AI model in 2025 (50 in the US). France is also relatively more exposed to defence production, and among the larger EU member states has relatively favourable demographics.



Appendix 1

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